

Stochastic Analysis

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Chapter 1

Stochastic Processes in Continuous Time

1.1 Basic definitions

A *stochastic process* is a mathematical model for the occurrence, at each moment after some initial time, of a random phenomena. The randomness is captured by the introduction of a measurable space $(\Omega, \mathcal{F})$, the *sample space* on which a probability measure can be placed. A *stochastic process* is a family $X = (X_t)_{t \in I}$ of random variables on $(\Omega, \mathcal{F})$ taking values in measurable space $(S, \mathcal{S})$, the *state space*. Here, I a finite or infinite time interval, i.e. $I = [0, T]$, for some $T > 0$ or $I = [0, \infty)$. In this course, we shall only consider stochastic processes taking values in $\mathbb{R}$ or $\mathbb{R}^d$. That is, $X_t : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ or $X_t : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^d, \mathcal{B}^d)$ where $\mathcal{B}$ and $\mathcal{B}^d$ denote the Borel σ -fields on $\mathbb{R}$ and $\mathbb{R}^d$, respectively. The Borel σ -field on $[0, t]$ is denoted $\mathcal{B}_t$. If the index set is not relevant, we denote a stochastic process by (X_t) instead of $(X_t)_{t \in I}$. For each $\omega \in \Omega$ we call $X(\omega) := (X_t(\omega))_{t \in I}$ a *trajectory* or *sample path* of X . Stochastic processes can be equal in different ways, listed below.

Definition 1.1.1 Let $X = (X_t)_{t \in I}$ and $Y = (Y_t)_{t \in I}$ be stochastic processes defined on the same probability space.

- X and Y are called *indistinguishable* if

$$\mathbb{P}[X_t = Y_t \text{ for all } t \in I] = 1.$$

- X and Y are called *modifications* if

$$\mathbb{P}[X_t = Y_t] = 1 \quad \text{for all } t \in I.$$

The definitions of indistinguishable and modification automatically require the sets $\{\omega : X_t(\omega) = Y_t(\omega), t \in I\}$, respectively, $\{\omega : X_t(\omega) = Y_t(\omega)\}$ to be measurable, which does not need to be the case in general. Moreover, it does not make sense to ask whether X and Y are modifications or

indistinguishable unless they are defined on the same probability space. They do not have to be defined on the same space to have the same finite dimensional distributions.

Definition 1.1.2 *The stochastic processes X and Y have the same finite dimensional distributions if*

$$\mathbb{P}[X_{t_i} \in B_i \text{ for all } i = 1, \dots, n] = \mathbb{P}[Y_{t_i} \in B_i \text{ for all } i = 1, \dots, n]$$

for all $0 \leq t_1 \leq \dots \leq t_n$, all $n \in \mathbb{N}$ and all $B_i \in \mathcal{B}$ ($i = 1, \dots, n$).

Proposition 1.1.3 *i) If X and Y are indistinguishable, then they are modifications.*

ii) If X and Y are modifications, then they have the same finite dimensional distributions.

PROOF:

i) For any $t \in I$,

$$\mathbb{P}[X_t = Y_t] \geq \mathbb{P}[X_t = Y_t \text{ for all } t \in I] = 1.$$

ii) Let $N_t = \{X_t \neq Y_t\}$. Then $\mathbb{P}(N_t) = 0$ for all $t \in I$. For all $B \in \mathcal{B}^d$,

$$\begin{aligned} \mathbb{P}[(X_{t_1}, \dots, X_{t_d}) \in B] &= \mathbb{P}[(X_{t_1}, \dots, X_{t_d}) \in B \setminus N_1 \cup \dots \cup N_d] \\ &= \mathbb{P}[(Y_{t_1}, \dots, Y_{t_d}) \in B \setminus N_1 \cup \dots \cup N_d] \\ &= \mathbb{P}[(Y_{t_1}, \dots, Y_{t_d}) \in B] \end{aligned}$$

□

Exercise 1 *Prove that the converse implications do not hold in general.*

Let us recall that a function f is called left-continuous if $\lim_{t_n \uparrow t} f(t_n) = f(t)$ and right-continuous if $\lim_{t_n \downarrow t} f(t_n) = f(t)$. The next result shows that the distinction between modifications and indistinguishability is not necessary if the trajectories of a stochastic process are either all left-continuous or all right-continuous.

Proposition 1.1.4 *If X and Y are modifications and all trajectories are left-continuous (or all trajectories are right-continuous), then X and Y are indistinguishable.*

PROOF: Under the assumptions on the trajectories,

$$A := \{X_t = Y_t \text{ for all } t \in I\} = \{X_t = Y_t \text{ for all } t \in I \cap \mathbb{Q}\}.$$

In particular, $\{X_t = Y_t \text{ for all } t \in I \cap \mathbb{Q}\} \in \mathcal{F}$ and

$$\begin{aligned} \mathbb{P}(A) &= 1 - \mathbb{P}(X_t \neq Y_t \text{ for some } t \in I \cap \mathbb{Q}) \\ &\geq 1 - \sum_{t \in I \cap \mathbb{Q}} \mathbb{P}(X_t \neq Y_t) \\ &= 1. \end{aligned}$$

□

Strictly speaking, a stochastic process is a function of a pair of variables (ω, t) . It is thus reasonable to ask for some joint measurability properties.

Definition 1.1.5 A stochastic process X defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called measurable, if the mapping $(t, \omega) \mapsto X_t(\omega)$ from $(I \times \Omega, \mathcal{B} \otimes \mathcal{F})$ to $(\mathbb{R}, \mathcal{B})$ is measurable.

It is an immediate consequence of Fubini's theorem that the trajectories of a measurable stochastic process are Borel-measurable functions of time and that the same holds for the function $t \mapsto \mathbb{E}X_t$ provided the components of X have defined expectations.

In order to model the flow of information, we introduce the notion of a filtration.

Definition 1.1.6 A family of σ -fields $\mathcal{F}_s \subset \mathcal{F}_t \subset \mathcal{F}$ ($s \leq t$) on $(\Omega, \mathcal{F})$ is called a filtration. We set $\mathcal{F}_\infty := \sigma(\cup_{t \geq 0} \mathcal{F}_t)$. A probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ equipped with a filtration is called a filtered probability space or stochastic basis.

The concept of measurability of a stochastic process introduced above is a weak one. The use of filtrations opens up the possibility of introducing more advanced concepts of measurability.

Definition 1.1.7 • A stochastic process X defined on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ is called

- progressively measurable with respect to $(\mathcal{F}_t)$ if for all $t \in I$ the mapping $(\omega, s) \mapsto X_s(\omega)$ from $(\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t)$ to $(\mathbb{R}, \mathcal{B})$ is measurable;
- adapted to the filtration $(\mathcal{F}_t)$ if the mapping $\omega \mapsto X_t(\omega)$ from $(\Omega, \mathcal{F}_t)$ to $(\mathbb{R}, \mathcal{B})$ is measurable for all $t \in I$.
- The progressive σ -field on $\Omega \times [0, \infty)$ generated by a filtration $(\mathcal{F}_t)$ is the smallest σ -field such that all progressive processes (considered as mappings on $\Omega \times [0, \infty)$) are measurable.
- The predictable σ -field on $\Omega \times [0, \infty)$ generated by a filtration $(\mathcal{F}_t)$ is the smallest σ -field such that all left-continuous adapted processes (considered as mappings on $\Omega \times [0, \infty)$) are measurable.

Exercise 2 (*)

i) Show that the progressive σ -field is the σ -field on $\Omega \times [0, \infty)$ is given by the set

$$\{A \subseteq \Omega \times [0, \infty) : A \cap (\Omega \times [0, t]) \in \mathcal{F}_t \otimes \mathcal{B}_t\}.$$

In other words, show that a process is progressively measurable if and only if it is measurable w.r.t. the progressive σ -field.

(ii) Show that the σ -fields generated by the following sets of functions coincide:

- a) the set of adapted left-continuous functions
- b) the set of all elementary functions, that is functions of the form

$$X_t(\omega) = e_{-1}(\omega) + \sum_{j=0}^n e_j(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

where $0 = t_0 < t_1 < \dots < t_n$ and e_j is $\mathcal{F}_{t_j}$ -measurable and e_{-1} is $\mathcal{F}_0$ -measurable and bounded.

c) the space of predictable processes of the form: for $A \in \mathcal{F}_s$ with $s > 0$, respectively, $A \in \mathcal{F}_0$

$$X_r(\omega) = \mathbf{1}_{A \times (s, t]}(\omega, r), \quad X_r(\omega) = \mathbf{1}_{A \times 0}(\omega, r)$$

d) the space of continuous functions.

If we endow the set $\Omega \times [0, T]$ with the progressive σ -field $\mathcal{P}$, then a real-valued stochastic process X is $\mathcal{P}$ - $\mathcal{B}$ -measurable if and only if it is progressively measurable in the sense of the above definition. Progressive measurability guarantees that time-integrals of stochastic processes are measurable and, as we will see later, suitably stopped processes have good measurability properties as well.

Exercise 3 let $X = (X_t)_{t \geq 0}$ be a real-valued progressively measurable process such that for every $t \geq 0$,

$$\int_0^t |X_s(\omega)| ds < \infty \quad \text{for all } \omega \in \Omega$$

or $X \geq 0$. Prove that the process

$$Y_t(\omega) := \int_0^t X_s(\omega) ds$$

is progressively measurable. Hint: Use Tonelli's (Fubini's) theorem.

Let us now recall the following result from Analysis III. If $X_n : (\tilde{\Omega}, \tilde{\mathcal{F}}) \rightarrow (E, \mathcal{E})$ where (E, d) is a metric space, equipped with its Borel- σ -algebra $\mathcal{E}$, is measurable for all $n \in \mathbb{N}$ and

$$X_n(\omega) \rightarrow X(\omega)$$

for all $\omega \in \tilde{\Omega}$, then X is measurable as well. Indeed, let $A \subseteq E$ be closed and $A_n := \{x \in E : d(x, A) < 1/n\}$. Then A_n is open and

$$X^{-1}(A) = \bigcap_{n \in \mathbb{N}} \bigcup_{k=1}^{\infty} \bigcap_{m=k}^{\infty} X_m^{-1}(A_n) \in \tilde{\mathcal{F}}.$$

We notice that this result does not hold in general if E is not a metric space.

Lemma 1.1.8 Let X be a progressively measurable stochastic process. Then X is measurable and adapted.

PROOF: In order to show measurability, define for any $t \in I$ the process

$$X^t : (\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t) \rightarrow (\mathbb{R}, \mathcal{B}), \quad X^t(\omega, s) := X(\omega, s).$$

Then X^t is measurable by assumption. From this we will conclude that the process

$$\tilde{X}^t(s, \omega) := X_{s \wedge t}(\omega) : (\Omega \times [0, \infty), \mathcal{F}_t \otimes \mathcal{B}_t) \rightarrow (\mathbb{R}, \mathcal{B})$$

is measurable as well and hence also measurable w.r.t $\mathcal{F} \otimes \mathcal{B}$. Since

$$X_s(\omega) = \lim_{t \rightarrow \infty} X_{s \wedge t}(\omega)$$

we conclude that X is $\mathcal{F} \otimes \mathcal{B}$ -measurable as the limit of measurable functions. Using Fubini's theorem this also shows that $X_t : (\Omega, \mathcal{F}_t) \rightarrow (\mathbb{R}, \mathcal{B})$ is measurable, and hence that X is adapted. $\square$

Exercise 4 Show that a measurable process does not have to be progressively measurable and that an adapted process does not need to be measurable.

We now state a partial inverse of the above result without proof.

Proposition 1.1.9 *If X is adapted and measurable, then it has a progressively measurable modification.*

The preceding result will not be used in this course. We are only interested in stochastic processes that have continuous sample paths, for which a stronger result can easily be established.

Proposition 1.1.10 *If X is adapted and all trajectories are right-continuous (or all trajectories are left-continuous), then X is progressively measurable.*

PROOF: Let us assume that all trajectories are right-continuous. We need to prove that the mapping $(\omega, s) \mapsto X_s(\omega)$ from $(\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t)$ to $(\mathbb{R}, \mathcal{B})$ is measurable. To this end, we define (left-continuous!) processes

$$X_s^n(\omega) := X_{\frac{k}{2^n}t}(\omega), \quad \frac{k-1}{2^n}t < s \leq \frac{k}{2^n}t, \quad k = 1, \dots, 2^n.$$

Since X is adapted, this defines measurable mappings

$$F^n : (\Omega \times [0, t], \mathcal{F}_t \otimes \mathcal{B}_t) \rightarrow (\mathbb{R}, \mathcal{B})$$

whose limit as $n \rightarrow \infty$ is X , restricted to $(\Omega \times [0, t])$, due to the right-continuity of the sample paths. Hence, X is progressively measurable. $\square$

Definition 1.1.11 *A stochastic process X is called continuous, if all trajectories are continuous. It is called càdlàg - which stands for “continue à droite, limites à gauche” - (respectively, càglàd) if all trajectories are right (left)-continuous with left (right) limits.*

Exercise 5 (*) *Prove that a càdlàg function $f : [0, \infty) \rightarrow \mathbb{R}$ has at most countably many jumps.*

The most important example of a continuous stochastic process for us is Brownian motion.

Definition 1.1.12 *A stochastic process B on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called (standard) Brownian motion (BM) if $B_0 = 0$ and B has*

- *independent increments: for all $0 \leq s_1 \leq s_2 \leq \dots \leq s_n \leq s < t < \infty$,*

$$B_t - B_s \perp (B_{s_1}, \dots, B_{s_n})$$

- *stationary increments: for all $0 \leq s < t < \infty$,*

$$B_t - B_s \sim N(0, t - s)$$

- *almost surely continuous sample paths*

The standard example of a càdlàg process is the Poisson process.

Definition 1.1.13 *A stochastic process N on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called Poisson process with parameter $\lambda > 0$ if*

- $N_0 = 0$
- N is a càdlàg process

- N has independent increments: for all $0 \leq s_1 \leq s_2 \leq \dots \leq s_n \leq s < t < \infty$,

$$N_t - N_s \perp (N_{s_1}, \dots, N_{s_n})$$

- N has Poisson distributed increments: for all $0 \leq s < t < \infty$,

$$\mathbb{P}(N_t - N_s = k) = \frac{(\lambda(t-s))^k}{k!} e^{-\lambda(t-s)}$$

More generally, a stochastically continuous process X is called a Lévy process if $X_0 = 0$ and if X is a càdlàg process with stationary independent increments.

Our focus will be on Brownian motion and integrals with respect to Brownian motions. Brownian motion is continuous by definition; it will turn out that the stochastic integrals can be chosen to be continuous. Hence almost all processes considered in this course will be continuous.

1.2 σ -fields and stopping times

Let $(\Omega, \mathcal{G}, (\mathcal{G}_t), \mathbb{P})$ be a filtered probability space and let

$$\mathcal{N} := \{A \subset \Omega : \text{there exists } B \in \mathcal{G}, A \subset B, \mathbb{P}(B) = 0\}$$

be the set of all *null sets*. The *completion* of $\mathcal{G}$ is given by $\tilde{\mathcal{G}} := \mathcal{G} \vee \mathcal{N}$. We can extend the probability measure $\mathbb{P}$ on $(\Omega, \mathcal{G})$ to a probability measure $\tilde{\mathbb{P}}$ on $(\Omega, \tilde{\mathcal{G}})$ by¹

$$\tilde{\mathbb{P}}(A) = \mathbb{P}(B) \quad \text{for } A \in \mathcal{G} \text{ such that } A \Delta B \in \mathcal{N}.$$

The probability space $(\Omega, \tilde{\mathcal{G}}, \tilde{\mathbb{P}})$ is called the *completion* of $(\Omega, \mathcal{G}, \mathbb{P})$. A filtration $(\mathcal{G}_t)$ is called *complete* if $\mathcal{N} \in \mathcal{G}_0$ and hence $\mathcal{N} \in \mathcal{G}_t$ for all $t \geq 0$. We call $(\mathcal{G}_t)$ *right-continuous* if

$$\mathcal{G}_t = \bigcap_{s>t} \mathcal{G}_s =: \mathcal{G}_{t+}.$$

Definition 1.2.1 We say that a filtration $(\mathcal{G}_t)$ satisfies the usual conditions if it is complete and right-continuous.

The usual conditions are almost always assumed as they provide additional regularity. For instance, we shall see that martingales w.r.t. filtrations that satisfy the usual conditions have right continuous modifications if (and only if) their expected values are right-continuous in the time variable. They also guarantee that first entrance times are stopping times and that optional and stopping times coincide (see below).

Definition 1.2.2 For a stochastic process X we denote by $\tilde{\mathcal{F}}_t^X := \sigma(X_s, 0 \leq s \leq t)$ the filtration generated by X and put

$$\tilde{\mathcal{F}}_\infty^X := \sigma(X_t : t \in I).$$

The completion of $(\tilde{\mathcal{F}}_t^X)$ is defined by $\mathcal{F}_t^X := \tilde{\mathcal{F}}_t^X \vee \mathcal{N}$ for all $t \in [0, \infty]$.

¹For two sets A and B the symmetric difference $A \Delta B$ is defined by $A \Delta B = (A \setminus B) \cup (B \setminus A)$.

Exercise 6 Give an example that shows that $(\tilde{\mathcal{F}}_t^X)$ is not right-continuous in general (we will see that the completion of the Brownian filtration is right-continuous).

In what follows we put $\bar{I} = [0, \infty]$ if $I = [0, \infty)$ and $[0, T]$ else.

Definition 1.2.3 Let $(\mathcal{F}_t)$ be a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$.

- The mapping $\tau : \Omega \rightarrow \bar{I}$ is called a stopping time if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \in I$. We define

$$\mathcal{F}_\tau := \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for all } t \in I\}$$

- The mapping $\tau : \Omega \rightarrow \bar{I}$ is called an optional time if $\{\tau < t\} \in \mathcal{F}_t$ for all $t \in I$. We define

$$\mathcal{F}_{\tau+} := \{A \in \mathcal{F} : A \cap \{\tau < t\} \in \mathcal{F}_t \text{ for all } t \in I\}$$

Exercise 7 i) Show that every stopping is an optional time and that the two concepts coincide if the filtration is right-continuous.

ii) Show that for any stopping time τ the set $\mathcal{F}_\tau$ is a σ -field.

iii) Show that for any stopping time τ the set $\{\tau \leq s\}$ belongs to $\mathcal{F}_\tau$ for all $s \in I$; in particular, τ is $\mathcal{F}_\tau$ -measurable.

Lemma 1.2.4 Let $I = [0, \infty)$ and let $X : \Omega \rightarrow \mathbb{R}$ be right-(or left-)continuous and adapted. Let $\Gamma \subset \mathbb{R}$ be non-empty, and consider the first entrance time

$$\tau_\Gamma := \inf\{t \geq 0 : X_t \in \Gamma\} \quad (\inf \emptyset := \infty).$$

- If Γ is open, then τ_Γ is an optional time.
- If X is continuous and Γ is closed, then τ is a stopping time.

PROOF: Let Γ be open and X be right- (or left-) continuous. Then, by the right/left-continuity of X

$$\{\tau_\Gamma < t\} = \bigcup_{0 \leq s < t} \{X_s \in \Gamma\} = \bigcup_{s \in [0, t) \cap \mathbb{Q}} \{X_s \in \Gamma\},$$

the latter being a countable union. Since X is adapted and Γ is measurable, $\{X_s \in \Gamma\} \in \mathcal{F}_s \subseteq \mathcal{F}_t$. As a result, $\{\tau_\Gamma < t\} \in \mathcal{F}_t$.

Let us now assume that Γ is closed set $d(x, \Gamma) := \inf\{|x - y| : y \in \Gamma\}$. Then the mapping $x \mapsto d(x, \Gamma)$ is continuous, $\Gamma = \{x : d(x, \Gamma) = 0\}$ and $s \mapsto d(X_s(\omega), \Gamma)$ is continuous. Hence

$$\begin{aligned} \{\tau_\Gamma \leq t\} &= \{\omega \in \Omega : \inf_{s \leq t} d(X_s(\omega), \Gamma) = 0\} \\ &= \{\omega \in \Omega : \inf_{s \in \mathbb{Q}, s \leq t} d(X_s(\omega), \Gamma) = 0\} \\ &= \bigcap_{n \in \mathbb{N}} \bigcup_{s \in \mathbb{Q}, s \leq t} \{\omega : d(X_s(\omega), \Gamma) < \frac{1}{n}\}. \end{aligned}$$

By the above

$$\bigcup_{s \in \mathbb{Q}, s \leq t} \{\omega : d(X_s(\omega), \Gamma) < \frac{1}{n}\} \in \mathcal{F}_t.$$

This shows that $\{\tau_\Gamma \leq t\} \in \mathcal{F}_t$. □

If Γ is open and X is right-continuous, then we know that the process spends a non-trivial amount of time in Γ once it entered the set and we can approximate the process at any such time from the right. The situation is different if Γ is closed. In this case, let us assume that X is only right-continuous and only defined on $[0, t]$ for $t \notin \mathbb{Q}$. Then we can approximate $d(X_t(\omega), \Gamma)$ only from the left but not from the right; hence right-continuity of the function $s \mapsto d(X_s(\omega), \Gamma)$ is of little use here. Now, if the function $s \mapsto d(X_s(\omega), \Gamma)$ jumps to zero in t , then

$$\inf_{s \leq t} d(X_s(\omega), \Gamma) = 0 \neq \inf_{s \in \mathbb{Q}, s \leq t} d(X_s(\omega), \Gamma) = 0.$$

The following exercise shows how to overcome this problem.

Exercise 8 *Let $X : \Omega \rightarrow \mathbb{R}$ be right-(or left-)continuous and adapted. Prove that the first entrance time to an open set is a stopping time if the filtration satisfies the usual conditions. Give an example that shows that the first entrance time to an open set is not necessarily a stopping time if the filtration does not satisfy the usual conditions.*

The following properties are more or less obvious, except perhaps the last two that can be proved as an exercise.

Lemma 1.2.5 *i) If τ is an optional time and $\theta \in \mathbb{R}_+$, then $\tau + \theta$ is optional.*

ii) The sum, minimum and maximum of stopping times is a stopping time.

iii) The sum of optional times is an optional time.

iv) The infimum, supremum, limit inferior and limit superior of a sequence of optional times is an optional time.

v) The supremum of a sequence of stopping times is a stopping time.

vi) If T and S are stopping times and $A \in \mathcal{F}_S$, then $A \cap \{S \leq T\} \in \mathcal{F}_T$. In particular, if $S \leq T$, then $\mathcal{F}_S \subset \mathcal{F}_T$.

vii) If T and S are stopping times, then $\mathcal{F}_{T \wedge S} = \mathcal{F}_T \cap \mathcal{F}_S$ and each of the events $\{T < S\}$, $\{S < T\}$, $\{T \leq S\}$, $\{S \leq T\}$ and $\{T = S\}$ belongs to $\mathcal{F}_T \cap \mathcal{F}_S$.

The following result shows the usefulness of progressive measurability.

Proposition 1.2.6 *Let X be a progressively measurable stochastic process and τ be a stopping time. Then the random variable*

$$X_\tau(\omega) := X_{\tau(\omega)}(\omega)$$

defined on $\{\tau < \infty\}$ (and only there!) is $\mathcal{F}_\tau$ -measurable. Moreover, the stopped process

$$X^\tau = (X_{\tau \wedge t})$$

is progressively measurable.

PROOF: We first need to show that for any $B \in \mathcal{B}$ and all $t \in I$

$$\{X_\tau \in B\} \cap \{\tau \leq t\} \in \mathcal{F}_t.$$

This event equals

$$\{X_{\tau \wedge t} \in B\} \cap \{\tau \leq t\} = \{X_t^\tau \in B\} \cap \{\tau \leq t\}.$$

Hence it is enough to prove progressive measurability of the stopped process. To this end, we notice that the mapping

$$(s, \omega) \mapsto X_s(\omega) \quad \text{from } ([0, t] \times \Omega, \mathcal{B}_t \otimes \mathcal{F}_t) \text{ to } (\mathbb{R}, \mathcal{B})$$

is measurable by assumption. The mapping

$$(s, \omega) \mapsto (\tau(\omega) \wedge s, \omega) \quad \text{from } ([0, t] \times \Omega, \mathcal{B}_t \otimes \mathcal{F}_t) \text{ to } ([0, t] \times \Omega, \mathcal{B}_t \otimes \mathcal{F}_t)$$

is measurable because τ is a stopping time. Hence, the mapping $(s, \omega) \mapsto X_{\tau(\omega) \wedge s}(\omega)$ is progressively measurable as the composition of suitably measurable mappings.

□

1.3 Martingales in continuous time

In this section we extend some well-known results from discrete-time martingale theory to continuous time. Throughout, we fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$.

Definition 1.3.1 A stochastic process $M = (M_t)$ is called a martingale (with respect to $(\mathcal{F}_t)$) if

- M is $(\mathcal{F}_t)$ -adapted
- $M_t \in L^1(\mathbb{P})$ for every $t \in I$
- $\mathbb{E}[M_t | \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$.

The process is called a sub- respectively, super martingale if the equality in the third item is replaced by $\geq$, respectively, $\leq$.

From the theory of conditional expectation it is easy to see that an integrable, $(\mathcal{F}_t)$ -adapted process M is a martingale if and only if $\int_A M_t d\mathbb{P} = \int_A M_s d\mathbb{P}$ for all $A \in \mathcal{F}_s$ and all $s \leq t$.

1.3.1 Fundamental inequalities and a first convergence result

This section establishes a series of fundamental inequalities for (sub)martingales in continuous time. The following results are well know from discrete-time martingale theory ($I = \mathbb{N}$).

Theorem 1.3.2 • (Doob's maximal inequality) If $(M_n, \mathcal{F}_n)_{n \in \mathbb{N}}$ is a non-negative sub-martingale (or a martingale), then

$$\lambda^p \mathbb{P}(\sup_{m \leq n} |M_m| \geq \lambda) \leq \mathbb{E}|M_n|^p.$$

If $M_n \in L^p$ for $p > 1$, then

$$\mathbb{E}[\sup_{m \leq n} |M_m|^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}|M_n|^p.$$

- (Martingale convergence) Let $(M_n, \mathcal{F}_n)_{n \in \mathbb{N}}$ be an L^p bounded (super)martingale, i.e.

$$\sup_n \mathbb{E}|M_n|^p < \infty.$$

Then there exists a random variable M_∞ such that

$$\mathbb{E}|M_\infty|^p < \infty \text{ and } M_t \rightarrow M_\infty \text{ a.s.}$$

If $p > 1$ we also have convergence in L^p (this is not true for $p = 1$).

- (Optional sampling) Let (M_n) be a uniformly integrable discrete time $(\mathcal{F}_n)$ -martingale, and let σ, τ be two stopping times. Then

$$\mathbb{E}(M_\tau | \mathcal{F}_\sigma) = M_{\tau \wedge \sigma} \quad \text{a.s.}$$

Analogous results hold for sub- and supermartingales.

We now show that the preceding results carry over to continuous time martingales under continuity conditions on the same paths.

Theorem 1.3.3 (Doob's maximal inequality) Let M be a right-continuous martingale (or a non-negative sub-martingale) and put

$$M_t^* := \sup_{0 \leq s \leq t} |M_s|.$$

Then, M_t^* is measurable and for any $\lambda \geq 0$ and $t \geq 0$

$$\lambda^p \mathbb{P}(M_t^* \geq \lambda) \leq \mathbb{E}|M_t|^p.$$

Furthermore, if $p > 1$, then

$$\mathbb{E}|M_t^*|^p \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}|M_t|^p.$$

PROOF: By right-continuity we can write

$$M_t^* = \sup_{s \in [0, t] \cap \mathbb{Q}} (|M_s| \vee |M_t|).$$

In particular, the running maximum is measurable as a countable supremum of measurable functions.

We put

$$S_n := \left\{ s_{nk} := \frac{t}{2^n} k : 0 \leq k \leq 2^n \right\}.$$

Then $t \in S_n$ and $u \leq t$ for all $u \in S_n$. We consider the discrete-time process

$$M_v^{n,*} := \sup_{u \in S_n, u \leq v} |M_u|, \quad v \in S_n.$$

By the discrete time version of Doob's maximal inequality,

$$\lambda^p \mathbb{P}(M_t^{n,*} \geq \lambda) \leq \mathbb{E}|M_t^n|^p = \mathbb{E}|M_t|^p$$

and

$$\mathbb{E}|M_t^{n,*}|^p \leq \left(\frac{p}{p-1} \right)^p \mathbb{E}|M_t|^p.$$

By right-continuity $M_t^{n,*} \uparrow M_t^*$ and the result follows from Fatou's lemma. $\square$

Corollary 1.3.4 *If (M_t) is a right-continuous (non-negative sub-) martingale. Then almost every sample path $M(\omega)$ is bounded on any bounded time interval.*

Next, we prove the optional sampling theorem for continuous martingales. In particular, stopped martingales are martingales.

Theorem 1.3.5 (Doob's Optional Sampling Theorem) *Let (M_t) be a uniformly integrable right-continuous martingale (e.g. M is L^p -bounded for $p > 1$). Let τ, σ be finite stopping times. Then*

$$\mathbb{E}[M_\tau | \mathcal{F}_\sigma] = M_{\tau \wedge \sigma} \quad \text{and} \quad \mathbb{E}[M_\tau | \mathcal{F}_{\sigma+}] = M_{\tau \wedge \sigma} \quad \text{a.s.}$$

In particular, the stopped process $M^\tau = (M_{t \wedge \tau})$ is a martingale. Similar results hold for sub-martingales.

PROOF: Let M be a martingale. The proof uses discrete time optional sampling along with the fact that M can be approximated by discrete time martingales, due to its right-continuity. For any $n \in \mathbb{N}$, let

$$S_n := \left\{ s_{n,k} := \frac{k}{2^n} : k \in \mathbb{N} \right\}.$$

Next, we introduce the discretised random times

$$\tau_n := \sum_{1 \leq k \leq n2^n} s_{n,k} \mathbf{1}_{(s_{n,k-1}, s_{n,k}]}(\tau) + \infty \cdot \mathbf{1}_{\{\tau > n\}}, \quad \sigma_n := \sum_{1 \leq k \leq n2^n} s_{n,k} \mathbf{1}_{(s_{n,k-1}, s_{n,k}]}(\sigma) + \infty \cdot \mathbf{1}_{\{\sigma > n\}}.$$

Then, for all $u \in S_n$,

$$\{\tau_n \leq u\} = \bigcup_{k: s_{n,k} \leq u} \{\tau \leq s_{n,k}\} \in \mathcal{F}_u.$$

Hence τ_n is an $(\mathcal{F}_u)_{u \in S_n}$ -stopping time and $\tau_n \downarrow \tau$ a.s. Analogous results hold for σ_n . Since (M_t) is uniformly integrable and right-continuous, we thus get from Fatou's lemma that

$$\begin{aligned} \mathbb{E}|M_{t \wedge \tau}| &= \mathbb{E} \left[\liminf_{n \rightarrow \infty} |M_{t \wedge \tau_n}| \right] \\ &\leq \liminf_{n \rightarrow \infty} \mathbb{E}|M_{t \wedge \tau_n}| \\ &\leq \mathbb{E}|M_t| < \infty. \end{aligned}$$

In particular, the stopped process is integrable. Using similar arguments as in the proof of Proposition 1.2.6 we can show that it is also adapted.

The martingale property of the stopped process thus follows from the main statement of the theorem, which we now prove. Since M is uniformly integrable, so is the discrete time martingale $\{(M_u, \mathcal{F}_u)\}_{u \in S_n}$ and hence it is closed by a random variable M_∞ . By discrete optional sampling

$$\int_A M_{\tau_n} d\mathbb{P} = \int_A M_{\tau_n \wedge \sigma_n} d\mathbb{P} \tag{1.1}$$

for all $A \in \mathcal{F}_{\sigma_n}$ and hence all $A \in \mathcal{F}_{\sigma+}$ (since $\sigma_n > \sigma$ on the set $\{\sigma < \infty\}$ implies $\mathcal{F}_{\sigma+} \subset \mathcal{F}_{\sigma_n}$). By right-continuity and the convergence of the stopping times to τ , respectively σ from above

$$M_{\tau_n \wedge \sigma_n} \rightarrow M_{\tau \wedge \sigma}, \quad M_{\tau_n} \rightarrow M_\tau \quad \text{a.s.}$$

Thus, if we were allowed to interchange the limit $n \rightarrow \infty$ and the integration on both sides of equation (1.1), the result would follow. By Vitali's convergence theorem interchanging limit and integration is possible if the integrands form families of uniformly integrable random variables, which we now show.

By the de la Vallée-Poussin theorem a family (g_t) of real-valued random variables is uniformly integrable if and only if there exists a nonnegative nondecreasing convex function $f : [0, \infty) \rightarrow [0, \infty)$ such that $\frac{f(t)}{t} \rightarrow \infty$ as $t \rightarrow \infty$ and

$$L := \sup_{t \in [0, \infty]} \mathbb{E}f(|g_t|) < \infty.$$

Since M is uniformly integrable by assumption, such a function exists for M . The convexity of f yields that $f(|M|)$ is a sub-martingale. Hence, by the discrete time stopping theorem

$$f(|M_{\tau_n \wedge N}|) \leq \mathbb{E}[f(|M_N|) | \mathcal{F}_{\tau_n \wedge N}]$$

for each $n, N \in \mathbb{N}$. Hence, using that $\mathbf{1}_{\{\tau_n \leq N\}} \in \mathcal{F}_{\tau_n \wedge N}$ we get for each $n, N \in \mathbb{N}$ that

$$\begin{aligned} \mathbb{E}[f(|M_{\tau_n}|) \mathbf{1}_{\{\tau_n \leq N\}}] &= \mathbb{E}[f(|M_{\tau_n \wedge N}|) \mathbf{1}_{\{\tau_n \leq N\}}] \\ &\leq \mathbb{E}[\mathbb{E}[f(|M_N|) | \mathcal{F}_{\tau_n \wedge N}] \mathbf{1}_{\{\tau_n \leq N\}}] \\ &\leq \mathbb{E}[f(|M_N|) \mathbf{1}_{\{\tau_n \leq N\}}] \\ &\leq \sup_t \mathbb{E}[f(|M_t|)] < \infty =: L \end{aligned}$$

where $L < \infty$ because M is uniformly integrable. Hence,

$$\mathbb{E}[f(|M_{\tau_n}|) \mathbf{1}_{\{\tau_n < \infty\}}] \leq L,$$

and, accounting for that fact that the event $\{\tau = \infty\}$ may have positive probability,

$$\mathbb{E}[f(|M_{\tau_n}|)] \leq 2L.$$

This implies uniform integrability of the family (M_{τ_n}) ; similar considerations apply to $(M_{\tau_n \wedge \sigma_n})$. $\square$

Remark 1.3.6 *We emphasize that the last step of the above proof used the fact that the discrete martingale (M_n) is closed by a random variable M_∞ so that $M_{\tau_n} = M_\infty$ on $\{\tau_n = \infty\}$. We did not assume/use that the continuous-time martingale is closed for which we have no proof yet; see also Exercise 12.*

Exercise 9 (*) *Suppose that (X_t) is a right-continuous sub-martingale and that σ, τ are two bounded stopping times with $\sigma \leq \tau$ a.s. Then X_τ and X_σ are integrable and*

$$\mathbb{E}[X_\tau | \mathcal{F}_\sigma] \geq X_\sigma.$$

The following result shows that the right continuity assumption of the filtration is in some sense without loss of generality.

Corollary 1.3.7 *A right continuous $(\mathcal{F}_t)$ -martingale M is also an $(\mathcal{F}_{t+})$ -martingale.*

PROOF: It is clear that M is $(\mathcal{F}_{t+})$ -adapted. To establish the martingale property, we apply the previous theorem to $0 \leq s < t < \infty$. $\square$

We close this section with the continuous time version of the martingale convergence theorem.

Theorem 1.3.8 (Martingale convergence) *Let M be a right-continuous L^p bounded (sub)martingale where $p > 1$. Then there exists a random variable M_∞ such that*

$$\mathbb{E}|M_\infty|^p < \infty \quad \text{and} \quad M_t \rightarrow M_\infty \text{ a.s. and in } L^p.$$

If $p = 1$, then almost sure convergence (but not convergence in L^1) holds if M is continuous.

PROOF: Since M is L^p -bounded, so is the discrete-time (sub)-martingale $\{(M_n, \mathcal{F}_n)\}_{n \in \mathbb{N}}$. If $p > 1$, the discrete martingale convergence theorem yields a random variable M_∞ such that

$$M_n \rightarrow M_\infty \quad \text{a.s. and in } L^p.$$

Moreover, by the continuous-time maximal inequality,

$$\begin{aligned} \mathbb{P}(\sup_{t \geq m} |M_t - M_m| > \lambda) &= \limsup_{l \rightarrow \infty, l \in \mathbb{N}} \mathbb{P}(\sup_{l \geq t \geq m} |M_t - M_m| > \lambda) \\ &\leq \lambda^{-p} \limsup_{l \rightarrow \infty, l \in \mathbb{N}} \mathbb{E}|M_l - M_m|^p \\ &\leq \varepsilon \end{aligned}$$

for all $m \in \mathbb{N}$ large enough using discrete-time L^p -convergence. This proves a.s. convergence of M_t to M_∞ as $t \rightarrow \infty$ as

$$\mathbb{P}(\sup_{t \geq m} |M_t - M_\infty| > \lambda) \leq \mathbb{P}(|M_m - M_\infty| > \lambda/2) + \mathbb{P}(\sup_{t \geq m} |M_t - M_m| > \lambda/2).$$

The proof of convergence in L^p is very similar.

Let us consider the case $p = 1$ and assume that M is continuous. Then

$$\tau_n := \inf\{t \geq 0 : |M_t| \geq n\}$$

is a stopping time by Lemma 1.2.4. Hence the stopped process M^{τ_n} is a bounded, hence L^p -bounded ($p > 1$) martingale (from optional sampling) and so it follows from the above that

$$M_t \rightarrow M_\infty \text{ a.s. on } \{\tau_n = \infty\}.$$

Hence we are finished if we can prove that

$$\mathbb{P}\left(\bigcup_n \{\tau_n = \infty\}\right) = 1.$$

To this end, let $b := \sup_{t \geq 0} \mathbb{E}|M_t|$. This quantity is finite by assumption. By Doob's maximal inequality

$$\mathbb{P}\left(\sup_{0 \leq t \leq T} |M_t| > \lambda\right) \leq \frac{b}{\lambda}.$$

By monotone convergence,

$$\mathbb{P}(\sup_{t \geq 0} |M_t| > \lambda) = \lim_{n \rightarrow \infty} \mathbb{P}(\sup_{0 \leq t \leq n} |M_t| > \lambda) \leq \frac{b}{\lambda}$$

and so

$$\mathbb{P}(\tau_n < \infty) \leq \mathbb{P}(\sup_{t \geq 0} |M_t| \geq n) \leq \frac{b}{n-1}.$$

Hence

$$\mathbb{P}(\tau_n = \infty) = 1 - \mathbb{P}(\tau_n < \infty) \rightarrow 1.$$

□

1.3.2 Upcrossing lemma and submartingale convergence

In what follows we establish a stronger convergence result for sub- and supermartingales. To this end, we first recall Doob's up-crossing lemma for discrete time super-martingales.

Lemma 1.3.9 (Discrete time up-crossing lemma) *Let $(M_t, \mathcal{F}_t)_{t \geq 0}$ be a super-martingale and let $F = \{0 \leq t_1 < t_2 < \dots < t_n\} \subset [0, \infty)$ be a finite set of time points. Let $U_F(\alpha, \beta, M)$ be the number of up-crossings of the interval $[\alpha, \beta]$ by the restricted path $(M_{t_i})_{i=1}^n$. Then*

$$\mathbb{E}[U_F(\alpha, \beta, M)] \leq \frac{\mathbb{E}(M_{t_n} - \alpha)^-}{\beta - \alpha} \leq \frac{\mathbb{E}(M_{t_n}^-) + \alpha^+}{\beta - \alpha}.$$

For the case of a sub-martingale

$$\mathbb{E}[U_F(\alpha, \beta, M)] \leq \frac{\mathbb{E}(M_{t_n} - \alpha)^+}{\beta - \alpha} \leq \frac{\mathbb{E}(M_{t_n}^+) + |\alpha|}{\beta - \alpha}.$$

For continuous time super-martingales we put for any $I \subset [0, \infty)$

$$U_I(\alpha, \beta, M) := \sup\{U_F(\alpha, \beta, M) : F \subset I \text{ finite}\}.$$

Proposition 1.3.10 (Continuous time up-crossing lemma) *Let $(M_t, \mathcal{F}_t)_{t \geq 0}$ be a super-martingale and let $I \subset [0, T]$. Then*

$$\mathbb{E}[U_{I \cap \mathbb{Q}}(\alpha, \beta, M)] \leq \sup_{t \in [0, T] \cap \mathbb{Q}} \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha}.$$

If M is right-continuous, then

$$\mathbb{E}[U_I(\alpha, \beta, M)] \leq \sup_{t \leq T} \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha}.$$

If M is a right-continuous submartingale, then

$$\mathbb{E}[U_I(\alpha, \beta, M)] \leq \sup_{t \leq T} \frac{\mathbb{E}(M_t - \alpha)^+}{\beta - \alpha} \leq \sup_{t \leq T} \frac{\mathbb{E}(M_t^+) + |\alpha|}{\beta - \alpha}.$$

PROOF: Let $\{F_n\}$ be an increasing sequence of finite sets whose union is $I \cap \mathbb{Q}$. Then each finite set $F \subset I \cap \mathbb{Q}$ belongs to some F_n . Hence

$$U_{I \cap \mathbb{Q}}(\alpha, \beta, M) = \sup_n U_{F_n}(\alpha, \beta, M).$$

By the discrete time up-crossing lemma we know that $U_{F_n}(\alpha, \beta, M)$ is measurable, hence $U_{I \cap \mathbb{Q}}(\alpha, \beta, M)$ is measurable. Since $\{U_{F_n}(\alpha, \beta, M)\}_{n \in \mathbb{N}}$ is an increasing sequence it follows from monotone convergence that

$$\mathbb{E}[U_{I \cap \mathbb{Q}}(\alpha, \beta, M)] = \sup_n \mathbb{E}[U_{F_n}(\alpha, \beta, M)].$$

Let s_n be the largest element in F_n . Since $(M_t - \alpha)^-$ is a sub-martingale² we get for any $t \in I \cap \mathbb{Q}$ with $t \geq s_n$ that (note that we do not assume that $s_n \in I \cap \mathbb{Q}$)

$$\mathbb{E}[U_{F_n}(\alpha, \beta, M)] \leq \frac{\mathbb{E}(M_{s_n} - \alpha)^-}{\beta - \alpha} \leq \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha} \leq \sup_{t \in I \cap \mathbb{Q}} \frac{\mathbb{E}(M_t - \alpha)^-}{\beta - \alpha}.$$

This proves the first assertion. The case of a right-continuous super/sub-martingale is left as an exercise. $\square$

Exercise 10 (*)

- a) Prove the up-crossing lemma for a right-continuous super-martingale.
- b) Conclude that almost every sample path $M(\omega)$ of a right-continuous super-martingale admits left-limits everywhere on $(0, \infty)$.

The following result is more or less immediate corollaries of the upcrossing lemma (for right-continuous) submartingales. The proof of the first result is left as an exercise.

Proposition 1.3.11 (Submartingale convergence) *Let $(M_t, \mathcal{F}_t)_{t \geq 0}$ be a right-continuous submartingale that satisfies $\sup_{t \geq 0} \mathbb{E}(M_t^+) < \infty$. Then there exists an integrable random variable M_∞ such that*

$$M_n \rightarrow M_\infty \quad \mathbb{P}\text{-a.s.}$$

Exercise 11 (*) Prove the above proposition.

Hint. Use upcrossing on the time interval $[0, n]$, then monotone convergence and show that $\sup_{t \geq 0} \mathbb{E}(M_t^+) < \infty$ is equivalent to the seemingly stronger condition $\sup_t \mathbb{E}|M_t| < \infty$.

The following two exercises are very useful; the results should be known from Stochastic II. The first exercise requires Vitali's theorem.

Exercise 12 (*) Prove that the following three conditions are equivalent for a non-negative right-continuous sub-martingale (M_t) .

- a) (M_t) is a uniformly integrable family of random variables
- b) (M_t) converges in L^1 as $t \rightarrow \infty$
- c) (M_t) converges a.s. to some $M_\infty \in L^1$ and $(M_t)_{0 \leq t \leq \infty}$ is a sub-martingale

Exercise 13 (*) Prove that the following four conditions are equivalent for a right-continuous martingale (M_t) .

- a) (M_t) is a uniformly integrable family of random variables

²If X is a sub-martingale, then X^+ is a sub-martingale. Hence this follows from the fact that $X^- = (-X)^+$.

- b) (M_t) converges in L^1 as $t \rightarrow \infty$
- c) (M_t) converges a.s. to some $M_\infty \in L^1$ and $(M_t)_{0 \leq t \leq \infty}$ is a martingale
- d) there exists $Y \in L^1$ such that $M_t = \mathbb{E}[Y|\mathcal{F}_t]$ a.s., for every $t \geq 0$.

Moreover, if d) holds, and M_∞ is the random variable from c), then $\mathbb{E}[Y|\mathcal{F}_\infty] = M_\infty$.

So far we have seen that right-continuity of (super-) martingales is a very useful property when it comes to convergence. The following section establishes sufficient conditions for the existence of right-continuous modifications.

1.3.3 Regularization and right-continuous modifications

We close this section with a useful result on the path regularities of (sub/super-)martingales. This result is the main reason why one typically assumes that a filtration satisfies the usual conditions.

Theorem 1.3.12 *Let M be an $(\mathcal{F}_t)$ -super-martingale and assume that $(\mathcal{F}_t)$ satisfies the usual conditions. Then M has a right continuous modification if and only if the function $t \mapsto \mathbb{E}M_t$ is right continuous. If the right continuous modification exists it is also a super-martingale and can be chosen to have left limits everywhere.*

The proof requires the following result. It shows that any super-martingale almost surely admits right and left limits everywhere.

Proposition 1.3.13 *Let M be a super-martingale. Then there exists a set $\Omega^* \in \mathcal{F}_\infty$ of measure one such that for all $\omega \in \Omega^*$ the following limits exist and are finite for all $t \in [0, \infty)$:*

$$\lim_{s \downarrow t, s \in \mathbb{Q}} M_s(\omega), \quad \lim_{s \uparrow t, s \in \mathbb{Q}} M_s(\omega).$$

PROOF: By the up-crossing lemma we know that the following event has measure zero for every $n \in \mathbb{N}$:

$$A_{\alpha, \beta}^n := \{\omega \in \Omega : U_{[0, n] \cap \mathbb{Q}}(\alpha, \beta, M(\omega)) = \infty\}.$$

The same is true for the event

$$A^n := \bigcup_{\alpha < \beta, \alpha, \beta \in \mathbb{Q}} A_{\alpha, \beta}^n.$$

This events contains an event that contains the event

$$\left\{ \omega \in \Omega : \liminf_{s \uparrow t, s \in \mathbb{Q}} M_s(\omega) < \limsup_{s \uparrow t, s \in \mathbb{Q}} M_s(\omega), \text{ for some } t \in [0, n] \right\}.$$

As a result, the left limit exists for all $t \geq 0$ and all $\omega \notin \bigcup_n A^n$. Similar arguments apply to the right limits.

We still need to prove that the limits are finite, i.e. that the sample paths are bounded on compact time intervals. To this end, let $t \in [0, \infty)$ and $n > t$. Let $\{Q_{n, m}\}_{m \in \mathbb{N}}$ be a sequence of finite sets of rational numbers that increase to $Q_n := [0, n] \cap \mathbb{Q}$, all containing 0 and n . Then $(M_s)_{s \in Q_{n, m}}$ is a discrete time super-martingale. Hence (assuming w.l.o.g. that it starts in 0),

$$\mathbb{P}\left(\max_{s \in Q_{n, m}} |M_s| > \lambda\right) \leq \frac{\mathbb{E}|M_n|}{\lambda}.$$

By the monotone convergence theorem for sets this shows that $\mathbb{P}(\sup_{s \in Q_n} |M_s| < \infty) = 1$. $\square$

For a given super-martingale M we now define $M_{t+}(\omega) := \lim_{s \downarrow t, s \in \mathbb{Q}} M_s(\omega)$ if this limit exists and is finite and zero otherwise. The process (M_{t+}) satisfies the following properties.

Lemma 1.3.14 *Let M be a super-martingale.*

- *The process (M_{t+}) is integrable and $\mathbb{E}[M_{t+}|\mathcal{F}_t] \leq M_t$ a.s. If the mapping $t \mapsto \mathbb{E}M_t$ is right-continuous, then this inequality is an equality.*
- *The process (M_{t+}) is an $(\mathcal{F}_{t+})$ -super-martingale and a martingale, if M is a martingale.*

Review: Backward martingales

The proof of the preceding lemma uses the following result on backward martingales that is known from Stochastic II.

Lemma 1.3.15 (Backwards martingale convergence) *Let (n_k) be a sequence of decreasing numbers and $(\mathcal{G}_{n_k})$ be a decreasing sequence of σ -fields and*

$$\mathcal{G}_\infty := \bigcap \mathcal{G}_{n_k}.$$

Let (X_{n_k}) be a $(\mathcal{G}_{n_k})$ -backwards super-martingale, that is

$$X_{n_{k+1}} \geq \mathbb{E}[X_{n_k}|\mathcal{G}_{n_{k+1}}].$$

Then the following holds:

- *The process (X_{n_k}) converges almost surely to a limit X_∞ .*
- *The process (X_{n_k}) is uniformly integrable if and only if it is L^1 bounded. In this case*

$$X_{n_k} \rightarrow X_\infty \text{ a.s. and in } L^1 \text{ and } X_\infty \geq \mathbb{E}[X_{n_0}|\mathcal{G}_\infty].$$

- *If X is a backwards martingale, then uniform integrability is automatically satisfied and*

$$X_\infty = \mathbb{E}[X_{n_0}|\mathcal{G}_\infty].$$

The uniform integrability of a backwards super-martingale (X_{n_k}) is equivalent to its L^1 -boundedness. Since

$$\mathbb{E}[|X_{n_k}|] = \mathbb{E}[X_{n_k}] + 2\mathbb{E}[X_{n_k}^-] \leq \mathbb{E}[X_{n_0}] + 2\mathbb{E}[X_{n_k}^-]$$

the latter follows if $\sup_k \mathbb{E}[X_{n_k}^-] < \infty$. To verify this condition, the following lemma is often useful.

Lemma 1.3.16 *Let $(M_t)_{t \geq 0}$ be a super-martingale with respect to a filtration $(\mathcal{F}_t)$. Then for every $0 \leq t \leq s$,*

$$\mathbb{E}[M_s^-] \leq \mathbb{E}[M_t^-].$$

PROOF: Fix $t \leq s$. For $K > 0$, we define the truncated variable

$$\phi_K(x) := (-x) \wedge K = \max(-x, -K) \cdot (-1)$$

so that $\phi_K(x) \uparrow x^-$ as $K \rightarrow \infty$. The mapping $x \mapsto \phi_K(x)$ is convex and of linear growth. Hence conditional expectations of $\phi_K(M_s)$ are well defined and the conditional Jensen inequality for convex functions gives

$$\mathbb{E}[\phi_K(M_s)] \leq \mathbb{E}[\phi_K(M_t)].$$

Since $\phi_K(M_t) \uparrow M_t^-$ pointwise as $K \rightarrow \infty$, and similarly $\phi_K(M_s) \uparrow M_s^-$ monotone convergence yields

$$\mathbb{E}[M_s^-] = \lim_{K \rightarrow \infty} \mathbb{E}[\phi_K(M_s)] \leq \lim_{K \rightarrow \infty} \mathbb{E}[\phi_K(M_t)] = \mathbb{E}[M_t^-].$$

□

The following corollary is key to the proof of Lemma 1.3.14.

Corollary 1.3.17 *Let M be a super-martingale and (q_n) be a decreasing sequence of real numbers that converge to t from above. Then the backwards super-martingale (M_{q_n}) is uniformly integrable.*

PROOF: It is enough to prove that $\sup_n \mathbb{E}[M_{q_n}^-] < \infty$. Since $q_n \geq t$ for all $n \in \mathbb{N}$ this follows from the previous lemma as

$$\mathbb{E}[M_{q_n}^-] \leq \mathbb{E}[M_t^-] < \infty.$$

□

Remaining Proofs.

Proof of Lemma 1.3.14. Let $q_n \downarrow t$ be a decreasing sequence of rational numbers. Then (M_{q_n}) is a backwards super-martingale. By Corollary 1.3.17 it is uniformly integrable. Hence the limit M_{t+} is integrable and $M_{q_n} \rightarrow M_{t+}$ in L^1 . Uniform integrability also allows to interchange limit and integration to obtain from $\mathbb{E}[M_{q_n}|\mathcal{F}_t] \leq M_t$ that

$$\mathbb{E}[M_{t+}|\mathcal{F}_t] \leq M_t, \quad \text{a.s.}$$

Moreover, L^1 convergence implies convergence of the expected values. Hence, $\mathbb{E}M_{q_n} \rightarrow \mathbb{E}M_{t+}$. Hence, if the mapping $t \mapsto \mathbb{E}M_t$ is right-continuous, then $\mathbb{E}M_{t+} = \lim_{n \rightarrow \infty} \mathbb{E}M_{q_n} = \mathbb{E}M_t$. Along with the previous inequality this implies that $\mathbb{E}[M_{t+}|\mathcal{F}_t] = M_t$ (cf. Exercise 14 below applied to $Y = M_t, X = M_{t+}$ and $\mathcal{G} = \mathcal{F}_t$).

To prove the second statement, let $s < t$ and let $\{q_n\}$ be a sequence of rational numbers decreasing to s . W.l.o.g. $q_n \leq t$. By the tower property of conditional expectations

$$\mathbb{E}[M_{t+}|\mathcal{F}_{q_n}] = \mathbb{E}[\mathbb{E}[M_{t+}|\mathcal{F}_t]|\mathcal{F}_{q_n}] \leq \mathbb{E}[M_t|\mathcal{F}_{q_n}] \leq M_{q_n}$$

with equality if M is a martingale. The right side converges to M_{s+} by definition; the left side converges to $\mathbb{E}[M_{t+}|\mathcal{F}_{s+}]$ by the backwards martingale convergence theorem. □

Exercise 14 *Let X, Y be integrable random variables on $(\Omega, \mathcal{F})$. Let $\mathcal{G} \subset \mathcal{F}$ be another σ -field. Let Y be $\mathcal{G}$ -measurable. Show that $\mathbb{E}[X|\mathcal{G}] \leq Y$ together with $\mathbb{E}X = \mathbb{E}Y$ implies $\mathbb{E}[X|\mathcal{G}] = Y$ a.s.*

We are now ready to prove the main result of this section.

PROOF OF THEOREM 1.3.12. By Proposition 1.3.13 and because the filtration is complete there exists a set $\Omega^* \in \mathcal{F}_0(!)$ of full measure on which the limits $M_{t\pm}$ exist for every $t \in [0, \infty)$. We set $\tilde{M}_t(\omega) = M_{t+}(\omega)$ on Ω^* and $\tilde{M}_t(\omega) = 0$ otherwise. Since M_{t+} is $\mathcal{F}_{t+}$ -measurable, $\tilde{M}_t$ is $\mathcal{F}_{t+}$ -measurable as well because the two variables differ only on a set of measure zero and the filtration is complete.

In view of the right-continuity of the filtration ($\mathcal{F}_t = \mathcal{F}_{t+}$), and it follows that $\tilde{M}_t$ is actually $\mathcal{F}_t$ -measurable because

$$\tilde{M}_t = M_{t+} = \mathbb{E}[M_{t+} | \mathcal{F}_{t+}] = \mathbb{E}[M_{t+} | \mathcal{F}_t].$$

Now, if the process of expected values is right-continuous, then Lemma 1.3.14 yields that $\tilde{M}_t = M_t$ a.s. for all $t \geq 0$. Hence $\tilde{M}$ is a modification of M .

To see the super-martingale property we take $s < t$, a set $A \in \mathcal{F}_s$ and a sequence of rational numbers $\{q_n\}$ that decreases to s ; w.l.o.g. $q_n < t$ for all $n \in \mathbb{N}$. From the super-martingale property of M and uniform integrability of the backwards super-martingale (M_{q_n}) we have that

$$\int_A \tilde{M}_t d\mathbb{P} = \int_A M_t d\mathbb{P} \leq \int_A M_{q_n} d\mathbb{P} \rightarrow \int_A \tilde{M}_s d\mathbb{P}$$

and so $\tilde{M}$ is a super-martingale. It follows from Exercise 10 that $\tilde{M}$ has left limits everywhere on a set of full measure. The converse implication is left as an exercise. $\square$

Exercise 15 *Prove missing converse implication in the proof of Theorem 1.3.12.*

Hint. Use backwards martingale convergence along with $|x| = 2(-x)\mathbf{1}_{\{-x>0\}} + x$.

Chapter 2

Brownian Motion

Gaussian processes form a class of stochastic processes used in several branches in pure and applied mathematics. The most important Gauss process is the Brownian motion; we recall the definition of Brownian motion.

Definition 2.0.1 *A stochastic process $B = (B_t)_{t \geq 0}$ defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called a (standard) Brownian motion, if*

i) $B_0 = 0$ a.s.

ii) for all $n \in \mathbb{N}$ and $0 \leq t_0 < t_1 < \dots < t_n$, we have

$$(B_{t_1} - B_{t_0}, \dots, B_{t_n} - B_{t_{n-1}}) \sim \mathcal{N}(0, \Sigma_{t_1, \dots, t_n}), \quad \Sigma_{t_1, \dots, t_n} := \begin{pmatrix} t_1 & & & 0 \\ & t_2 - t_1 & & \\ & & \ddots & \\ 0 & & & t_n - t_{n-1} \end{pmatrix}$$

i.e. the increments $\Delta B_{t_k} := B_{t_k} - B_{t_{k-1}}, k = 1, \dots, n$ are independent and $\mathcal{N}(0, t_k - t_{k-1})$ -distributed;

iii) the sample paths $t \mapsto B_t(\omega)$ are continuous $\mathbb{P}$ -almost surely.

The second property says that Brownian motion has stationary, normally distributed, independent increments. In what follows we provide three different constructions of Brownian motion.

2.1 Brownian motion as a Gauss process

Definition 2.1.1 *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A random variable $f : \Omega \rightarrow \mathbb{R}$ is called Gaussian if*

$$\mathbb{P}(f \in B) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_B e^{-\frac{1}{2\sigma^2}(x-m)^2} dx, \quad B \in \mathcal{B}$$

for some $m \in \mathbb{R}$ and $\sigma > 0$. A random vector

$$f = (f_1, \dots, f_n) : \Omega \rightarrow \mathbb{R}^n$$

is called Gaussian if

$$\langle f, a \rangle := \sum_{i=1}^n a_i f_i$$

is Gaussian for all $a = (a_1, \dots, a_n) \in \mathbb{R}^n$. We call

$$m = (m_1, \dots, m_n), \quad m_i := \mathbb{E}f_i$$

the mean vector and

$$\sigma = (\sigma_{ij}), \quad \sigma_{ij} := \mathbb{E}[(f_i - m_i)(f_j - m_j)]$$

the covariance (matrix). If $n = 1$, then we put $m = \mathbb{E}f$ and $\sigma^2 = \mathbb{E}[(f - m)^2]$.

We recall that probability measures are uniquely determined by their characteristic functions and that the characteristic function of a d -dimensional Gaussian random variable X with mean vector m and covariance matrix C is given by

$$\phi_X(\theta) = \exp\left(i\langle \theta, m \rangle - \frac{1}{2}\theta^t C \theta\right), \quad \theta \in \mathbb{R}^d.$$

Exercise 16 Show that if f, g are Gaussian random variables with the same mean vector and covariance matrix, then they have the same distributions.

Definition 2.1.2 A stochastic process X is called Gaussian if its finite dimensional distributions are Gaussian. The mean process and the covariance function are given by, respectively,

$$m_t := \mathbb{E}X_t, \quad \Gamma(s, t) := \mathbb{E}[(X_t - m_t)(X_s - m_s)].$$

If B is Brownian motion, then $m_t = 0$ by definition and for $s \leq t$, by the independence of the increments,

$$\begin{aligned} \Gamma(s, t) &= \mathbb{E}[B_s B_t] \\ &= \mathbb{E}[B_s(B_t - B_s)] + \mathbb{E}B_s^2 \\ &= \mathbb{E}[B_s]\mathbb{E}[B_t - B_s] + s = s. \end{aligned}$$

Hence

$$\Gamma(s, t) = s \wedge t.$$

We know how to extend a consistent family of finite-dimensional distributions to a stochastic process.

Theorem 2.1.3 (Daniel-Kolmogorov) Let S be a Polish space equipped with its Borel σ -field $\mathcal{S}$. Let

$$\{Q^J : J \subset I, |J| < \infty\}$$

be a consistent family of finite-dimensional distributions Q^J on $(S^J, \mathcal{S}^J)$. Then, there exists a unique probability measure $\mathbb{P}$ on $(S^I, \mathcal{S}^I)$ that has all the Q^J as its finite dimensional distributions.

Let $(S^I, \mathcal{S}^I, \mathbb{P})$ be a probability space as in the above theorem and $\pi = (\pi_t)_{t \in I}$ be the canonical process defined by $\pi_t(\omega) = \omega_t$ for all $t \in I$ where $\omega = (\omega_t)_{t \in I}$. Then the finite dimensional distributions of π are given by the family of probability measures $\{Q^J : J \subset I, |J| < \infty\}$. With this, we are now ready to prove the existence of Gauss processes and to show that a Gauss process is uniquely (in distribution) determined by its covariance function.

Theorem 2.1.4 (*Existence of Gaussian processes*) Let

$$\Gamma : I \times I \rightarrow \mathbb{R}$$

be positive, semi-definite and symmetric, i.e.

$$\sum_{i,j=1}^n \Gamma(t_i, t_j) a_i a_j \geq 0, \quad \Gamma(s, t) = \Gamma(t, s)$$

for all $t, s, t_1, \dots, t_n \in I$, $a_1, \dots, a_n \in \mathbb{R}$ and $n \in \mathbb{N}$. Then, there exists a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a Gaussian process X on it such that

$$\mathbb{E}X_t = 0, \quad \mathbb{E}X_s X_t = \Gamma(s, t).$$

PROOF: We construct a consistent family of probability measures with the desired properties. To this end, let γ_n be a standard Gauss measure on $\mathbb{R}^n$ and

$$C = (\Gamma(t_i, t_j))_{i,j=1}^n.$$

The matrix C is symmetric, positive semi-definite and by the Cholesky decomposition there exists a matrix A such that

$$C = AA^t.$$

Let $\mu_n := \gamma_n \circ A^{-1}$. Let e_i be the i th unit vector in $\mathbb{R}^n$. Since every linear combination of normal random variables is normal

$$\int \langle x, e_i \rangle \mu_n(dx) = \int \langle Ax, e_i \rangle \gamma_n(dx) = 0$$

and

$$\begin{aligned} \int \langle x, e_i \rangle \langle x, e_j \rangle \mu_n(dx) &= \int \langle Ax, e_i \rangle \langle Ax, e_j \rangle \gamma_n(dx) \\ &= \int \langle x, A^t e_i \rangle \langle x, A^t e_j \rangle \gamma_n(dx) \\ &= \langle A^t e_i, A^t e_j \rangle \\ &= \langle e_i, C e_j \rangle \end{aligned}$$

This defines a consistent family of Gaussian measures (exercise). Hence, there exists a unique probability measure ν on the measure space

$$(\Omega, \mathcal{F}) = (\mathbb{R}^I, \mathcal{B}^I)$$

such that the coordinate process $X_t(\omega) = \omega(t)$ has the desired properties. $\square$

Exercise 17 Prove that the family of measures defined in the previous proof is consistent.

Example 2.1.5 (*Brownian bridge*) Let $I = [0, 1]$. We want to construct a Gaussian process X that starts and ends in 0. One possibility is to choose a Brownian motion B and to define

$$X_t := B_t - tB_1, \quad t \in [0, 1];$$

we will see an alternative construction later. Then, for all $0 \leq t_1 \leq \dots \leq t_d < 1$,

$$(X_{t_1}, \dots, X_{t_d}) = A(B_{t_1}, \dots, B_{t_d}, B_1)^t, \quad A = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 & -t_1 \\ 0 & 1 & 0 & \dots & 0 & -t_2 \\ \dots & \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & \dots & 1 & -t_d \end{pmatrix}.$$

Hence, X is a Gaussian process; its covariance function can easily be computed: for each $0 \leq s \leq t \leq 1$

$$\begin{aligned} \mathbb{E}[X_s X_t] &= \mathbb{E}[(B_s - sB_1)(B_t - tB_1)] \\ &= \mathbb{E}B_s B_t - t\mathbb{E}B_s B_1 - s\mathbb{E}B_t B_1 + st\mathbb{E}B_1^2 \\ &= s(1 - t). \end{aligned}$$

Example 2.1.6 (Fractional Brownian motion) There exists a unique Gauss process X with the following covariance function:

$$\Gamma(s, t) = \frac{1}{2} (t^{2H} + s^{2H} - |t - s|^{2H})$$

where $H \in (0, 1)$ is the Hurst coefficient. The process is called fractional Brownian motion with Hurst coefficient H . The case $H = 1/2$ corresponds to Brownian motion. It is not entirely trivial to show that Γ satisfies the properties of a covariance function.

Up to now we have constructed stochastic processes with certain finite-dimensional distributions. In the case of Gaussian processes this can be done through the covariance structure. Unfortunately, we cannot guarantee that the probability measure on $(S^I, \mathcal{S}^I)$ constructed via Daniel-Kolmogorov is concentrated on the set of continuous functions. To the contrary, the set $C(I, \mathbb{R})$ of continuous real-valued functions on I is usually not an element of $\mathcal{S}^I$ as shown by the following lemma.

Lemma 2.1.7 • Let $(E, \mathcal{B})$ be a measurable space and I be a non-empty index set. For any set $B \in \mathcal{B}^I$ there exists a countable set $S \subset I$ such that for any pair $(x, y) \in E^I \times B$ the following implication holds:

$$x(t) = y(t) \text{ for all } t \in S \text{ implies } x \in B.$$

That is, any set of functions $B \in \mathcal{S}^I$ is determined by the values on a countable set $J \subset I$.

• Let E be a metric space that contains at least two elements and $\mathcal{B}$ be a σ -field on E . Then

$$C(\mathbb{R}_+, E) \notin \mathcal{B}^{\mathbb{R}_+}.$$

PROOF: To see the first assertion let us recall that the σ -field $\mathcal{B}^I$ is generated by the sets

$$\text{pr}_J^{-1}(B_0); \quad J \subset I \text{ finite}; \quad B_0 \in \mathcal{B}^J.$$

Let $\mathcal{B}^a$ be the class of all sets $B \subseteq E^I$ for which there exists a countable set $S \subseteq I$ such that, for any pair $x \in E^I$ and $y \in B$ the above implication holds. Then $\mathcal{B}^a$ is a σ -field. The properties of a σ -field are evident except, perhaps, stability w.r.t. complements, which can be seen as follows. Let $B \in \mathcal{B}^a$ and S be the corresponding countable index set. Let $x \in E^I$ and $y \in B^c$ with $x = y$ on S . Then $x \notin B$ as otherwise $y \in B$ (note that $y \in E^I$). Hence $x \in B^c$ and so $B^c \in \mathcal{B}^a$. Now, choosing $S = J$ we see that $\text{pr}_J^{-1}(B_0) \in \mathcal{B}^a$ for all $B_0 \in \mathcal{B}^J$. This shows that $\mathcal{B}^a = \mathcal{B}^I$.

As for the second assertion, let us assume to the contrary that $C(\mathbb{R}_+, E) \in \mathcal{B}^{\mathbb{R}^+}$ then there exists a countable set S such that any mapping x that coincides with $y \in C$ on S is continuous. Adding the set of rational numbers to S if necessary, we may assume that $S \subseteq \mathbb{R}_+$ dense and so y is uniquely determined by its values on S . Since E contains two points $e_1 \neq e_2$ we can now choose $t_0 \notin S$ such that

$$x(t_0) = e_1; \quad x(t) = e_2 \quad \text{for } t \neq t_0; \quad y \equiv e_2.$$

Then $x = y$ on S but $x \neq y$ on $\mathbb{R}_+$ a contradiction. $\square$

The following remark sheds additional light on the preceding result.

Remark 2.1.8 *Let E be a complete separable metric space equipped with the Borel- σ -field $\mathcal{E} = \mathcal{B}(E)$. If I is a countable set, then $\mathcal{E}^I = \mathcal{B}(E)^I = \mathcal{B}(E^I)$. The above lemma shows that this is not true for uncountable index sets. The intuitive reason is that σ -algebras are only stable w.r.t. countable but not uncountable operations.*

In view of the preceding lemma an additional argument is required when constructing continuous stochastic processes. The next theorem provides a sufficient condition that guarantees that a stochastic process has a continuous modification. In this case, we shall always work with this modification.

Theorem 2.1.9 (Kolmogorov-Centsov) *Let X be a stochastic process on $(\Omega, \mathcal{F}, \mathbb{P})$ such that*

$$\mathbb{E}[|X_t - X_s|^\alpha] \leq C|t - s|^{1+\beta}, \quad 0 \leq s \leq t \leq T$$

for some $\alpha, \beta, C > 0$. Then there exists a continuous modification $\tilde{X}$ that satisfies

$$\mathbb{P}\left(\sup_{0 < t-s < h(\omega)} \frac{|\tilde{X}_t(\omega) - \tilde{X}_s(\omega)|}{|t-s|^\gamma} \leq \delta\right) = 1$$

for some $\delta > 0$, where h is a positive random variable and $\gamma \in (0, \frac{\beta}{\alpha})$.

PROOF: Let us take $T = 1$. We establish “increasing degrees of continuity”.

- It follows from Tschebyscheff’s inequality and the moment condition on X that the mapping $t \mapsto X_t(\omega)$ is continuous in probability:

$$\mathbb{P}(|X_t - X_s| \geq \varepsilon) \leq \frac{\mathbb{E}|X_t - X_s|^\alpha}{\varepsilon^\alpha} \leq C\varepsilon^{-\alpha}|t - s|^{1+\beta}.$$

- Next, we establish uniform continuity on a set of full measure. To this end, we apply the above inequality with $t := \frac{k}{2^n}$, $s := \frac{k-1}{2^n}$ and $\varepsilon = 2^{-\gamma n}$. Then

$$\mathbb{P}\left(|X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\right) \leq C2^{-n(1+\beta-\alpha\gamma)}.$$

Thus,

$$\mathbb{P}\left(\max_{1 \leq k \leq 2^n} |X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\right) = \mathbb{P}\left(\bigcup_{k=1}^{2^n} |X_{\frac{k}{2^n}} - X_{\frac{k-1}{2^n}}| \geq 2^{-\gamma n}\right) \leq C2^{-n(\beta-\alpha\gamma)}$$

By the Borel-Cantelli lemma (recalled below) this yields the existence of a set Ω^* of full measure such that for all $\omega \in \Omega^*$,

$$\max_{1 \leq k \leq 2^n} |X_{\frac{k}{2^n}}(\omega) - X_{\frac{k-1}{2^n}}(\omega)| \leq 2^{-\gamma n}, \quad n \geq n^*(\omega) \quad (2.1)$$

where n^* is some integer-valued random variable. In order to show uniform continuity on Ω^* , we set

$$D := \bigcup_{i=1}^{\infty} D_n; \quad D_n := \left\{ \frac{k}{2^n}, k = 0, \dots, 2^n \right\}$$

and fix $\omega \in \Omega^*$, $n \geq n^*(\omega)$, $m > n$ and $0 < t - s < 2^{-n}$. We show below that for all $t, s \in D_m$

$$|X_t(\omega) - X_s(\omega)| \leq 2 \sum_{j=n+1}^m 2^{-\gamma j}. \quad (2.2)$$

For all $s, t \in D$ with $0 < t - s < h(\omega) := 2^{-n^*(\omega)}$, we select $n > n^*(\omega)$ such that

$$2^{-(n+1)} \leq t - s < 2^{-n}.$$

For any such s, t we get from (2.2) that

$$|X_t(\omega) - X_s(\omega)| \leq 2 \sum_{j=n+1}^{\infty} 2^{-\gamma j} \leq 22^{-\gamma(n+1)} \sum_{j=0}^{\infty} 2^{-\gamma j} \leq \delta |t - s|^{\gamma}, \quad \delta := \frac{2}{1 - 2^{-\gamma}}. \quad (2.3)$$

- We now define the continuous modification. For $\omega \notin \Omega^*$ we set $\tilde{X}_t(\omega) \equiv 0$. For $\omega \in \Omega^*$ and $t \in D$ we set $\tilde{X}_t(\omega) = X_t(\omega)$. For $\omega \in \Omega^*$ and $t \in D^c$ we choose a sequence $\{s_n\} \subset D$ that converges to t . By the uniform continuity of $X(\omega)$ on D , the sequence $\{X_{s_n}(\omega)\}$ has a limit $\tilde{X}_t(\omega)$ that depends on t but not on the choice of the sequence $\{s_n\}$. The process $\tilde{X}$ satisfies almost surely the path-wise Hölder continuity (2.3). Moreover, since $\{X_{s_n}\}$ converges in probability to X_t and a.s. to $\tilde{X}_t$, we conclude that $X_t = \tilde{X}_t$ a.s.
- It remains to establish (2.2). We proceed by induction. For $m = n + 1$ the assertion is correct by the choice of Ω^* . Let us therefore assume it is also correct for $m = n + 1, \dots, M - 1$ and let $s < t, s, t \in D_M$. Let

$$t' := \max\{u \in D_{M-1} : u \leq t\}; \quad s' := \max\{u \in D_{M-1} : u \geq s\}.$$

Then $s \leq s' \leq t' \leq t$, and $s' - s < 2^{-M}$, $t - t' < 2^{-M}$. By (2.1)

$$|X_{s'}(\omega) - X_s(\omega)| \leq 2^{-\gamma M}, \quad |X_{t'}(\omega) - X_t(\omega)| \leq 2^{-\gamma M}$$

and by the induction hypothesis, since $t', s' \in D_{M-1}$,

$$|X_{t'}(\omega) - X_{s'}(\omega)| \leq 2 \sum_{j=n+1}^{M-1} 2^{-\gamma j}.$$

□

The proof of the preceding theorem used the Borell-Cantelli-Lemma.

Lemma 2.1.10 (Borell-Cantelli) *Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of events defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.*

- (i) *If $\sum_n \mathbb{P}[A_n] < \infty$, only finitely many A_n occur: $\mathbb{P}[\text{only finitely many } A_n \text{ occur}] = 1$.*

(ii) If $\sum_n \mathbb{P}[A_n] = \infty$ and the events $A_1, A_2, \dots$ are independent, infinitely many A_n occur:

$$\mathbb{P}[\text{infinitely many } A_n \text{ occur}] = 1.$$

Since the covariance function $\Gamma(s, t) = s \wedge t$ is symmetric and positively semi-definite, we know that there exists a centered Gaussian process with this covariance function. This process satisfies the following properties, all derived entirely(!) from the covariance function:

- $\mathbb{E}X_t^2 = \Gamma(t, t) = t$ for all $t \geq 0$; in particular, $X_0 = 0$ a.s.
- the increments $X_t - X_s$ follow a normal distribution with mean zero and variance $\mathbb{E}(X_t - X_s)^2 = \mathbb{E}X_t^2 - 2\mathbb{E}X_tX_s + \mathbb{E}X_s^2 = t - 2s \wedge t + s = |t - s|$; hence, the increments have a stationary distribution.
- any two increments are uncorrelated: for any $0 \leq u < v \leq s < t$ we have $\mathbb{E}[(X_v - X_u)(X_t - X_s)] = v - v - u + u = 0$; likewise, any finite number of increments are uncorrelated, hence independent.
- for any $t_1 \leq t_2 \leq \dots \leq t_n \leq s \leq t$,

$$\begin{aligned} \mathbb{P}(X_{t_1} \in A_1, \dots, X_{t_n} \in A_n, X_t - X_s \in A) &= \mathbb{P}((X_{t_1}, X_{t_2} - X_{t_1}, \dots, X_{t_n} - X_{t_{n-1}}) \in C, X_t - X_s \in A) \\ &= \mathbb{P}((X_{t_1}, X_{t_2} - X_{t_1}, \dots, X_{t_n} - X_{t_{n-1}}) \in C) \mathbb{P}(X_t - X_s \in A) \\ &= \mathbb{P}(X_{t_1} \in A_1, \dots, X_{t_n} \in A_n) \mathbb{P}(X_t - X_s \in A) \end{aligned}$$

where $C := \{(y_1, \dots, y_n) \in \mathbb{R}^n : y_1 \in A_1, y_1 + y_2 \in A_2, \dots, y_1 + \dots + y_n \in A_n\}$; hence

$$X_t - X_s \perp\!\!\!\perp X_{t_1}, \dots, X_{t_n}.$$

Thus, the process X satisfies the desired properties, except for the continuity of the sample paths. This can be guaranteed by the Kolmogorov-Centsov theorem.

Corollary 2.1.11 (Existence of Brownian motion) *Let X be a continuous Gaussian process with mean zero and covariance function $\Gamma(s, t) = s \wedge t$. Then a continuous modification B exists. In particular, a standard Brownian motion exists.*

PROOF: Since $X_t - X_s \sim N(0, t - s)$ there exists a constant C_n such that

$$\mathbb{E}|X_t - X_s|^{2n} = C_n |t - s|^n.$$

Thus, by Kolmogorov-Centsov, for each $T > 0$ there exists a continuous modification B^T of X on $[0, T]$. We need to show that a continuous modification exists on $[0, \infty)$. To this end, let

$$\Omega_T := \{\omega : B_t^T(\omega) = X_t(\omega) \text{ for all } t \in [0, T] \cap \mathbb{Q}\}; \quad \tilde{\Omega} := \bigcap_{T=1}^{\infty} \Omega^T.$$

Then $\mathbb{P}(\tilde{\Omega}) = 1$ and for all $\omega \in \tilde{\Omega}$ and any $T_1 \leq T_2 \in \mathbb{N}$,

$$B_t^{T_1}(\omega) = B_t^{T_2}(\omega), \quad t \in [0, T_1] \cap \mathbb{Q}$$

and by continuity of B^{T_1} and B^{T_2} on $[0, T_1]$ we get $B_t^{T_1}(\omega) = B_t^{T_2}(\omega)$ for all $t \in [0, T_1]$. Thus, we can define $B_t(\omega)$ as the respective common values for $\omega \in \tilde{\Omega}$ and $B_t(\omega) = 0$ for $\omega \in \tilde{\Omega}^c$. $\square$

The Kolmogorov-Centsov theorem shows that Brownian sample paths are even Hölder continuous with exponent $0 \leq \alpha < \frac{1}{2}$.

Proposition 2.1.12 $0 \leq \alpha < \frac{1}{2}$, almost every trajectory of a Brownian motion belongs to C^α .

Exercise 18 Show that the processes $(B_t^2 - t)_{t \geq 0}$ and $\left(e^{\alpha B_t - \frac{\alpha^2}{2}t}\right)_{t \geq 0}$ ($\alpha \in \mathbb{R}$) are martingales. In particular, a Brownian motion admits an exponential moment.

2.2 The Wiener-Lévy construction

The sample space for the Brownian motion constructed in the previous section was $\mathbb{R}^{[0, \infty)}$ the space of all real-valued functions on $[0, \infty)$. In this section, we provide a second construction of Brownian motion by Fourier expansion of the trajectories. In this case, the sample space will be a space Ω that is rich enough to carry a countable collection of independent, standard normal random variables.

To simplify the notation we restrict our construction to the time interval $[0, 1]$ and consider the space $L^2[0, 1]$ equipped with the usual inner product $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$. We recall that $\{\phi^n\} \subset L^2$ is called a complete orthonormal system (CONS) if $\langle \phi^n, \phi^m \rangle = \delta_{n,m}$ and its span is dense in L^2 . In this case, any $f \in L^2$ admits the representation

$$f = \sum_{n=0}^{\infty} \langle f, \phi^n \rangle \phi^n,$$

that is, $f_m := \sum_{n=0}^m \langle f, \phi^n \rangle \phi^n \rightarrow f$ in L^2 . In particular, $\|f_m\|_2^2 = \sum_{n=0}^m \langle f, \phi^n \rangle^2 \rightarrow \|f\|_2^2 = \sum_{n=0}^{\infty} \langle f, \phi^n \rangle^2$. Moreover, the Parseval identity holds:

$$\begin{aligned} \langle f, g \rangle &= \frac{1}{4} (\|f + g\|^2 - \|f - g\|^2) \\ &= \frac{1}{4} \left(\sum_{n=0}^{\infty} \langle f + g, \phi^n \rangle^2 - \sum_{n=0}^{\infty} \langle f - g, \phi^n \rangle^2 \right) \\ &= \sum_{n=0}^{\infty} \langle f, \phi^n \rangle \langle g, \phi^n \rangle. \end{aligned}$$

We now show that one can use Fourier expansions of this type for the construction of Brownian motion. To this end, let $0 \leq s, t \leq 1$, $f = \mathbf{1}_{[0,t]}$ and $g = \mathbf{1}_{[0,s]}$. Then

$$s \wedge t = \langle f, g \rangle = \sum_{n=0}^{\infty} \langle f, \phi^n \rangle \langle g, \phi^n \rangle.$$

Let us now fix a sequence $\{Z_n\}$ of i.i.d. $N(0,1)$ distributed random variables and assume for the moment that the following definition makes sense:

$$X_t := \sum_{n=0}^{\infty} Z_n \langle \mathbf{1}_{[0,t]}, \phi^n \rangle = \sum_{n=0}^{\infty} Z_n \int_0^t \phi^n(x) dx.$$

Then, heuristically $\mathbb{E}X_t = 0$ and

$$\mathbb{E}[X_t X_s] = \sum_{n,m=0}^{\infty} \mathbb{E}[Z_n Z_m] \langle \mathbf{1}_{[0,t]}, \phi^n \rangle \langle \mathbf{1}_{[0,s]}, \phi^m \rangle = s \wedge t.$$

This suggests that X is a Brownian motion on $[0, 1]$.

We now make this construction rigorous. We first notice that we can choose *any* CONS for our construction. We use the *Haar basis*; it can be constructed by translations and scalings of the base function

$$H(t) = \begin{cases} 1, & 0 \leq t < \frac{1}{2} \\ -1, & \frac{1}{2} \leq t < 1 \\ 0, & \text{else} \end{cases}.$$

Putting $H_0(t) = \mathbf{1}_{[0,t]}$ and $H_1(t) = H(t)$, we can then define a sequence of functions $\{H_n\}$ via

$$H_n(t) = 2^{j/2} H(2^j t - k), \quad n = 2^j + k, \quad 0 \leq k < 2^j.$$

The function H_n is supported on the right-open interval $I_n = [k2^{-j}, (k+1)2^{-j})$ and has integral zero and norm one in the Hilbert space $L^2(\mathbb{R})$. We leave it as an exercise to show that this system forms an CONS of $L^2[0,1]$ (the set of indicator functions on dyadic intervals is dense; definition of integral through integrals of step functions).

The description of a Brownian motion X uses the functions $\int_0^t H_n(x) dx$ for which we now derive a convenient representation. Let us put

$$G(t) := 2 \int_0^t H(x) dx = \begin{cases} 2t, & 0 \leq t < \frac{1}{2} \\ 2(1-t), & \frac{1}{2} \leq t < 1 \\ 0, & \text{else} \end{cases}.$$

For $n \geq 0$, let $G_0(t) = t$, $G_1(t) = G(t)$, and

$$G_n(t) = G(2^j t - k), \quad n = 2^j + k, \quad 0 \leq k < 2^j.$$

Then, $0 \leq G_n \leq 1$ on $[0,1]$. Moreover,

$$\begin{aligned} \int_0^t H_n(x) dx &= \int_0^t 2^{j/2} H(2^j x - k) dx \\ &= \int_{-k}^{2^j t - k} H(s) 2^{j/2} 2^{-j} ds \quad (s = 2^j x - k) \\ &= \int_0^{2^j t - k} H(s) 2^{-j/2} ds \quad (H = 0 \text{ on } \mathbb{R}_-) \\ &= \lambda_n G_n(t), \quad \lambda_n := 2^{-\frac{j}{2}-1}. \end{aligned}$$

Remark 2.2.1 We notice that for $n = 2^j + k$ and $n' = 2^{j'} + k'$ if $k \neq k'$, then $G_n(t)G_{n'}(t) = 0$. This important result will be used repeatedly in what follows.

Before we state and prove the main result of this section, we establish the following useful estimate on large values of sequences of normal random variables.

Lemma 2.2.2 Let $\{Z_n\}$ be a sequence of i.i.d. $N(0,1)$ distributed random variables. Then there exists an a.s. finite random variable C such that almost surely, for all $n \geq 2$,

$$|Z_n| \leq C \sqrt{\log n}.$$

PROOF: For any $x \geq 1$,

$$\begin{aligned}\mathbb{P}(|Z_n| \geq x) &= 2 \int_x^\infty \frac{1}{\sqrt{2\pi}} \exp(-t^2/2) dt \\ &\leq \sqrt{\frac{2}{\pi}} \int_x^\infty t \exp(-t^2/2) dt \\ &= \sqrt{\frac{2}{\pi}} \left[-\exp(-t^2/2) \right]_x^\infty \\ &= \sqrt{\frac{2}{\pi}} \exp(-x^2/2).\end{aligned}$$

Hence, for all $\alpha > 1$ and $n \geq 2$,

$$\mathbb{P}(|Z_n| \geq \sqrt{2\alpha \log n}) \leq \sqrt{\frac{2}{\pi}} n^{-\alpha}.$$

In particular, $\sum_n \mathbb{P}(|Z_n| \geq \sqrt{2\alpha \log n}) < \infty$ and so it follows by the Borel-Cantelli lemma that

$$|Z_n| \leq \sqrt{2\alpha \log n}, \quad \text{for all } n \geq n(\omega).$$

In particular,

$$C(\omega) := \sup_{n \geq 2} \frac{|Z_n(\omega)|}{\sqrt{\log n}} < \infty \quad \text{a.s.}$$

□

Theorem 2.2.3 *Let $\{Z_n\}$ be a sequence of i.i.d. $N(0,1)$ distributed random variables. Then,*

$$X_t := \sum_{n=1}^{\infty} \lambda_n Z_n G_n(t), \quad t \in [0, 1]$$

converges uniformly on $[0, 1]$ almost surely and X is a Brownian motion.

PROOF: We first prove absolute uniform convergence of the series. This implies that X is a.s. continuous. Let $J, M \in \mathbb{N}$. For $M \geq 2^J$ and $t \in [0, 1]$ in view of the preceding lemma and using Remark 2.2.1 in the third step,

$$\begin{aligned}\sum_{n=M}^{\infty} \lambda_n |Z_n| G_n(t) &\leq C \sum_{n=M}^{\infty} \lambda_n \sqrt{\log n} G_n(t) \\ &\leq C \sum_{j=J}^{\infty} \sum_{k=0}^{2^j-1} 2^{-j/2-1} \sqrt{j+1} G_{2^j+k}(t) \quad (n \leq 2^{j+1}; \log n \leq \log(2)(j+1) \leq j+1) \\ &\leq C \sum_{j=J}^{\infty} 2^{-j/2-1} \sqrt{j+1} \quad (G \leq 1, \text{ only one of the } G\text{'s in the second sum is nonzero}) \\ &\rightarrow 0 \quad \text{as } J \rightarrow \infty, \text{ uniformly in } t \in [0, 1].\end{aligned}$$

Next, we prove that X has the correct covariance structure. To this end, we first notice that the above series forms a Cauchy sequence in L^2 . In fact, for any $m \leq n$, and $2^{j_0} \leq m$ using the same arguments as above (expanding the sum as a double sum, then using that only one of the G 's in the second sum is different from zero) it holds that

$$\mathbb{E} \left[\sum_{k=m+1}^n \lambda_k Z_k G_k(t) \right]^2 = \sum_{k=m+1}^n \lambda_k^2 G_k^2(t) \leq \sum_{j \geq j_0} 2^{-j-1} \rightarrow 0$$

for $j_0 \rightarrow \infty$. Since X_s exists as an L^2 limit,

$$\mathbb{E}[(X_t \pm X_s)^2] = \sum_{k=0}^{\infty} \lambda_k^2 (G_k(t) \pm G_k(s))^2$$

and by polarisation

$$\begin{aligned} \mathbb{E}[X_t X_s] &= \frac{1}{4} \mathbb{E}[(X_t + X_s)^2] - \frac{1}{4} \mathbb{E}[(X_t - X_s)^2] \\ &= \sum_{k=0}^{\infty} \lambda_k^2 G_k(t) G_k(s) \\ &= \sum_{k=0}^{\infty} \langle \mathbf{1}_{[0,t]}, H_k \rangle \langle \mathbf{1}_{[0,s]}, H_k \rangle \\ &= \langle \mathbf{1}_{[0,t]}, \mathbf{1}_{[0,s]} \rangle \quad (\text{Parseval identity}) \\ &= s \wedge t. \end{aligned}$$

It remains to prove that X is a centered Gaussian process. This is left as an exercise. $\square$

Exercise 19 *Prove that the process X is Gaussian with $\mu(t) = 0$ and $\text{cov}(X_s, X_t) = s \wedge t$.*

2.3 Weak convergence and the Wiener measure

So far we have established the existence of a probability measure $\mathbb{P}$ on $(\mathbb{R}^{[0,T]}, \mathcal{B}^{[0,T]})$ such that the canonical process has the distributional properties of a Brownian motion and that there exists a modification with a.s. continuous sample paths. It is thus natural to ask if we can work right away on the set

$$\Omega := C([0, \infty), \mathbb{R}) \subset \mathbb{R}^{[0, \infty)}$$

of all real-valued continuous functions on $[0, \infty)$. In other words, can we construct a measure on the set of continuous functions such that the canonical process

$$X_t : \Omega \rightarrow \mathbb{R}, \quad \omega \mapsto \omega(t), \quad t \geq 0$$

is a Brownian motion? To do this, we introduce on Ω the metric

$$d(f, g) := \sum_{n=1}^{\infty} \frac{1}{2^n} \max_{0 \leq t \leq n} (|f(t) - g(t)| \wedge 1) =: \sum_{n=1}^{\infty} \frac{1}{2^n} d_n(f, g).$$

Theorem 2.3.1 *The space (Ω, d) is a complete, separable metric space. The metric d induces the topology of uniform convergence on Ω .*

PROOF: Each d_n is a complete metric on $(C([0, n], \mathbb{R}), \|\cdot\|_{\infty})$. Let (f_N) be a Cauchy sequence in (Ω, d) . Then there exists a function g_n such that $d_n(f_N, g_n) \rightarrow 0$ as $N \rightarrow \infty$. For $x \leq n \leq m$ we have that $g_n(x) = g_m(x)$ and so there exists $g \in \Omega$ whose restriction to $[0, n]$ coincides with g_n , for each $n \in \mathbb{N}$. Then, $d(f_N, g) \rightarrow 0$. Hence Ω is complete. Moreover, by Weierstrass' theorem, the countable class of polynomials with rational coefficients is dense in $(C([0, n], \mathbb{R}), \|\cdot\|_{\infty})$, for each $n \in \mathbb{N}$ and is hence dense in (Ω, d) as well. $\square$

In view of the previous theorem, we can equip $C([0, \infty), \mathbb{R}) \subset \mathbb{R}^{[0, \infty)}$ with its Borel- σ -field $\mathcal{B}(C)$. The following exercise shows that $\mathcal{B}(C)$ is generated by the continuous functions as well as by the cylinder sets (hence projection mappings).

Exercise 20 (*) Let us consider the following σ -algebras on $C = C[0, 1]$:

- the Borel sigma-algebra $\mathcal{B}(C)$;
- the sigma-algebra $\sigma(C)$ generated by continuous functions $f : C[0, 1] \rightarrow \mathbb{R}$, i.e.

$$\sigma(C) := \sigma \left(\bigcup_{f \in C} \{f^{-1}(B), B \in \mathcal{B}(\mathbb{R})\} \right);$$

- the sigma-algebra $\sigma(\mathcal{Z})$ generated by the system $\mathcal{Z}$ of all cylinder sets

$$Z = \{x \in C : x(t_j) \in A_j, j = 1, \dots, n\},$$

with $n \in \mathbb{N}, 0 \leq t_1 < t_2 < \dots < t_n \leq 1$ and $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$.

Show that

$$\sigma(C) = \mathcal{B}(C) = \sigma(\mathcal{Z}).$$

In particular, the mapping $X : (\Omega, \mathcal{F}) \rightarrow (S, \mathcal{B}(S))$, $\omega \mapsto X(\omega)$ is measurable if and only if for all $t_i \in [0, 1]$ the coordinate evaluation mapping $\omega \mapsto X_{t_i}(\omega)$ is measurable.

Hint. To show that $\mathcal{B}(S) \subset \sigma(\mathcal{Z})$ show first that every closed ε -ball is in $\sigma(\mathcal{Z})$ and then use separability of the metric space S to conclude. To show that $\mathcal{B}(S) \subset \sigma(C(S))$ let $A \subset S$ be closed and find a function $h \in C(S)$ such that $A = \{h = 0\}$.

Let B be a Brownian motion on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Then there exists a measurable set $\bar{\Omega}$ of full measure such that $B(\bar{\omega}) \in C([0, \infty), \mathbb{R})$ for all $\bar{\omega} \in \bar{\Omega}$. Let $\bar{\mathcal{A}} := \mathcal{F}|_{\bar{\Omega}}$ and $\bar{\mathbb{P}} := \mathbb{P}|_{\bar{\Omega}}$. Then $B : \bar{\Omega} \rightarrow C([0, \infty), \mathbb{R})$ is $(\bar{\mathcal{A}}, \mathcal{B}(C))$ -measurable,

$$\bar{\mathbb{P}} := \bar{\mathbb{P}} \circ B^{-1}$$

is a measure on $C([0, \infty), \mathbb{R})$, and the canonical process X on $C([0, \infty), \mathbb{R})$ is a Brownian motion. Further details can be found in my Stochastic II lecture notes.

Definition 2.3.2 Let $\bar{\mathbb{P}}$ be the probability measure on $\Omega = C([0, \infty), \mathbb{R})$ w.r.t. which the canonical process is a Brownian motion. Then $\bar{\mathbb{P}}$ is called the Wiener measure. The triple $(\Omega, \mathcal{B}(C), \bar{\mathbb{P}})$ is called the Wiener space. This is the canonical model when working with Brownian motion.

In what follows we are going to show that there exists a sequence of measures $\{\mathbb{P}_n\}$ on $(C, \mathcal{B}(C))$ - namely the distributions of a sequence of rescaled and linearly interpolated random walks - that converges to the Wiener measure. This provides a third construction of Brownian motion.

2.3.1 Prohorov's theorem and tightness of probability measures

The goal of this section is to establish a general condition that guarantees that a family of probability measures on a Polish space is weakly relatively sequentially compact, i.e. has a weakly convergent subsequence. We are particularly interested in families of probability measures on the space $C([0, 1], \mathbb{R})$. This result will be used to provide an alternative proof to Donsker's invariance principle.

We recall that a sequence of probability measures $(\mathbb{P}_n)_{n \in \mathbb{N}}$ on a metric space $(E, \mathcal{E})$ converges weakly to some probability measure $\mathbb{P}$ on $(E, \mathcal{E})$, written $\mathbb{P}_n \Rightarrow \mathbb{P}$, if

$$\int_E g(x) \mathbb{P}_n(dx) \rightarrow \int_E g(x) \mathbb{P}(dx) \quad \text{for all bounded and continuous function } g : E \rightarrow \mathbb{R}.$$

Definition 2.3.3 Let $(E, \mathcal{B}(E))$ be a metric space with Borel σ -field.

- A family of probability measures $\mathcal{P}$ on $(E, \mathcal{B}(E))$ is called
 - a) relatively (sequentially) compact, if every $(\mathbb{P}_n) \subset \mathcal{P}$ contains a subsequence, which converges weakly to some probability measure $\mathbb{P}$ on $(E, \mathcal{B}(E))$, i.e. if there exists a subsequence such that

$$\int_E g(x) \mathbb{P}_{n_k}(dx) \longrightarrow \int_E g(x) \mathbb{P}(dx)$$

for all bounded and continuous $g : E \rightarrow \mathbb{R}$.

- b) tight, if for every $\varepsilon > 0$ there exists a compact set K such that $\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(K^c) < \varepsilon$.
- A family of random variables $\mathcal{X}$ taking values in $(E, \mathcal{B}(E))$ is called tight, if for every $\varepsilon > 0$ there exists a compact set $K \subset E$ such that $\sup_{X \in \mathcal{X}} \mathbb{P}(X \in K^c) < \varepsilon$.

Theorem 2.3.4 (Prohorov) A family of probability measures $\mathcal{P}$ on a Polish space (E, d) is tight, if and only if it is weakly relatively sequentially compact.

The proof of Prohorov's theorem is involved. The fact that weakly relative sequentially compact implies tightness is relatively easy and established by the following lemma.

Lemma 2.3.5 If a family of probability measures $\mathcal{P}$ on a Polish space $(E, \mathcal{B}(E))$ is weakly relatively sequentially compact, then it is tight.

PROOF: Let $\{x_n\} \subset E$ be dense and for any $n \in \mathbb{N}$, let

$$A_{n,N} := \bigcup_{i=1}^N B_{1/n}(x_i).$$

Then $A_{n,N} \uparrow E$ as $N \rightarrow \infty$ for any $n \in \mathbb{N}$. We claim that

$$\delta := \sup_n \inf_N \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(A_{n,N}^c) = 0.$$

In fact, there exists $n \in \mathbb{N}$ such that for every $N \in \mathbb{N}$ there exists $\mathbb{P}_N \in \mathcal{P}$ such that $\mathbb{P}_N(A_{n,N}^c) \geq \delta/2$. Let $(\mathbb{P}_{N_k})$ converge weakly to $\mathbb{P}$. By the Portmanteau theorem

$$\mathbb{P}(A_{n,N}^c) \geq \liminf_k \mathbb{P}_{N_k}(A_{n,N}^c) \geq \liminf_k \mathbb{P}_{N_k}(A_{n,N_k}^c) \geq \delta/2.$$

At the same time $A_{n,N}^c \downarrow \emptyset$ so $\mathbb{P}(A_{n,N}^c) \rightarrow 0$, hence $\delta = 0$.

Let us now fix $\varepsilon > 0$. We have just shown that for any $n \in \mathbb{N}$ we can choose $N'_n \in \mathbb{N}$ such that $\mathbb{P}(A_{n,N'_n}^c) < \frac{\varepsilon}{2^n}$ for any $\mathbb{P} \in \mathcal{P}$. By construction, the set

$$A := \bigcap_{n \in \mathbb{N}} A_{n,N'_n}$$

is totally bounded, hence relatively compact. Denoting by $\bar{A}$ the closure of A tightness now follows from

$$\mathbb{P}((\bar{A})^c) \leq \mathbb{P}(A^c) \leq \sum_{n=1}^{\infty} \mathbb{P}(A_{n,N'_n}^c) \leq \varepsilon, \quad \mathbb{P} \in \mathcal{P}.$$

□

The converse implication in Prohorov's theorem is much harder to prove. The proof for $E = \mathbb{R}$ is simple and given below. We are particularly interested in the case $E = C([0, T], \mathbb{R})$. This space is not locally compact hence we cannot apply the standard Riesz representation theorem (see below proof). But the space is Polish (complete, separable, metric space). For Polish spaces we first prove the result for compact spaces and then for general ones.

The case $E = \mathbb{R}$

If $E = \mathbb{R}$, then the proof of Prohorov's theorem is essentially a corollary of Helly's selection theorem.

Proposition 2.3.6 (Helly's selection theorem) *Let V be the class of all right-continuous, increasing and bounded function $F : \mathbb{R} \rightarrow \mathbb{R}$. Let $\{F_n\} \subset V$ be uniformly bounded. Then there exists $F \in V$ and a subsequence $\{F_{n_k}\}$ such that*

$$F_{n_k}(x) \rightarrow F(x) \quad \text{for all } x \in \mathbb{R} \text{ in which } F \text{ is continuous.}$$

PROOF: The proof is based on a diagonal argument. Since the set of rational numbers is countable, we may write $\mathbb{Q} = \{q_n\}$. The theorem of Bolzano-Weierstrass guarantees that the sequence $\{F_n(q_1)\}$ converges along a subsequence $\{n_k^1\}$. This sequence has a further subsequence $\{n_k^2\}$ such that $\{F_{n_k^2}(q_2)\}$ converges; etc. Iteratively, we obtain subsequences $\{n_k^1\} \supseteq \{n_k^2\} \supseteq \{n_k^3\} \supseteq \dots$ such that $\{F_{n_k^l}(q_l)\}$ converges for every $l \in \mathbb{N}$. Let $n_k := n_k^k$. Then $\{F_{n_k}(q)\}$ converges, for every $q \in \mathbb{Q}$. Let $\tilde{F}(q) := \lim_k F_{n_k}(q)$ and

$$F(x) := \inf\{\tilde{F}(q) : q \in \mathbb{Q}, q > x\}.$$

Since $\tilde{F}$ is increasing, F is right-continuous and increasing. If F is continuous in x , then for any $\varepsilon > 0$ we can choose, $q_1, q_2 \in \mathbb{Q}$ with $q_1 \leq x \leq q_2$ and

$$\tilde{F}(q_1) \geq F(x) - \varepsilon; \quad \tilde{F}(q_2) \leq F(x) + \varepsilon.$$

Thus, by construction,

$$\limsup_k F_{n_k}(x) \leq \lim_k F_{n_k}(q_2) = \tilde{F}(q_2) \leq F(x) + \varepsilon$$

and so $\limsup_k F_{n_k}(x) \leq F(x)$; likewise $\liminf_k F_{n_k}(x) \geq F(x)$. □

We emphasize that Helly's theorem states that the set V is sequentially compact; it does *not* state that the set of distribution functions is sequentially compact as we do not know the values of F at $\pm\infty$. This is precisely the reason why we need tightness of a family of probability measures to obtain the convergence of their distribution functions to a distribution function along a subsequence.

Corollary 2.3.7 *Any tight family of probability measures $\mathcal{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is weakly relatively sequentially compact.*

PROOF: Now, let $\mathcal{P}$ be a tight sequence of probability measures on $\mathbb{R}$ and $\{\mathbb{P}_n\} \subseteq \mathcal{P}$. Let F_n be the distribution function of $\mathbb{P}_n$ and let F be bounded, right-continuous, increasing function such that $F_{n_k}(x) \rightarrow F(x)$ for all $x \in \mathbb{R}$ in which F is continuous. We need to show that F is a distribution function, that is, that $F(-\infty) = 0$ and $F(+\infty) = 1$.

By the tightness, there exists for all $\varepsilon > 0$ a constant $K > 0$ such that $F_n(x) < \varepsilon$ for all $x < -K$ and all $n \in \mathbb{N}$. Let $x < -K$ be a point of continuity of F , e.g. a rational number. Then, $F_{n_k}(x) \rightarrow F(x)$ and hence

$$0 = \liminf_k F_{n_k}(-\infty) \geq \liminf_k F_{n_k}(x) - \varepsilon = F(x) - \varepsilon \geq F(-\infty) - \varepsilon \geq -\varepsilon.$$

Hence $F(-\infty) = 0$. Likewise, we obtain that $F(+\infty) \geq 1$. At the same time, by construction,

$$F(+\infty) \leq \liminf_n F_n(+\infty) = 1.$$

Thus, F is a distribution function and it follows from the Portemanteau theorem that $\mathbb{P}_n \rightarrow \mathbb{P}$. $\square$

Example 2.3.8 *This example illustrates that pointwise limits of sequences of distribution functions do not need to be distribution functions. In fact, let F be a distribution function and let*

$$F_n(x) := F(x + n); \quad G_n(x) := F(x - n).$$

Then, $F_n \rightarrow \tilde{F} \equiv 1$ and $G_n \rightarrow \tilde{G} \equiv 0$ pointwise, which are obviously not distribution functions.

It is not difficult to show that the set V is a metric space under the Lévy metric

$$L(F, G) := \inf\{\varepsilon > 0 : F(x - \varepsilon) - \varepsilon \leq G(x) \leq F(x + \varepsilon) + \varepsilon \text{ for all } x \in \mathbb{R}\}.$$

Furthermore, if (F_n) is a sequence of distribution functions and F is also a distribution function with corresponding probability measures μ_n and μ , then $\mu_n \rightarrow \mu$ weakly if and only if $L(F_n, F) \rightarrow 0$.

Exercise 21 (*)

- Prove that L is a metric on the space of distribution functions.
- Let (F_n) and F be distribution functions with corresponding probability measures μ_n and μ . Prove $\mu_n \rightarrow \mu$ weakly if and only if $L(F_n, F) \rightarrow 0$.

In particular, the set $\mathcal{P}(\mathbb{R})$ of probability measures on $\mathbb{R}$ is metrizable and complete by Helly's theorem. The next subsection establishes a similar result for probability measures on Polish spaces.

The case of Polish spaces.

In what follows we prove Prohorov's theorem for general Polish spaces (E, d) . We start with the following result that shows that every probability measure on a Polish space is tight.

Proposition 2.3.9 (Tightness on Polish spaces) *Let (E, d) be a Polish space. Then every Borel probability measure μ on E is tight, i.e. for every $\varepsilon > 0$ there exists a compact set $K \subset E$ such that*

$$\mu(K) \geq 1 - \varepsilon.$$

PROOF: Since E is Polish, it is separable. Let $(x_k)_{k \in \mathbb{N}}$ be a dense sequence in E . For each $m \in \mathbb{N}$ the open balls $B(x_k, 2^{-m})$, $k \in \mathbb{N}$, cover E . In particular, there exists $n_m \in \mathbb{N}$ such that

$$\mu\left(\bigcup_{k=1}^{n_m} B(x_k, 2^{-m})\right) > 1 - \varepsilon 2^{-m}.$$

Let

$$U_m := \bigcup_{k=1}^{n_m} B(x_k, 2^{-m}), \quad K := \bigcap_{m=1}^{\infty} \overline{U}_m.$$

Then K is totally bounded (since $K \subset \overline{U}_m$ and each $\overline{U}_m$ is covered by finitely many balls) and closed, hence compact, and it is easy to see that

$$\mu(K^c) < \varepsilon.$$

□

We now recall that continuous functions on compact metric spaces can be approximated by Lipschitz continuous functions. For instance, if K is a compact subset of some metric space X and $f \in C(K)$, then the functions

$$f_n(x) := \inf_{y \in K} (f(y) + n d(x, y))$$

are Lipschitz with constant n and converge uniformly to f on K . This allows us to prove the following two results.

Proposition 2.3.10 *Let (K, d) be a compact metric space. Then $C(K)$ is complete and separable.*

PROOF: The proof of completeness is standard. Separability follows from the above approximation result as follows. Since K is a compact metric, it is separable. Let

$$D = \{x_1, x_2, x_3, \dots\} \subset K$$

be a dense subset. For any $N \in \mathbb{N}$ and any rational numbers $q_1, \dots, q_N \in \mathbb{Q}$ satisfying

$$|q_i - q_j| \leq L d(x_i, x_j),$$

we define the function

$$g(x) := \min_{1 \leq i \leq N} (q_i + L d(x, x_i)).$$

Each such function is Lipschitz continuous with constant L . Let $\mathcal{A}_L$ denote the set of all these. Then $\mathcal{A}_L$ is countable. Since any continuous function on K can be uniformly approximated by Lipschitz functions it is enough to prove that $\mathcal{A}_L$ is dense in the class of Lipschitz continuous functions with constant L .

To this end, let $f : K \rightarrow \mathbb{R}$ be Lipschitz with constant L and let $\varepsilon > 0$. Choose $\delta := \varepsilon/(3L)$. Since D is dense and K is compact, we can find N such that $\{x_1, \dots, x_N\}$ is a δ -net, i.e., for every $x \in K$ there exists i with

$$d(x, x_i) < \delta.$$

We now choose $q_i \in \mathbb{Q}$ such that

$$|q_i - f(x_i)| < \varepsilon/3 \quad \text{for } i = 1, \dots, N$$

and define

$$g(x) := \min_{1 \leq i \leq N} (q_i + L d(x, x_i)).$$

We claim that $\|f - g\|_\infty < \varepsilon$. To prove an upper bound let $x \in K$ and choose i with $d(x, x_i) < \delta$. Then

$$g(x) \leq q_i + L d(x, x_i) \leq f(x_i) + \frac{\varepsilon}{3} + L\delta \leq f(x) + \frac{2\varepsilon}{3}$$

where the second inequality follows from the Lipschitz continuity of f and the definition of $\delta > 0$. To establish a lower bound, we note that for any i ,

$$q_i \geq f(x_i) - \frac{\varepsilon}{3}.$$

Thus, by Lipschitz continuity of f ,

$$q_i + L d(x, x_i) \geq f(x_i) - \frac{\varepsilon}{3} + L d(x, x_i) \geq f(x) - \frac{\varepsilon}{3}.$$

Taking the minimum over i yields

$$g(x) \geq f(x) - \frac{\varepsilon}{3}.$$

□

Proposition 2.3.11 (Bounded Lipschitz functions are convergence-determining) *Let (E, d) be a Polish space. Let μ_n, μ be probability measures on E . Then $\mu_n \Rightarrow \mu$ if*

$$\int f d\mu_n \rightarrow \int f d\mu \quad \text{for all bounded Lipschitz functions } f.$$

PROOF: Let $f \in C_b(E)$, fix $\varepsilon > 0$ and let $M := \|f\|_\infty$. By the tightness of μ , there exists a compact set $K \subset E$ such that

$$\mu(K) \geq 1 - \varepsilon.$$

The function

$$\varphi(x) := \max(0, 1 - d(x, K))$$

is Lipschitz continuous, $0 \leq \varphi \leq 1$, and $\varphi = 1$ on K . Hence, the function $g := f\varphi$ has compact support, belongs to $C_b(E)$, $g = f$ on K , and

$$|f - g| \leq 2M \mathbf{1}_{K^c}.$$

As a result,

$$\left| \int f d\mu - \int g d\mu \right| \leq 2M \mu(K^c) \leq 2M\varepsilon \quad \text{and} \quad \left| \int f d\mu_n - \int g d\mu_n \right| \leq 2M \mu_n(K^c).$$

Since g has compact support we may w.l.o.g. assume that it is Lipschitz continuous. We now write

$$\left| \int f d\mu_n - \int f d\mu \right| \leq A_n + B_n + C_n,$$

where

$$A_n := \left| \int (f - g) d\mu_n \right|, \quad B_n := \left| \int g d\mu_n - \int g d\mu \right|, \quad C_n := \left| \int (g - f) d\mu \right|.$$

We know that $C_n \leq 2M\varepsilon$, that $B_n \rightarrow 0$, due to the Lipschitz continuity of g and that $A_n \leq 2M\mu_n(K^c)$. Hence it remains to control $\mu_n(K^c)$. For this, define

$$\psi_m(x) := \min(1, m d(x, K)).$$

Then ψ_m is bounded Lipschitz and $\psi_m \uparrow \mathbf{1}_{K^c}$ pointwise. Thus for each m large enough,

$$\int \psi_m d\mu_n \rightarrow \int \psi_m d\mu \leq \mu(K^c) + \varepsilon.$$

Thus,

$$\limsup_n \lim_m \int \psi_m d\mu_n = \limsup_{n \rightarrow \infty} \mu_n(K^c) \leq \mu(K^c) + \varepsilon.$$

□

The next result shows that weak convergence of probability measures on Polish spaces is metrizable (cf. the case $E = \mathbb{R}$ where a similar result was obtained).

Theorem 2.3.12 *Let (E, d) be a Polish space and let $\mathcal{P}(E)$ be the space of probability measures on E . Let*

$$d_{BL}(\mu, \nu) := \sup_{f \in \text{BL}_1(E)} \left| \int f d\mu - \int f d\nu \right|,$$

where $\text{BL}_1(E)$ denotes the set of all functions $f : E \rightarrow \mathbb{R}$ such that $\|f\|_\infty \leq 1$ and $\text{Lip}(f) \leq 1$. Then d_{BL} metrizes the weak convergence of probability measures.

PROOF: The preceding result shows that $\mu_n \Rightarrow \mu$ if $d_{BL}(\mu_n, \mu) \rightarrow 0$. Conversely, suppose that $\mu_n \Rightarrow \mu$ and fix $\varepsilon > 0$. By the easy implication of Prohorov's theorem our probability measures are tight: there exists a compact set $K \subset E$ such that

$$\mu_n(K) \geq 1 - \varepsilon, \quad \mu(K) \geq 1 - \varepsilon \quad \text{for all sufficiently large } n.$$

Let $f \in \text{BL}_1(E)$. Then

$$\int f d\mu_n - \int f d\mu = \int_K f d\mu_n - \int_K f d\mu + R_n \quad \text{where} \quad |R_n| \leq 2\varepsilon.$$

Since the family $\text{BL}_1(E)$ is equicontinuous and uniformly bounded, by Arzelà–Ascoli and Proposition 2.3.10 the restriction $\text{BL}_1(E)|_K$ is totally bounded in the sup norm. Hence there exist $f_1, \dots, f_m \in \text{BL}_1(E)$ such that for every $f \in \text{BL}_1(E)$ there exists i with

$$\sup_{x \in K} |f(x) - f_i(x)| < \varepsilon.$$

Thus,

$$\left| \int_K f d\mu_n - \int_K f d\mu \right| \leq \left| \int_K f_i d\mu_n - \int_K f_i d\mu \right| + 2\varepsilon.$$

Taking supremum over $f \in \text{BL}_1(E)$ gives

$$d_{BL}(\mu_n, \mu) \leq \max_{1 \leq i \leq m} \left| \int f_i d\mu_n - \int f_i d\mu \right| + 4\varepsilon.$$

Since each $f_i \in C_b(E)$ and $\mu_n \Rightarrow \mu$, we have

$$\int f_i d\mu_n \rightarrow \int f_i d\mu, \quad i = 1, \dots, n.$$

Hence the maximum term tends to zero, and therefore

$$\limsup_{n \rightarrow \infty} d_{BL}(\mu_n, \mu) \leq 4\varepsilon.$$

□

In a first step we now prove the complicated implication of Prohorov's theorem for probability measures on compact sets. The proof uses the Riesz representation theorem that we state without proof.

Theorem 2.3.13 (Riesz–Markov–Kakutani Representation Theorem) *Let X be a compact Hausdorff space and*

$$L : C(X) \rightarrow \mathbb{R}$$

be a positive linear functional, i.e. $L(f) \geq 0$ whenever $f \geq 0$. Then there exists a unique regular Borel measure μ on X such that

$$L(f) = \int_X f d\mu \quad \text{for all } f \in C(X).$$

Moreover,

$$\|L\| = \mu(X).$$

In particular, if $L(1) = 1$, then μ is a probability measure.

Proposition 2.3.14 (Sequential compactness on compact metric spaces) *Let K be a compact metric space. Then every sequence of probability measures $(\mu_n) \subset \mathcal{P}(K)$ admits a subsequence (μ_{n_k}) and a probability measure $\mu \in \mathcal{P}(K)$ such that*

$$d_{BL}(\mu_{n_k}, \mu) \rightarrow 0.$$

Equivalently, $\mathcal{P}(K)$ is weakly sequentially compact.

PROOF: Since K is compact metric, it follows from Proposition 2.3.10 that the space $C(K)$ is separable in the uniform norm. As a subset of the metric space $C(K)$ the set $BL_1(K)$ is also separable; we fix a countable dense family

$$(f_j)_{j \in \mathbb{N}} \subset BL_1(K).$$

For each fixed j , the sequence $\int f_j d\mu_n$ lies in $[-1, 1]$, hence admits a convergent subsequence. By a standard diagonal argument we find a subsequence (μ_{n_k}) such that for every $j \in \mathbb{N}$,

$$\int_K f_j d\mu_{n_k} \rightarrow \ell_j \in \mathbb{R}.$$

Next, we show that the extracted subsequence (μ_{n_k}) is Cauchy in d_{BL} . Since the set $BL_1(K)$ is bounded, equicontinuous and closed, it follows from the Arzelà–Ascoli theorem that it is compact, hence totally bounded in $\|\cdot\|_\infty$. Hence, for any $\varepsilon > 0$ there exists a finite set $F_\varepsilon = \{f_1, \dots, f_J\} \subset BL_1(K)$ such that for every $f \in BL_1(K)$, there exists $j \in \{1, \dots, J\}$ with

$$\|f - f_j\|_\infty \leq \varepsilon.$$

Thus, for any k, m ,

$$\left| \int f d(\mu_{n_k} - \mu_{n_m}) \right| \leq \left| \int f_j d(\mu_{n_k} - \mu_{n_m}) \right| + 2\varepsilon.$$

Since

$$\max_{1 \leq j \leq J} \left| \int f_j d(\mu_{n_k} - \mu_{n_m}) \right| \rightarrow 0.$$

and $\varepsilon > 0$ was arbitrary we conclude that

$$d_{BL}(\mu_{n_k}, \mu_{n_m}) \rightarrow 0.$$

We now identify the limit via Riesz–Markov.¹ For this, we define a linear functional on $BL_1(K)$ by

$$L(f) := \lim_{k \rightarrow \infty} \int f d\mu_{n_k}.$$

The functional is well-defined on $BL_1(K)$ and satisfies

$$|L(f)| \leq \|f\|_\infty, \quad L(f) \geq 0 \text{ if } f \geq 0, \quad L(1) = 1.$$

The operator extends to the class $BL(K)$ of all Lipschitz continuous functions and it is linear on this space. Since $BL(K)$ is dense in $C(K)$, the functional extends uniquely to a positive linear functional L on $C(K)$. Since K is compact Hausdorff, the Riesz–Markov–Kakutani representation theorem yields a unique probability measure $\mu \in \mathcal{P}(K)$ such that

$$L(f) = \int_K f d\mu, \quad f \in C(K).$$

In particular, for all $f \in BL_1(K)$,

$$\int f d\mu_{n_k} \rightarrow \int f d\mu.$$

Taking supremum over $\|f\|_{BL_1} \leq 1$, using again the total boundedness for $BL_1(K)$,

$$d_{BL}(\mu_{n_k}, \mu) = \sup_{\|f\|_{BL} \leq 1} \left| \int f d\mu_{n_k} - \int f d\mu \right| \rightarrow 0.$$

□

We are now ready to prove Prohorov’s theorem on Polish spaces. The key difference is that for general Polish spaces we cannot apply the standard Riesz–Markov–Kakutani theorem which only works in locally compact topological spaces.

Theorem 2.3.15 *Let (E, d) be a Polish space and let $(\mu_n) \subset \mathcal{P}(E)$ be tight. Then there exists a subsequence (μ_{n_k}) and a probability measure $\mu \in \mathcal{P}(E)$ such that*

$$d_{BL}(\mu_{n_k}, \mu) \rightarrow 0.$$

¹For this, the compactness (or at least local compactness) of K is required; compactness is not needed for the Arzelà–Ascoli theorem.

PROOF: Since (μ_n) is tight, for each $m \in \mathbb{N}$, there exists a compact set $K_m \subset E$ such that

$$\mu_n(K_m) \geq 1 - 2^{-m} \quad \text{for all } n \in \mathbb{N}.$$

We may w.l.o.g. assume that $K_m \subset K_{m+1}$. For fixed m we consider the restricted measures (these are NOT probability measures)

$$\mu_n^{(m)} := \mu_n|_{K_m}.$$

Although the $\mu_n^{(m)}$ are not probability measures we can still use the same arguments as in the proof of the previous proposition to show that, for each $m \in \mathbb{N}$, the sequence $(\mu_n^{(m)})$ admits a subsequence converging in d_{BL} on K_m . Again by a standard diagonalization argument there exists a subsequence (μ_{n_k}) such that for every $m \in \mathbb{N}$, the restriction $\mu_{n_k}|_{K_m}$ converges in d_{BL} on K_m . Thus for each fixed $m \in \mathbb{N}$, there exists a finite Borel measure $\mu^{(m)}$ on K_m such that

$$d_{BL}(\mu_{n_k}|_{K_m}, \mu^{(m)}) \rightarrow 0.$$

Since $K_{m_1} \subset K_{m_2}$ whenever $m_1 < m_2$ and both limits agree on K_{m_1} , the family $(\mu^{(m)})$ is consistent.

To construct the global limit for $f \in BL(E)$ we set

$$L_m(f) := \int_{K_m} f d\mu^{(m)}$$

and claim that the sequence $(L_m(f))$ is Cauchy. Indeed, let $m < n$. Then, by consistency

$$L_n(f) = \int_{K_n} f d\mu^{(n)} = \int_{K_m} f d\mu^{(n)} + \int_{K_n \setminus K_m} f d\mu^{(n)} = L_m(f) + \int_{K_n \setminus K_m} f d\mu^{(n)}$$

and since $|f| \leq 1$,

$$\begin{aligned} |L_n(f) - L_m(f)| &\leq \mu^{(n)}(K_n \setminus K_m) \\ &= \lim_{k \rightarrow \infty} \mu_{n_k}(K_n \setminus K_m) \\ &\leq \limsup_{k \rightarrow \infty} \mu_{n_k}(K_m^c) \\ &\leq 2^{-m}. \end{aligned}$$

Since $(L_m(f))$ is Cauchy we can define the limit

$$L(f) := \lim_{m \rightarrow \infty} L_m(f).$$

Then L is linear, non-negative, normalized, and continuous under bounded convergence on compact sets. By the Riesz–Markov representation theorem for Radon measures on Polish spaces in the form of Proposition 2.3.17 below,² there exists a unique probability measure μ such that

$$L(f) = \int_X f d\mu \quad \text{for all } f \in BL(E).$$

It remains to prove the convergence of the integrals. For this, we fix $f \in BL(E)$. For any $m \in \mathbb{N}$,

$$\int f d\mu_{n_k} = \int_{K_m} f d\mu_{n_k} + \int_{K_m^c} f d\mu_{n_k}.$$

²We emphasize that we assume that E but not necessarily locally compact. Hence we cannot use the standard Riesz representation theorem.

Since $|f| \leq 1$,

$$\left| \int_{K_m^c} f d\mu_{n_k} \right| \leq \mu_{n_k}(K_m^c) \leq 2^{-m}.$$

Also:

$$\int_{K_m} f d\mu_{n_k} \rightarrow \int_{K_m} f d\mu^{(m)} = L_m(f).$$

Thus:

$$\limsup_{k \rightarrow \infty} \left| \int f d\mu_{n_k} - L(f) \right| \leq 2^{-m}.$$

Letting $m \rightarrow \infty$, we obtain:

$$\int f d\mu_{n_k} \rightarrow \int f d\mu.$$

□

The proof of the main theorem requires the following auxiliary result.

Lemma 2.3.16 (Vanishing of decreasing closed sets under weak limits) *Let C be a compact metric space and let (μ_n) be a sequence of probability measures on C . Let $(F_k)_{k \in \mathbb{N}}$ be a decreasing sequence of closed subsets of C such that*

$$F_{k+1} \subset F_k \quad \text{and} \quad \bigcap_{k=1}^{\infty} F_k = \emptyset.$$

Then

$$\lim_{k \rightarrow \infty} \limsup_{n \rightarrow \infty} \mu_n(F_k) = 0.$$

In particular, there exists a subsequence (n_k) such that

$$\mu_{n_k}(F_k) \rightarrow 0.$$

PROOF: We proceed by contradiction and assume to the contrary that there exists $\varepsilon > 0$ such that for every k there exists $n(k)$ with

$$\mu_{n(k)}(F_k) \geq \varepsilon.$$

Since C is compact metric, the space of probability measures on C is weakly sequentially compact. Hence, passing to a subsequence if necessary, we may assume that $\mu_{n(k)} \Rightarrow \nu$ weakly for some probability measure ν on C . We now apply the Portmanteau theorem. For any k_0 , since F_{k_0} is closed, Portmanteau gives

$$\limsup_{j \rightarrow \infty} \mu_{n(j)}(F_{k_0}) \leq \nu(F_{k_0}).$$

Now use monotonicity: for $j \geq k_0$ we have $F_{n(j)} \subset F_{k_0}$ and so

$$\mu_{n(j)}(F_{n(j)}) \leq \mu_{n(j)}(F_{k_0}).$$

In particular,

$$\limsup_{j \rightarrow \infty} \mu_{n(j)}(F_{k_0}) \geq \varepsilon$$

and so $\nu(F_{k_0}) \geq \varepsilon$. Since k_0 was arbitrary,

$$\nu(F_k) \geq \varepsilon$$

for all $k \in \mathbb{N}$. On the other hand, $\bigcap_k F_k = \emptyset$ which, since C is compact, implies $\nu(F_k) \downarrow \nu(\emptyset) = 0$.³

□

Proposition 2.3.17 *Let X be a Polish space and let $(\mu_n)_{n \in \mathbb{N}}$ be a sequence of probability measures on X . Assume:*

1. **Tightness:** *For every $\varepsilon > 0$, there exists a compact set $K \subset X$ such that*

$$\sup_n \mu_n(X \setminus K) < \varepsilon.$$

2. **Pointwise convergence on $C_b(X)$:** *There exists a bounded, positive, linear functional $T : C_b(X) \rightarrow \mathbb{R}$ such that*

$$\int_X f d\mu_n =: T_n(f) \rightarrow T(f) \quad \text{for all } f \in C_b(X).$$

Then there exists a unique probability measure μ on X such that

$$T(f) = \int_X f d\mu \quad \text{for all } f \in C_b(X).$$

PROOF: Step 1. Construction of the candidate probability measure μ . For a compact set $K \subset X$, we set

$$\mu(K) := \inf \{T(f) : f \in C_b(X), f \geq \mathbf{1}_K\}.$$

This function is well-defined by Urysohn's lemma,⁴ monotone, finitely sub-additive and $\mu(\emptyset) = 0$. One can show that μ extends uniquely to a probability measure on the Borel sets in the usual way.⁵

We now show that $\mu_n(K) \rightarrow \mu(K)$ for any compact set K . One inequality is easy: for any $f \in C_b(X)$ with $f \geq \mathbf{1}_K$ we have that

$$\limsup \mu_n(K) \leq \limsup \mu_n(f) = T(f).$$

Taking the infimum over all such function gives

$$\limsup \mu_n(K) \leq \mu(K).$$

³We recall that probability measures on compact sets are continuous from above.

⁴If X is topological space where any two disjoint closed sets can be separated by disjoint open neighborhoods. Let $A, B \subset X$ be disjoint closed sets. Then there exists a continuous function

$$f : X \rightarrow [0, 1]$$

such that

$$f(x) = 0 \quad \text{for all } x \in A, \quad f(x) = 1 \quad \text{for all } x \in B.$$

⁵For open sets $U \subset X$ we set $\mu(U) := \sup \{\mu(K) : K \subset U, K \text{ compact}\}$ and for arbitrary sets A we set $\mu^*(A) := \inf \{\mu(U) : A \subset U, U \text{ open}\}$. Then μ^* is an outer measure which extends uniquely to a measure on the Borel- σ -field.

To obtain the converse inequality we define the Lipschitz cutoff functions

$$f_m(x) := \max\{1 - m d(x, K), 0\}.$$

and the sets

$$U_m := \{x \in X : d(x, K) < 1/m\}, \quad A_m := U_m \setminus K.$$

By tightness we can fix $\varepsilon > 0$ and choose a compact set $C \subset X$ such that for every $n \in \mathbb{N}$,

$$\left| \int f_m d\mu_n - \mu_n(K) \right| \leq \mu_n(A_m \cap C) + \varepsilon.$$

The inequality still holds if we replace A_m by its closure. It then follows from the previous lemma that

$$\sup_n \left| \int f_m d\mu_n - \mu_n(K) \right| \rightarrow 0.$$

Since $\lim_{n \rightarrow \infty} \int f_m d\mu_n = T(f_m)$ for every fixed $m \in \mathbb{N}$, it follows that

$$\lim_{m \rightarrow \infty} T(f_m) = \lim_{n \rightarrow \infty} \mu_n(K).$$

In particular,

$$\mu(K) \leq \liminf_m T(f_m) = \liminf_n \mu_n(K).$$

Step 2. μ is the desired measure. We first show that it is enough to identify T on the set $C_c(X)$.

In fact, let $f \in C_b(X)$ and $\varepsilon > 0$. By tightness, we can choose a compact set $K \subset X$ such that

$$\sup_n \mu_n(X \setminus K) < \varepsilon$$

and a function $\varphi \in C_c(X)$ such that $0 \leq \varphi \leq 1$, $\varphi|_K = 1$. Set $g := f\varphi \in C_c(X)$. Then

$$|T_n(f) - T_n(g)| \leq \|f\|_\infty \varepsilon \quad \text{and} \quad |T(f) - \lim_{n \rightarrow \infty} T_n(g)| \leq \|f\|_\infty \varepsilon.$$

To identify T on $C_c(X)$, let $g \in C_c(X)$, and let $K = \text{supp}(g)$, which is compact. Then

$$T_n(g) = \int_X g d\mu_n = \int_K g d\mu_n.$$

Since $\mu_n(K) \rightarrow \mu(K)$, and by standard approximation of continuous functions on a fixed compact set (using linear combinations of indicators of small sets and uniform continuity of g), it follows that

$$\int_K g d\mu_n \rightarrow \int_K g d\mu.$$

Hence,

$$T(g) = \int_X g d\mu \quad \text{for } g \in C_c(X).$$

For a general continuous function $f \in C_b(X)$ we use again tightness to approximate f by a continuous function with compact support to show that $T(f) = \int f d\mu$. Finally, two probability measures agree if they agree on $C_b(X)$. $\square$

Weak uniqueness of accumulation points

In any metric space a sequence converges if and only if any subsequence contains a further subsequence that converges to the same limit. The following lemma establishes a corresponding result for probability measures on general Borel spaces.

Lemma 2.3.18 *Let $(\mathbb{P}_n)_{n \in \mathbb{N}}$ and $\mathbb{P}$ be probability measures on a metric Borel space $(E, \mathcal{B}(E))$. Then $\mathbb{P}_n \Rightarrow \mathbb{P}$ if and only if every subsequence of $(\mathbb{P}_n)_{n \in \mathbb{N}}$ contains a further subsequence that converges weakly to $\mathbb{P}$.*

PROOF: The necessity is obvious. For sufficiency, suppose that $\mathbb{P}_n \Rightarrow \mathbb{P}$ does not hold. Then there exists a subsequence $(\mathbb{P}_{n_j}) \subset (\mathbb{P}_n)$, a bounded continuous function f and some $\varepsilon > 0$ such that

$$\left| \int_E f(x) \mathbb{P}_{n_j}(dx) - \int_E f(x) \mathbb{P}(dx) \right| > \varepsilon$$

for all $j \in \mathbb{N}$. Hence, no further subsequence of $\{\mathbb{P}_{n_j}\}_{j \in \mathbb{N}}$ can converge weakly to $\mathbb{P}$. $\square$

Corollary 2.3.19 *If $(\mathbb{P}_n)_{n \in \mathbb{N}}$ is a tight sequence of probability measures on a Polish space $(E, \mathcal{B}(E))$ and if each convergent subsequence converges to $\mathbb{P}$, then the entire sequence converges weakly to $\mathbb{P}$.*

PROOF: By Prohorov's theorem, $(\mathbb{P}_n)_{n \in \mathbb{N}}$ is relatively sequentially compact, i.e. every subsequence contains a convergent further subsequence, which by assumption converges to $\mathbb{P}$. The result now follows from Lemma 2.3.18. $\square$

2.3.2 Weak convergence of probability measures on $C([0, 1], \mathbb{R})$

We are now going to prove convergence results for sequences of probability measures on $C([0, 1], \mathbb{R})$. Since $C([0, 1], \mathbb{R})$ equipped with the supremum-norm is a Polish space, Prohorov's theorem is applicable in this setting. In particular, the weak convergence $\mathbb{P}_n \Rightarrow \mathbb{P}$ of a sequence of probability measures on $C([0, 1], \mathbb{R})$ implies the tightness of the sequence. It also implies convergence of the finite-dimensional distributions. In fact, for any finite set $J \subset [0, 1]$ the projection mapping

$$\pi_J : C([0, 1], \mathbb{R}) \rightarrow \mathbb{R}^{|J|}, \quad x \mapsto (x(t))_{t \in J}$$

is continuous. Hence, for any bounded continuous $g : \mathbb{R}^{|J|} \rightarrow \mathbb{R}$ it holds that

$$\int g(y) (\mathbb{P}_n \circ \pi_J^{-1})(dy) = \int g(\pi_J(x)) \mathbb{P}_n(dx) \xrightarrow{n} \int g(\pi_J(x)) \mathbb{P}(dx) = \int g(y) (\mathbb{P} \circ \pi_J^{-1})(dy). \quad (2.4)$$

The following result proves the converse implication. Hence a sequence of probability measures on $C([0, 1], \mathbb{R})$ converges weakly if and only if it is tight and the finite-dimensional distributions converge.

Theorem 2.3.20 *Let $(\mathbb{P}_n)_{n \in \mathbb{N}}$ and $\mathbb{P}$ be probability measures on $C([0, 1], \mathbb{R})$. If the finite-dimensional distributions of the $\mathbb{P}_n$ converge weakly to that of $\mathbb{P}$ and if $(\mathbb{P}_n)_{n \in \mathbb{N}}$ is tight, then $\mathbb{P}_n \Rightarrow \mathbb{P}$.*

PROOF: If the sequence $(\mathbb{P}_n)$ is tight, then it is relatively sequentially compact. To prove the weak convergence, we would like to apply Corollary 2.3.19, for which we have to prove that all weakly convergent subsequences converge to $\mathbb{P}$.

To this end, let $(\mathbb{P}_{n_k})$ be a weakly convergent subsequence so that $\mathbb{P}_{n_k} \Rightarrow \mathbb{Q}$ for some probability measure $\mathbb{Q}$ on $C([0, 1], \mathbb{R})$. We need to show that $\mathbb{P} = \mathbb{Q}$. By assumption,

$$\mathbb{P}_n \circ \pi_J^{-1} \Rightarrow \mathbb{P} \circ \pi_J^{-1} \quad \text{for all } J \subseteq [0, 1] \text{ finite.}$$

Moreover, it follows from (2.4) that $\mathbb{P}_{n_k} \circ \pi_J^{-1} \Rightarrow \mathbb{Q} \circ \pi_J^{-1}$. Hence

$$\mathbb{P} \circ \pi_J^{-1} = \mathbb{Q} \circ \pi_J^{-1} \quad \text{for all finite subsets } J \subseteq [0, 1].$$

Since a probability distribution on $C([0, 1]; \mathbb{R})$ is completely characterized by its finite-dimensional distributions (by Kolmogorov's consistency theorem), we conclude that $\mathbb{P} = \mathbb{Q}$. □

The following counterexample shows that the convergence of the finite-dimensional distributions of the $(\mathbb{P}_n)$ is in general not sufficient in order to have $\mathbb{P}_n \Rightarrow \mathbb{P}$.

Example 2.3.21 For every $n \in \mathbb{N}$, let us define a function $z_n : [0, 1] \rightarrow \mathbb{R}$ by

$$z_n(t) := nt \cdot \mathbb{1}_{[0, \frac{1}{n}]}(t) + (2 - nt)\mathbb{1}_{(\frac{1}{n}, \frac{2}{n}]}(t), \quad t \in [0, 1].$$

Then $\|z_n\|_\infty = 1$ for all $n \in \mathbb{N}$ and $z_n(t) \rightarrow 0$ for all $t \in [0, 1]$. Let

$$\mathbb{P}_n := \delta_{z_n}, \quad n \in \mathbb{N} \quad \text{and} \quad \mathbb{P} := \delta_0.$$

To show that the sequence $(\mathbb{P}_n)$ does not converge weakly to $\mathbb{P}$ we introduce the closed set $B := \{z \in C([0, 1]) : \|z\|_\infty = 1\}$. Then $\mathbb{P}(B) = 0$ and by the Portemanteau theorem the weak convergence $\mathbb{P}_n \Rightarrow \mathbb{P}$ would give us that

$$1 = \limsup_n \mathbb{P}_n(B) \leq \mathbb{P}(B),$$

which does not hold. To show that we nonetheless have convergence of the finite dimensional distributions, let $0 = t_0 < t_1 < \dots < t_m \leq 1$. Then for all $n \in \mathbb{N}$ with $\frac{2}{n} < t_1$ we have

$$\pi_{\{t_0, \dots, t_m\}}(z_n) = (0, \dots, 0)$$

and so,

$$\mathbb{P}_n \circ \pi_{\{t_0, \dots, t_m\}}^{-1}(A) = \mathbb{P} \circ \pi_{\{t_0, \dots, t_m\}}^{-1}(A) \quad \text{for all } A \in \mathcal{B}(\mathbb{R}^{|J|}), \quad n > \frac{2}{t_1}.$$

Thus, $\mathbb{P}_n \circ \pi_J^{-1}(A) \rightarrow \mathbb{P} \circ \pi_J^{-1}(A)$ for all $J \subset [0, 1]$ finite and $A \in \mathcal{B}(\mathbb{R}^{|J|})$. That is, all finite-dimensional distributions converge.

In what follows we convert the classical Arzelà-Ascoli compactness criterion into a condition for the tightness of probability measures on $C(K, S)$ where (K, d) and (S, ϱ) are metric spaces. To this end, we recall that the *modulus of continuity* of a function $x \in C(K, S)$ is given by

$$w(x, h) := \sup\{\varrho(x_t, x_s) : d(t, s) \leq h\}, \quad h > 0.$$

The function w is continuous for fixed $h > 0$ and hence a measurable function of x . In what follows we need the following version of the Arzelà-Ascoli theorem.

Theorem 2.3.22 (Arzelà, Ascoli) *Let K and S be metric spaces where K is compact and S is complete, and let $D \subseteq K$ be dense. Then a set $A \subseteq C(K, S)$ is relatively compact if and only if $\{x_t : x \in A\}$ is relatively compact in S for any $t \in D$ and*

$$\lim_{h \rightarrow 0} \sup_{x \in A} w(x, h) = 0.$$

In this case, even the set $\bigcup_{t \in K} \{x_t : x \in A\}$ is relatively compact in S .

We are now ready to prove the following tightness condition in $C(K, S)$, due to Prohorov.

Theorem 2.3.23 (Tightness in $C(K, S)$) *Let K and S be metric spaces where K is compact and S is separable and complete. Let $\{X_n\}$ be a sequence of random elements in $C(K, S)$. Then $X_n \xrightarrow{d} X$ if and only if the finite-dimensional distributions converge and*

$$\lim_{h \rightarrow 0} \sup_n \mathbb{E}[w(X_n, h) \wedge 1] = 0. \quad (2.5)$$

PROOF: Since $C(K, S)$ is separable and complete, Prohorov's theorem shows that tightness and relative compactness are equivalent for $\{X_n\}$. Moreover, weak convergence is equivalent to tightness plus convergence of the finite-dimensional distributions. It is thus enough to prove that tightness implies (2.5) and that (2.5) implies tightness if the finite-dimensional distributions converge.

- Let us first assume that $X_n \xrightarrow{d} X$, hence tightness of the sequence $\{X_n\}$ and choose a compact set $B \subseteq C(K, S)$ such that $\sup_n \mathbb{P}(X_n \in B^c) < \varepsilon$. By the theorem of Arzelà-Ascoli we may choose $h > 0$ so small that $w(x, h) \leq \varepsilon$ for all $x \in B$. But then $\sup_n \mathbb{P}(w(X_n, h) > \varepsilon) < \varepsilon$, and (2.5) follows because ε was arbitrary.
- Next, let us assume convergence of the finite-dimensional distributions along with (2.5). Using that L^1 -convergence implies convergence in measure we may, for any $\varepsilon > 0$ choose a sequence $h_1, h_2, \dots > 0$ so small that

$$\sup_n \mathbb{P}(w(X_n, h_k) > 2^{-k}) \leq 2^{-(k+1)} \varepsilon, \quad k \in \mathbb{N}.$$

Next, let $\{t_n\} \subset K$ be dense. Convergence of the finite-dimensional distributions implies that the corresponding families of finite-dimensional distributions are tight. This allows us to choose compact sets $C_1, C_2, \dots \subseteq S$ such that

$$\sup_n \mathbb{P}(X_n(t_k) \in C_k^c) \leq 2^{-(k+1)} \varepsilon, \quad k \in \mathbb{N}.$$

Let us then define the set

$$B := \bigcap_{k \in \mathbb{N}} \left\{ x \in C(K, S) : x(t_k) \in C_k, w(x, h_k) \leq 2^{-k} \right\}.$$

Then, B is relatively compact, due to the Arzelà-Ascoli theorem. Moreover, $\sup_n \mathbb{P}(X_n \in B^c) \leq \varepsilon$.

□

Exercise 22 (Centsov Tightness Condition) (*)

- A sequence of probability measures $\{\mu_n\}$ on $C([0, 1], \mathbb{R})$ is tight if and only if the following two conditions hold. (i) for all $\varepsilon > 0$ there exists M such that $\sup_n \mu_n(|x(0)| > M) < \varepsilon$ and (ii) for all $\varepsilon, \eta > 0$ there exists $\delta > 0$ such that $\sup_n \mu_n(w(x, \delta) \geq \eta) < \varepsilon$ where w denotes the modulus of continuity.
- Let $\{X^n\}$ be a sequence of continuous stochastic processes on $(\Omega, \mathcal{F}, \mathbb{P})$ that satisfies the following conditions:

- $\sup_m \mathbb{E}[|X_0^m|^\nu] := M < \infty$.
- $\sup_m \mathbb{E}|X_t^m - X_s^m|^\alpha \leq C_T |t - s|^{1+\beta}$ for all $T > 0$ and $0 \leq s \leq t \leq T$.

Follow the technique of the Kolmogorov-Centsov Theorem to prove that the sequence of probability measures $\{\mathbb{P} \circ (X^m)^{-1}\}$ is tight.

2.3.3 Donsker's invariance principle

Armed with the above results we can now give an alternative proof of Donsker's invariance principle. As a byproduct we obtain a third construction of Brownian motion, through the weak convergence of a family of probability measures in $C([0, 1], \mathbb{R})$ to the Wiener measure.

To this end, we consider a sequence $(X_n)_{n \in \mathbb{N}}$ of $\mathbb{R}$ -valued i.i.d. random variables on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\mathbb{E}(X_n) = 0$ and $\mathbb{E}(X_n^2) = 1$ and introduce the continuous time rescaled random walks

$$S_n(t) := \frac{1}{\sqrt{n}} \sum_{k=1}^{\lfloor nt \rfloor} X_k, \quad n \in \mathbb{N}$$

where $\lfloor x \rfloor := \max\{k \in \mathbb{Z} : k \leq x\}$ denotes the Gauss bracket of $x \in \mathbb{R}$. Let B be a Brownian motion. By the classical central limit theorem

$$(S_n(t_1) - S_n(t_0), \dots, S_n(t_N) - S_n(t_{N-1})) \xrightarrow{d} (B(t_1) - B(t_0), \dots, B(t_N) - B(t_{N-1})).$$

By linear interpolation we can then define the continuous process

$$S_n^*(t) := S_n(t) + \frac{tn - \lfloor tn \rfloor}{\sqrt{n}} X_{\lfloor tn \rfloor + 1}.$$

By Tschebyscheff's inequality

$$\mathbb{P}[|S_n^*(t) - S_n(t)| \geq \varepsilon] \rightarrow 0$$

and so it follows from Slutsky's theorem⁶ that

$$(S_n^*(t_1) - S_n^*(t_0), \dots, S_n^*(t_N) - S_n^*(t_{N-1})) \xrightarrow{d} (B(t_1) - B(t_0), \dots, B(t_N) - B(t_{N-1})).$$

As a result, the sequence of the linear interpolated rescaled random walks $\{S_n^*\}$ converges weakly to Brownian motion (i.e. the sequence of laws converges weakly to the Wiener measure) if we can prove that the sequence $\{S_n^*\}$ is tight. The argument uses the following maximum inequality for random walks.

⁶We recall that $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{d} Y$ does not imply that $X_n + Y_n \xrightarrow{d} X + Y$. Slutsky's theorem states that this is true if Y is a deterministic constant.

Proposition 2.3.24 (Ottaviani) *Let $\{\xi_n\}$ be a sequence of i.i.d. random variables with mean 0 and variance 1 and put $T_n := \sum_{i=1}^n \xi_i$. Let $\{T_n^*\}$ be the sequence of the running maxima of $\{T_n\}$. Then*

$$\mathbb{P}[T_n^* \geq 2r\sqrt{n}] \leq \frac{\mathbb{P}[|T_n| \geq r\sqrt{n}]}{1 - r^{-2}}, \quad r > 1, \quad n \in \mathbb{N}.$$

In particular,

$$\lim_{r \rightarrow \infty} r^2 \limsup_{n \rightarrow \infty} \mathbb{P}[T_n^* \geq r\sqrt{n}] = 0.$$

PROOF: Let $A_k := \{T_{k-1}^* \leq 2r, |T_k| > 2r\}$ the event where the random walk exceeds the level $2r$ for the first time at time $k \in \mathbb{N}$. Then the following holds:

$$\begin{aligned} \mathbb{P}[|T_n| > r] &= \sum_{k=1}^n \mathbb{P}(|T_n| > r, A_k) + \mathbb{P}(|T_n| > r, A_n^c) \\ &\geq \sum_{k=1}^n \mathbb{P}(|T_n| > r, A_k) \\ &\geq \sum_{k=1}^n \mathbb{P}(|T_n - T_k| \leq r, A_k) \\ &\geq \min_{1 \leq k \leq n} \mathbb{P}(|T_n - T_k| \leq r) \mathbb{P}\left(\bigcup_{k=1}^n A_k\right) \quad \text{- independence of } \{|T_n - T_k| \leq r\}, A_k \\ &= \min_{1 \leq k \leq n} \mathbb{P}(|T_k| \leq r) \mathbb{P}(T_n^* > 2r). \end{aligned}$$

Let $c := r\sqrt{n}$. Then it follows from Tschebyscheff's inequality that

$$\min_{1 \leq k \leq n} \mathbb{P}[|T_k| \leq c] \geq \min_{1 \leq k \leq n} (1 - kc^{-2}) \geq (1 - r^{-2}).$$

This shows the first assertion. The second assertion follows from the central limit theorem along with the Portmanteau theorem and the already established fact (see proof of Lemma (2.2.2)) that for any $N(0, 1)$ random variable Z and $r > 1$,

$$\mathbb{P}[|Z| > r] \leq \sqrt{\frac{2}{\pi}} e^{-r^2/2}.$$

□

Theorem 2.3.25 (Donsker's invariance principle) *Let B be a Brownian motion. Under the above assumption, restricted to any compact interval (and hence on $[0, \infty)$)*

$$S_n^* \xrightarrow{d} B.$$

PROOF: The statement follows as soon as we can prove the tightness of the sequence $\{S_n^*\}$. For this we apply the previous result to the sequence

$$T_{[nt]} = \sqrt{n} S_n(t) \quad \text{so that} \quad T_{[nh]}^* = \sqrt{n} \sup_{0 \leq r \leq h} |S_n(r)|.$$

In particular,

$$T_{[nh]}^* \geq r\sqrt{[nh]} \Leftrightarrow \sup_{0 \leq r \leq h} S_n(r) \geq r\sqrt{\frac{[nh]}{n}} \approx r\sqrt{h}.$$

We hence obtain from Ottaviani's lemma that

$$\lim_{r \rightarrow \infty} r^2 \lim_{n \rightarrow \infty} \mathbb{P} \left[\sup_{0 \leq t \leq h} |S_n(t)| \geq r\sqrt{h} \right] = 0.$$

Setting $r = \varepsilon(\sqrt{h})^{-1}$ and using the independence of the increments of the sums, this shows that

$$\lim_{h \rightarrow 0} h^{-1} \limsup_{n \rightarrow \infty} \sup_{t \in [0,1]} \mathbb{P} \left[\sup_{0 \leq r \leq h} |S_n^*(t+r) - S_n^*(t)| > \varepsilon \right] = 0.$$

Now (2.5) follows by dividing $[0, 1]$ into h^{-1} subintervals $[t_{k-1}, t_k]$ of length h :

$$\begin{aligned} & \limsup_{n \rightarrow \infty} \mathbb{P} \left[\sup_{0 \leq t \leq 1, r \leq h} |S_n^*(t+r) - S_n^*(t)| > \varepsilon \right] \\ & \leq \sum_{k=1}^{h^{-1}} \limsup_{n \rightarrow \infty} \mathbb{P} \left[\sup_{r \leq h} |S_n^*(t_{k-1}+r) - S_n^*(t_{k-1})| > \varepsilon \right] \\ & = h^{-1} \limsup_{n \rightarrow \infty} \sup_{t \in [0,1]} \mathbb{P} \left[\sup_{0 \leq r \leq h} |S_n^*(t+r) - S_n^*(t)| > \varepsilon \right] \rightarrow 0. \end{aligned}$$

□

2.4 The Brownian filtration

The natural filtration $\mathcal{F}^X$ of a right-continuous process is not right-continuous in general. A simple counter example is given by the process

$$X_t = tZ, \quad t \geq 0$$

where Z is some non-trivial random variable. In this case $\mathcal{F}_0^X = \{\emptyset, \Omega\}$ while $\mathcal{F}_\varepsilon^X = \sigma(Z)$ for all $\varepsilon > 0$. Hence

$$\mathcal{F}_{0+}^X \neq \mathcal{F}_0^X.$$

However it can be shown that the completion of the natural filtration of a Feller Markov process, including all Lévy processes (stochastically continuous càdlàg processes starting in zero that have stationary, independent increments) and in particular Brownian motion, is always right-continuous.

In what follows we prove this result for Brownian motion. To this end, we need a slight modification of the definition of the Brownian motion that includes a filtration.

Definition 2.4.1 *Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be a filtered probability space. An adapted process B is called a Brownian motion w.r.t. the filtration $(\mathcal{F}_t)$ if*

- $B_0 = 0$ a.s.
- $B_t - B_s \perp \mathcal{F}_s$ for all $0 \leq s \leq t < \infty$
- $B_t - B_s \sim N(0, t-s)$
- B has almost surely continuous sample paths.

Exercise 23 Let B be a Brownian motion in the sense of the original definition and let

$$\mathcal{F}_t^B := \sigma(B_s : 0 \leq s \leq t)$$

be the filtration generated by B . Then B is a Brownian motion w.r.t. the filtration $(\mathcal{F}_t^B)$.

The filtration $\mathcal{F}^B$ generated by a Brownian motion B is not right-continuous. Its completion

$$\mathcal{F}_t := \mathcal{F}_t^B \vee \mathcal{N} \quad (2.6)$$

however, is, as shown by the next result.

Remark 2.4.2 We notice that a Brownian motion remains a BM with respect to the completed filtration: completion of σ -fields does not affect the assumptions of Definition 2.4.1.

To prove the next result, we recall that by the monotone class theorem, if some property holds for all elements from a $\cap$ -stable generator of a σ -field, then it holds for all elements from the σ -field. Since $\mathcal{F}^B$ is generated by the projection mappings it is the smallest σ -field with respect to which all bounded functions depending on finitely many values $B_{t_1}, \dots, B_{t_d}$ are measurable. Hence, if we can prove that a property holds for all bounded measurable functions depending only on finitely many coordinates, then it holds for all events in $\mathcal{F}^B$.

Proposition 2.4.3 The filtration $(\mathcal{F}_t)$ introduced in (2.6) satisfies the usual conditions. In particular, it is right-continuous.

PROOF: Completeness is clear. To prove the right-continuity we fix $d \in \mathbb{N}$, $0 \leq t_1 < \dots < t_d$ and a bounded and continuous function $f : \mathbb{R}^d \rightarrow \mathbb{R}$. We claim that the conditional expectation

$$\mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+}]$$

is $\mathcal{F}_t$ -measurable. Indeed, let

$$t_k < t < t + \frac{1}{n} < t_{k+1}$$

and $p_n(x_1, \dots, x_{d-k})$ be the density of the increments $(B_{t_{k+1}} - B_{t+1/n}, \dots, B_{t_d} - B_{t+1/n})$. Then,

$$\begin{aligned} & \mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+1/n}] \\ &= \int_{\mathbb{R}^{d-k}} f(B_{t_1}, \dots, B_{t_k}, B_{t+1/n} + x_1, \dots, B_{t+1/n} + x_{d-k}) p_n(x_1, \dots, x_{d-k}) dx_1 \dots dx_{d-k}. \end{aligned}$$

The densities p_n converge to the density p of $(B_{t_{k+1}} - B_t, \dots, B_{t_d} - B_t)$. By dominated convergence, using the continuity of f and B ,

$$\begin{aligned} & \mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+1/n}] \\ & \rightarrow \int_{\mathbb{R}^{d-k}} f(B_{t_1}, \dots, B_{t_k}, B_t + x_1, \dots, B_t + x_{d-k}) p(x_1, \dots, x_{d-k}) dx_1 \dots dx_{d-k}, \end{aligned}$$

which is $\mathcal{F}_t$ -measurable for any $n \in \mathbb{N}$. At the same time, by Lemma 1.3.15 (backwards martingale convergence)

$$\mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+1/n}] \rightarrow \mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_t] \quad \text{a.s.}$$

Since $\mathcal{N} \in \mathcal{F}_t$ we see that $\mathbb{E}[f(B_{t_1}, \dots, B_{t_d}) | \mathcal{F}_{t+}]$ is indeed $\mathcal{F}_t$ -measurable. The reason we need completeness is that we only have a.s. convergence.⁷ Since for any open interval (a, b) the indicator function $\mathbf{1}_{(a,b)}$ can be approximated pointwise by an increasing sequence of continuous functions it follows from dominated convergence that the $\mathcal{F}_t$ -measurability carries over to all functions of the form

$$f = \mathbf{1}_{(a_1, b_1) \times \dots \times (a_d, b_d)}.$$

Hence, it follows from the monotone class theorem that

$$\mathbb{E}[\mathbf{1}_A | \mathcal{F}_{t+}] \text{ is } \mathcal{F}_t\text{-measurable}$$

for any $A \in \mathcal{F}^B$. In particular, for $A \in \mathcal{F}_{t+}$,

$$\mathbf{1}_A = \mathbb{E}[\mathbf{1}_A | \mathcal{F}_{t+}] \text{ is } \mathcal{F}_t\text{-measurable.}$$

Hence $\mathcal{F}_{t+} = \mathcal{F}_t$. □

2.5 Sample path properties

In this section we establish three important sample path properties of Brownian motion: the nowhere differentiability, the law of iterated logarithm and the nowhere monotonicity.

2.5.1 Nowhere differentiability

By construction almost every Brownian sample path is α -Hölder continuous for $\alpha \in (0, 1/2)$. We now show that a typical Brownian path is nowhere differentiable. To this end, we recall that the upper and lower right Dini derivatives of a continuous function $f : [0, \infty) \rightarrow \mathbb{R}$ are given by

$$D^+ f(t) := \limsup_{h \downarrow 0} \frac{f(t+h) - f(t)}{h} \quad \text{and} \quad D_+ f(t) := \liminf_{h \downarrow 0} \frac{f(t+h) - f(t)}{h}.$$

Theorem 2.5.1 *Let B be a standard Brownian motion. A typical sample path $B(\omega)$ is nowhere differentiable. More precisely, the set*

$$\{\omega \in \Omega : \text{for each } t \in [0, \infty) \text{ either } D^+ B_t(\omega) = \infty \text{ or } D_+ B_t(\omega) = -\infty\}$$

contains an event F with $\mathbb{P}(F) = 1$.

PROOF: It is enough to consider the time interval $[0, 1]$. For fixed integers $j, k \geq 1$ we define the set

$$A_{j,k} = \bigcup_{t \in [0,1]} \bigcap_{h \in [0,1/k]} \{\omega \in \Omega : |B_{t+h}(\omega) - B_t(\omega)| \leq jh\}.$$

The union of the sets $A_{j,k}$ collects all those ω for which the path $t \mapsto B_t(\omega)$ has a locally Lipschitz continuous increment somewhere in $[0, 1]$. In particular,

$$\{\omega \in \Omega : -\infty < D_+ B_t(\omega) \leq D^+ B_t(\omega) < \infty \text{ for some } t \in [0, 1]\} \subset \bigcup_{j=1}^{\infty} \bigcup_{k=1}^{\infty} A_{j,k}.$$

⁷We recall that an a.s. limit of a sequence of measurable functions is measurable only if the underlying filtration is complete!

We will show that for any pair (j, k) there exists a set C of $\mathbb{P}$ -measure zero such that $A_{j,k} \subseteq C$. Since there are only countably many pairs $(j, k) \in \mathbb{N} \times \mathbb{N}$ this shows that there exists a set N of measure zero, such that

$$\{\omega \in \Omega : -\infty < D_+ B_t(\omega) \leq D^+ B_t(\omega) < \infty \text{ for some } t \in [0, 1]\} \subseteq N.$$

Let us fix a sample path $\omega \in A_{j,k}$. Our first goal is to show that the assumption $\omega \in A_{j,k}$ provides information about the size of the Brownian increments over *neighboring* intervals of the same length.

Indeed, by the definition of the set $A_{j,k}$ there exists $t \in [0, 1)$ such that

$$|B_{t+h}(\omega) - B_t(\omega)| \leq jh \quad \text{for all } h \in [0, 1/k]. \quad (2.7)$$

We now fix an integer $n \geq 4k$ and divide the interval $[0, 1]$ into equi-distant subintervals of length $\frac{1}{n}$. Then $\frac{i-1}{n} \leq t \leq \frac{i}{n}$ for some integer $1 \leq i \leq n$, and using that $n \geq 4k$,

$$\frac{i+v}{n} - t \leq \frac{v+1}{n} \leq \frac{1}{k} \quad \text{for } v = 1, 2, 3. \quad (2.8)$$

As a result, the quantities

$$h_1 := \frac{i}{n} - t \leq \frac{1}{n} \leq \frac{1}{k}, \quad \text{and} \quad h_2 := \frac{i+1}{n} - t \leq \frac{2}{n} \leq \frac{1}{k}$$

satisfy $h_1, h_2 \leq \frac{1}{k}$ and are hence “admissible” in equation (2.7). Using that $h_2 \leq 2/n$ and $h_1 \leq 1/n$ we see that the increment of the Brownian motion on the interval that contains t satisfies

$$|B_{(i+1)/n}(\omega) - B_{i/n}(\omega)| \leq |B_{(i+1)/n}(\omega) - B_t(\omega)| + |B_{i/n}(\omega) - B_t(\omega)| \leq \frac{2j}{n} + \frac{j}{n} = \frac{3j}{n}.$$

The assumption $\omega \in A_{j,k}$ also provides information about the size of the Brownian increment over the next two intervals. Applying (2.8) and (2.7) to h_2 and

$$h_3 := \frac{i+2}{n} - t \leq \frac{3}{n} \leq \frac{1}{k}$$

yields

$$\begin{aligned} |B_{(i+2)/n}(\omega) - B_{(i+1)/n}(\omega)| &\leq |B_{(i+2)/n}(\omega) - B_t(\omega)| + |B_{(i+1)/n}(\omega) - B_t(\omega)| \\ &\leq \frac{3j}{n} + \frac{2j}{n} = \frac{5j}{n} \end{aligned}$$

and applying it to h_3 and

$$h_4 := \frac{i+3}{n} - t \leq \frac{4}{n} \leq \frac{1}{k}$$

yields

$$\begin{aligned} |B_{(i+3)/n}(\omega) - B_{(i+2)/n}(\omega)| &\leq |B_{(i+3)/n}(\omega) - B_t(\omega)| + |B_{(i+2)/n}(\omega) - B_t(\omega)| \\ &\leq \frac{4j}{n} + \frac{3j}{n} = \frac{7j}{n}. \end{aligned}$$

With

$$C_i^{(n)} := \bigcap_{v=1}^3 \left\{ \omega \in \Omega : |B_{(i+v)/n}(\omega) - B_{(i+v-1)/n}(\omega)| \leq \frac{2v+1}{n} j \right\} \quad (2.9)$$

we therefore conclude that $A_{j,k} \subset \bigcup_{i=1}^n C_i^{(n)}$, for all $n \geq 4k$. Moreover, the random variables

$$\sqrt{n}(B_{(i+v)/n} - B_{(i+v-1)/n}) =: Z_v \quad (v = 1, 2, 3)$$

are independent and $\mathcal{N}(0, 1)$ distributed. Since

$$\mathbb{P}(|Z_v| \leq \varepsilon) \leq \varepsilon$$

it follows from (2.9) that

$$\mathbb{P}[C_i^{(n)}] \leq \frac{105j^3}{n^{3/2}} \quad \left(105 = \prod_{\nu=1}^3 (2\nu + 1)\right)$$

and hence that $\mathbb{P}[\bigcup_{i=1}^n C_i^{(n)}] = \mathcal{O}(n^{-1/2})$. Since

$$A_{j,k} \subseteq C := \bigcap_{n=4k}^{\infty} \bigcup_{i=1}^n C_i^{(n)}$$

the assertion follows from

$$\mathbb{P}[C] \leq \inf_{n \geq 4k} \mathbb{P}[\bigcup_{i=1}^n C_i^{(n)}] = 0.$$

□

Exercise 24 *Prove that a Brownian sample path is a.s. nowhere Hölder continuous with exponent $\alpha > \frac{1}{2}$.*

Hint. Follow the proof of nowhere differentiability.

2.5.2 The law of iterated logarithm

Brownian sample paths are unbounded. The law of iterated logarithm gives the exact growth rate of Brownian motion at 0 and ∞ . It requires the following auxiliary result on the tail of the normal distribution.

Lemma 2.5.2 *Let $X \sim \mathcal{N}(0, 1)$. For every $x > 0$, we have*

$$\frac{1}{\sqrt{2\pi}} \frac{1}{x + 1/x} e^{-x^2/2} \leq \mathbb{P}(X \geq x) \leq \frac{1}{\sqrt{2\pi}} \frac{1}{x} e^{-x^2/2}.$$

PROOF: Let φ be the density of the standard normal distribution. By partial integration ($u = 1/t, v' = t\varphi(t)$)

$$\mathbb{P}(X \geq x) = \int_x^{\infty} \frac{1}{t} (t\varphi(t)) dt = -\frac{1}{t} \varphi(t) \Big|_x^{\infty} - \int_x^{\infty} \frac{1}{t^2} \varphi(t) dt \leq \frac{1}{x} \varphi(x).$$

By analogy

$$\mathbb{P}(X \geq x) \geq \frac{1}{x} \varphi(x) - \frac{1}{x^2} \int_x^{\infty} \varphi(t) dt = \frac{1}{x} \varphi(x) - \frac{1}{x^2} \mathbb{P}(X \geq x),$$

from which we deduce the second inequality. □

Theorem 2.5.3 (Law of iterated logarithm) *For a Brownian motion B in $\mathbb{R}$ we have a.s.*

$$\limsup_{t \rightarrow 0} \frac{B_t}{\sqrt{2t \log \log(1/t)}} = \limsup_{t \rightarrow \infty} \frac{B_t}{\sqrt{2t \log \log t}} = 1.$$

PROOF: The inversion $t \mapsto tB_{1/t}$ converts the two formulas onto one another, so we only need to prove the result for $t \rightarrow \infty$.

- To prove “ $\leq$ ” we set $t_n = \alpha^n$ for some $\alpha > 1$ (we later let $\alpha \downarrow 1$) and put

$$f(t) := 2\alpha^2 \log \log t.$$

Then

$$\frac{f(t_n)}{2\alpha} = \alpha (\log(n \log \alpha)) = \alpha \log n + \alpha \log \log \alpha.$$

Let us also put

$$B_{a,b} = \{B_t\}_{a \leq t \leq b}$$

and denote by B^* the running maximum of the Brownian motion. Since $\mathbb{P}[B_t^* > a] = 2\mathbb{P}[B_t > a]$ it follows from the above lemma that up to a multiplicative constant

$$\begin{aligned} \mathbb{P}\left(\sup_{t_n, t_{n+1}} B_{t_n, t_{n+1}} > \sqrt{t_n f(t_n)}\right) &\leq \mathbb{P}\left(t_{n+1}^{-1/2} B_{t_{n+1}}^* > \sqrt{f(t_n)/\alpha}\right) \quad (\text{using that } t_{n+1} = \alpha t_n) \\ &= \mathbb{P}\left(B_1^* > \sqrt{f(t_n)/\alpha}\right) \\ &\leq \sqrt{\frac{\alpha}{f(t_n)}} e^{-f(t_n)/2\alpha} \\ &= (\log \alpha)^{-\alpha} \sqrt{\frac{\alpha}{f(t_n)}} n^{-\alpha} \\ &\leq n^{-\alpha} \quad \text{for } n \text{ large enough.} \end{aligned}$$

Since $\alpha > 1$, it follows from the Borel-Cantelli lemma that

$$\mathbb{P}\left(\sup_{t_n, t_{n+1}} B_{t_n, t_{n+1}} > \sqrt{t_n f(t_n)} \text{ infinitely often}\right) = 0.$$

Since the function $t \mapsto \sqrt{t f(t)}$ is non-decreasing this shows that a.s. for all $n \geq n^*(\omega)$

$$\sup_{t_n \leq t \leq t_{n+1}} \frac{B_t}{\sqrt{t f(t)}} \leq \sup_{t_n \leq t \leq t_{n+1}} \frac{B_t}{\sqrt{t_n f(t_n)}} \leq 1$$

and hence that

$$\limsup_{t \rightarrow \infty} \frac{B_t}{\sqrt{t f(t)}} \leq 1 \quad \text{a.s.}$$

Letting $\alpha \downarrow 1$ yields

$$\limsup_{t \rightarrow \infty} \frac{B_t}{\sqrt{2t \log \log t}} \leq 1.$$

- To prove “ $\geq$ ” we consider again the sequence $\{t_n\}$ and prove that for $\eta > 0$ we can find a large enough α such that

$$\limsup_{n \rightarrow \infty} \frac{B_{t_n}}{\sqrt{2t_n \log \log t_n}} \geq 1 - \eta \quad \text{a.s.} \quad (2.10)$$

We first note that the previously obtained upper bound applied to the Brownian motion $-B$ yields

$$\liminf_{t \rightarrow \infty} \frac{B_t}{\sqrt{2t \log \log t}} \geq -1 \quad (2.11)$$

and set

$$\beta := \frac{\alpha}{\alpha - 1}, \quad g(t) := \frac{2}{\beta^2} \log \log t.$$

We may assume w.l.o.g. that $\beta g(t_n) \geq 1$ and note that

$$e^{-\beta g(t_n)/2} = n^{-1/\beta} (\log \alpha)^{-1/\beta}, \quad \frac{t_n g(t_n)}{2 t_n \log \log t_n} = \beta^{-2}. \quad (2.12)$$

Since $t_n - t_{n-1} = \beta t_n$ it follows from the scaling property of Brownian motion that

$$\mathbb{P}(B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)}) = \mathbb{P}(B_1 > \sqrt{\beta g(t_n)}).$$

We now apply the lower bound in Lemma 2.5.2 to $x = \sqrt{\beta g(t_n)} \geq 1$. Since $(x + 1/x)^{-1} \geq \frac{1}{2x}$ for $x \geq 1$ we get that

$$\begin{aligned} \mathbb{P}(B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)}) &= \mathbb{P}(B_1 > \sqrt{\beta g(t_n)}) \\ &\geq \frac{1}{\sqrt{2\pi}} \frac{1}{2} \frac{1}{\sqrt{\beta g(t_n)}} e^{-\beta g(t_n)/2} \\ &= \frac{1}{\sqrt{2\pi}} \frac{1}{2} (\log \alpha)^{-1/\beta} \sqrt{\frac{1}{\beta g(t_n)}} n^{-1/\beta} \\ &\geq n^{-\varepsilon} n^{-1/\beta} \\ &\geq n^{-1} \end{aligned}$$

for $\varepsilon \in (0, 1 - 1/\beta)$ and $n \in \mathbb{N}$ large enough. Since the events

$$\{B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)}\}_{n \in \mathbb{N}}$$

are independent (this is important!) it follows from the Borel-Cantelli lemma that

$$\mathbb{P}(B_{t_n} - B_{t_{n-1}} > \sqrt{t_n g(t_n)} \text{ infinitely often}) = 1.$$

This allows us to bound first $B_{t_{n-1}}$ and then $\frac{B_{t_n}}{\sqrt{2 t_n \log \log t_n}}$ from below using (2.11) and the fact that

$$\frac{t_n \log \log t_n}{t_{n-1} \log \log t_{n-1}} \rightarrow \alpha.$$

In fact, by (2.11)

$$B_{t_{n-1}} > -(1 + \varepsilon) \alpha^{-1/2} \sqrt{2 t_n \log \log t_n} \quad \text{for almost all } n \in \mathbb{N} \text{ a.s.}$$

and by the second equation in (2.12)

$$\limsup_{n \rightarrow \infty} \frac{B_{t_n}}{\sqrt{2 t_n \log \log t_n}} \geq \frac{1}{\beta} - (1 + \varepsilon) \alpha^{-1/2} = \frac{\alpha - 1}{\alpha} - (1 + \varepsilon) \alpha^{-1/2}.$$

Choosing first α large and then ε small we obtain (2.10). This prove the desired lower bound.

□

2.5.3 Nowhere monotonicity

Next, we prove that a typical Brownian sample path (starting in zero) changes its sign infinitely often on any small time interval. From this we conclude that the Brownian sample paths are almost surely nowhere monotone. Our first result is Blumenthal's 0-1 law.

Proposition 2.5.4 (Blumenthal's 0-1 law) *Let B be a Brownian motion and $(\mathcal{F}_t)$ be its canonical filtration. The σ -field $\mathcal{F}_{0+} := \bigcap_{t>0} \mathcal{F}_t$ is trivial in the sense that*

$$\mathbb{P}(A) \in \{0, 1\} \quad \text{for all } A \in \mathcal{F}_{0+}.$$

PROOF: Let $A \in \mathcal{F}_{0+}$, let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and continuous and let $0 < t_1 < \dots < t_d$. By continuity and dominated convergence

$$\mathbb{E}[\mathbf{1}_A f(B_{t_1}, \dots, B_{t_d})] = \lim_{\varepsilon \rightarrow 0} \mathbb{E}[\mathbf{1}_A f(B_{t_1} - B_\varepsilon, \dots, B_{t_d} - B_\varepsilon)].$$

If $\varepsilon < t_1$, then the above increments are independent of $\mathcal{F}_\varepsilon$ and hence of $\mathcal{F}_{0+}$. Thus,

$$\mathbb{E}[\mathbf{1}_A f(B_{t_1}, \dots, B_{t_d})] = \lim_{\varepsilon \rightarrow 0} \mathbb{E}[\mathbf{1}_A f(B_{t_1} - B_\varepsilon, \dots, B_{t_d} - B_\varepsilon)] = \mathbb{P}(A) \mathbb{E}[f(B_{t_1}, \dots, B_{t_d})].$$

This shows that $\mathcal{F}_{0+} \perp \sigma(B_{t_1}, \dots, B_{t_d})$ and hence

$$\mathcal{F}_{0+} \perp \sigma(B_t, t > 0).$$

Since B_0 is deterministic,

$$\sigma(B_t, t > 0) = \sigma(B_t, t \geq 0)$$

and so $\mathcal{F}_{0+} \perp \mathcal{G}$ for any sub- σ -field $\mathcal{G} \subset \sigma(B_t, t \geq 0)$. Choosing $\mathcal{G} = \mathcal{F}_{0+}$ we see that

$$\mathcal{F}_{0+} \perp \mathcal{F}_{0+}$$

and hence $\mathcal{F}_{0+}$ is trivial. □

Corollary 2.5.5 *For any $\varepsilon > 0$,*

$$\sup_{0 \leq s \leq \varepsilon} B_s > 0, \quad \inf_{0 \leq s \leq \varepsilon} B_s < 0.$$

PROOF: Let $\varepsilon_n \downarrow 0$ and set $A := \bigcap_n \{\sup_{0 \leq s \leq \varepsilon_n} B_s > 0\}$. Since this is a monotone decreasing intersection, $A \in \mathcal{F}_{0+}$ and

$$\mathbb{P}(A) = \lim_{n \rightarrow \infty} \mathbb{P}(\sup_{0 \leq s \leq \varepsilon_n} B_s > 0).$$

At the same time, by the symmetry of Brownian motion

$$\mathbb{P}(\sup_{0 \leq s \leq \varepsilon_n} B_s > 0) \geq \mathbb{P}(B_{\varepsilon_n} > 0) = \frac{1}{2},$$

hence $\mathbb{P}(A) > 0$. Therefore, by Blumenthal's 0-1 law, $\mathbb{P}(A) = 1$. The proof for the infimum is similar. □

It follows from the above corollary that Brownian sample paths are almost surely nowhere monotone. In particular, they cannot be viewed as (random) distribution functions. Later, we show that Brownian sample paths also cannot be represented as the difference of monotone functions. Hence, we cannot integrate w.r.t. Brownian sample paths in the sense of real analysis.

Chapter 3

Quadratic Variation and the Basic Itô Formula

Let B be a Brownian motion on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Our goal is to define an integral of the form

$$\int_0^T B_t(\omega) dB_t(\omega).$$

Since $t \mapsto B_t(\omega)$ is a uniformly continuous function on $[0, T]$, the naive approach would be to take a partition $0 = t_0 < t_1, \dots, < t_n = T$ of $[0, T]$ and to discretise the sample paths. Both

$$\phi_1(t, \omega) = \sum_j B_{t_j}(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

and

$$\phi_2(t, \omega) = \sum_j B_{t_{j+1}}(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

“reasonably” approximate the sample path $t \rightarrow B_t(\omega)$.¹ Since ϕ_1 and ϕ_2 are piecewise constant the integrals

$$\int_0^T \phi_i(t, \omega) dB_t(\omega) \quad (i = 1, 2)$$

make sense if we define $\int_s^u dB_t = B_u - B_s$. However,

$$\mathbb{E} \left[\int_0^T \phi_1(t, \omega) dB_t(\omega) \right] = \sum_j \mathbb{E} [B_{t_j} (B_{t_{j+1}} - B_{t_j})] = 0$$

¹We shall see later that only the first approximation is indeed “reasonable”.

while, using the martingale property of Brownian motion ($\mathbb{E}[B_{t_{j+1}}|\mathcal{F}_{t_j}] = B_{t_j}$)

$$\begin{aligned}
\mathbb{E}\left[\int_0^T \phi_2(t, \omega) dB_t(\omega)\right] &= \sum_j \mathbb{E}[B_{t_{j+1}}(B_{t_{j+1}} - B_{t_j})] \\
&= \sum_j \mathbb{E}[(B_{t_{j+1}} - B_{t_j})^2 + B_{t_{j+1}}B_{t_j} - B_{t_j}^2] \\
&= \sum_j \mathbb{E}[(B_{t_{j+1}} - B_{t_j})^2] + \sum_j [\mathbb{E}[B_{t_j}\mathbb{E}[B_{t_{j+1}}|\mathcal{F}_{t_j}]] - \mathbb{E}B_{t_j}^2] \\
&= T.
\end{aligned}$$

This suggests that we cannot define the above integral in the usual sense. In what follows we explore the reasons why this naive construction fails.

3.1 Quadratic variation and the Stieltjes Integral

Let us fix a *continuous* function $X : [0, T] \rightarrow \mathbb{R}^n$. We denote the components of X by X^i ($i = 1, \dots, n$). Our goal is to clarify under which conditions we can define an integral with respect to X analogous to the usual Riemann integral. To this end, we first introduce the concept of quadratic variation. Subsequently we introduce the Stieltjes integral.

Definition 3.1.1 A partition Π of the time interval $[0, T]$ is a set of points $0 = t_0 < t_1 < \dots < t_n = T$. The mesh is

$$|\Pi| := \max_i |t_i - t_{i-1}|.$$

Definition 3.1.2 • The (total) variation of the function $X^i : [0, T] \rightarrow \mathbb{R}$ along the partition $\Pi = \{t_0, \dots, t_n\}$ of $[0, T]$ is defined by

$$V_T(X^i, \Pi) := \sum_{i=1}^n |X^i(t_j) - X^i(t_{j-1})|$$

The (total) variation on $[0, T]$ is defined by

$$V_T(X^i) := \sup \{V_T(X^i, \Pi) : \Pi \text{ is a partition of } [0, T]\}.$$

If $V_T(X^i) < \infty$, then X^i is of bounded/finite variation on $[0, T]$.

- The co-variation process of the functions $X^j, X^k : [0, T] \rightarrow \mathbb{R}$ along the partition $\Pi = \{t_0, \dots, t_n\}$ of $[0, T]$ is given by

$$V_t^2(X^j, X^k, \Pi) = \sum_{t_{i-1}, t_i \in \Pi, t_i \leq t} (X^j(t_i) - X^j(t_{i-1}))(X^k(t_i) - X^k(t_{i-1})).$$

The quadratic variation process of X^i along Π is given by

$$V_t^2(X^i, \Pi) = V_t^2(X^i, X^i, \Pi), \quad t \in [0, T].$$

- We say that X is of continuous quadratic variation on $[0, T]$ if the limit

$$\langle X^j, X^k \rangle_t = \lim_{n \rightarrow \infty} V_t^2(X^j, X^k, \Pi^n)$$

exists for all $j, k = 1, \dots, n$ along a sequence of partitions $\{\Pi^n\}$ with $|\Pi^n| \rightarrow 0$ as $n \rightarrow \infty$ and the map $t \mapsto \langle X, Y \rangle_t$ is continuous. The quadratic variation process $\langle X^i \rangle$ of the function X^i is defined by

$$\langle X^i \rangle := \langle X^i, X^i \rangle.$$

Notice that we require the quadratic variation to exist only along a given sequence of partition. For the special case of Brownian sample paths we shall show that the choice of partition is irrelevant as long as the mesh sizes converge to zero sufficiently fast. In general, this need not be the case, though.

Remark 3.1.3 *Strictly speaking, we should have defined our co-variation processes along a partition Π as*

$$V_t^2(X^j, X^k, \Pi) = \sum_{t_{i-1}, t_i \in \Pi} (X^j(t_i \wedge t) - X^j(t_{i-1} \wedge t))(X^k(t_i \wedge t) - X^k(t_{i-1} \wedge t)).$$

In other words, when computing quantities like $V_t^2(X^j, X^k, \Pi)$ we assume that $t \in \Pi$. Since we only consider continuous processes, a simple 3ε -argument shows that the two definitions result in the same limit if $|\Pi| \rightarrow 0$. Our slightly sloppy definition saves us a lot of notation. In the sequel we follow the convention that $t \in \Pi$ when computing (co-)variations.

Every monotone real valued function defined on a bounded interval is of bounded variation, and every real-valued function of bounded variation defined on a bounded interval is bounded. The quadratic variation process $\langle X \rangle$ of a continuous real valued function X is increasing and satisfies $\langle X \rangle_0 = 0$. If $X \in \mathcal{C}^1$ with $X_0 = 0$, then

$$X_t = \int_0^t X'_s ds \quad \text{and} \quad V_t(X) = \int_0^t |X'_s| ds.$$

The next proposition shows that X has quadratic variation zero if it is of bounded variation. We deduce that all monotone processes are of trivial quadratic variation.

Proposition 3.1.4 *If $X : [0, T] \rightarrow \mathbb{R}$ is continuous and of bounded variation, then its quadratic variation process is trivial, i.e., identically equal to zero*

PROOF: For any partition $\Pi^n = (t_0^n, \dots, t_{m_n}^n)$ we have that

$$\begin{aligned} \sum_{t_{i-1}^n, t_i^n \in \Pi^n, t_i^n \leq t} (X(t_i^n) - X(t_{i-1}^n))^2 &\leq \sup_i |X(t_i^n) - X(t_{i-1}^n)| \cdot \sum_{t_{i-1}^n, t_i^n \in \Pi^n, t_i^n \leq t} |X(t_i^n) - X(t_{i-1}^n)| \\ &\leq \sup_i |X(t_i^n) - X(t_{i-1}^n)| V_T(X). \end{aligned}$$

Since X is continuous and hence uniformly continuous on compact sets

$$\lim_{n \rightarrow \infty} \sup_i |X(t_i^n) - X(t_{i-1}^n)| = 0 \quad \text{if} \quad |\Pi^n| \rightarrow 0.$$

□

The following properties are easily verified.

Exercise 25 *Let $X, Y : [0, T] \rightarrow \mathbb{R}$ be continuous and of continuous quadratic variation with respect to a common sequence of partitions $\{\Pi_n\}$. Then the following holds:*

- (i) The quadratic variation $\langle X, Y \rangle$ exists and is continuous (along $\{\Pi_n\}$) if and only if $\langle X + Y \rangle$ exists and is continuous.
- (ii) $\langle \cdot, \cdot \rangle$ defines a (not necessarily positive semi-definite) bilinear form along $\{\Pi_n\}$.
- (iii) The polarization identity holds along $\{\Pi_n\}$:

$$\langle X, Y \rangle = \frac{1}{2} (\langle X + Y \rangle - \langle X \rangle - \langle Y \rangle).$$

In particular, co-variation processes are differences of two non-decreasing processes and hence of bounded variation (cf. Exercise 26).

- (iv) The Cauchy-Schwarz inequality holds along $\{\Pi_n\}$:

$$|\langle X, Y \rangle| \leq \sqrt{\langle X \rangle \langle Y \rangle}.$$

The polarization identity along with the Cauchy-Schwarz inequality yields the following corollary.

Corollary 3.1.5 *Let X be continuous with continuous variation and A be of bounded variation. Then*

$$\langle X + A \rangle = \langle X \rangle.$$

PROOF: Since A is of finite variation $\langle A \rangle = 0$ Cauchy-Schwarz yields $|\langle X, A \rangle| = 0$. Hence $\langle X + A \rangle = \langle X \rangle$, due to the polarization identity. $\square$

Exercise 26 *Prove that a real-valued function is of bounded variation if and only if it can be represented as the difference of two non-decreasing functions. In particular, (co-)variation processes are of bounded variation.*

Bounded variation functions can be linked to measure theory and hence integrals w.r.t. bounded variation functions are well defined. To see this, we recall basic facts from measure theory (Analysis III). Let $\mathcal{X}$ be a set and $\mathcal{A}$ be a σ -algebra in $\mathcal{X}$, and let $\mu : \mathcal{A} \rightarrow \mathbb{R}$ be a measure on $\mathcal{A}$, i.e. a countably additive function. The measure can be decomposed as $\mu = \mu^+ - \mu^-$ (Hahn-Jordan decomposition) where the measures $\mu^\pm$ are concentrated on disjoint sets $\mathcal{X}^\pm$ of $\mathcal{X}$ with

$$\mu^+(X) \geq 0, \quad \mu^-(X) \leq 0 \quad \text{for any } X \in \mathcal{X}^\pm$$

and at least one of the measures is finite. The *total variation* of μ is given by $|\mu| := \mu^+ + \mu^-$ and the *total variation norm* is given by $\|\mu\| := |\mu|(\mathcal{X})$.

A measure on the Borel-field $\mathcal{B}(\mathcal{X})$ is called a Borel measure. For a *bounded* Borel measure $\mu : \mathcal{B} \rightarrow \mathbb{R}$ on the real line we can define the distribution function $F_\mu(t) := \mu((-\infty, t))$. If the measure is non-negative, then F_μ is increasing; else it follows from the Hahn-Jordan decomposition that F_μ can be represented as the difference of two increasing functions. Since μ is bounded, it follows that F_μ is of bounded variation. Conversely, it follows from the preceding exercise that we can associated with every function $X : [0, T] \rightarrow \mathbb{R}$ of bounded variation a bounded *signed* measure $\mu : \mathcal{B}_T \rightarrow \mathbb{R}$. Indeed, since X is of bounded variation, $X = X^+ - X^-$ where $X^\pm$ are non-decreasing functions. We can associated a non-negative measure $\mu^\pm : \mathcal{B}_T \rightarrow \mathbb{R}_+$ with $X^\pm$ via $\mu^\pm([a, b)) := X^\pm(b) - X^\pm(a)$ and so $\mu := \mu^+ - \mu^-$ is a signed measure.

Remark 3.1.6 *In view of the above, we can define integrals of (e.g. bounded or continuous) functions with respect to functions of bounded variation. In particular, the integral $\int_0^T f(t)d\langle X \rangle_t$ is well defined for any continuous $f : [0, T] \rightarrow \mathbb{R}$ if X has continuous quadratic variation.*

Another approach to define this integral is the Riemann-Stieltjes integral, defined in terms of Riemann-Stieltjes sums. To this end, let

$$X, Y : [0, T] \rightarrow \mathbb{R}$$

be real-valued functions and $\Pi = \{t_0 < t_1 < \dots < t_n\}$ be a partition of $[0, T]$. Let $\xi = (\xi_1, \dots, \xi_n)$ be a vector of points $\xi_k \in [t_{k-1}, t_k]$. We call

$$S(X, Y, \Pi, \xi) := \sum_{k=1}^n Y(\xi_k) (X(t_k) - X(t_{k-1}))$$

a *Riemann-Stieltjes (RS) sum*. A sequence $\{S(X, Y, \Pi^n, \xi^n)\}_{n \in \mathbb{N}}$ is called a *RS sequence* if $|\Pi^n| \rightarrow 0$ as $n \rightarrow \infty$. If the sequence converges to some (and hence unique - the argument is the same as for Riemann sums) limit, then the limit is called the *Riemann-Stieltjes integral of Y with respect to X* . The limit is denoted

$$\int_0^T Y(s) dX(s).$$

The limit satisfies the same properties (linearity, ...) as the Riemann integral. Moreover, if Y is bounded, then $|\int Y dX| \leq \|Y\|_\infty V_T(X)$. The following lemma is easily established.

Lemma 3.1.7 *Let $X_1, X_2, Y : [0, T] \rightarrow \mathbb{R}$ be real-valued functions. If the integrals $\int Y dX_i$ exists for $i = 1, 2$, then the integral $\int Y d(X_1 + X_2)$ exist and*

$$\int Y d(X_1 + X_2) = \int Y dX_1 + \int Y dX_2.$$

The RS integral exists whenever Y is continuous and X is of bounded variation as shown by the following theorem.

Theorem 3.1.8 *Let $Y : [0, T] \rightarrow \mathbb{R}$ be continuous and $X : [0, T] \rightarrow \mathbb{R}$ be of bounded variation. Then the RS integral $\int Y dX$ exists.*

PROOF: Since X is of bounded variation, it can be decomposed into the difference of two non-decreasing functions: $X = X_+ - X_-$ where $X_\pm : [0, T] \rightarrow \mathbb{R}$ is non-decreasing. The existence of the integrals $\int Y dX_\pm$ can be established in exactly the same manner as the existence of the Riemann integral. The existence of the integral $\int Y dX$ then follows from the preceding lemma. $\square$

The analysis of this section suggests that a Brownian sample path is typically not of bounded variation. We analyse the quadratic variation of Brownian motion in the following section.

3.2 Quadratic Variation of Brownian motion

We are now going to compute the quadratic variation of Brownian motion. It turns out that the quadratic variation can be computed as a *deterministic* limit for almost every Brownian sample path. This will allow us to compute integrals with respect to Brownian motion for a class of integrands that

are smooth functions of the Brownian motion using *deterministic* arguments. In a later chapter we extend the definition of the integral to a larger class of integrands using *probabilistic* arguments.

Theorem 3.2.1 *Let $\{\Pi^n\}$ be a sequence of partitions of $[0, T]$ such that $\sum_n |\Pi^n| < \infty$. Then*

$$\lim_{n \rightarrow \infty} V_t^2(B(\omega), \Pi^n) = t, \quad \text{for all } t \in [0, T] \quad \mathbb{P}\text{-a.s.}$$

PROOF: In proving the assertion we shall assume that t belongs to the partitions Π^n . The arguments can be made precise using a standard ε -argument. In a first step we prove L^2 -convergence. To this end, observe that independence of the increments along with $\mathbb{E}[(B_{t_i} - B_{t_{i-1}})^2] = t_i - t_{i-1}$ yields

$$\begin{aligned} \mathbb{E} \left[\sum_{t_i \in \Pi^n} (B_{t_i} - B_{t_{i-1}})^2 - t \right]^2 &= \mathbb{E} \left[\sum_{t_i \in \Pi^n} \{ (B_{t_i} - B_{t_{i-1}})^2 - (t_i - t_{i-1}) \} \right]^2 \\ &= \sum_{t_i \in \Pi^n} \mathbb{E} \left[(B_{t_i} - B_{t_{i-1}})^2 - (t_i - t_{i-1}) \right]^2 \\ &= \sum_{t_i \in \Pi^n} \text{Var} \left((B_{t_i} - B_{t_{i-1}})^2 \right). \end{aligned}$$

Since

$$B_{t_i} - B_{t_{i-1}} \sim N(0, t_i - t_{i-1}) \quad \text{it follows that} \quad \text{Var} \left((B_{t_i} - B_{t_{i-1}})^2 \right) = 2(t_i - t_{i-1})^2.$$

Hence

$$\begin{aligned} \mathbb{E} \left[\sum_{t_i \in \Pi^n} (B_{t_i} - B_{t_{i-1}})^2 - t \right]^2 &= 2 \sum_{t_i \in \Pi^n} (t_i - t_{i-1})^2 \\ &\leq 2 \cdot |\Pi^n| \cdot t \end{aligned}$$

which converges to zero as $n \rightarrow \infty$. In particular, the quadratic variation $\langle B(\omega) \rangle_t$ exists for every $t \in [0, T]$ as an L^2 limit. This convergence holds without any additional assumptions on the partitions.

L^2 -convergence implies almost sure convergence along a subsequence but we actually need to show convergence of the entire sequence. For this we write

$$\mathbb{E} \left[V_t^2(B, \Pi^n) \right] = \mathbb{E} \left[\sum_{t_i \in \Pi^n} (B_{t_i} - B_{t_{i-1}})^2 \right] = \sum_{t_i \in \Pi^n} \mathbb{E} \left[(B_{t_i} - B_{t_{i-1}})^2 \right] = t.$$

Thus, by Tchebysheff's inequality we have for every $\varepsilon > 0$ that

$$\mathbb{P} \left[|V_t^2(B, \Pi^n) - t| > \varepsilon \right] \leq \frac{\text{Var}(V_t^2(B, \Pi^n))}{\varepsilon^2}.$$

Since a Brownian sample path has independent increments

$$\text{Var}(V_t^2(B, \Pi^n)) = \sum_{t_i \in \Pi^n} \text{Var} \left((B_{t_i} - B_{t_{i-1}})^2 \right) = 2 \sum_{t_i \in \Pi^n} (t_i - t_{i-1})^2 \leq 2T |\Pi^n|$$

so

$$\mathbb{P} \left[|V_t^2(B, \Pi^n) - t| > \varepsilon \right] \leq \frac{2T}{\varepsilon^2} |\Pi^n|.$$

Since $\sum_n |\Pi^n| < \infty$ it follows from the lemma by Borel and Cantelli that

$$\mathbb{P} \left[|V_t^2(B, \Pi^n) - t| > \varepsilon \text{ infinitely often} \right] = 0.$$

This shows that, for every $t \in [0, T]$ there exists a set N_t of measure zero such that

$$\lim_{n \rightarrow \infty} V_t^2(B(\omega), \Pi^n) = \langle B(\omega) \rangle_t = t \quad \text{for all } \omega \notin N_t.$$

Let us now put $\mathcal{N} := \bigcup_{t \in [0, T] \cap \mathbb{Q}} N_t$; this set has measure zero and

$$\lim_{n \rightarrow \infty} V_t^2(B(\omega), \Pi^n) = \langle B(\omega) \rangle_t = t \quad \text{for all } t \in [0, T] \cap \mathbb{Q}; \quad \omega \notin \mathcal{N}.$$

For any $t \notin [0, T] \cap \mathbb{Q}$ there are sequences $\{t_n\}, \{s_n\} \in [0, T] \cap \mathbb{Q}$ such that $t_n \uparrow t$ and $s_n \downarrow t$. Moreover, there exists a random variable n^* such that for any $\omega \notin \mathcal{N}$ and all $n \geq n^*(\omega)$,

$$t_n - \varepsilon \leq V_{t_n}^2(B(\omega), \Pi^n) \leq V_t^2(B(\omega), \Pi^n) \leq V_{s_n}^2(B(\omega), \Pi^n) \leq s_n + \varepsilon.$$

This proves the result. $\square$

Remark 3.2.2 *We emphasize that we used the same sequence of partition of every ω when computing the quadratic variation of Brownian sample paths. If we were allowed to choose different partitions for different sample paths, then we could obtain any quadratic variation, including the zero process. For instance, on can prove that for any continuous function $f : [0, 1] \rightarrow \mathbb{R}$ there exists a sequence of partitions $\{\Pi_n\}$ with mesh sizes going to zero, such that $V_t^2(f, \Pi_n) \rightarrow 0$ for all $t \in [0, 1]$.*

3.2.1 The basic Itô formula

For a given partition Π of the interval $[0, T]$ and a given continuous function $X : [0, T] \rightarrow \mathbb{R}^n$ of continuous quadratic variation we denote by

$$\Delta_i X^j := X_{t_{i+1}}^j - X_{t_i}^j$$

the increments of X^j along Π . If X were of bounded variation, then for every continuously differentiable function f on $\mathbb{R}^n$ we would have

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s.$$

If $\langle X \rangle \neq 0$, i.e., X is of unbounded variation, then we need a generalization that accounts for the “many small fluctuations generating the quadratic variation”. The Itô formula provides such a generalization. For its proof we first state without proof the following result on the weak convergence of signed measures².

Proposition 3.2.3 *Let $\{\mu_n\}$ be a sequence of signed measures on $[a, b]$. The sequence converges weakly to a signed measure μ if and only if $\sup_n \|\mu_n\| < \infty$ and every subsequence of the sequence of distribution functions $\{F_{\mu_n}\}$ contain a further subsequence that converges to F_μ at all points excepting points of an at most countable set.*

Corollary 3.2.4 *Let $\{\Pi^n\}$ be a sequence of partitions of $[0, T]$ with $|\Pi^n| \rightarrow 0$ and $g : [0, T] \rightarrow \mathbb{R}$ continuous. Then*

$$\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n, t_i \leq t} g(t_i) \Delta_i X^j \Delta_i X^k = \int_0^t g(s) d\langle X^j, X^k \rangle_s \quad \text{for all } t \in [0, \infty).$$

²For the special case of probability measures the theorem simplifies to the well known Portemonteau theorem that states that a sequence of probability measures $\{\mu_n\}$ converges weakly to a probability measure μ if and only if the associated sequence of distribution functions $\{F_{\mu_n}\}$ converges to F_μ at any point of continuity of F_μ .

PROOF: We can define signed measures μ_n and μ on the real line by

$$\mu_n(dt) := \sum_{t_i \in \Pi^n} \delta_{t_i}(dt) \Delta_i X^j \Delta_i X^k \quad \text{and} \quad \mu(dt) := d\langle X^j, X^k \rangle_t.$$

The associated distribution functions are

$$\mu_n([0, t]) = \sum_{t_i \leq t} \Delta_i X^j \Delta_i X^k \quad \text{and} \quad \mu([0, t]) = \langle X^j, X^k \rangle_t.$$

By assumption $\mu_n([0, t]) \rightarrow \mu([0, t])$ for every $t \in [0, T]$. Hence, the assertion follows from the above proposition along with the continuity of g and the fact that

$$\sum_{t_i \leq t} g(t_i) \Delta_i X^j \Delta_i X^k = \int_0^t g(s) \mu_n(ds) \quad \text{and} \quad \int_0^t g(s) d\langle X^j, X^k \rangle_s = \int_0^t g(s) \mu(ds).$$

□

Remark 3.2.5 *There is another way to prove Corollary 3.2.4 that does not require Proposition 3.2.3. The proof of Exercise 25(i) shows that we can decompose the measure μ_n as $\mu_n = \mu_n^+ - \mu_n^-$ where*

$$\mu_n^+(dt) := \sum_{t_i \in \Pi^n} \delta_{t_i}(dt) (\Delta_i(X^j + X^k))^2 \quad \text{and} \quad \mu_n^-(dt) := \sum_{t_i \in \Pi^n} \delta_{t_i}(dt) ((\Delta_i X^j)^2 + (\Delta_i X^k)^2)$$

are non-negative measures and hence probability measures after suitable scaling. Again by Exercise 25(i) we know that all the distribution functions converges pointwise. Convergence in $t = T$ shows that the total masses converge. Hence, the assertion follows from the Portemonteau theorem applied to the scaled measures. This argument works because we know that not only that the distribution function of μ_n but also the distribution functions associated with $\mu_n^\pm$ converge.

In what follows $C^{1,2}(\mathbb{R}^m \times \mathbb{R}^n)$ denotes the class of all continuous functions $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$ whose first the partial derivatives on $\mathbb{R}^m$ and whose first and second partial derivatives exist on $\mathbb{R}^n$ exist and are continuous. The notation $F \in C^{1,2}([0, T] \times \mathbb{R}^n)$ indicates that $F \in C^{1,2}((0, T) \times \mathbb{R}^n)$ and that all partial derivatives extend continuously to $[0, T] \times \mathbb{R}^n$. We drop the domain of the functions if there is no risk of confusion.

We are now ready to state and prove a first version of Itô's formula.

Theorem 3.2.6 (*Basic Itô formula*) *Let $X : [0, T] \rightarrow \mathbb{R}^n$ be continuous, of continuous QV, and let $A : [0, T] \rightarrow \mathbb{R}^m$ be continuous and of bounded variation along a sequence of partitions $\{\Pi^n\}$. Let $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$ be of class $C^{1,2}$. Then*

$$\begin{aligned} F(A_t, X_t) &= F(A_0, X_0) + \int_0^t \nabla_a F(A_s, X_s) dA_s + \int_0^t \nabla_x F(A_s, X_s) dX_s \\ &\quad + \frac{1}{2} \sum_{j,k=1}^n \int_0^t \frac{\partial^2}{\partial x^j \partial x^k} F(A_s, X_s) d\langle X^j, X^k \rangle_s \end{aligned} \tag{3.1}$$

with

$$\int_0^t \nabla_x F(A_s, X_s) dX_s := \lim_n \sum_{t_i} \nabla_x F(A_{t_i}, X_{t_i}) \Delta_i X.$$

*In particular, the integral $\int_0^t \nabla_x F(A_s, X_s) dX_s$ exists as a Riemann-Stieltjes-type integral.*³

³Do note that this is not(!) a Riemann-Stieltjes integral in the sense of the usual definition. We do not have the choice of sample points of the integrand as we always take the value of the integrand at the left end points of our intervals.

PROOF: We prove the theorem for $n = m = 1$; the general case follows from straightforward modifications. Let us put $F_a := \nabla_a F$, $F_x := \nabla_x F$ and $F_{xx} := \frac{\partial^2 F}{\partial x^2}$ and let $\{\Pi_n\}$ be a sequence of partitions of the interval $[0, T]$ whose mesh sizes tend to zero. Assuming for simplicity that $t \in \Pi^n$ we can write

$$\begin{aligned} F(A_t, X_t) - F(A_0, X_0) &= \sum_{t_i \in \Pi^n} \{F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_i})\} \\ &= \sum_{t_i \in \Pi^n} \{F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_{i+1}})\} \\ &\quad + \sum_{t_i \in \Pi^n} \{F(A_{t_i}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_i})\}. \end{aligned}$$

We study the two sums separately.

- As for the first sum we notice that because F is continuously differentiable in its first argument the mean value theorem yields $\tilde{a}_i \in (A_{t_i}, A_{t_{i+1}})^4$ such that

$$F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_{i+1}}) = F_a(\tilde{a}_i, X_{t_{i+1}}) \Delta_i A.$$

Since F_a, A and X are continuous, so is the function $g(t) := F_a(A_t, X_t)$, and the definition of the Riemann-Stieltjes integral yields

$$\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} g(\xi_i) \Delta_i A = \int_0^t g(s) dA_s$$

where $t_i \leq \xi_i \leq t_{i+1}$. Furthermore, by continuity of A , if $|\Pi_n|$ is small enough, then

$$\max_i |g(t_{i+1}) - F_a(\tilde{a}_i, X_{t_{i+1}})| < \varepsilon.$$

This yields that

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} \{F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_{i+1}})\} &= \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} F_a(\tilde{a}_i, X_{t_{i+1}}) \Delta_i A \\ &= \int_0^t F_a(A_s, X_s) dA_s. \end{aligned}$$

- Regarding the second sum a Taylor expansion w.r.t. the second variable yields

$$F(A_{t_i}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_i}) = F_x(A_{t_i}, X_{t_i}) \Delta_i X + \frac{1}{2} F_{xx}(A_{t_i}, X_{t_i}) (\Delta_i X)^2 + R_{i,n}.$$

The remainder $R_{i,n}$ (in Lagrange form) is given by

$$R_{i,n} = \frac{1}{2} (F_{xx}(A_{t_i}, \tilde{x}_i) - F_{xx}(A_{t_i}, X_{t_i})) (\Delta_i X)^2 \quad \text{for some } \tilde{x}_i \in (X_{t_i}, X_{t_{i+1}}).$$

Corollary 3.2.4 applied to the continuous function $g(t) = F_{xx}(A_t, X_t)$ yields

$$\lim_{n \rightarrow \infty} \frac{1}{2} \sum_{t_i \in \Pi^n} F_{xx}(A_{t_i}, X_{t_i}) (\Delta_i X)^2 = \frac{1}{2} \int_0^t F_{xx}(A_s, X_s) d\langle X \rangle_s.$$

⁴We use the following convention: for a non-monotone function g by $a \in (g_{t_i}, g_{t_{i+1}})$ we mean that $a \in (g_{t_i}, g_{t_{i+1}})$ if $g_{t_i} \leq g_{t_{i+1}}$ and that $a \in (g_{t_{i+1}}, g_{t_i})$ else.

Thus, we have established convergence of $\sum_{t_i} F_x(A_{t_i}, X_{t_i}) \Delta_i X$ if we can show that

$$\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} R_{i,n} = 0.$$

To this end, we observe that X is uniformly continuous on $[0, T]$ so $\max_i |X_{t_{i+1}} - X_{t_i}| \leq \delta_n \rightarrow 0$. Hence the remainder term can be estimated from above by

$$|R_{i,n}| \leq \max\{|F_{xx}(A_{t_i}, x) - F_{xx}(A_{t_i}, y)| : |x - y| \leq \delta_n, t_i \in \Pi^n\} (\Delta_i X)^2.$$

Continuity of F_{xx} and A yields $\max\{|F_{xx}(A_t, x) - F_{xx}(A_t, y)| : |x - y| \leq \delta_n; t \in [0, T]\} \rightarrow 0$ as $n \rightarrow \infty$. Since $\sum (\Delta_i X) \rightarrow \langle X \rangle_T$ this shows that

$$\sum_i |R_{i,n}| \rightarrow 0.$$

□

A special case of the preceding formula is $A_t = t$. In this case the mapping f is of the form $f(t, X_t)$.

Corollary 3.2.7 *Let $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ be of class $C^{1,2}$ and $X : [0, T] \rightarrow \mathbb{R}$ be continuous with continuous quadratic variation $\langle X \rangle$. Then*

$$f(t, X_t) - f(0, X_0) = \int_0^t f_t(s, X_s) ds + \int_0^t f_x(s, X_s) dX_s + \frac{1}{2} \int_0^t f_{xx}(s, X_s) d\langle X \rangle_s.$$

Another immediate result is the product rule which is obtained by applying the Itô formula to the function $f(x, y) = xy$.

Corollary 3.2.8 (*Product Rule*) *Let (X, Y) be continuous with continuous quadratic variation. Then*

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \langle X, Y \rangle_t.$$

The Itô formula also allows us to evaluate the integral of smooth functions of BM w.r.t. BM.

Example 3.2.9 *Let B be a standard Brownian motion and $F : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable. Then a.s.*

$$F(B_t) = F(0) + \int_0^t F'(B_s) dB_s + \frac{1}{2} \int_0^t F''(B_s) ds. \quad (3.2)$$

If $F(x) = x^n$, then

$$X_t^n = X_0^n + n \int_0^t X_s^{n-1} dX_s + \frac{n(n-1)}{2} X_s^{n-2} d\langle X \rangle_s.$$

In particular, we obtain a closed form solution for the integral of a Brownian path with respect to itself: for almost all $\omega \in \Omega$

$$B_t^2 = B_0^2 + 2 \int_0^t B_s dB_s + \int_0^t d\langle B \rangle_s = 2 \int_0^t B_s dB_s + t.$$

The following example is of particular interest in mathematical finance.

Example 3.2.10 (*Geometric Brownian Motion*) Given a BM B , and constants $\mu \in \mathbb{R}$ and $\sigma > 0$ the geometric Brownian motion is given by the process

$$S_t = S_0 \exp\left(\sigma B_t + \left(\mu - \frac{1}{2}\sigma^2\right)t\right).$$

In order to write this process in differential form we apply Itô's formula to the exponential function $x \mapsto e^x$ and to the process

$$X_t = \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t.$$

In view of Corollary 3.1.5 we have that $\langle X \rangle_t = \sigma^2 t$ so Itô's formula yields

$$\begin{aligned} S_t = S_0 \exp(X_t) &= S_0 + \int_0^t S_u dX_u + \frac{1}{2} \int_0^t S_u d\langle X \rangle_u \\ &= S_0 + \int_0^t S_u \left[\left(\mu - \frac{1}{2}\sigma^2\right) du + \sigma dB_u \right] + \frac{1}{2} \int_0^t \sigma^2 S_u du \\ &= S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u dB_u. \end{aligned}$$

In differential form the geometric Brownian motion can be written as

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t.$$

Thus, if we assume that a stock price grows at an average rate μ and that the growth rates fluctuate around their average rate according to σB_t , then the price dynamics follows a geometric Brownian motion. This is the assumption underlying the Black-Scholes option pricing model that we briefly consider in the next section.

3.2.2 The quadratic variation of Itô integrals

If $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable and $G' = g$, then it follows from the Itô formula that the Itô integral

$$Y_t := \int_0^t g(X_s) dX_s$$

is well defined and it is easy to show that the mapping $t \mapsto Y_t$ is continuous. The latter uses the mean-value theorem of RS integrals. For many applications it will be important to identify the quadratic variation of Y . To this end, we first prove the following result.

Proposition 3.2.11 *Let X be real-valued, continuous and of continuous quadratic variation and $f \in C^1$. Then the mapping $t \mapsto f(X_t)$ is of continuous quadratic variation and*

$$\langle f(X) \rangle_t = \int_0^t (f'(X_s))^2 d\langle X \rangle_s.$$

PROOF: We apply the Taylor formula in order to obtain

$$f(X_{t_{i+1}}) - f(X_{t_i}) = f'(X_{t_i})\Delta_i X + (f'(\tilde{x}_i) - f'(X_{t_i}))\Delta_i X$$

for some $\tilde{x}_i \in (X_{t_i}, X_{t_{i+1}})$. Taking the squares and summing over all points in time yields

$$\sum_{t_i} (f(X_{t_{i+1}}) - f(X_{t_i}))^2 = \sum_{t_i} (f'(X_{t_i}))^2 (\Delta_i X)^2 + \sum_{t_i} R_{i,n}$$

where

$$R_{i,n} = 2f'(X_{t_i}) (f'(\tilde{x}_i) - f'(X_{t_i})) (\Delta_i X)^2 + (f'(\tilde{x}_i) - f'(X_{t_i}))^2 (\Delta_i X)^2.$$

Thus, continuity of f' and X guarantee that

$$\sum_{t_i} R_{i,n} \leq C_n \sum_{t_i} (\Delta_i X)^2$$

for some sequence $C_n \rightarrow 0$. Hence the assertion follows because $\sum_{t_i} (\Delta_i X)^2 \rightarrow \langle X \rangle_t$. $\square$

As a corollary to the preceding proposition we obtain a formula for the quadratic variation of Itô integrals.

Corollary 3.2.12 *Let X be real-valued, continuous and of continuous quadratic variation and $g \in C^1$ and $G' = g$. Then the process*

$$Y_t := \int_0^t g(X_s) dX_s$$

has quadratic variation

$$\langle Y \rangle_t = \int_0^t g^2(X_s) d\langle X \rangle_s.$$

PROOF: By Itô's formula

$$Y_t = \int_0^t g(X_s) dX_s = G(X_t) - G(X_0) - \frac{1}{2} \int_0^t g'(X_s) d\langle X \rangle_s.$$

The integral on the r.h.s. is of bounded variation because the process $\langle X \rangle$ is increasing; specifically,

$$V_T \left(\int_0^\cdot g'(X_s) d\langle X \rangle_s, \Pi^n \right) \leq \|g'\|_\infty \langle X \rangle_T.$$

In particular, its quadratic variation is zero and so

$$\langle Y \rangle = \langle G(X) \rangle.$$

Hence, the assertion follows from the preceding proposition. $\square$

Example 3.2.13 *The quadratic variation process of the geometric Brownian motion*

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t$$

is given by the process

$$t \mapsto \int_0^t \sigma^2 S_u^2 du.$$

The preceding example suggests that if we can observe a geometric Brownian motion with unknown drift μ and unknown volatility σ^2 continuously over a small time interval, then we can infer the unknown σ^2 from our observations. In order to make this more precise, let us assume that we are given the discrete observations S_{t_i} ($i = 0, \dots, n$) where we assume for simplicity that $t_{i+1} - t_i = \delta$. The logarithmic increments can then be represented as

$$R_i := \ln \frac{S_{t_i}}{S_{t_{i-1}}} = \left(\mu - \frac{1}{2} \sigma^2 \right) \delta + \sigma \sqrt{\delta} Z_i$$

where $Z_1, \dots, Z_n$ are independent standard normal random variables. In particular, $R_1, \dots, R_n$ are i.i.d. normal random variables with mean $(\mu - \frac{1}{2} \sigma^2) \delta$ and variance $\sigma^2 \delta$.

Definition 3.2.14 For a given family $X_1, \dots, X_n$ random variables with common density function $f(\cdot, \theta)$ with unknown parameter θ the likelihood function is given by the function

$$L(\theta) := f(X_1, \dots, X_n, \theta).$$

The maximizer $\hat{\theta}$ of the function $L(\theta)$ (assuming it is unique) is called the maximum likelihood estimator (MLE) of θ , given $X_1, \dots, X_n$. The MLE is called consistent if $\hat{\theta} \rightarrow \theta$ a.s. if $n \rightarrow \infty$.

Exercise 27 (i) Let $X_1, \dots, X_n$ independent normal random variables with unknown mean μ and unknown volatility σ^2 . Show that the MLE is consistent and given by⁵

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i; \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2.$$

(ii) Let $S_{t_0}, \dots, S_{t_n}$ be the realizations of a geometric Brownian motion process with fixed sample frequency $\delta = t_i - t_{i-1}$ ($i = 1, \dots, n$) and unknown (μ, σ^2) . Show that

$$f(S_1, \dots, S_n, \mu, \sigma) = \left(\prod_{i=1}^n \frac{1}{S_i} \right) \frac{1}{\sigma^n (2\pi\delta)^{\frac{n}{2}}} \exp \left(-\frac{1}{2\sigma^2\delta} \sum_{i=1}^n (\ln S_{t_i} - \ln S_{t_{i-1}} - (\mu - \frac{1}{2}\sigma^2)\delta)^2 \right).$$

(iii) Show that the MLE for geometric Brownian motion is given by

$$\hat{\mu} = \frac{1}{2}\hat{\sigma}^2 + \frac{1}{\delta}\bar{R} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{\delta} \frac{1}{n} \sum_{i=1}^n (R_i - \bar{R})^2$$

where $R_i := \ln \frac{S_{t_i}}{S_{t_{i-1}}}$ and $\bar{R} := \frac{1}{n} \sum_{i=1}^n R_i$.

In order to see if the estimators for the unknown parameters of the geometric Brownian motion obtained in the previous exercise are consistent, let us first notice that since $n\delta = t_n - t_0$,

$$\frac{1}{\delta}\bar{R} = \left(\mu - \frac{1}{2}\sigma^2 \right) + \sigma \frac{B_{t_n} - B_{t_0}}{t_n - t_0} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{\delta} \frac{1}{n} \sum_{i=1}^n (R_i)^2 - \frac{1}{\delta} (\bar{R})^2.$$

If we hold the sampling frequency δ fix and increase the number of observation to infinity, then

$$\frac{1}{t_n - t_0} (B_{t_n} - B_{t_0}) = \frac{1}{n\delta} \sum_{i=1}^n (B_{t_i} - B_{t_{i-1}}) \rightarrow 0 \text{ a.s.}$$

by the independence of Brownian increments and the strong law of large numbers. Furthermore, again by the LLN, $\hat{\sigma}^2 \rightarrow \sigma^2$ and hence $\hat{\mu} \rightarrow \mu$ a.s. However, holding δ fix and increase the number of observations to infinity requires to observe the process S on an arbitrarily long time interval.

If we only observe S on a finite time interval, say $[0, T]$ and increase instead the sampling frequency, i.e. let $\delta \rightarrow 0$, the situation is very different. In this case,

$$\frac{1}{\delta}\bar{R} = \left(\mu - \frac{1}{2}\sigma^2 \right) + \sigma \frac{B_T - B_{t_0}}{T - t_0},$$

which does not longer converge because now T is a fixed finite constant. Thus, the MLE for the drift μ is no longer consistent. But the MLE for the volatility still is. This shows that it is much more difficult to estimate the drift of a geometric BM from a single finite sample path than to estimate the volatility.

Exercise 28 Prove that $\hat{\sigma}^2 \rightarrow \sigma^2$ a.s. and hence that $\hat{\mu} \rightarrow \mu + \sigma \frac{B_T - B_{t_0}}{T - t_0}$.

⁵The estimator $\hat{\sigma}$ is consistent but not unbiased, that is, its expected value does not equal σ .

3.3 Quadratic variation of continuous local martingales

So far, we proved that integrals of smooth functions of Brownian motion w.r.t. Brownian sample paths exist almost surely. The proof heavily relied on the path-wise existence of the quadratic variation of Brownian motion. In this section, we consider the quadratic variation of continuous local martingales of which BM is a particular example. We prove that the quadratic variation exists as a limit in probability. This will later allow us to analyse more general stochastic integrals.

3.3.1 Local martingales

We start with the definition a local martingale. Loosely speaking an adapted stochastic process is a local martingale if it is a martingale when suitably stopped.

Definition 3.3.1 *A stochastic process $M = (M_t)$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ is called a local martingale if there exists a localising sequence of stopping times $\{\tau_n\}$, that is, a sequence of stopping times $\{\tau_n\}$ such that*

- $\tau_n \uparrow \infty$ a.s.
- the stopped process $M^{\tau_n} = M_{\tau_n \wedge \cdot}$ is a martingale w.r.t. $(\mathcal{F}_t)$ for every $n \in \mathbb{N}$.

The space of all continuous local martingales is denoted $\mathcal{M}_{loc}$; the space of continuous local martingales starting in zero is denoted $\mathcal{M}_{loc}^0$.

Localising sequences are not unique. For instance, if $\{\tau_n\}$ is a localising sequence, then so is $\{\tau_n \wedge n\}$.

Remark 3.3.2 *If M is a continuous local martingale with localizing sequences $\{\tau_n\}$, we may set*

$$\sigma_n := \inf\{t \geq 0 : M_t = n\} \wedge n; \quad \mu_n := \tau_n \wedge \sigma_n.$$

Since M is continuous $\sigma_n \uparrow \infty$ and so $\mu_n \uparrow \infty$. Moreover, since M^{τ_n} is a continuous martingale and σ_n is a bounded stopping time it follows from optional sampling that

$$M^{\mu_n} = (M^{\tau_n})^{\sigma_n}$$

is a bounded, hence uniformly integrable martingale. Thus, whenever we are working with continuous local martingales, we may w.l.o.g. assume that the martingale is uniformly integrable after stopping.

Obviously, any martingale is a local martingale: in this case we can take $\tau_n \equiv +\infty$ for all $n \in \mathbb{N}$. But not every local martingale is a (true) martingale; even uniform integrability is not enough. We give an example in Chapter 4.

Let us give two conditions that guarantee that a local martingale is a true martingale.

Lemma 3.3.3 *Let M be a local martingale such that the running maximum $M_t^* \in L^1$ for all $t \geq 0$. Then M is a martingale.⁶ In particular, every bounded local martingale is a martingale.*

PROOF: Let $\{\tau_n\}$ be a localising sequence and $s \leq t$. Then $\mathbb{E}[M_{t \wedge \tau_n} | \mathcal{F}_s] = M_{s \wedge \tau_n}$. Since $|M_{t \wedge \tau_n}| \leq |M_t^*|$ dominated convergence yields the desired result. $\square$

⁶Warning: The condition $\sup_t \mathbb{E}|M_t| < \infty$ is not sufficient for M to be a martingale.

It turns out that continuous local martingales are martingales if and only if the stopped processes have certain uniform integrability properties. For this we define what it means for a local martingale to be of “class DL”.

Definition 3.3.4 Let $\mathcal{T}$ be the class of all finite-valued stopping times and $\mathcal{T}_T$ be the class of all stopping times bounded by $T > 0$. A stochastic process X is said to be of class D, respectively of class DL if the set

$$\{X_\tau : \tau \in \mathcal{T}\}, \quad \text{respectively,} \quad \{X_\tau : \tau \in \mathcal{T}_T\}$$

is uniformly integrable for all $T > 0$.

Each uniformly integrable martingale is of class D, in particular of class DL. Conversely, if a local martingale M is of class DL, then it is a martingale and a uniformly integrable martingale if it is of class D. In fact, for any localising sequence $\{\tau_n\}$ we have that for any $s \leq t$ that

$$M_{s \wedge \tau_n} = \mathbb{E}[M_{t \wedge \tau_n} | \mathcal{F}_s].$$

The stopping times $s \wedge \tau_n, t \wedge \tau_n$ ($n \in \mathbb{N}$) are uniformly bounded. Since M is of class DL, the sequences $\{M_{s \wedge \tau_n}\}_{n \in \mathbb{N}}$ and $\{M_{t \wedge \tau_n}\}_{n \in \mathbb{N}}$ are hence uniformly integrable. Hence, they converge to M_s respectively M_t both a.s. and in L^1 and so the limit passes through the above conditional expectation. This shows the martingale property. If the local martingale M is of class D, it is a-fortiori uniformly integrable and hence a uniformly integrable martingale. We actually have the following slightly stronger result.

Proposition 3.3.5 A continuous local martingale M is a martingale if and only if it is of class DL.

PROOF: If M is a continuous martingale, then Doob’s optional sampling theorem yields that

$$M_\tau = \mathbb{E}[M_T | \mathcal{F}_\tau] \quad \text{for all } \tau \in \mathcal{T}_T.$$

In particular $\{M_\tau : \tau \in \mathcal{T}_T\}$ is uniformly integrable. The converse has already been shown. $\square$

Exercise 29 Let X be a non-negative local martingale with $\mathbb{E}X_0 < \infty$. Show that X is a supermartingale.

Exercise 30 Let X be a right continuous submartingale. Show that under any of the following conditions the set

$$\{X_\tau : \tau \in \mathcal{T}_T\}$$

is uniformly integrable for all $T \geq 0$.

- $X_t \geq 0$ a.s. for all $t \geq 0$.
- X admits the Doob-Meyer decomposition $X_t = M_t + A_t$ where M is a martingale and A is an increasing adapted process.

We say that a stopping time τ reduces a process M if the stopped process M^τ is a uniformly integrable martingale.

Exercise 31 Let M, N be local martingales and let σ, τ be stopping times. Show the following results

- If τ reduces M and $\sigma \leq \tau$ a.s., then σ reduces M .

- The sum $M + N$ is a local martingale.
- If σ and τ both reduce M , then so does the $\sigma \vee \tau$.
- The processes M^τ and $M^\tau \mathbf{1}_{\{\tau > 0\}}$ are local martingales.
- Let X be a càdlàg process and let $\{\tau_n\}$ be a sequence of stopping times such that $\tau_n \uparrow \infty$ a.s. and such that $X^{\tau_n} \mathbf{1}_{\{\tau_n > 0\}}$ is a local martingale for each $n \in \mathbb{N}$. Then X is a local martingale.

Hint. Use the Optional Sampling Theorem.

Let X be a stochastic process. We say that a property holds locally if there exists a sequence of stopping times $\{\tau_n\}$ such that $\tau_n \uparrow \infty$ a.s. such that $X^{\tau_n} \mathbf{1}_{\{\tau_n > 0\}}$ has this property for all $n \in \mathbb{N}$. It follows from the preceding exercise that a process that is locally a local martingale is also a local martingale. The following exercise discusses other examples of local properties.

Exercise 32 • Let X be a process that is locally a square integrable martingale. Then X is a local martingale.

- Let X be an adapted, càdlàg process and let $\{\tau_n\}$ be a sequence of stopping times such that $\tau_n \uparrow \infty$ a.s. If $X^{\tau_n} \mathbf{1}_{\{\tau_n > 0\}}$ is a martingale for each $n \in \mathbb{N}$, then X is a local martingale.

3.3.2 The quadratic variation process

If M is a local martingale, then M^2 is a local sub-martingale, and hence it has increasing expected values. The next theorem states that we can compensate for the increase in expected values by subtracting the quadratic variation of the trajectories of M , which exist as limits in probability. In the case of Brownian motion they exist as almost sure limits.

Theorem 3.3.6 Let M be a continuous local square integrable ($\mathbb{E}M_t^2 < \infty$ for all $t \geq 0$) martingale with respect to a filtration that satisfies the usual conditions. Let $M_0 = 0$. Then there exists an almost surely unique continuous process $\langle M \rangle$ (called the bracket process, compensator or quadratic variation process) such that

- $\langle M \rangle_0 = 0$ and $\langle M \rangle$ is non-decreasing
- $M^2 - \langle M \rangle$ is a continuous local martingale (a martingale if M is)
- for any sequence of partitions $\{\Pi^n\}$ of $[0, t]$ with $|\Pi^n| \rightarrow 0$,

$$V_t^2(M(\omega), \Pi^n) \rightarrow \langle M \rangle_t \text{ uniformly on compact time intervals in probability (ucp)}$$

The proof is given below. It follows from the proof that if M is a square integrable continuous martingale, then $M^2 - \langle M \rangle$ is a continuous martingale and

$$\mathbb{E}M_t^2 = \mathbb{E}\langle M \rangle_t, \quad t \geq 0.$$

Using Theorem 3.3.6 along with the (uniform) continuity of M (on compact time intervals) yields the following result.

Corollary 3.3.7 *Let A be a process of bounded variation, and let M be a continuous local martingale starting in zero. Then*

$$V_t^2(M(\omega) + A(\omega), \Pi^n) \rightarrow \langle M \rangle_t \text{ ucp.}$$

Thus, we can write

$$\langle M + A \rangle = \langle M \rangle.$$

We also have the following result on the bracket process of stopped local martingales.

Corollary 3.3.8 *Let M be a continuous square integrable local martingale and τ be a stopping time. Then the bracket of the stopped process M^τ is given by*

$$\langle M^\tau \rangle_t = \langle M \rangle_{\tau \wedge t}.$$

PROOF: By the optional sampling theorem, the process $t \mapsto M_{\tau \wedge t}^2 - \langle M \rangle_{\tau \wedge t}$ is a local martingale. Hence the assertion follows by the uniqueness of the bracket process. $\square$

The following exercise encourages you to review several important results on uniform integrable martingales and their long run behavior that have been established in Section 1.

Exercise 33 *Let M be a continuous, square integrable local martingale and assume that $\mathbb{E}\langle M \rangle_\infty < \infty$. Then the following holds.*

- i) M is a martingale and both M and M^2 are uniformly integrable.
- ii) The limit M_∞ exists a.s. and $\mathbb{E}M_\infty^2 = \mathbb{E}\langle M \rangle_\infty$.
- iii) We may take the right-continuous modification of

$$Z_t := \mathbb{E}[M_\infty^2 | \mathcal{F}_t] - M_t^2$$

which is a right-continuous (by appropriate choice of the modification), non-negative supermartingale and satisfies $\mathbb{E}[Z_t] \rightarrow 0$ for $t \rightarrow \infty$

Hint. You may use without proof that for any continuous local martingale M and any stopping time τ it holds that $\mathbb{E}[M_\tau^2] \leq \mathbb{E}\langle M \rangle_\tau$ where $M_\infty^2 := \liminf_{t \rightarrow \infty} M_t^2$. Also, recall that an L^2 -bounded family of random variables is uniformly integrable.

The covariation process and orthogonality of martinagles

For (right-)continuous, local square integrable martingales M and N we define their crossvariation $\langle M, N \rangle$ through the polarisation identity:

$$\langle M, N \rangle := \frac{1}{4} (\langle M + N \rangle - \langle M - N \rangle). \quad (3.3)$$

Clearly, the $(M, N) \mapsto \langle M, N \rangle$ is bilinear and symmetric. It is not difficult to show following result. The proof is left as an exercise.

Proposition 3.3.9 *Let M, N be continuous, square integrable local martingales. Let $\{\Pi^n\}$ be any sequence of partitions of $[0, t]$ with $|\Pi^n| \rightarrow 0$. Then*

$$\sum_{i=1}^m (M_{t_{i+1}^n} - M_{t_i^n})(N_{t_{i+1}^n} - N_{t_i^n}) \rightarrow \langle M, N \rangle_t \quad \text{ucp.}$$

Remark 3.3.10 *For $0 \leq s < t$ the following defines a symmetric positive definite bilinear form as well:*

$$\langle M, N \rangle_s^t := \langle M, N \rangle_t - \langle M, N \rangle_s.$$

Hence it follows from the usual Cauchy-Schwarz inequality that a.s.

$$|\langle M, N \rangle_t - \langle M, N \rangle_s| \leq \sqrt{\langle M \rangle_t - \langle M \rangle_s} \sqrt{\langle N \rangle_t - \langle N \rangle_s}.$$

This implies that

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |\langle M, N \rangle_s| \right] \leq \left(\mathbb{E} \sqrt{\langle M \rangle_t} \right) \left(\mathbb{E} \sqrt{\langle N \rangle_t} \right). \quad (3.4)$$

It is easy to see that

$$MN - \langle M, N \rangle = \frac{(M + N)^2 - \langle M + N \rangle}{4} - \frac{(M - N)^2 - \langle M - N \rangle}{4},$$

which is the difference of two continuous local martingales and hence a continuous local martingale. We call M, N orthogonal if $\langle M, N \rangle \equiv 0$. Since for any $t \geq s$,

$$\mathbb{E}[(M_t - M_s)(N_t - N_s) | \mathcal{F}_s] = \mathbb{E}[M_t N_t - M_s N_s | \mathcal{F}_s] = \mathbb{E}[\langle M, N \rangle_t - \langle M, N \rangle_s | \mathcal{F}_s]$$

we see that orthogonality of square integrable martingales means their product is martingale and their increments are conditionally uncorrelated.

The converse is also true as shown by the following lemma. A continuous local martingale w.r.t. a filtration that satisfies the usual conditions is constant if it is of bounded variation. Hence, if the filtration satisfies the usual conditions, then the above equality shows that orthogonality of M, N is equivalent to the product MN being a martingale.

Lemma 3.3.11 *Let M be a continuous local martingale with respect to a filtration that satisfies the usual conditions starting in 0. If M is of bounded variation (that is, $V_t(M(\omega)) < \infty$ a.s. for all $t \geq 0$), then $\mathbb{P}[M_t = 0 \text{ for all } t \geq 0] = 1$.*

PROOF: By the continuity of the sample paths, it is enough to prove that $\mathbb{P}[M_t = 0] = 1$ for all $t \geq 0$. To this end, we proceed in three steps.

Let us first assume that M is a continuous square integrable martingale and satisfies $V_t(M(\omega)) \leq C$ for almost all $\omega \in \Omega$. Then, for any partition $0 = t_1 < \dots < t_n = t$,

$$\mathbb{E} M_t^2 = \mathbb{E} \left[\sum_i \Delta_i M \right]^2 = \sum_i \mathbb{E} [\Delta_i M]^2 \leq C \mathbb{E} \left[\sup_i |\Delta_i M| \right]$$

By continuity, $\sup_i |\Delta_i M| \rightarrow 0$ a.s. if $|\Pi^n| \rightarrow 0$. Moreover, by Doob's maximal inequality

$$\mathbb{E} \left[\sup_{n \in \mathbb{N}, t_i \in \Pi^n, t_i \leq t} |\Delta_i M|^2 \right] \leq 2 \mathbb{E} \left[\sup_{0 \leq u \leq t} |M_u|^2 \right] \leq C \mathbb{E} |M_t|^2 < \infty.$$

In particular, $\sup_{n \in \mathbb{N}, t_i \in \Pi^n} |\Delta_i M| \in L^2$. Hence, by dominated convergence $\mathbb{E}M_t^2 = 0$ and $\mathbb{P}(M_t = 0) = 1$.

Let us now assume that M is a continuous square integrable martingale of bounded (yet not uniformly bounded) variation and put

$$\tau^n := \inf\{t \geq 0 : V_t(M(\omega)) > n\}.$$

Then

$$\{t \leq \tau^n\} = \{V_t(M(\omega)) \leq n\} \in \mathcal{F}_t.$$

Thus τ^n is an optional time and hence a stopping time because the filtration satisfies the usual conditions. By optional sampling, the stopped process M^{τ^n} is a martingale and by the first step $\mathbb{E}M_{\tau^n \wedge t}^2 = 0$. Using that $\tau^n \uparrow \infty$ (on the set where V_t is bounded, $\tau^n = \infty$ for $n \geq n^*(\omega)$, and on the set where V_t is unbounded $\tau_n \uparrow \infty$) we obtain from $\mathbb{E}|M_t^*|^2 < \infty$ that $\mathbb{E}|M_t|^2 = 0$.

Let us finally assume that M is a continuous local martingale with localising sequence $\{\tau^n\}$. We may w.l.o.g. assume that M^{τ^n} is a bounded martingale; else we can define

$$\sigma^n := \tau^n \wedge \inf\{t \geq 0 : |M_t| \geq n\},$$

which is again a localising sequence. Hence $\mathbb{P}(M_{t \wedge \sigma^n} = 0) = 1$. Since $\sigma^n \uparrow \infty$ a.s. dominated convergence yields $\mathbb{P}(M_t = 0) = 1$. $\square$

Exercise 34 Let M be a continuous local martingale with continuous quadratic variation and τ be a stopping time. If $\langle M \rangle_\tau = 0$, then $\mathbb{P}[X_{\tau \wedge t} = 0, t \in [0, \infty)] = 1$.

Some useful martingale properties

In what follows we will repeatedly use the following property of a square integrable martingale M : for every $0 \leq s \leq t \leq u < v < \infty$,

$$\begin{aligned} \mathbb{E}[(M_v - M_u)^2 | \mathcal{F}_t] &= \mathbb{E}[M_v^2 - M_u^2 | \mathcal{F}_t] \\ \mathbb{E}[(M_v - M_u)(M_t - M_s)] &= 0. \end{aligned} \tag{3.5}$$

In particular

$$\mathbb{E}[M_t(M_v - M_u)] = 0, \quad 0 \leq t \leq u \leq v.$$

Hence, when squaring the sums of martingale increments and taking the expectation, we can neglect the cross-product terms. In particular,

$$\mathbb{E}[(M_v - M_u)^2 | \mathcal{F}_t] = \mathbb{E}[\langle M \rangle_v - \langle M \rangle_u | \mathcal{F}_t].$$

Thus, the terms

$$M_v^2 - \langle M \rangle_v - (M_u^2 - \langle M \rangle_u) \quad \text{and} \quad (M_v^2 - M_u^2) - (\langle M \rangle_v - \langle M \rangle_u)$$

have the same conditional expectation given $\mathcal{F}_t$, namely zero. In what follows, for a given partition $\Pi = \{t_0 < t_1 < \dots < t_m\}$,

$$\Delta_i M := M_{t_i} - M_{t_{i-1}}, \quad \text{for } i = 1, \dots, m$$

3.3.3 Proof of Theorem 3.3.6 for bounded martingales

In this section we prove Theorem 3.3.6 under the additional assumption that M is a continuous martingale (not just a local martingale) and uniformly bounded; that is we assume in this section that $|M_t| \leq K$, for some $K > 0$. We start with the following auxiliary result.

Lemma 3.3.12 *Let M be a martingale bounded by K with $M_0 = 0$. For any partition $\Pi = \{0 = t_0 < t_1 < \dots < t_m = t\}$ of $[0, t]$,*

$$\mathbb{E} \left[\left[V_t^2(M, \Pi) \right]^2 \right] \leq 8K^4.$$

PROOF: We have that

$$\mathbb{E} \left[V_t^2(M, \Pi) \right]^2 = \mathbb{E} \left[\sum_{i=1}^m (\Delta_i M)^4 \right] + 2 \mathbb{E} \left[\sum_{k=1}^{m-1} \sum_{j=k+1}^m (\Delta_j M)^2 (\Delta_k M)^2 \right]$$

Using (3.5) (resulting in a telescope sum) and the fact that $M_0 = 0$ we get

$$\mathbb{E} \left[\sum_{i=1}^m (\Delta_i M)^4 \right] \leq 4K^2 \mathbb{E} \left[\sum_{i=1}^m (\Delta_i M)^2 \right] \leq 4K^2 \mathbb{E} M_t^2 \leq 4K^4.$$

Furthermore, for $k \leq m-1$,

$$\mathbb{E} \left[\sum_{j=k+1}^m (\Delta_j M)^2 | \mathcal{F}_{t_k} \right] = \mathbb{E} \left[\sum_{j=k+1}^m (M_{t_j}^2 - M_{t_{j-1}}^2) | \mathcal{F}_{t_k} \right] \leq 2K^2.$$

Hence, we obtain for the second term that

$$\begin{aligned} 2 \mathbb{E} \left[\sum_{k=1}^{m-1} \sum_{j=k+1}^m (\Delta_j M)^2 (\Delta_k M)^2 \right] &= 2 \mathbb{E} \left[\sum_{k=1}^{m-1} (\Delta_k M)^2 \mathbb{E} \left[\sum_{j=k+1}^m (\Delta_j M)^2 | \mathcal{F}_{t_k} \right] \right] \\ &\leq 4K^2 \mathbb{E} \left[\sum_{k=1}^{m-1} (\Delta_k M)^2 \right] \leq 4K^4. \end{aligned}$$

□

Corollary 3.3.13 *Let M be a bounded continuous martingale starting in zero, and let $V_t^4(M, \Pi)$ denote the sum of the fourth moments of the increments along the partition Π of $[0, t]$. Then,*

$$\lim_{|\Pi^n| \rightarrow 0} \mathbb{E} V_t^4(M, \Pi^n) = 0.$$

PROOF: For a given partition Π let

$$m_t(M, \Pi) := \sup \{ |M_r - M_s| : 0 \leq r < s \leq t, |s - r| \leq |\Pi| \}.$$

Since M is uniformly continuous on $[0, t]$, we get $\lim_{|\Pi^n| \rightarrow 0} m_t(M, \Pi^n) = 0$ a.s. Since M is uniformly bounded, the convergence also holds in L^1 and for any power of m . For any partition Π^n ,

$$V_t^4(M, \Pi^n) \leq m_t^2(M, \Pi^n) V_t^2(M, \Pi^n).$$

By Hölder's inequality,

$$\mathbb{E} V_t^4(M, \Pi^n) \leq \sqrt{\mathbb{E} [m_t^4(M, \Pi^n)]} \sqrt{\mathbb{E} [V_t^2(M, \Pi^n)]^2}.$$

Thus, the result follows from the fact that $V_t^2(M, \Pi^n)$ is L^2 bounded (by the preceding lemma). □

Construction of the bracket process

For bounded, continuous martingales we are now going to construct the process $\langle M \rangle$ as an L^2 limit (in the set of set of continuous functions) of quadratic increments along an *increasing* sequence of partitions $\{\Pi^n\}$, $\Pi^n = \{t_0^n < t_1^n < \dots < t_{m_n}^n\}$. Specifically, let

$$A_t^n := \sum_{t_i^n \leq t} (M_{t_i^n} - M_{t_{i-1}^n})^2 + (M_t - M_{t_{k_n(t)}^n})^2,$$

where $k_n(t) := \max\{i : t_i^n \leq t\}$ and put

$$N_t^n := M_t^2 - A_t^n.$$

Writing M_t^2 as a telescope sum of differences of squares and using the the fact that

$$M_t^2 - M_s^2 - (M_t - M_s)^2 = 2M_s(M_t - M_s)$$

we see that

$$N_t^n = 2 \sum_{i=1}^{k_n(t)} M_{t_{i-1}^n} (M_{t_i^n} - M_{t_{i-1}^n}) + 2M_{t_{k_n(t)}^n} (M_t - M_{t_{k_n(t)}^n}).$$

From this, it is not difficult to see that N^n is a continuous martingale. Hence, for any two partitions $\Pi^n \subseteq \Pi^m$, the difference $(N^n - N^m)$ is a continuous martingale and it follows from Doob's maximal inequality that

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |N_s^n - N_s^m|^2 \right] \leq C \mathbb{E} [|N_t^n - N_t^m|^2]. \quad (3.6)$$

Since $\Pi^n \subseteq \Pi^m$ we can write

$$\dots < t_{i-1}^n = t_{j_{n_i}}^m < t_{j_{n_i+1}}^m < \dots < t_{j_{n_{i+1}}}^m = t_i^n < \dots$$

Moreover, assuming w.l.o.g. that t belongs to both Π^n and Π^m) and putting $\Delta_j^m M = M_{t_j^m} - M_{t_{j-1}^m}$ the variable N_t^n can be expressed as

$$N_t^n = 2 \sum_i \sum_{j=n_i+1}^{n_{i+1}} M_{t_{i-1}^n} \Delta_j^m M$$

while N_t^m can be expressed as

$$N_t^m = 2 \sum_i \sum_{j=n_i+1}^{n_{i+1}} M_{t_{j-1}^m} \Delta_j^m M.$$

As a result,

$$N_t^n - N_t^m = 2 \sum_i \sum_{j=n_i+1}^{n_{i+1}} (M_{t_{i-1}^n} - M_{t_{j-1}^m}) \Delta_j^m M.$$

By direct computation using the fact that we take increments over non-overlapping intervals, we can verify that the cross terms vanish if the take the expectation of the square of the sum, and an

application of the Cauchy-Schwarz inequality yields

$$\begin{aligned} \mathbb{E}[|N_t^n - N_t^m|^2] &= 4\mathbb{E}\left[\sum_t \sum_{j=n_i+1}^{n_{i+1}} (M_{t_{i-1}^n} - M_{t_{j-1}^m})^2 (\Delta_j^m M)^2\right] \\ &\leq 4\mathbb{E}\left[\sup_{i \in \{1, \dots, m_n\}, j \in \{n_i+1, \dots, n_{i+1}\}} |M_{t_{i-1}^n} - M_{t_{j-1}^m}|^2 \sum_j (\Delta_j^m M)^2\right] \\ &\leq 4\sqrt{\mathbb{E}\left[\sup_{i \in \{1, \dots, m_n\}, j \in \{n_i+1, \dots, n_{i+1}\}} |M_{t_{i-1}^n} - M_{t_{j-1}^m}|^4\right]} \sqrt{\mathbb{E}\left[\sum_j (\Delta_j^m M)^2\right]^2}. \end{aligned}$$

Boundedness along with continuity of M implies that the first term tends to zero if $n, m \rightarrow \infty$. The second term is bounded, due to Lemma 3.3.12. In view of (3.6) this proves that

$$\lim_{m, n \rightarrow \infty} \mathbb{E}\left[\sup_{0 \leq s \leq t} |N_s^n - N_s^m|^2\right] = 0.$$

In particular, $\{N^m\}$ forms a Cauchy sequence in $L^2(\Omega, C([0, T], \mathbb{R}))$ for all $T \geq 0$, where $C([0, T], \mathbb{R})$ is equipped with the supremum norm. Hence there exists $N \in L^2(\Omega, C([0, T], \mathbb{R}))$ such that

$$N^m \rightarrow N \quad \text{in } L^2(\Omega, C([0, T], \mathbb{R})).$$

The martingale property of N will follow from more general results proved in below; see the proof of Theorem 4.2.2. If we now define

$$\langle M \rangle_t := M_t^2 - N_t,$$

then $M^2 - \langle M \rangle$ is a continuous martingale and $A_t^n \rightarrow \langle M \rangle_t$ in L^2 uniformly on compact time intervals. This proves the existence of the bracket process for bounded martingales.

Uniqueness of the bracket process

To see the uniqueness of the bracket process, let us assume that there exist two continuous, non-decreasing processes A, A' starting in zero such that $M^2 - A$ and $M^2 - A'$ are continuous local martingales. Then $A - A'$ is a continuous local martingale of bounded variation (as the difference of two non-decreasing processes) starting in zero. Hence it follows from Lemma 3.3.11 that $A = A'$ a.s.

3.3.4 Proof of Theorem 3.3.6

Let us now consider the general case of a continuous local martingale M that is not uniformly bounded. Let $\{\tau^n\}$ be a localising sequence and put

$$M_t^n := M_{\tau^n \wedge t}.$$

We can w.l.o.g. assume that M^n is bounded; else we can consider the localising sequence

$$\sigma^n := \tau^n \wedge \inf\{t \geq 0 : |M_t| \geq n\},$$

Let $\langle M^n \rangle$ be the process constructed in the previous section. By construction and the uniqueness of the bracket process we have for all $n \geq m$ that

$$\langle M^n \rangle \mathbf{1}_{[0, \tau^m]} = \langle M^m \rangle \mathbf{1}_{[0, \tau^m]}.$$

The sequence $\langle M^n \rangle_t$ is increasing in $n \in \mathbb{N}$ because $\tau^n \uparrow \infty$. Hence, we can define the limit

$$\langle M \rangle_t := \lim_{n \rightarrow \infty} \langle M^n \rangle_t.$$

The limit is continuous as a pointwise limit of continuous functions, and finite on $[0, \tau^m]$ for all $m \in \mathbb{N}$ because $\langle M \rangle \mathbf{1}_{[0, \tau^m]} = \langle M^m \rangle \mathbf{1}_{[0, \tau^m]}$. Moreover,

$$(M^2 - \langle M \rangle)_{\tau^m \wedge \cdot} = ((M^m)^2 - \langle M^m \rangle)_{\tau^m \wedge \cdot}.$$

is a martingale because the term on the right-hand side is a martingale without stopping and hence a martingale after stopping by the optional sampling theorem. This proves that $M^2 - \langle M \rangle$ is a local martingale. The facts that $\langle M \rangle_0 = 0$ and that $\langle M \rangle$ is non-decreasing follow from construction. Uniqueness of such a process follows again from Lemma 3.3.11.

To prove the convergence in probability of $V_t^2(M, \Pi^m)$ to $\langle M \rangle_t$ for any $t \geq 0$ we recall that the following holds:

•

$$V_t^2(M^n, \Pi^m) \rightarrow \langle M^n \rangle_t \quad \text{uniformly in } L^2 \text{ and hence in probability as } m \rightarrow \infty$$

•

$$M^n \equiv M \text{ on } [0, \tau^n] \quad \text{and hence} \quad V^2(M^n, \Pi^m) \equiv V^2(M, \Pi^m) \text{ on } [0, \tau^n] \text{ for all } m \in \mathbb{N}.$$

Since $\{\tau^n\}$ is a localizing sequence, we know that

$$\lim_{n \rightarrow \infty} \mathbb{P}[\tau^n < t] = 0.$$

In particular, for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that⁷

$$\mathbb{P}[\tau^n \leq t] \leq \varepsilon \quad \text{for all } n \geq n(\varepsilon).$$

Hence, for $n \geq n(\varepsilon)$

$$\mathbb{P}[\langle M \rangle_t = \langle M^n \rangle_t \text{ on } [0, t]] \geq 1 - \varepsilon$$

and

$$\mathbb{P}[V^2(M^n, \Pi^m) \equiv V^2(M, \Pi^m) \text{ on } [0, t] \text{ for all } m \in \mathbb{N}] \geq 1 - \varepsilon.$$

As a result, there exists $m_0 = m(\varepsilon/3, n(\varepsilon/3)) \in \mathbb{N}$ such that for all $m \geq m_0$ the set

$$\begin{aligned} \left\{ \sup_{0 \leq t \leq T} |V_t^2(M, \Pi^m) - \langle M \rangle_t| < \varepsilon \right\} &\supseteq \left\{ \sup_{0 \leq t \leq T} |V_t^2(M, \Pi^m) - V_t^2(M^{n(\varepsilon/3)}, \Pi^m)| < \varepsilon/3 \right\} \\ &\quad \cap \left\{ \sup_{0 \leq t \leq T} |V_t^2(M^{n(\varepsilon/3)}, \Pi^m) - \langle M^{n(\varepsilon/3)} \rangle_t| < \varepsilon/3 \right\} \\ &\quad \cap \left\{ \sup_{0 \leq t \leq T} |\langle M^{n(\varepsilon/3)} \rangle_t - \langle M \rangle_t| < \varepsilon/3 \right\} \end{aligned}$$

has measure at least $1 - \varepsilon$. This proves the desired convergence in probability of the quadratic variation to the bracket process $\langle M \rangle$. $\square$

Remark 3.3.14 Notice that the process $\langle M \rangle$ can be defined as an L^2 -limit for bounded martingales but only as a limit in probability for unbounded, respectively local martingales.

⁷This is the reason why we can only expect convergence in probability.

Chapter 4

The Stochastic Integral

Let us fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a filtration $(\mathcal{F}_t)$ that satisfies the usual conditions and a continuous local $(\mathcal{F}_t)$ -martingale M . Our goal is to define an integral process

$$t \mapsto \int_0^t f_t(\omega) dM_t(\omega)$$

for a large class of processes f using probabilistic arguments. If $M = B$ and $f_t(\omega) = f(t, B_t(\omega))$ for a sufficiently smooth function f , then this integral can be defined ω -by- ω as shown in the previous section. It was Itô's fundamental insight that one can generally *not* proceed in a pathwise way to define stochastic integrals. Instead, one needs to take a functional-analytic point of view, applying Hilbert spaces. As a warm-up we first introduce the stochastic integral w.r.t. Brownian motion. Subsequently, we show how to extend the main ideas to continuous local martingales.

4.1 Stochastic integration with respect to Brownian motion

We start with the stochastic integral w.r.t. Brownian motion. The definition of the stochastic integral is obvious for *elementary functions*.

Definition 4.1.1 *We call a process $\phi = (\phi_t)$ simple/elementary if it is of the form*

$$\phi_t(\omega) = \sum_j e_j(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

where $0 = t_0 < t_1 < t_2 < \dots$, where e_j is $\mathcal{F}_{t_j}$ -measurable and where $\sup_{i, \omega} |e_j(\omega)| < \infty$. The class of all such functions is denoted $\mathcal{L}_0$.

For any elementary function ϕ and every $T > 0$ we put¹

$$I_T(\phi)(\omega) = \int_0^T \phi_t(\omega) dB_t := \sum_{t_j \leq T} e_j(\omega) [B_{t_{j+1}} - B_{t_j}](\omega).$$

¹We use again our convention on the sums along partitions and assume w.l.o.g. that the end point is an element of the partition. What we really mean is this: $I_T(\phi)(\omega) = \int_0^T \phi_t(\omega) dB_t := \sum_{t_j} e_j(\omega) [B_{T \wedge t_{j+1}} - B_{T \wedge t_j}](\omega)$.

The key insight is that the norm of an elementary function equals the norm of the stochastic integral. This is known as the *Itô isometry for elementary processes*.

Lemma 4.1.2 (*Itô isometry*) *Let ϕ be an elementary function. Then*

$$\mathbb{E} \left[\left(\int_0^T \phi_t(\omega) dB_t \right)^2 \right] = \mathbb{E} \left[\int_0^T \phi_t^2(\omega) dt \right].$$

PROOF: Let $\Delta_j B := B_{t_{j+1}} - B_{t_j}$ and fix $i \leq j \in \mathbb{N}$. Since e_i and e_j are measurable with respect to $\mathcal{F}_{t_i}$ and $\mathcal{F}_{t_j}$, respectively, we have for $i = j$ that

$$\begin{aligned} \mathbb{E}[e_i e_j \Delta_i B \Delta_j B] &= \mathbb{E}[\mathbb{E}[e_j^2 (\Delta_j B)^2 | \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[e_j^2 \mathbb{E}[(\Delta_j B)^2 | \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[e_j^2] (t_{j+1} - t_j) \end{aligned}$$

while for $i < j$ we obtain

$$\begin{aligned} \mathbb{E}[e_i e_j \Delta_i B \Delta_j B] &= \mathbb{E}[\mathbb{E}[e_i e_j \Delta_i B \Delta_j B | \mathcal{F}_{t_j}]] \\ &= \mathbb{E}[e_i \Delta_i B e_j \mathbb{E}[\Delta_j B | \mathcal{F}_{t_j}]] \\ &= 0. \end{aligned}$$

Hence

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^T \phi_t(\omega) dB_t \right)^2 \right] &= \sum_{i,j} \mathbb{E}[e_i e_j \Delta_i B \Delta_j B] \\ &= \sum_j \mathbb{E}[e_j^2] (t_{j+1} - t_j) \\ &= \mathbb{E} \left[\left(\int_0^T \phi_t^2(\omega) dt \right) \right]. \end{aligned}$$

□

Construction of the Itô integral

The second fundamental insight is that the stochastic integral for simple processes is a martingale. More precisely, it belongs to the class $\mathcal{M}_2^{c,0}$ of all square integrable, continuous martingales that start in zero. The mapping $\langle \cdot, \cdot \rangle$ introduced in (3.3) defines a bilinear form on $\mathcal{M}_2^{c,0}$.

Proposition 4.1.3 *For any $\phi \in \mathcal{L}_0$ the integral process $I(\phi)$ belongs to $\mathcal{M}_2^{c,0}$.*

PROOF: By definition the integral process $I(\phi)$ starts in 0 and is adapted. Square integrability has already been established in the preceding lemma. The martingale property can be seen as follows: for all $s > t$,

$$\begin{aligned} \mathbb{E}[I_s(\phi) | \mathcal{F}_t] &= \mathbb{E} \left[\int_0^t \phi_u dB_u | \mathcal{F}_t \right] + \mathbb{E} \left[\int_t^s \phi_u dB_u | \mathcal{F}_t \right] \\ &= \int_0^t \phi_u dB_u + \mathbb{E} \left[\int_t^s \phi_u dB_u | \mathcal{F}_t \right] \quad (\text{by the definition of the integral}) \\ &= \int_0^t \phi_u dB_u + \mathbb{E} \left[\sum_{t \leq t_j < t_{j+1} \leq s} e_j \Delta_j B | \mathcal{F}_t \right]. \end{aligned}$$

Since e_j is $\mathcal{F}_{t_j}$ -adapted we see that $\mathbb{E}[e_j \Delta_j B | \mathcal{F}_t] = \mathbb{E}[e_j \mathbb{E}[\Delta_j B | \mathcal{F}_{t_j}] | \mathcal{F}_t] = 0$, and so

$$\mathbb{E}[I_s(\omega) | \mathcal{F}_t] = \int_0^t \phi_u dB_u = I_t(\omega).$$

□

So far, we have defined (pointwise for every $\omega \in \Omega$) a linear mapping

$$I : \mathcal{L}_0 \rightarrow \mathcal{M}_2^{c,0}.$$

In what follows we extend this mapping to a larger domain. We recall from Chapter 1 that the time integral of a progressively measurable process yields a progressively measurable process. In particular, the expected value of the time integral of the squared process is well defined.

Definition 4.1.4 Let $\mathcal{L}_2$ be the class of all jointly measurable and $(\mathcal{F}_t)$ -adapted functions $f_t(\omega) : [0, \infty) \times \Omega \rightarrow \mathbb{R}$ such that

$$\|f\|_{L_T^2(\lambda \otimes \mathbb{P})} := \mathbb{E} \left[\int_0^T f_t^2(\omega) dt \right] < \infty$$

for all $T > 0$. We equip the space $\mathcal{L}_2$ with the (semi-)norm

$$\|f\|_{L^2(\lambda \otimes \mathbb{P})} := \sum_{n=1}^{\infty} 2^{-n} \max\{1, \|f - g\|_{L_n^2(\lambda \otimes \mathbb{P})}\}.$$

Proposition 4.1.5 The inclusion $\mathcal{L}_0 \subseteq \mathcal{L}_2$ is dense.

PROOF: Let $f \in \mathcal{L}_2$ and $T > 0$. We need to find a sequence $\{\phi^n\}$ of simple processes such that

$$\|f - \phi^n\|_{L_T^2(\lambda \otimes \mathbb{P})} < \varepsilon \quad \text{for all } n \geq n_0.$$

Then $d(f, \phi^n) < \varepsilon$ if T is large enough. To this end, we proceed in three steps.

- (i) Let us assume that f is uniformly bounded in (t, ω) and that the function $t \mapsto f_t(\omega)$ is a.s. continuous. Then the sequence of adapted, measurable processes

$$\phi_t^n(\omega) = \sum_j f_{t_j}(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t), \quad t \leq T$$

converges to $f \lambda \otimes \mathbb{P}$ -a.s. and in L_T^2 by dominated convergence.

- (ii) Let us now drop the assumption of continuity and assume only that f is uniformly bounded in (t, ω) and in addition that f is progressively measurable. Let

$$f_t^m(\omega) := m \int_{(t-1/m)^+}^t f_s(\omega) ds = m [K_t(\omega) - K_{(t-1/m)^+}(\omega)]$$

where

$$K_t(\omega) := \int_0^{t \wedge T} f_s(\omega) ds.$$

By Exercise 3 the uniformly bounded, continuous process K is progressively measurable, hence jointly measurable and adapted. By Step (i) there exist sequences $\{\phi^{m,n}\}$ of simple processes for every $m \in \mathbb{N}$ such that

$$\phi^{m,n} \rightarrow f^m \quad \text{in } L_T^2 \quad \text{as } n \rightarrow \infty.$$

Since f is bounded it is hence enough to prove that $f^m \rightarrow f$ $\lambda \otimes \mathbb{P}$ -a.s. Then L_T^2 -convergence follows from dominated convergence. To this end, let²

$$\mathcal{A} := \{(t, \omega) \in [0, T] \times \Omega : \lim f_t^m(\omega) = f_t(\omega)\}^c \in \mathcal{B}_T \otimes \mathcal{F}_T$$

and let

$$\mathcal{A}_\omega := \{t \in [0, T] : (t, \omega) \in \mathcal{A}\}.$$

By Fubini's theorem $\mathcal{A}_\omega \in \mathcal{B}_T$. Moreover, for all $t \in \mathcal{A}_\omega$

$$f_t^m(\omega) = m \int_{(t-1/m)^+}^t f_s(\omega) ds \rightarrow f_t(\omega).$$

By the Lebesgue differentiation theorem this implies that $\lambda(\mathcal{A}_\omega^c) = 0$ and hence that

$$(\lambda \otimes \mathbb{P})(\mathcal{A}) = 0.$$

- (iii) Let us now assume that f is uniformly bounded but drop the assumption of progressive measurability. In this case, we do not know a priori that the continuous process

$$F_t := \int_0^t f_s ds$$

is adapted, hence progressively measurable by continuity. However, by Proposition 1.1.9 we do now that f has a progressively measurable modification g . Let

$$G_t := \int_0^t g_s ds.$$

Then G is progressively measurable. It is also a modification of F . Indeed, by Fubini

$$\mathbb{E} \left[\int_0^t |f_s - g_s| ds \right] = \int_0^t \mathbb{E}[|f_s - g_s|] ds = 0.$$

Hence

$$\int_0^t |f_s - g_s| ds = 0 \quad \text{a.s.}$$

and so $G_t = F_t$ a.s. for all $t \geq 0$. Since G is a modification of F and our filtration satisfies the usual conditions, we conclude that F is adapted. Since adaptedness and continuity imply progressive measurability, we can repeat verbatim the argument in (ii).

- (iv) Let f be progressively measurable. We can define bounded approximating functions by:

$$h_t^n(\omega) = \begin{cases} -n & \text{if } f_t(\omega) < -n \\ f_t(\omega) & \text{if } -n \leq f_t(\omega) \leq n \\ n & \text{if } f_t(\omega) > n \end{cases}.$$

□

²Notice that for a sequence of real-valued measurable mappings $\{g_n\}$ the set $\mathcal{A} := \{\lim g_n = 0\}$ is measurable because $\mathcal{A} = \bigcap_m \bigcup_k \bigcap_{n \geq k} \{|g_n| < 1/m\}$ and $\{|g_n| < 1/m\}$ is measurable.

Remark 4.1.6 Let us denote by $\mathcal{L}^*$ the subset of all (equivalence classes of) progressively measurable processes in $\mathcal{L}_2$. The proof of the preceding lemma actually shows that $\mathcal{L}_0 \subset \mathcal{L}^* \subset \mathcal{L}_2$ dense.

The next result shows that the stochastic integral for simple processes extends uniquely to the set $\mathcal{L}_2$.

Proposition 4.1.7 The mapping $I : \mathcal{L}_0 \rightarrow \mathcal{M}_2^{c,0}$ extends uniquely to a continuous mapping

$$J : \mathcal{L}_2 \rightarrow \mathcal{M}_2^{c,0}$$

such that $I = J$ on $\mathcal{L}_0$ and such that J has almost surely continuous sample paths.

PROOF: Let us fix $f \in \mathcal{L}_2$ and a sequence of elementary processes $\{\phi^n\}$ that approximates f in the L^2 -sense.

- **EXISTENCE.** By the Itô isometry for simple processes, the sequence $\{I_t(\phi^n)\}$ forms a Cauchy sequence in $L^2(\mathbb{P})$ for any $t \geq 0$. By the completeness of $L^2(\mathbb{P})$, this shows that there exists a random variable $Y_t \in L^2(\mathbb{P})$ such that $I_t(\phi^n) \rightarrow Y_t$ in $L^2(\mathbb{P})$.
- **UNIQUENESS.** To show that the random variables Y_t are uniquely defined, let $\{\psi^n\}$ be another sequence that approximates f in the L^2 -sense. Then the isometry for elementary functions yields

$$\mathbb{E} \left[\left(\int_0^t \phi_s^n dB_s - \int_0^t \psi_s^n dB_s \right)^2 \right] = \mathbb{E} \left[\int_0^t (\phi_s^n - \psi_s^n)^2 ds \right] \rightarrow 0.$$

- **CONTINUITY OF THE SAMPLE PATHS.** We defined the process (Y_t) separately for each $t \geq 0$. We now prove that this process has a continuous modification.

We know already that $I(\phi^n)$ is a square integrable $(\mathcal{F}_t)$ -martingale with continuous sample paths. Hence $I(\phi^n) - I(\phi^m)$ is also an $(\mathcal{F}_t)$ martingale with continuous sample paths so Doob's maximal inequality yields

$$\mathbb{P} \left[\sup_{0 \leq t \leq T} |I_t(\phi^n) - I_t(\phi^m)| > \varepsilon \right] \leq \frac{1}{\varepsilon^2} \mathbb{E}[|I_T(\phi^n) - I_T(\phi^m)|^2] \rightarrow 0 \quad \text{as } n, m \rightarrow \infty.$$

Hence we can choose a subsequence (n_k) such that

$$\mathbb{P} \left[\sup_{0 \leq t \leq T} |I_t(\phi^{n_k}) - I_t(\phi^{n_{k+1}})| > 2^{-k} \right] \leq 2^{-k}.$$

Thus, the Borel-Cantelli lemma shows that

$$\mathbb{P} \left[\sup_{0 \leq t \leq T} |I_t(\phi^{n_k}) - I_t(\phi^{n_{k+1}})| > 2^{-k} \text{ infinitely often} \right] = 0.$$

In particular, for almost all $\omega \in \Omega$, the sequence $\{I(\phi^{n_k})\}_{k \in \mathbb{N}}$ forms a Cauchy sequence in the space $C[0, T]$ of continuous real-valued functions on $[0, T]$, equipped with the supremum norm.³ Since this space is complete, there exists a limit process M that has almost surely continuous sample paths such that

$$I(\phi^{n_k}) \rightarrow M \quad \text{in } C[0, T] \text{ a.s.}$$

³We recall that a sequence $\{x_n\}$ in a metric space is a Cauchy sequence if $\sum_n d(x_n, x_{n+1}) < \infty$.

In particular, $I_t(\phi^{n_k}) \rightarrow M_t$ a.s. for all $t \in [0, T]$. This shows that

$$M_t = Y_t \quad \mathbb{P}\text{-a.s.} \quad \text{for all } t \in [0, T].$$

The above arguments assumed a finite time interval $[0, T]$. In particular, the null-set $\mathcal{N}_T$ depends on T . To finish the proof, we just consider a sequence of compact time intervals $[0, N]$ for $N \in \mathbb{N}$ to obtain a stochastic process $(M_t)_{t \geq 0}$ with almost surely continuous sample paths on $(\bigcup_n \mathcal{N}_N)^c$. We define $J(f) = M$.

- **MARTINGALE PROPERTY.** We know that $\mathbb{E}[I_s(\phi^n)|\mathcal{F}_t] = I_t(\phi^n)$, $0 \leq t \leq s < \infty$. In particular,

$$\lim_{n \rightarrow \infty} \mathbb{E}[I_s(\phi^n)|\mathcal{F}_t] = J_t(f) \quad \text{in } L^2.$$

At the same time Jensen's inequality yields

$$\begin{aligned} \mathbb{E} \left[(\mathbb{E}[I_s(\phi^n)|\mathcal{F}_t] - \mathbb{E}[J_s(f)|\mathcal{F}_t])^2 \right] &= E \left[(\mathbb{E}[I_s(\phi^n) - J_s(f)|\mathcal{F}_t])^2 \right] \\ &\leq E \left[\mathbb{E}[(I_s(\phi^n) - J_s(f))^2 | \mathcal{F}_t] \right] \\ &= \mathbb{E}[(I_s(\phi^n) - J_s(f))^2] \\ &\rightarrow 0. \end{aligned}$$

- **CONTINUITY OF J .** This follows from the construction of the stochastic integral along with the Itô isometry for simple processes.

□

The process

$$J_t(f) =: \int_0^t f_s dB_s, \quad 0 \leq t < \infty$$

is called the *stochastic integral* (or Itô integral) of f with respect to Brownian motion. The following properties of the Itô integral are easily verified for elementary functions and hence for all $f \in \mathcal{L}_2$:

- (i) For any $0 \leq c \leq T$ we have $\int_0^T f dB = \int_0^c f dB + \int_c^T f dB$.
- (ii) Itô integrals are linear: For any constant c and all $f, g \in \mathcal{L}_2$ we have $\int (cf + g) dB = c \int f dB + \int g dB$.
- (iii) Expected values of Itô integrals are zero: $\mathbb{E}[\int f dB] = 0$.
- (iv) Itô integrals are adapted: The random variable $\int_0^t f dB$ is $\mathcal{F}_t$ -measurable.

The following exercise shows that the representation of Itô integrals is a.s. unique.

Exercise 35 Let $f, g \in \mathcal{L}_2$ and C, D be constants. Let

$$C + \int_0^T f dB = D + \int_0^T g dB.$$

Show that $C = D$ and that $f = g$ dt $\otimes$ $\mathbb{P}$ -a.s.

Exercise 36 Prove that the following stochastic processes are martingales:

$$(i) \quad X_t = e^{t/2} \cos(B_t),$$

$$(ii) \quad X_t = (B_t + t) \exp(-B_t - \frac{t}{2})$$

Exercise 37 (i) Prove directly from the construction of the Itô integral that

$$\int_0^t s dB_s = tB_t - \int_0^t B_s ds.$$

(ii) Show that the integral $\int B_s dB_s$ has finite first and second moments.

The Itô isometry

As an immediate corollary from the construction of the stochastic integral we obtain an isometry for functions belonging to the class $\mathcal{L}_2$.

Lemma 4.1.8 (Itô isometry) For every $f \in \mathcal{L}_2$ and any $t > 0$,

$$\mathbb{E} \left[\left(\int_0^t f_s dB_s \right)^2 \right] = \mathbb{E} \left[\int_0^t f_s^2 ds \right].$$

More generally, for any $f, g \in \mathcal{L}_2$ and any $0 \leq s \leq t < \infty$,

$$\mathbb{E} \left[\int_s^t f_u dB_u \int_s^t g_u dB_u \middle| \mathcal{F}_s \right] = \mathbb{E} \left[\int_s^t f_u g_u du \middle| \mathcal{F}_s \right].$$

PROOF: The first assertion follows from the construction of the stochastic integral and the Itô isometry for simple processes.

Using that $ab = \frac{1}{4}((a+b)^2 - (a-b)^2)$ it is enough to prove the second assertion for the case $f = g$. To this end, let $\{\phi^n\} \subset \mathcal{L}_0$ be such

$$\|\phi^n - f\|_{L_t^2(\lambda \otimes \mathbb{P})}^2 \rightarrow 0 \quad \text{for all } t \geq 0.$$

Let us then put

$$F_n := \int_s^t \phi_u^n dB_u \in L^2(\mathbb{P}) \quad \text{and} \quad F := \int_s^t f_u dB_u \in L^2(\mathbb{P}).$$

The first assertion yields the L^2 -convergence of F_n to F :

$$\|F_n - F\|_{L^2(\mathbb{P})}^2 = \mathbb{E} \left[\left| \int_s^t \phi_u^n dB_u - \int_s^t f_u dB_u \right|^2 \right] = \mathbb{E} \left[\int_0^t |\phi_u^n - f_u|^2 du \right] \rightarrow 0.$$

As a result, the lemma below implies that

$$\mathbb{E}[F_n^2 | \mathcal{F}_s] \rightarrow \mathbb{E}[F^2 | \mathcal{F}_s] \quad \text{in } L^1(\mathbb{P})$$

and

$$\mathbb{E} \left[\int_s^t (\phi_s^n)^2 ds \middle| \mathcal{F}_s \right] \rightarrow \mathbb{E} \left[\int_s^t f_s^2 ds \middle| \mathcal{F}_s \right] \quad \text{in } L^1(\mathbb{P}).$$

Now, the assertion follows from by the Itô isometry for simple processes from which we deduce that

$$\mathbb{E}[F_n^2 | \mathcal{F}_s] = \mathbb{E} \left[\int_s^t (\phi_s^n)^2 ds \middle| \mathcal{F}_s \right].$$

□

The preceding proof used the following well-known result from probability theory.

Lemma 4.1.9 *Let f_n, f be square integrable random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{G} \subset \mathcal{F}$ be a sub- σ -field. If $f_n \rightarrow f$ in L^2 , then*

$$\mathbb{E}[(f_n)^2|\mathcal{G}] \rightarrow \mathbb{E}[f^2|\mathcal{G}] \quad \text{in } L^1.$$

PROOF: Using the Hölder inequality for conditional expectations

$$|\mathbb{E}[f_n^2 - f^2|\mathcal{G}]| \leq \mathbb{E}[|f_n + f||f_n - f|\mathcal{G}] \leq \sqrt{\mathbb{E}[|f_n + f|^2|\mathcal{G}]} \sqrt{\mathbb{E}[|f_n - f|^2|\mathcal{G}]}$$

An application of Jensen's inequality yields the desired L^1 -convergence:

$$\mathbb{E}|\mathbb{E}[f_n^2 - f^2|\mathcal{G}]| \leq \sqrt{\mathbb{E}[|f_n + f|^2]} \sqrt{\mathbb{E}[|f_n - f|^2]} \rightarrow 0 \text{ as } n \rightarrow \infty.$$

□

The following result is an immediate consequence of Lemma 4.1.8. It shows that if we subtract the covariation of an Itô integral from the square of the integral, then we get a martingale. This identifies the bracket process of Itô integrals!

Corollary 4.1.10 *For any $f \in \mathcal{L}_2$ the following is a continuous martingale:*

$$\left(\left(\int_0^t f_s dB_s \right)^2 - \int_0^t f_s^2 ds \right)_{t \geq 0}.$$

PROOF: Let

$$M_t := \int_0^t f_u dB_u.$$

Using Lemma 4.1.8, the martingale property of the stochastic integral and the orthogonality of martingale increments,

$$\begin{aligned} & \mathbb{E} \left[M_t^2 - \int_0^t f_u^2 du \middle| \mathcal{F}_s \right] \\ &= \mathbb{E} \left[(M_t - M_s)^2 - \int_s^t f_u^2 du + 2(M_t - M_s)M_s + M_s^2 - \int_0^s f_u^2 du \middle| \mathcal{F}_s \right] \\ &= M_s^2 - \int_0^s f_u^2 du \end{aligned}$$

□

Exercise 38 *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be C^1 everywhere and C^2 outside finitely many points $z_1, \dots, z_n$ and let $|g''(x)| \leq M$ for $x \notin \{z_1, \dots, z_n\}$. Let B_t be one dimensional Brownian motion. Prove that*

$$g(B_t) = g(0) + \int_0^t g'(B_s) dB_s + \frac{1}{2} \int_0^t g''(B_s) ds.$$

Hint: Show that there exist functions $f_k \in C^2$ with bounded second derivative such that $f_k \rightarrow g$ uniformly, $f'_k \rightarrow g'$ uniformly, and $f''_k \rightarrow g''$ outside $\{z_1, \dots, z_n\}$. Then apply Itô to f_k and let $k \rightarrow \infty$.

Exercise 39 (Tanaka's formula and local time) What happens if we try to apply Itô's formula to $g(B_t)$ when $g(x) = |x|$? In this case g is not C^2 at $x = 0$. Instead, for any $\varepsilon > 0$ we modify $g(x)$ near $x = 0$ to $g_\varepsilon(x)$ as follows:

$$g_\varepsilon(x) = \begin{cases} |x| & \text{if } |x| \geq \varepsilon \\ \frac{1}{2} \left(\varepsilon + \frac{x^2}{\varepsilon} \right) & \text{if } |x| < \varepsilon \end{cases}.$$

(i) Apply Exercise 38 to show that

$$g_\varepsilon(B_t) = g_\varepsilon(B_0) + \int_0^t g'_\varepsilon(B_s) dB_s + \frac{1}{2\varepsilon} |\{s \in [0, t]; B_s \in (-\varepsilon, \varepsilon)\}|$$

where $|F|$ denotes the Lebesgue measure of the set F .

(ii) Prove that

$$\int_0^t g'_\varepsilon(B_s) 1_{(-\varepsilon, \varepsilon)}(B_s) dB_s = \int_0^t \frac{B_s}{\varepsilon} 1_{(-\varepsilon, \varepsilon)}(B_s) dB_s \rightarrow 0$$

in $L^2(P)$ as $\varepsilon \rightarrow 0$.

(iii) By letting $\varepsilon \rightarrow 0$ prove that

$$|B_t| = |B_0| + \int_0^t \text{sign}(B_s) dB_s + L_t(\omega), \quad (4.1)$$

where (the limit is taken in $L^2(\mathbb{P})$)

$$L_t = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\varepsilon} |\{s \in [0, t]; B_s \in (-\varepsilon, \varepsilon)\}| = \lim_{\varepsilon \rightarrow 0} \frac{1}{2\varepsilon} |s \in [0, t] : |B_t| < \varepsilon|$$

and

$$\text{sign}(x) = \begin{cases} +1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases}.$$

The process L_t is called the local time for Brownian motion at 0 and (4.1) is the Tanaka formula for Brownian motion.

4.2 Stochastic integration w.r.t. continuous local martingales

We proceed to define the stochastic integral with respect to continuous martingales. Throughout, we fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ satisfying the usual conditions.

4.2.1 The space $\mathcal{M}$

In a first step we are going to define the stochastic integral for L^2 -bounded martingales. A martingale M is called L^2 -bounded if

$$\sup_{t \geq 0} \mathbb{E} M_t^2 < \infty.$$

In particular, an L^2 -bounded martingale is uniformly integrable.

Definition 4.2.1 The space of all L^2 -bounded, continuous martingales starting in zero is denoted $\mathcal{M}$.

For any $M \in \mathcal{M}$ the limit

$$\lim_{t \rightarrow \infty} M_t =: M_\infty$$

exists a.s. and in L^2 . Moreover, $(M_t)_{t \in [0, \infty]}$ is a martingale w.r.t. $(\mathcal{F}_t)_{t \in [0, \infty]}$ where $\mathcal{F}_\infty := \bigvee_{t \geq 0} \mathcal{F}_t$ (the fact that M extends to the time interval $[0, \infty]$ will be used repeatedly in what follows), and

$$M_t = \mathbb{E}[M_\infty \mid \mathcal{F}_t].$$

In particular, any element $M \in \mathcal{M}$ can be identified with the random variable M_∞ . Furthermore, we can define the bracket process $\langle M \rangle$ as a limit in probability. The following result shows that the bracket process introduces a Hilbert space structure on $\mathcal{M}$.

Theorem 4.2.2 1. *The space $\mathcal{M}$ is a real Hilbert space with scalar product*

$$\langle M, N \rangle_{\mathcal{M}} := \lim_{t \rightarrow \infty} \mathbb{E}[|M_t N_t|] = \mathbb{E}[|M_\infty N_\infty|].$$

and norm

$$\|M\|_{\mathcal{M}} := \mathbb{E}[M_\infty^2]^{1/2} \stackrel{3}{=} \mathbb{E}[\langle M \rangle_\infty]^{1/2}.$$

2. *The following norm on $\mathcal{M}$ is equivalent to $\|\cdot\|_{\mathcal{M}}$:*

$$\|M\| := \mathbb{E} \left[\sup_{t \geq 0} |M_t|^2 \right]^{1/2}.$$

3. *For $M \in \mathcal{M}$ the limit $\langle M \rangle_\infty := \lim_{t \rightarrow \infty} \langle M \rangle_t$ exists almost surely and*

$$\|M\|_{\mathcal{M}}^2 = \mathbb{E}[\langle M \rangle_\infty].$$

PROOF:

1. The second equality follows from L^2 -convergence $M_t^n \rightarrow M_\infty^n$ as $t \rightarrow \infty$. To show the Hilbert space property we need to prove that $\mathcal{M}$ is complete. To this end, let $\{M^n\}_{n \in \mathbb{N}}$ be a Cauchy sequence in $\mathcal{M}$. Then, $\{M_\infty^n\}_{n \in \mathbb{N}}$ is a Cauchy sequence in $L^2(\Omega, \mathcal{F}_\infty, \mathbb{P})$ by the definition of $\|\cdot\|_{\mathcal{M}}$ and hence the limit

$$M_\infty = \lim_{n \rightarrow \infty} M_\infty^n$$

exists in $L^2(\Omega, \mathcal{F}_\infty, \mathbb{P})$. As a result, it follows from Doob's L^2 -inequality that

$$\mathbb{E}[\sup_t |M_t^n - M_t^m|^2] \leq 4\mathbb{E}[|M_\infty^n - M_\infty^m|^2] \rightarrow 0 \quad \text{for } n, m \rightarrow \infty.$$

Hence there exists a subsequence (w.l.o.g. the original sequence) and a stochastic process $\bar{M}$ with continuous sample paths such that

$$M_t^n \rightarrow \bar{M}_t \quad \text{uniformly in } t \geq 0 \text{ a.s.}$$

Then $\bar{M}$ is adapted (using the usual conditions) and, using the above established L^2 -convergence,

$$\bar{M}_t = \lim_{n \rightarrow \infty} M_t^n = \lim_{n \rightarrow \infty} \mathbb{E}[M_\infty^n \mid \mathcal{F}_t] = \mathbb{E}[M_\infty \mid \mathcal{F}_t].$$

This shows that $\bar{M}$ is a continuous L^2 -bounded martingale.

2. Recalling that our martingale is defined for all $t \in [0, \infty]$, this follows from

$$\mathbb{E}\left[\sup_{t \in [0, \infty]} M_t^2\right] \leq 4\mathbb{E}[M_\infty^2] \leq 4\mathbb{E}\left[\sup_{t \in [0, \infty]} M_t^2\right].$$

This first inequality follows from Doob's maximal inequality; the second is trivial.

3. Exercise.

□

Exercise 40 *Prove 3.) of the above theorem.*

Remark 4.2.3 *The proof of the first item of the above theorem provides the missing martingale argument in the proof Theorem 3.3.6 for bounded martingales.*

4.2.2 Stochastic integration w.r.t. L^2 -bounded martingales

In this section we define the stochastic integral $h \cdot M := \int h dM$ with respect to a continuous martingale $M \in \mathcal{M}$. We proceed by analogy to Brownian motion. For a simple/elementary (left-continuous) process

$$h_t(\omega) = \sum_j h_j(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

where $0 = t_0 < t_1 < t_2 < \dots$, $h_j \in \mathcal{F}_{t_j}$ and $\sup_j |h_j| \leq C$ we define the stochastic integral $h \cdot M$ of h w.r.t. M as

$$h \cdot M := \left(\int_0^t h_s(\omega) dM_s \right)_{t \geq 0} := \left(\sum_{j=0}^{\infty} h_j(\omega) [M_{t_{j+1} \wedge t}(\omega) - M_{t_j \wedge t}(\omega)] \right)_{t \geq 0}.$$

The class of all simple (left-continuous) processes is again denoted $\mathcal{L}_0$. The proof of the following lemma is similar to the Brownian case and hence left as an exercise. In particular, stochastic integrals of simple processes w.r.t. L^2 -bounded continuous martingales are martingales and their quadratic variation is given by the integral of the squared integrand with respect to the bracket of the martingale.

Lemma 4.2.4 *For any elementary process h and any martingale $M \in \mathcal{M}$ the following holds:*

1. $h \cdot M \in \mathcal{M}$
2. $\langle h \cdot M \rangle_t = \int_0^t h_s^2 d\langle M \rangle_s$.
- 3.

$$\|h \cdot M\|_{\mathcal{M}} = \mathbb{E}[(h \cdot M)_\infty^2]^{1/2} = \mathbb{E}\left[\int_0^\infty |h_s|^2 d\langle M \rangle_s\right]^{1/2}.$$

PROOF: Clearly, $H \cdot M_t$ is continuous and adapted and vanishes at zero. It is L^2 -bounded as

$$\begin{aligned} \mathbb{E}[(H \cdot M)_t^2] &= \mathbb{E}\left[\sum_{i=0}^{n-1} h_i^2 (M_{t_{i+1} \wedge t} - M_{t_i \wedge t})^2\right] \\ &= \mathbb{E}\left[\sum_{i=0}^{n-1} h_i^2 (M_{t_{i+1} \wedge t}^2 - M_{t_i \wedge t}^2)\right] \\ &= \mathbb{E}\left[\int_0^t H_s^2 d\langle M \rangle_s\right] \\ &\leq \mathbb{E}\left[\int_0^\infty H_s^2 d\langle M \rangle_s\right] < \infty. \end{aligned}$$

The martingale property is easy to show. Hence $H \cdot M \in \mathcal{M}$ and

$$\mathbb{E}[(H \cdot M)_\infty^2] = \mathbb{E}\left[\int_0^\infty H_s^2 d\langle M \rangle_s\right] \quad \text{as } t \rightarrow 0.$$

□

To extend the stochastic integrals to a larger class of integrands we put

$$\bar{\Omega} := \Omega \times [0, \infty), \quad \bar{\mathcal{F}} := \mathcal{F} \otimes \mathcal{B}.$$

For $M \in \mathcal{M}$ the associated *Doleans measure* $\mathbb{P}_M$ on $(\bar{\Omega}, \bar{\mathcal{F}})$ is defined by

$$\mathbb{P}_M(A) := \mathbb{E} \int_0^\infty \mathbf{1}_A(s, \omega) d\langle M \rangle_s.$$

The Doleans measure is a finite measure with total mass $\mathbb{E}[\langle M \rangle_\infty]$; see Theorem 4.2.2(3.). We define

$$\mathcal{L}(M) \subset \mathcal{L}^2(\bar{\Omega}, \bar{\mathcal{F}}, \mathbb{P}_M)$$

as the Hilbert space of all measurable, adapted processes f such that

$$\|f\|_M := \|f\|_{L^2(\mathbb{P}_M)} < \infty.$$

Remark 4.2.5 *We will be working with several subspaces of $\mathcal{L}(M)$ in what follows. We denote the corresponding spaces of equivalence classes using the letter “ L ”. That is, $L(M)$ is the space of equivalence classes of $\mathcal{L}(M)$, L_0 is the space of equivalence classes of $\mathcal{L}_0$ etc. In particular, $\|\cdot\|_M$ is a norm on $L(M)$ (but only a semi-norm on $\mathcal{L}(M)$).*

In view of 3.) of the above lemma the stochastic integral defines an isometry

$$I : L_0 \rightarrow \mathcal{M}; \quad I(h) := h \cdot M$$

between the spaces L_0 and $\mathcal{M}$. An isometry from some metric space to a complete metric space can always be extended uniquely to an isometry on the completion of the first space. We denote the completion of L_0 in the Hilbert space $L(M)$ by

$$\bar{L}(M) := \overline{L_0}^{\|\cdot\|_M}.$$

We will then simply define the stochastic integral $h \cdot M$ for $h \in \bar{L}(M)$ as the image of the unique extension of our integral to a (linear) isometry from $\bar{L}(M)$ to $\mathcal{M}$ which we again denote by $h \cdot M$. In particular, the mapping $h \mapsto h \cdot M$ is linear.

It is of course of interest to have a more explicit description - or at least to identify a large subset - of the space $\bar{L}(M)$.

Remark 4.2.6 *It is definitely not true (in general) that $\bar{L}(M) = L(M)$. If $L'(M)$ denotes the closed subspace of $L(M)$ consisting of all progressive processes in $L(M)$ (more precisely: consisting of all equivalence classes containing some progressive process) then $L'(M)$ contains L_0 (since left continuous adapted processes are progressive) and therefore also its completion $\bar{L}(M)$. We emphasize that an equivalence class of $L(M)$ may well contain both progressive and non-progressive processes: just think of the extreme case in which $M \equiv 0$. Then $\mathbb{P}_M = 0$ and all processes in $\mathcal{L}(M)$ are equivalent.*

If $\langle M \rangle$ is absolutely continuous w.r.t. Legesque measure, then a straightforward extension of the arguments given in the proof of Proposition 4.1.5 yields the following result.

Proposition 4.2.7 *If $\langle M \rangle$ is absolutely continuous w.r.t. Legesque measure, then L_0 is dense in the subset of all measurable, adapted processes in $L(M)$.*

Predictable integrands

We are now going to show that any predictable process in $\mathcal{L}(M)$, that is, any process in $\mathcal{L}(M)$ that is measurable w.r.t. the predictable σ -algebra on $\mathcal{B} \otimes \mathcal{F}$, can be approximated by simple functions with respect to the semi-norm $\|\cdot\|_M$.⁴ In principle, we know already from Exercise 2(ii) that this is indeed true but we did not work with the Doleans measure there. The following result shows that the use of the Doleans measure does not alter anything. The proof uses the following vector version of the monotone class theorem.

Monotone Class Theorem (Vector version) *Let $\mathcal{D}$ be a Dynkin system and $\mathcal{H}$ be a set of functions from Ω to $\mathbb{R}$ that satisfies the following properties:*

- $\mathcal{H}$ is a vector space
- if $D \in \mathcal{D}$, then $\mathbf{1}_D \in \mathcal{H}$
- $\mathcal{H}$ is closed with respect to bounded increasing limits

Then $\mathcal{H}$ contains all bounded $\sigma(\mathcal{D})$ -measurable functions.

Proposition 4.2.8 *Let $M \in \mathcal{M}$ and $h \in \mathcal{L}(M)$ be predictable. Then there exists a sequence of simple processes $\{h^n\}$ such that $\|h - h^n\|_M \rightarrow 0$ as $n \rightarrow \infty$. In particular,*

$$h \cdot M \quad \text{exists for any predictable } h \in \mathcal{L}(M).$$

PROOF: Let $\mathcal{R}$ be the class of all bounded, predictable processes that can be approximated as claimed. Then $\mathcal{R}$ is a vector space that contains all constant processes as well as the family of indicators

$$A \times (s, t], \quad A \in \mathcal{F}_s.$$

Since this family generates the predictable σ -field the above monotone class theorem states that $\mathcal{R}$ is equal to the set of all bounded predictable functions provided we can show that $\mathcal{R}$ is closed with

⁴Recall that the predictable σ -algebra on $\mathcal{B} \otimes \mathcal{F}$ is contained in the progressive one and is generated by the processes in $\mathcal{L}_0$, that the predictable σ -algebra is the smallest σ -algebra w.r.t. which all left-continuous processes are measurable and that the predictable σ -algebra is contained in the progressive σ -algebra; cf. Exercise 2.

respect to limits of increasing, nonnegative and uniformly bounded elements of $\mathcal{R}$. To show this, take a sequence of non-negative processes g^n from $\mathcal{R}$ such that $g^n \uparrow g$ where g is bounded. Being the pointwise limit of predictable processes g is also predictable. Further, we have $\|g^n - g\|_M \rightarrow 0$ by dominated convergence and it follows from the usual “triangle trick” that $g \in \mathcal{R}$. This shows that any bounded predictable process can be approximated as claimed. To complete the proof, let h be as in the proposition and define $h^{(N)} := h \mathbf{1}_{|h| \leq N}$. Then $h^{(N)} \in \mathcal{R}$ and $\|h^{(N)} - h\|_M \rightarrow 0$ by dominated convergence. Again the triangle trick shows that the proposition holds for h . $\square$

Progressive integrands

The following proposition shows that progressively measurable processes can also be approximated by simple processes, at least when restricted to finite time intervals.

Proposition 4.2.9 *Let A be an adapted, continuous, increasing process starting in zero (think of A as $\langle M \rangle$) such that $\mathbb{E}A_T < \infty$ for all $T > 0$. Let h be a progressively measurable process that satisfies*

$$\mathbb{E} \int_0^T h_t^2 dA_t < \infty,$$

for each $T > 0$. Then there exists a sequence $\{h^n\}$ of simple processes such that

$$\sup_{T>0} \lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |h_t^n - h_t|^2 dA_t = 0.$$

PROOF: Using a standard truncation argument, we may assume w.l.o.g. that $h_t(\omega) \leq C$ for some $C > 0$ and all (t, ω) . Moreover, it suffices to construct, for each $T > 0$ a sequence $\{h^{n,T}\}$ of simple processes such that

$$\lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |h_t^{n,T} - h_t|^2 dA_t = 0.$$

In this case, we can find, for each $m \in \mathbb{N}$ an integer $n_m \in \mathbb{N}$ such that

$$\mathbb{E} \int_0^m |h_t^{n_m, m} - h_t|^2 dA_t \leq \frac{1}{m}$$

and the sequence $\{h^{n_m, m}\}$ has the desired properties. In what follows we may thus fix $T > 0$ and assume that $h_t = 0$ for all $t > T$ and all $\omega \in \Omega$.

Compared to the arguments given in Proposition 4.1.5 the idea is to first change the time to replace the dA_t -integral by a dt -integral so that Proposition 4.1.5 can be applied.

More precisely, let us consider the time change $T_s(\omega)$ defined by inverse of the strictly increasing function $A_s(\omega) + s$, that is

$$A_{T_s(\omega)} + T_s(\omega) = s.$$

Then $T_s \leq s$ and $\{T_s \leq t\} = \{A_t + t \geq s\} \in \mathcal{F}_t$ so T_s is a bounded stopping time for each $s \geq 0$. Taking s as our new variable, we define the filtration

$$\mathcal{G}_s = \mathcal{F}_{T_s}$$

and introduce the measurable time-changed process

$$g_s := h_{T_s} \quad \text{hence} \quad g_{A_s+s} = h_s.$$

This process is adapted to (G_s) because of the progressive measurability of h ; see Proposition 1.2.6. Furthermore, since $h \equiv 0$ on (T, ∞) and because h is bounded,

$$\mathbb{E} \int_0^\infty g_s^2 ds = \mathbb{E} \int_0^\infty \mathbf{1}_{\{T_s \leq T\}} h_{T_s}^2 ds = \mathbb{E} \int_0^{A_T+T} h_{T_s}^2 ds \leq C^2 (\mathbb{E} A_T + T) < \infty.$$

Thus, by Proposition Proposition 4.1.5 there exists for any $\varepsilon > 0$ and $R > 0$ a (G_s) -adapted simple process

$$g_s^\varepsilon = \sum_{j=1}^n \xi_{s_{j-1}} \mathbf{1}_{(s_{j-1}, s_j]}(s)$$

where each ξ_{s_j} is $\mathcal{F}_{T_{s_j}}$ -measurable and uniformly bounded, say by some constant $K > 0$, such that

$$\mathbb{E} \int_0^R |g^\varepsilon - g| ds < \frac{\varepsilon}{2}.$$

By choosing R large enough and setting $g_s^\varepsilon = 0$ for $s > R$, we obtain that

$$\mathbb{E} \int_0^\infty |g_s^\varepsilon - g_s|^2 ds < \varepsilon.$$

Reverting back to the original clock we set

$$h_t^\varepsilon := g_{A_t+t}^\varepsilon = \sum_{j=1}^n \xi_{s_{j-1}} \mathbf{1}_{(T_{s_{j-1}}, T_{s_j}]}(t).$$

By Lemma 1.2.5 vi), since $\xi_{s_{j-1}}$ is $\mathcal{F}_{T_{s_{j-1}}}$ -measurable

$$\xi_{s_{j-1}} \mathbf{1}_{\{T_{s_{j-1}} < t\}} \in \mathcal{F}_t. \quad (4.2)$$

Since

$$\xi_{s_{j-1}} \mathbf{1}_{(T_{s_{j-1}}, T_{s_j}]}(t) = \xi_{s_{j-1}} \mathbf{1}_{\{T_{s_{j-1}} < t\}} \mathbf{1}_{\{T_{s_j} \geq t\}} \in \mathcal{F}_t$$

we conclude that h^ε is adapted. Moreover, it approximates h because

$$\mathbb{E} \int_0^T |h_t^\varepsilon - h_t|^2 dA_t \leq \mathbb{E} \int_0^T |h_t^\varepsilon - h_t|^2 (dA_t + dt) \leq \mathbb{E} \int_0^\infty |g_t^\varepsilon - g_t|^2 dt < \varepsilon.$$

Since the process h^ε is not simple it remains to approximate the processes

$$\eta_t(\omega) := \xi_{s_{j-1}}(\omega) \mathbf{1}_{(T_{s_{j-1}}, T_{s_j}]}(t)$$

that “form the basis” for h^ε by simple ones. To simplify notation let $s_{j-1} = 1$ and $s_j = 2$ so $T_1 \leq T_2 \leq 2$.

Let

$$T_i^m(\omega) := \sum_{k=1}^{1+2^{m+1}} \frac{k}{2^m} \mathbf{1}_{[(k-1)/2^m, k/2^m)}(T_i(\omega)), \quad i = 1, 2$$

and define

$$\eta_t^m(\omega) := \xi_1(\omega) \mathbf{1}_{(T_1^m(\omega), T_2^m(\omega)]}(t)$$

We claim that η^m is a simple process and that it approximates the process η . Indeed, noticing that the random times $T_1^m(\omega)$ and $T_2^m(\omega)$ may be equal in which case $\eta_t^m(\omega) = 0$ we can represent η^m as

$$\begin{aligned}\eta_t^m(\omega) &= \sum_{k=1}^{2^{m+1}} \xi_1(\omega) \mathbf{1}_{((k-1)/2^m, k/2^m]}(t) \mathbf{1}_{\{T_1 \leq (k-1)/2^m < T_2\}}(\omega) \\ &= \sum_{k=1}^{2^{m+1}} \xi_1(\omega) \mathbf{1}_{\{T_1 \leq (k-1)/2^m\}} \mathbf{1}_{((k-1)/2^m, k/2^m]}(t) \mathbf{1}_{\{(k-1)/2^m < T_2\}}(\omega).\end{aligned}$$

Since T_1 and T_2 are stopping times and in view of (4.2),

$$\{(k-1)/2^m < T_2\} \in \mathcal{F}_{(k-1)/2^m} \quad \text{and} \quad \xi_1 \mathbf{1}_{\{T_1 \leq (k-1)/2^m\}} \in \mathcal{F}_{(k-1)/2^m}.$$

As a result, the process η^m is indeed simple. Furthermore,

$$\mathbb{E} \int_0^\infty |\eta_t^m - \eta_t|^2 dA_t \leq K^2 [\mathbb{E}(A_{T_2^m} - A_{T_2}) + \mathbb{E}(A_{T_1^m} - A_{T_1})] \rightarrow 0 \quad \text{as } m \rightarrow \infty.$$

□

Summary. So far we have shown that the stochastic integral $h \cdot M$ exists for all $M \in \mathcal{M}$ and all predictable/progressively measurable $\mathbb{P}_M$ -square integrable processes h and that $h \cdot M$ belongs to $\mathcal{M}$. Do note, though, that for a predictable process h with approximating sequence $\{h^n\}$ we have that

$$\lim_{n \rightarrow \infty} \mathbb{E} \int_0^\infty |h_t^n - h_t|^2 d\langle M \rangle_t = 0.$$

For progressively measurable (and also for predictable) processes we have that

$$\sup_{T>0} \lim_{n \rightarrow \infty} \mathbb{E} \int_0^T |h_t^n - h_t|^2 d\langle M \rangle_t = 0,$$

under the weaker requirement

$$\mathbb{E} \int_0^T h_s^2 d\langle M \rangle_s < \infty \quad \text{for all } T > 0.$$

We continue to work with predictable integrands.

4.2.3 Properties of the stochastic integral

In what follows, we establish properties for the bracket process of stochastic integrals. This requires some measure-theoretic preliminaries.

For a signed Borel measure ν on $\mathbb{R}$, we denote by $h \cdot |\nu|$ the integral of a function h w.r.t. the total variation measure $|\nu| := \nu^+ + \nu^-$ where $\nu^\pm$ denote the unique positive measures such that $\nu = \nu^+ - \nu^-$ (Hahn decomposition). We recall that

$$|h \cdot \nu| \leq |h| \cdot |\nu|.$$

Moreover, if $\nu \ll \mu$ for some positive measure μ with density $g := \frac{d\nu}{d\mu}$, then $\nu = g^+ d\mu - g^- d\mu$, and the uniqueness of the Hahn decomposition shows that

$$|\nu| = (g^+ + g^-) d\mu = |g| d\mu,$$

that is, $|\nu|$ is absolutely continuous w.r.t. μ as well with density

$$\frac{d|\nu|}{d\mu} = |g|. \quad (4.3)$$

For a function of bounded variation A with Jordan decomposition $A = A_1 - A_2$ we denote by dA the associated Lebesgue-Stieltjes measure and by $d|A|$ the total variation measure of dA .⁵

Thus, for every $M, N \in \mathcal{M}$ we can define the Lebesgue-Stieltjes integral (ω -by- ω) of h w.r.t. the total variation measure $d|\langle M, N \rangle|$, denoted

$$\int h d|\langle M, N \rangle| =: h \cdot |\langle M, N \rangle|.$$

Our first goal is to prove the Kunita-Watanabe inequality that estimates the co-variation of stochastic integrals in terms of the quadratic variation of the integrals involved. The proof requires the following result from which we shall deduce that the measure of $d\langle M_1, M_2 \rangle$ is almost surely absolutely continuous w.r.t. the measure $d\langle M_1 \rangle + d\langle M_2 \rangle$.

Lemma 4.2.10 *Let μ_i , $i = 1, 2$ be finite, non-negative Borel measures on $[0, T]$ and ν be a signed Borel measure on $[0, T]$ that satisfies⁶*

$$|\nu(I)| \leq \sqrt{\mu_1(I)\mu_2(I)}$$

for any half-open interval $I \subseteq [0, T]$. Then ν and hence $|\nu|$ is absolutely continuous w.r.t. $\mu := \frac{1}{2}(\mu_1 + \mu_2)$ and the density $g = \frac{d\nu}{d\mu}$ satisfies $|g| \leq \sqrt{f_1 f_2}$ μ -a.e. where $f_i = \frac{d\mu_i}{d\mu}$ for $i = 1, 2$.

PROOF: Absolute continuity follows from the binomial formula, which gives us for all $\alpha > 0$ that

$$|\nu(I)| \leq \sqrt{\mu_1(I)\mu_2(I)} = \sqrt{\alpha\mu_1(I)\frac{1}{\alpha}\mu_2(I)} \leq \frac{1}{2}\left(\alpha\mu_1(I) + \frac{1}{\alpha}\mu_2(I)\right).$$

In particular,

$$2\left|\int_0^T \varphi g d\mu\right| \leq \alpha \int_0^T |\varphi| f_1 d\mu + \alpha^{-1} \int_0^T |\varphi| f_2 d\mu,$$

first for all φ that are indicator functions of intervals, then for linear combinations and finally for all bounded, Borel measurable ones. This implies that μ -almost everywhere

$$2|g| \leq \alpha f_1 + \alpha^{-1} f_2.$$

Taking the unique $\alpha^* = \sqrt{f_2/f_1}$ that minimizes the right-hand side of the equation and plugging α^* back into this equations yields $|g| \leq \sqrt{f_1 f_2}$. $\square$

Corollary 4.2.11 *For $M_1, M_2 \in \mathcal{M}$, we have for each $s < t$ almost surely*

$$|\langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s| \leq \sqrt{\langle M_1 \rangle_t - \langle M_1 \rangle_s} \sqrt{\langle M_2 \rangle_t - \langle M_2 \rangle_s}.$$

⁵If we denote by V the total variation function of A with associated Lebesgue-Stieltjes measure dV , then it is not difficult to show that $dV = d|A|$.

⁶We do mean the absolute value of $\nu(I)$, not $|\nu|(I)$.

In particular, $\langle M_1, M_2 \rangle(\omega)$ and hence $|\langle M_1, M_2 \rangle|(\omega)$ is a.s. absolutely continuous w.r.t.

$$\mu_\omega := \frac{1}{2}(\langle M_1 \rangle(\omega) + \langle M_2 \rangle(\omega)).$$

If $f_{i,j}(\cdot, \omega)$ denotes the density of $d\langle M_i, M_j \rangle(\omega)$ w.r.t μ_ω , then a.s.

$$|f_{1,2}(\cdot, \omega)| \leq \sqrt{f_{1,1}(\cdot, \omega)f_{2,2}(\cdot, \omega)} \quad \mu_\omega\text{-almost everywhere.}$$

PROOF: The following (in)equality is a trivial consequence of non-negative bilinearity: for any $\alpha > 0$,

$$\begin{aligned} 0 &\leq \langle \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2, \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2 \rangle_t - \langle \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2, \alpha^{1/2} M_1 \pm \alpha^{-1/2} M_2 \rangle_s \\ &= \alpha \left(\langle M_1 \rangle_t - \langle M_1 \rangle_s \right) \pm 2 \left(\langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s \right) + \alpha^{-1} \left(\langle M_2 \rangle_t - \langle M_2 \rangle_s \right). \end{aligned}$$

From this we see that

$$2 \left| \langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s \right| \leq \alpha \left(\langle M_1 \rangle_t - \langle M_1 \rangle_s \right) + \frac{1}{\alpha} \left(\langle M_2 \rangle_t - \langle M_2 \rangle_s \right).$$

Minimizing the right side w.r.t. α and plugging the minimizer back into the equation yields the first assertion. The second assertion follows from the preceding lemma as

$$|d\langle M_1, M_2 \rangle|((s, t]) = |\langle M_1, M_2 \rangle_t - \langle M_1, M_2 \rangle_s|.$$

□

We are now ready to state and prove the Kunita-Watanabe (in)equality.⁷

Proposition 4.2.12 (*Kunita-Watanabe identity and inequality*) For $M, N \in \mathcal{M}$, $h \in \bar{L}(M)$ and $g \in \bar{L}(N)$ the following holds a.s.

- a) $((hg) \cdot \langle M, N \rangle)_t^2 \leq (|hg| \cdot |\langle M, N \rangle|)_t^2 \leq (h^2 \cdot \langle M \rangle)_t (g^2 \cdot \langle N \rangle)_t$
- b) $\langle h \cdot M, g \cdot N \rangle = (hg) \cdot \langle M, N \rangle$.
- c) $\mathbb{E}[|\langle h \cdot M, g \cdot N \rangle|] \leq \sqrt{\mathbb{E}[h^2 \cdot \langle M \rangle]} \sqrt{\mathbb{E}[g^2 \cdot \langle N \rangle]}.$

PROOF:

- a) The first inequality follows by definition. To see the second, let $d\xi := d\langle M, N \rangle$ and $d\hat{\xi} = d|\langle M, N \rangle|$. In view of Corollary 4.2.11 the measures $d\xi$ and $d\hat{\xi}$ are a.s. absolutely continuous w.r.t.

$$d\eta := \frac{1}{2} (d\langle M \rangle + d\langle N \rangle).$$

By the Radon-Nikodym theorem this implies the existence of functions $f_i(\cdot, \omega) : [0, \infty) \rightarrow \mathbb{R}$ for $i = 1, 2, 3$ such that a.s.

$$d\langle M \rangle(\omega) = f_1(\cdot, \omega) d\eta(\omega), \quad d\langle N \rangle(\omega) = f_2(\cdot, \omega) d\eta(\omega),$$

⁷Note that item b) in the following proposition follows directly from the isometry property of the stochastic integral in case $M = N$ (no matter if f and g agree or not) since an isometry between Hilbert spaces automatically preserves the inner product.

and

$$d\langle M, N \rangle(\omega) = f_3(\cdot, \omega) d\eta(\omega), \quad d|\langle M, N \rangle(\omega)| = |f_3|(\cdot, \omega) d\eta(\omega)$$

where the second equation in the last display follows from (4.3). As a result, for all $\alpha, \beta \in \mathbb{R}$ on a set $\Omega_{\alpha, \beta}$ of full measure, using the polarization identity

$$\begin{aligned} 0 &\leq \langle \alpha M + \beta N \rangle_t - \langle \alpha M + \beta N \rangle_s \\ &= \int_s^t (\alpha^2 f_1(u, \omega) + 2\alpha\beta f_3(u, \omega) + \beta^2 f_2(u, \omega)) d\eta_u(\omega); \quad 0 \leq s < t < \infty. \end{aligned}$$

Since the above inequality holds for all $0 \leq s < t < \infty$ a standard monotone class argument shows that there must exist - for every $\omega \in \Omega_{\alpha, \beta}$ - a set $T_{\alpha, \beta} \in \mathcal{B}$ with $\int_{T_{\alpha, \beta}} d\eta_t(\omega) = 0$ and

$$\alpha^2 f_1(t, \omega) + 2\alpha\beta f_3(t, \omega) + \beta^2 f_2(t, \omega) \geq 0 \quad \text{on} \quad T_{\alpha, \beta}^c.$$

This shows that there exists a set of full measure

$$\tilde{\Omega} := \bigcap_{\alpha, \beta \in \mathbb{Q}} \Omega_{\alpha, \beta}$$

and sets $T(\omega)$ such that $\int_{T(\omega)} d\eta_u(\omega) = 0$ for all $\omega \in \tilde{\Omega}$, and such that the above inequality holds for all $(\alpha, \beta) \in \mathbb{R}^2$ and all $t \notin T(\omega)$. In particular, for any such time, setting

$$\alpha := a|h_t(\omega)| \quad \text{and} \quad \beta := \text{sign}(f_3(t, \omega))|g_t(\omega)|$$

we see that

$$a^2|h_t(\omega)|^2 f_1(t, \omega) + 2a|h_t(\omega)||g_t(\omega)||f_3(t, \omega) + |g_t(\omega)|^2 f_2(t, \omega) \geq 0.$$

Integrating w.r.t. $d\eta$ we obtain

$$a^2 \int_0^t |h_s|^2 d\langle M \rangle_s + 2a \int_0^t |h_s g_s| d|\langle M, N \rangle|_s + \int_0^t |g_s|^2 d\langle N \rangle_s \geq 0.$$

Minimizing over a yields the desired result.

b) Exercise

c) This is just a combination of parts a) and b) using the Cauchy-Schwarz inequality.

□

Exercise 41 Let $\{M^n\}$ and M be continuous square-integrable martingales such that

$$\|M^n - M\|_{\mathcal{M}}^2 := \mathbb{E}[(M_\infty^n - M_\infty)^2] \longrightarrow 0.$$

Then

$$\langle M^n \rangle \xrightarrow{\text{ucp}} \langle M \rangle.$$

Exercise 42 Use the preceding exercise to prove part b) of the preceding proposition. First, show the claim for simple integrands, then approximate using (3.4).

It follows from the above proposition that the quadratic variation of the stochastic integral $h \cdot M$ equals the integral of $|h|^2$ with respect to the quadratic variation process of M . The Kunita-Watanabe (in)equality also allows us to characterize stochastic integrals.

Proposition 4.2.13 *Let $M \in \mathcal{M}$ and $h \in \bar{L}(M)$. Then the following holds.*

1. *The stochastic integral $h \cdot M$ is the unique martingale in $\mathcal{M}$ that satisfies for all $N \in \mathcal{M}$ that*

$$\langle h \cdot M, N \rangle_t = h \cdot \langle M, N \rangle_t; \quad 0 \leq t < \infty \quad \mathbb{P}\text{-a.s.}$$

2. *For any stopping time τ it holds that*

$$(\mathbf{1}_{[0, \tau]} h) \cdot M = (h \cdot M)^\tau = h \cdot M^\tau.$$

PROOF:

1. We already know that the integral satisfies the desired property. Let us then assume $X \in \mathcal{M}$ is another martingale satisfying $\langle X, N \rangle = h \cdot \langle M, N \rangle$ for all $N \in \mathcal{M}$. then $\langle h \cdot M - X, N \rangle = 0$ for all $N \in \mathcal{M}$. Setting $N = h \cdot M - X$ implies that

$$\langle h \cdot M - X, h \cdot M - X \rangle \equiv 0$$

which implies that

$$\|h \cdot M - X\|_{\mathcal{M}} = \mathbb{E}[\langle h \cdot M - X \rangle_\infty] = \mathbb{E}\left[\lim_{t \rightarrow \infty} \langle h \cdot M - X \rangle_t\right] = 0.$$

That is, $h \cdot M - X$ a.s. in $\mathcal{M}$.

2. For the first equality note that

$$\langle (h \cdot M)^\tau, N \rangle = \langle h \cdot M, N \rangle^\tau \stackrel{1.}{=} (h \cdot \langle M, N \rangle)^\tau = (\mathbf{1}_{[0, \tau]} h) \cdot \langle M, N \rangle \stackrel{1.}{=} \langle \mathbf{1}_{[0, \tau]} h \cdot M, N \rangle.$$

For the second equality observe that

$$\langle h \cdot M^\tau, N \rangle = h \cdot \langle M^\tau, N \rangle = \left(h \cdot \langle M, N \rangle \right)^\tau = \left(\langle h \cdot M, N \rangle \right)^\tau = \langle (h \cdot M)^\tau, N \rangle.$$

Thus, the assertion follows from (1.).

□

The following corollary shows that $N = h \cdot M$ implies $g \cdot N = (gh) \cdot M$. That is, the corollary tells us how to integrate a stochastic process w.r.t. stochastic integrals (which are themselves square integrable martingales).

Corollary 4.2.14 *Let $M \in \mathcal{M}$, $h \in \bar{L}(M)$ and $N = h \cdot M$. If $g \in \bar{L}(N)$, then $hg \in \bar{L}(M)$ and $g \cdot N = (hg) \cdot M$.*

PROOF: Using that $d\langle N \rangle_t = h_t^2 d\langle M \rangle_t$ we have that

$$\mathbb{E} \int_0^\infty h_t^2 g_t^2 d\langle M \rangle_t = \mathbb{E} \int_0^\infty g_t^2 d\langle N \rangle_t < \infty$$

and so $hg \in \bar{L}(M)$. For any $\tilde{N} \in \mathcal{M}$ the cross-variation formula for stochastic integrals yields

$$d\langle N, \tilde{N} \rangle = h d\langle M, \tilde{N} \rangle$$

and so

$$\langle (hg) \cdot M, \tilde{N} \rangle_t = \int_0^t h_s g_s d\langle M, \tilde{N} \rangle_s = \int_0^t g_s d\langle N, \tilde{N} \rangle_s = \langle g \cdot N, \tilde{N} \rangle_t.$$

□

4.2.4 Stochastic integration w.r.t. continuous local martingales

We are now going to extend our definition of stochastic integrals to continuous local martingales for integrands from the following set.

Definition 4.2.15 For a continuous local martingale $M \in \mathcal{M}_{loc}^0$ starting in zero, let

$$\mathcal{L}_{loc}(M) := \left\{ h \text{ progressive} : \int_0^t h_s^2 d\langle M \rangle_s < \infty \text{ a.s. for all } t \in (0, \infty) \right\}.$$

Remark 4.2.16 If the quadratic variation process of M is a.s. absolutely continuous as it is the case for Brownian motion, then progressive measurability is not required; it is enough to assume that h is measurable and adapted.

We are going to define the following localizing sequences of stopping times. For $M \in \mathcal{M}_{loc}^0$ and $h \in \mathcal{L}_{loc}(M)$ let $\{\tau_n\}$ be a sequence of stopping times such that $\tau_n \uparrow \infty$ and $M^{\tau_n} \in \mathcal{M}$ for all $n \in \mathbb{N}$. Further, let

$$\nu_n := n \wedge \inf \left\{ t \geq 0 : \int_0^t |h_s|^2 d\langle M \rangle_s \geq n \right\}$$

and let $\sigma_n := \tau_n \wedge \nu_n$. Then, $\sigma_n \rightarrow \infty$ a.s. and

$$M^n := M^{\sigma_n} \in \mathcal{M}, \quad h_t^n := h_t \mathbf{1}_{\{t \leq \sigma_n\}} \in \bar{L}(M^n).$$

Definition 4.2.17 For $M \in \mathcal{M}_{loc}^0$ and $h \in \mathcal{L}_{loc}(M)$ let

$$(h \cdot M)_t := \int_0^t h_s dM_s := (h^n \cdot M^n)_t \quad \text{on } \{t \leq \sigma_n\}$$

We need to make sure that the above definition makes sense. In other words, we need to show that stochastic integrals are determined locally by the local values of the integrator and integrand. This is achieved by the following corollary. The proof is left as an exercise.

Corollary 4.2.18 Let $M, N \in \mathcal{M}$ and $h \in \bar{L}(M)$, $g \in \bar{L}(N)$ and let τ be a stopping time. If, for $\mathbb{P}$ -a.e. ω ,

$$h_{t \wedge \tau}(\omega) = g_{t \wedge \tau}(\omega); \quad M_{t \wedge \tau}(\omega) = N_{t \wedge \tau}(\omega) \quad \text{for all } t \geq 0,$$

then

$$(h \cdot M)(\omega)_{t \wedge \tau} = (g \cdot N)(\omega)_{t \wedge \tau} \quad \text{for all } t \geq 0.$$

Exercise 43 Proof the above corollary.

Roughly speaking, all properties derived for stochastic integrals in case $M \in \mathcal{M}$ and $h \in \bar{\mathcal{L}}(M)$ which do not involve expected values carry over to the general case (and are easily proved). Those properties which do involve expected values will generally not carry over. For example, stochastic integrals w.r.t. local martingales cannot be expected to be martingales; in general they are only local martingales.

We close this section with two convergence results for stochastic integrals. This first characterizes the stochastic integral $h \cdot M$ in terms of a Lebesgue-Stieltjes integral.

Proposition 4.2.19 *Let $M \in \mathcal{M}_{loc}^0$ and h be adapted and continuous (hence predictable). Let $\Pi^n = \{t_i^n : i \in \mathbb{N}\}$ be a partition of $\mathbb{R}_+$ with $|\Pi^n| \rightarrow 0$. Then*

$$\lim_{n \rightarrow \infty} \sup_{s \leq t} \left| \sum_{k=0}^{\infty} h_{t_k^n} (M_{t_{k+1}^n \wedge s} - M_{t_k^n \wedge s}) - \int_0^s h_r dM_r \right| = 0$$

in probability for all $t \in [0, \infty)$.

PROOF: Let

$$h_s^n := \sum_{k=0}^{\infty} h_{t_k^n} \mathbf{1}_{(t_k^n, t_{k+1}^n]}(s).$$

Continuity of h allows us to choose a localizing sequence (τ_m) such that $h \mathbf{1}_{[0, \tau_m]}$ is uniformly bounded in (t, ω) , for all $m \in \mathbb{N}$. In particular, $h^n \mathbf{1}_{[0, \tau_m]}$ is a simple process for all $n, m \in \mathbb{N}$ and

$$h \mathbf{1}_{[0, \tau_m]} \in \mathcal{L}(M^m)$$

for all $m \in \mathbb{N}$ where $M^m := M^{\tau_m}$. Since $t \mapsto h \mathbf{1}_{[0, \tau_m]}$ is still adapted and continuous, hence predictable, it follows as in the proof of Proposition 4.2.8) that

$$\|h^n \mathbf{1}_{[0, \tau_m]} - h \mathbf{1}_{[0, \tau_m]}\|_{M^m} \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

and hence the corresponding stochastic integrals w.r.t. M^m converge ucp for all $m \in \mathbb{N}$. The result now follows from the definition of stochastic integrals w.r.t. local martingales; see also the last part of the proof of Theorem 3.3.6. $\square$

We emphasize that we did not require $h \in \mathcal{L}_{loc}(M)$ that is, integrability of h , in the above proposition. This assumption was “replaced” by stopping the integrands; they were bounded after stopping and the stopped integrands were still predictable because they were still adapted and continuous. In particular, we have the following corollary.

Corollary 4.2.20 *If h is adapted and continuous, then the stochastic integral $h \cdot M$ exists for all continuous local martingales M .*

The following “dominated convergence-type” result for stochastic integrals can be viewed as a generalization of Proposition 4.2.19.

Lemma 4.2.21 *Let $M \in \mathcal{M}_{loc}^0$ and $h, h^n \in \mathcal{L}_{loc}(M)$ for all $n \in \mathbb{N}$. Suppose that $h^n \rightarrow h$ uniformly in probability on compact sets (ucp). Then*

$$h^n \cdot M \rightarrow h \cdot M \quad \text{ucp.}$$

PROOF: We may w.l.o.g. assume that $h = 0$. Let

$$\tau_m := \inf \{s \geq 0 : |M_s| \geq m\}; \quad M_s^m := M_{s \wedge \tau_m}; \quad \tau_{m,n} := \inf \left\{ s \geq 0 : \int_0^s |h_u^n|^2 d\langle M^m \rangle_u \geq \frac{1}{m} \right\}.$$

Then, it holds for all $t, \delta > 0$ that (recall that $h = 0$)

$$\mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^s h_u^n dM_u \right| \geq \delta \right) \leq \mathbb{P}(\tau_{m,n} \wedge \tau_m \leq t) + \mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^{s \wedge \tau_{m,n} \wedge \tau_m} h_u^n dM_u \right| \geq \delta \right).$$

Using first Chebychev's inequality, then Doob's L^2 -maximal inequality and then that fact that $\|M\|_{\mathcal{M}}^2 = \mathbb{E}[\langle M \rangle_\infty]$ for any $M \in \mathcal{M}$ the last term can be estimated from above by

$$\mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^s h_u^n \mathbf{1}_{[0, \tau_{m,n}]} dM_u^m \right| \geq \delta \right) \leq \frac{4}{\delta^2} \mathbb{E} \left[\int_0^t (h_u^n \mathbf{1}_{[0, \tau_{m,n}]})^2 d\langle M^m \rangle_u \right] \leq \frac{4}{\delta^2 m}.$$

We notice that Doob's inequality applies because the stochastic integral is a proper martingale, due to the stopped integrand.

For given $\varepsilon, \delta > 0$, we now first choose $m \in \mathbb{N}$ such that

$$\mathbb{P}(\tau_m \leq t) \leq \varepsilon/3 \quad \text{and} \quad \frac{4}{\delta^2 m} \leq \varepsilon/3.$$

Having fixed a martingale M^m we now use the uniform ucp-convergence of the sequence (h^n) to deduce from

$$\int_0^s |h_u^n|^2 d\langle M^m \rangle_u \leq \left(\sup_{0 \leq u \leq s} |h_u^n|^2 \right) \langle M^m \rangle_s \xrightarrow{\mathbb{P}} 0$$

that we can choose $n_0 \in \mathbb{N}$ such that for all $n \geq n_0$ it holds that $\mathbb{P}(\tau_{m,n} \leq t) \leq \varepsilon/3$. For those $n \in \mathbb{N}$ we then obtain that

$$\mathbb{P} \left(\sup_{0 \leq s \leq t} \left| \int_0^s h_u^n dM_u \right| \geq \delta \right) \leq \varepsilon.$$

□

4.3 The Itô formula

Our next aim is to formulate and prove Itô's formula, the chain rule of stochastic calculus. An adapted process X is a semimartingale if it allows the decomposition

$$X_t = x + M_t + A_t$$

where M is a continuous local martingale starting in zero and A is a process of bounded variation, also starting in zero. By Lemma 3.3.11 this decomposition is unique. We start with the integration by parts formula.

Theorem 4.3.1 *Let X, Y be continuous semi-martingales. Then,*

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \langle X, Y \rangle_t.$$

PROOF: Let $\Pi^n = \{t_i^n : i \in \mathbb{N}\}$ be a partition of $[0, \infty)$. Then

$$\begin{aligned} X_t Y_t - X_0 Y_0 &= \sum_{k=0}^{\infty} (X_{t_{k+1}^n \wedge t} - X_{t_k^n \wedge t}) (Y_{t_{k+1}^n \wedge t} - Y_{t_k^n \wedge t}) \\ &\quad + \sum_{k=0}^{\infty} Y_{t_k^n \wedge t} (X_{t_{k+1}^n \wedge t} - X_{t_k^n \wedge t}) + \sum_{k=0}^{\infty} X_{t_k^n \wedge t} (Y_{t_{k+1}^n \wedge t} - Y_{t_k^n \wedge t}). \end{aligned}$$

The convergence of the last two sums to the corresponding integrals follows from our Riemann-type approximation result for stochastic integrals. The convergence of the first term follows by polarization and the previously established fact that the sum of quadratic increments of continuous local martingales converges in probability uniformly on compact time intervals to the bracket process. $\square$

We are now ready to state and proof the Itô formula for continuous semi-martingales. The proof uses the fact that the class of polynomials is dense in the set of continuous functions with compact supports (Weierstrass' approximation result).

Theorem 4.3.2 (*Itô formula*) Let $f \in C^2(\mathbb{R}^d)$ and let $X = (X^1, \dots, X^d)$ be a vector of continuous semi-martingales. Then, a.s.

$$f(X_t) = f(X_0) + \sum_{i=1}^d \int_0^t \frac{\partial f}{\partial x_i}(X_u) dX_u^i + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 f}{\partial x_i \partial x_j}(X_u) d\langle M^i, M^j \rangle_u$$

where $dX^i = dM^i + dA^i$ and

$$\langle M^i, M^j \rangle := \frac{1}{4} [\langle M^i + M^j \rangle - \langle M^i - M^j \rangle].$$

PROOF: Obviously, Itô's formula holds for constant F as well as for $F(x) = x_k$ for some $k \in \{1, \dots, d\}$. Moreover, it is straightforward to check that Itô's formula for $F(x) = x_k x_l$ for some $k, l \in \{1, \dots, d\}$ is the same as the product rule that we established above.

Further, if the formula holds for $F, G \in C^2(\mathbb{R}^d)$, then it holds for each linear combination of F and G , so we already know that the formula is true for all polynomials of order at most 2. In a first step we are now going to extend the formula to general polynomials; then we use Weierstrass' approximation theorem to conclude.

- If the formula holds for $F \in C^2$, then it holds for $G(x) = x_i F(x)$. Indeed, by first applying the product rule to G and then Itô's formula to $F(X)$ we get that

$$\begin{aligned} G(X_t) - G(X_0) &= X_t^i F(X_t) - X_0^i F(X_0) \\ &= \int_0^t X_s^i dF(X_s) + \int_0^t F(X_s) dX_s^i + \langle X^i, F(X) \rangle_t \\ &= \int_0^t \sum_k X_s^i \frac{\partial F(X_s)}{\partial x_k} dX_s^k + \frac{1}{2} \int_0^t \sum_{k,l} X_s^i \frac{\partial^2 F(X_s)}{\partial x_k \partial x_l} d\langle X^k, X^l \rangle_s \\ &\quad + \int_0^t F(X_s) dX_s^i + \langle X^i, F(X) \rangle_t. \end{aligned}$$

Applying Itô's formula again to $F(X)$ and recalling that the cross-variation of two processes vanishes if one process is of bounded variation yields that

$$\langle X^i, F(X) \rangle_t = \sum_k \int_0^t \frac{\partial F}{\partial x_k}(X_s) d\langle X^i, X^k \rangle_s.$$

Moreover,

$$\frac{\partial G}{\partial x_k} = x_i \frac{\partial F}{\partial x_k} + \delta_{i,k} F, \quad \frac{\partial^2 G}{\partial x_k \partial x_l} = x_i \frac{\partial^2 F}{\partial x_k \partial x_l} + \delta_{i,k} \frac{\partial F}{\partial x_k} + \delta_{i,l} \frac{\partial F}{\partial x_l}.$$

We conclude that Itô's formula holds for G and hence for all polynomials.

- By Weierstrass' approximation result F along with its first two derivatives can be uniformly approximated by polynomials p^n on compact sets. Thus, if X almost never leaves some compact set $K := [-k, k]^d$, then it follows from the preceding lemma that

$$\begin{aligned} p^n(X_t) - F(X_0) &= \sum_{i=1}^d \int_0^t \frac{\partial p^n}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 p^n}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \\ &\xrightarrow{n \rightarrow \infty} \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \end{aligned}$$

where the converge holds ucp, due to Lemma 4.2.21 and

$$p^n(X_t) \xrightarrow{n \rightarrow \infty} F(X_t).$$

- For a general continuous semimartingale X we may w.l.o.g. assume that $x = 0$ and define the sequence of stopping times

$$\tau^k := \inf\{t \geq 0 \mid X_t \notin [-k, +k]^d\}.$$

The stopped process $X^k := X^{\tau^k}$ is a bounded semimartingale. The Itô formula holds for all the stopped process; thus, there exist sets of full measure Ω_k such that for every $k \in \mathbb{N}$ and all $\omega \in \Omega_k$ the equation holds on $[0, \tau^k(\omega)]$.

The set $\Omega^* := \bigcap_k \Omega_k$ has full measure. Since $\tau^k \uparrow \infty$ the equation actually holds on $\bigcup_k [0, \tau^k(\omega)] = [0, \infty)$ for all $\omega \in \Omega^*$. More precisely, by Lemma 4.2.21 it holds for all $t \geq 0$ that

$$\begin{aligned} F(X_t^k) - F(X_0^k) &= \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x_i}(X_s^k) dX_s^{k,i} + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \\ &= \sum_{i=1}^d \int_0^{t \wedge \tau^k} \frac{\partial F}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^{t \wedge \tau^k} \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s \\ &\xrightarrow[k \rightarrow \infty]{ucp} \sum_{i=1}^d \int_0^t \frac{\partial F}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j=1}^d \int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s. \end{aligned}$$

□

Exercise 44 Let (B_t) and (W_t) be two independent one dimensional Brownian motions. Find dX_t for each of the following cases:

(i) $X_t = B_t \exp(W_t + t),$

(ii) $X_t = \sin(B_t + \int_0^t W_s ds)$

4.3.1 Examples and applications

In this section we consider some examples in a Brownian setting including exponential martingales and probabilistic representations of boundary value problems. In particular, we show that the 2-dimensional Brownian motion is recurrent while the 3-dimensional Brownian motion is transient. The latter allows us to construct a local martingale that is not a martingale.

We start with Burkholder-Davis-Gundy (BDG) inequality. This inequality relates the maximum of a local martingale with its quadratic variation.

Theorem 4.3.3 (BDG inequality) *For any $p \geq 2$ there exists a constant C_p such that for any $M \in \mathcal{M}_{loc}^0$ and any $t \geq 0$ the following holds:*

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |M_s|^p \right] \leq C_p \mathbb{E} \left[\langle M \rangle_t^{p/2} \right].$$

PROOF: By stopping it suffices to consider the case where M and its quadratic variation are bounded. Since the function $m \mapsto |m|^p$ is smooth for $p \geq 2$ it follows by Itô's formula that

$$|M_t|^p = p \int_0^t \text{sign}(M_s) |M_s|^{p-1} dM_s + \frac{p(p-1)}{2} \int_0^t |M_s|^{p-2} d\langle M \rangle_s$$

Since the stochastic integral is a true martingale starting in zero, it drops out when taking expected values. Thus, by Doob's maximal inequality there exists a constant $C > 0$ such that (recall that $M_t^* := \sup_{s \leq t} |M_s|$)

$$\begin{aligned} \mathbb{E} |M_t^*|^p &\leq C \mathbb{E} \left[\int_0^t |M_s|^{p-2} d\langle M \rangle_s \right] \\ &\leq C \mathbb{E} [|M_t^*|^{p-2} \langle M \rangle_t] \\ &\leq C (\mathbb{E} [|M_t^*|^p])^{1-2/p} \left(\mathbb{E} [\langle M \rangle_t^{p/2}] \right)^{2/p} \quad (\text{Hölder with } p' = \frac{p}{p-2}; q' = \frac{p}{2}) \end{aligned}$$

Dividing both sides by $(\mathbb{E} [|M_t^*|^p])^{1-2/p}$ and then raising both sides to the power $p/2$ yields the result. $\square$

Remark 4.3.4 *It follows from the BDG inequality that a local martingale M is a martingale if $\mathbb{E}[\langle M \rangle_\infty] < \infty$; in this case M^* and hence M is L^2 -bounded.*

Next, we introduce exponential martingales. Such martingales will be very important below when we introduce equivalent changes of measures and Girsanov's theorem.

Example 4.3.5 (Exponential martingale) *Let $L \in C[0, \infty)$ and $X_t = \int_0^t L_u dB_u$. The process*

$$\mathcal{E}(X)_t := e^{X_t - \frac{1}{2} \int_0^t L_u^2 du} = e^{X_t - \frac{1}{2} \langle X \rangle_t}$$

is called exponential martingale. Applying Itô's formula to $f(t, x) = e^{x - \frac{1}{2} \int_0^t L_u^2 du}$ gives, a.s.

$$\mathcal{E}(X)_t = 1 + \int_0^t f(u, X_u) L_u dB_u.$$

We now check that $\mathcal{E}(X)$ is indeed a martingale. We have that

$$\int_0^t \mathbb{E}|f(u, X_u)L_u|^2 du \leq \sup_{0 \leq u \leq t} L_u^2 \int_0^t \mathbb{E}e^{2X_u} du. \quad (4.4)$$

Hence, we need to compute $\mathbb{E}e^{2X_u}$. Since $X_u \sim N(0, \sigma^2(u))$ with $\sigma^2(u) = \int_0^u L_s^2 ds$ (check this as an exercise using the L^2 approximation of the stochastic integral; it is important that L is a deterministic function) we get that $\mathbb{E}e^{2X_u} = e^{2\sigma^2(u)}$. In view of (4.4) this shows that $f(\cdot, X)L \in \mathcal{L}(B)$ and hence that

$$(\mathcal{E}(X)_t - 1)_{t \geq 0} \in \mathcal{M}.$$

In differential form the process $\mathcal{E}(X)$ can be expressed as

$$d\mathcal{E}(X)_t = \mathcal{E}(X)_t dX_t.$$

Since this is similar to the exponential function, $\mathcal{E}(X)$ is also called the stochastic exponential of X .

We can write down the stochastic exponential for much more general local martingales. The verification of the martingale property relied heavily on the deterministic integrand, though. We shall consider more general situations in Chapter 5.

Another important application of Itô's formula is the link between Brownian motion and the Laplace operator. Specifically, Itô's formula allows for probabilistic representations of solutions to second order linear PDEs.

Example 4.3.6 (Probabilistic representation of boundary value problems) Suppose that the function

$$F \in C_b([0, T] \times \bar{\mathcal{D}}) \cap C^{1,2}((0, T) \times \mathcal{D})$$

solves the reverse heat equation

$$F_t(t, x) + \frac{1}{2} F_{xx}(t, x) = 0$$

on the bounded domain $\mathcal{D} \subset \mathbb{R}$ with bounded boundary condition h on $(T \times \mathcal{D}) \cup ((0, T) \times \partial\mathcal{D})$. Let B be a one-dimensional standard Brownian motion and let

$$\tau := \inf\{t : B_t \notin \mathcal{D}\} \wedge T.$$

By Itô's formula, we have that

$$F(\tau, B_\tau) = F(0, 0) + \int_0^\tau F_t(s, B_s) ds + \int_0^\tau F_x(s, B_s) dB_s + \frac{1}{2} \int_0^\tau F_{xx}(s, B_s) ds.$$

The stochastic integral is a (true) martingale on $[0, \tau]$ because $\mathcal{D}$ is bounded. Thus, $\mathbb{E} \int_0^\tau F_x(s, B_s) dB_s = 0$. Since F satisfies the reverse heat equation,

$$F(\tau, B_\tau) = F(0, 0) + \int_0^\tau F_{xx}(s, B_s) ds,$$

and since $F(\tau, B_\tau) = h(\tau, B_\tau)$ we conclude that

$$F(0, 0) = \mathbb{E}h(\tau, B_\tau).$$

Similarly, we can start the Brownian motion at time t in x . If we denote by $\mathbb{P}_{t,x}$ its resulting distribution and by $\mathbb{E}_{t,x}$ the expected value with respect to $\mathbb{P}_{t,x}$, then

$$F(t, x) = \mathbb{E}_{t,x} h(\tau, B_\tau).$$

Hence we obtain a probabilistic representation of the solution to the heat equation with boundary condition h .

Next, we study transience and recurrence properties of multi-dimensional Brownian motion. We call a standard multi/dimensional Brownian motion recurrent if any neighborhood of the origin is visited infinitely often and transient if this is not the case. As a byproduct we shall construct a local martingale that is not a martingale.

We use the following facts about Brownian motion that have been proved in Stochastic II. If W is a standard one-dimensional Brownian motion and W^* its running maximum, then

$$\mathbb{P}[W_t^* > a] = \mathbb{P}[|W_t| > a].$$

In particular, for any positive numbers $a, b > 0$ the stopping times

$$\tau_a := \inf\{t \geq 0 : W_t > a\}, \quad \tau_b := \inf\{t \geq 0 : W_t < -b\}, \quad \tau := \tau_a \wedge \tau_b.$$

are almost surely finite and $\mathbb{P}[\tau_a < \tau_b] = \frac{b}{a+b}$ and $\mathbb{P}[\tau_b < \tau_a] = \frac{a}{a+b}$.

Example 4.3.7 (2-dimensional Brownian motion is recurrent) Let $D \subset \mathbb{R}^d$ be open. A function $f \in C^2(D)$ is called harmonic if $\Delta f := \sum_{i=1}^d f_{x_i x_i} = 0$. In this case, if $0 \in D$ and B is a d -dimensional Brownian motion, it follows from Itô's formula that

$$f(B_t) = f(0) + \int_0^t \nabla f(B_s) dB_s$$

as long as B does not leave D . Let us now consider the special case where $f : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}$ be given by

$$f(x) = \ln|x|;$$

here, $|x| := \sqrt{\sum_{i=1}^2 x_i^2}$ denotes the Euclidean norm. Then,

$$f_{x_i} = \frac{x_i}{|x|^2}, \quad f_{x_i x_i} = \frac{1}{|x|^2} - \frac{2x_i^2}{|x|^4}.$$

Thus, f is harmonic on $\mathbb{R}^2 \setminus \{0\}$ and hence also on the open disc $D_{r,R} := \{x \in \mathbb{R}^2 : r < |x| < R\}$. We now show that if we start a Brownian motion at some point in the plane, then it will return to any neighborhood of the origin - and hence any other point - infinitely often. We shall see that this is not the case for 3-dimensional Brownian motions. Let

$$h(x) = \frac{\ln R - \ln|x|}{\ln R - \ln r}.$$

Then h is harmonic on $D_{r,R}$ and $h(x) = 0$ if $|x| = R$ and $h(x) = 1$ if $|x| = r$. Let $x \in D_{r,R}$ and

$$\mathbb{P}_x := \mathbb{P} \circ (x + B)^{-1}$$

be the law of a Brownian motion starting in x . Let

$$\tau_r := \inf\{t \geq |B_t + x| = r\}, \quad \tau_R := \inf\{t \geq |B_t + x| = R\}, \quad \tau := \tau_r \wedge \tau_R.$$

Then τ is a stopping time and $\mathbb{P}_x(\tau < \infty) = 1$. Moreover, $h(B)$ and $\nabla h(B)$ are $\mathbb{P}_x$ -a.s. bounded on $[0, \tau]$. Thus, by Itô's formula

$$h(x) = \mathbb{E}_x h(B_\tau) = \mathbb{P}_x(\tau_r < \tau_R).$$

In view of the previously recalled results on Brownian exit times, $\tau_R \rightarrow \infty$ a.s. for $R \rightarrow \infty$. Since $h(x) \rightarrow 1$ if $R \rightarrow \infty$ for all $|x| > r > 0$ it follows from dominated convergence that $\mathbb{P}_x(\tau_r < \infty) = 1$. That is, if we start a 2-dimensional Brownian motion at some point $x \neq 0$, then it will return to any neighbourhood of the origin infinitely often.

Exercise 45 (3-dimensional Brownian motion is transient) Show that the Brownian motion in dimension $d \geq 3$ is transient and that the sequence $\{\tau_{n^{-1}}\}$ converges to $+\infty$ a.s. as $n \rightarrow \infty$.

Hint. Proceed as in the preceding example but with

$$h(x) = \frac{R^{2-d} - |x|^{2-d}}{\ln R^{2-d} - r^{2-d}}.$$

Example 4.3.8 (A local martingale that is not a martingale) Related to the previous example/exercise we are now going to consider a local martingale that is not a strict martingale. To this end, let B be a 3-dimensional Brownian motion, let $f(x) = |x|^{-1}$ for $x \in \mathbb{R}^3 \setminus \{0\}$, and let τ_r be the stopping time introduced in the previous examples. Since f is harmonic it follows by Itô's formula that

$$f(x + B_{t \wedge \tau_r}) = x + \int_0^{t \wedge \tau_r} \nabla f(x + B_s) dB_s.$$

Hence the process $R_t = f(x + B_t)$ is a bounded martingale on $[0, \tau_r]$. Moreover, it follows from Exercise 45 that the sequence $\{\tau_{n^{-1}}\}$ defines a localising sequence. Thus, (R_t) is a local martingale. If the process (R_t) were a (true) martingale, then it would have constant expectation.

To see that this is not the case, we first recall that the sum of the squares of n independent $N(0, 1)$ distributed random variables is χ_n^2 -distributed. More generally, the sum S of n independent squared $N(\mu_i, 1)$ distributed random variables follows an $\chi_n^2(\lambda)$ distribution (non-central chi-square distribution) where $\lambda = \sum_i \mu_i^2$. Now, let

$$X := R_t^{-2}, \quad Y := \sum_{i=1}^3 \left(B_1^i + \frac{x^i}{\sqrt{t}} \right)^2.$$

Then

$$X = t \cdot Y \quad \text{in law and} \quad Y \sim \chi_3^2\left(\frac{|x|^2}{\sqrt{t}}\right)$$

Let $Z \sim \chi_3^2$. One can show (exercise) that

$$\mathbb{E}\left[\frac{1}{Y}\right] \leq \mathbb{E}\left[\frac{1}{Z}\right] = 1.$$

The equality follows from the fact that there is a closed form expression for the expected value of the inverse of a (standard, yet not generalized) chi-square distributed random variable. Thus, it follows from Cauchy-Schwarz that

$$\mathbb{E}[R_t] \leq (\mathbb{E}[R_t^2])^{1/2} \leq \frac{1}{\sqrt{t}}.$$

If (R_t) were a martingale, then this would contradict $\mathbb{E}R_0 = \frac{1}{|x|} > 0$.

Exercise 46 Let $X \sim \chi_n^2(\lambda)$ for some $\lambda > 0$ and $n > 2$ and let $Y \sim \chi_n^2$. Prove that $\mathbb{E}[\frac{1}{X}] \leq \mathbb{E}[\frac{1}{Y}]$.

Hint. Use that the noncentral chi-square density admits the Poisson mixture representation

$$f_\lambda(x) = \sum_{k=0}^{\infty} e^{-\lambda/2} \frac{(\lambda/2)^k}{k!} f_{n+2k}(x), \quad x > 0,$$

where f_m is the density of χ_m^2 and that for $m > 2$,

$$\mathbb{E}\left[\frac{1}{\chi_m^2}\right] = \frac{1}{m-2}.$$

4.3.2 The Feynman-Kac formula

In the previous subsection we illustrated how the solution to the heat equation with given terminal or boundary values can be represented stochastically. In this section we extend the result to a more general class of linear PDEs. Specifically, we consider the following Cauchy problem:

$$\begin{aligned} -\frac{\partial v}{\partial t} + kv &= \frac{1}{2}\Delta v + g \quad \text{on } [0, T) \times \mathbb{R}^d \\ v(T, x) &= f(x) \quad \text{on } \mathbb{R}^d \end{aligned} \tag{4.5}$$

where $f : \mathbb{R}^d \rightarrow \mathbb{R}, k : \mathbb{R}^d \rightarrow [0, \infty), g : [0, T] \times \mathbb{R}^d \rightarrow [0, \infty)$ are continuous.

Theorem 4.3.9 (Feynman-Kac formula) Suppose that $v \in C^{1,2}$ satisfies the Cauchy problem (4.5) and assume that

$$\max_{0 \leq t \leq T} |v(t, x)| + \max_{0 \leq t \leq T} |g(t, x)| \leq Ke^{a|x|^2}$$

for some $K > 0$ and $a < \frac{1}{2dT}$. Then it admits the representation

$$\begin{aligned} v(t, x) = & \mathbb{E}_x \left[f(B_{T-t}) \exp \left(- \int_0^{T-t} k(B_s) ds \right) \right. \\ & \left. + \int_0^{T-t} g(t+\theta, B_\theta) \exp \left(- \int_0^\theta k(B_s) ds \right) d\theta \right]; \quad \text{on } [0, T] \times \mathbb{R}^d. \end{aligned}$$

in terms of a d -dimensional Brownian motion $B = (B^1, \dots, B^d)$ starting in x .

PROOF: Using the fact that v satisfies our Cauchy problem and that the exponential term is of bounded variation we obtain from Itô's formula (for fixed $t \in [0, T]$) that

$$\begin{aligned} & d \left[v(t+\theta, B_\theta) \exp \left(- \int_0^\theta k(B_s) ds \right) \right] \\ &= \exp \left(- \int_0^\theta k(B_s) ds \right) \left[-g(t+\theta, B_\theta) d\theta + \sum_{i=1}^d \frac{\partial}{\partial x_i} v(t+\theta, B_\theta) dB_\theta^i \right]. \end{aligned}$$

Let $\tau_n := \inf\{t \geq 0 : |B_t| \geq n\sqrt{d}\}$ and $0 < r < T - t$ and integrate the above equation on $[0, r \wedge \tau_n]$. The resulting stochastic integrals are martingales so

$$\begin{aligned} v(t, x) = & \mathbb{E}_x \int_0^{r \wedge \tau_n} g(t + \theta, B_\theta) \exp\left(-\int_0^\theta k(B_s) ds\right) d\theta \\ & + \mathbb{E}_x \left[v(t + \tau_n, B_{\tau_n}) \exp\left(-\int_0^{\tau_n} k(B_s) ds\right) \mathbf{1}_{\{\tau_n \leq r\}} \right] \\ & + \mathbb{E}_x \left[v(t + r, B_r) \exp\left(-\int_0^r k(B_s) ds\right) \mathbf{1}_{\{\tau_n > r\}} \right]. \end{aligned}$$

The first term on the right-hand side converges to

$$\mathbb{E}_x \int_0^{T-t} g(t + \theta, B_\theta) \exp\left(-\int_0^\theta k(B_s) ds\right) d\theta \quad (4.6)$$

as $n \rightarrow \infty$ and $r \rightarrow T - t$ by monotone convergence because of the non-negativity of k and the exponential growth condition on g which guarantees that it has finite expectation.⁸

Using the growth condition on v the second term is dominated by

$$K e^{adn^2} \mathbb{P}_x[\tau_n \leq T] \leq K e^{adn^2} \sum_{j=1}^d \mathbb{P}_x[\max_{0 \leq t \leq T} |B^j| \geq n].$$

By Lemma 2.5.2 we know that $\int_x^\infty e^{-y^2/2} dy \leq \frac{1}{x} e^{-x^2/2}$. Using the fact that the running maximum W^* of a standard Brownian motion W satisfies $\mathbb{P}(W_t^* \geq x) = 2\mathbb{P}(W_t \geq x)$ we conclude that (shift by the starting value, rescale by time)

$$\mathbb{P}_x[\max_{0 \leq t \leq T} |B^j| \geq n] \leq 2\sqrt{\frac{T}{2\pi}} \frac{1}{n \mp x_j} e^{-(n \mp x_j)^2/2T}.$$

Since $0 < a < \frac{1}{2dT}$ the second term converges to 0 as $n \rightarrow \infty$. The third term converges again by dominated convergence to

$$\mathbb{E}_x \left[v(T - t, B_{T-t}) \exp\left(-\int_0^{T-t} k(B_s) ds\right) \right] = \mathbb{E}_x \left[f(B_{T-t}) \exp\left(-\int_0^{T-t} k(B_s) ds\right) \right]$$

as $n \rightarrow \infty$ and $r \rightarrow T - t$. □

Remark 4.3.10 A sufficient condition for the existence of a $C^{1,2}$ solution is that k is bounded and uniformly Hölder continuous, that g is Hölder continuous in the space variable uniformly in (t, x) and that g and f are of polynomial growth:

$$\max_{1 \leq t \leq T} |g(t, x)| + |f(x)| \leq L(1 + \|x\|^\nu), \quad \text{for some } \nu > 0.$$

The one-dimensional Cauchy problem 4.5 with $g = 0$ can be reduced to the heat equation as shown by the following exercise.

⁸Note that $\frac{1}{T}|B_T|^2 \sim \chi_d^2$ and recall that the moment generating function of a χ_n^2 -distributed random variable X is given by $\mathbb{E}[e^{tX}] = \frac{1}{(1-2t)^{n/2}}$ for $t < \frac{1}{2}$.

Exercise 47 Let u satisfy the one-dimensional heat equation

$$u_t = \frac{1}{2}u_{xx}$$

$$u(0, x) = \psi(x).$$

i) Solve the initial value problem

$$v_t = \left(\frac{1}{2}\beta^2 - \alpha\right)v + \beta v_x + \frac{1}{2}v_{xx}$$

$$v(0, x) = e^{-\beta x}\psi(x). \tag{4.7}$$

Hint. The solution can be expressed in terms of the solution to the heat equation.

ii) Use part (i) to solve, for $a > 0$ the initial value problem

$$w_t = \frac{a}{2}w_{xx} + bw_x + cw$$

$$w(0, x) = \Psi(x). \tag{4.8}$$

Hence, any solution to (4.8) can be traced back to a solution to the heat equation. Uniqueness of solutions to the heat equation thus yields uniqueness of solutions to (4.8).

Chapter 5

Representation Theorems and Change of Measure

In this chapter we return to the Brownian setting. Our guiding question is when continuous (semi-) martingales are integrals w.r.t. Brownian motion, respectively Brownian motions, respectively Brownian motions w.r.t. a different measure. For this, we fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ that carries a standard Brownian motion B and assume that $(\mathcal{F}_t)$ is the Brownian filtration augmented by the null sets.

We recall that $\mathcal{L}_2$ denotes the class of all measurable, $(\mathcal{F}_t)$ -adapted and $\mathbb{P} \otimes \lambda$ square integrable functions $f : [0, T] \times \Omega \rightarrow \mathbb{R}$ and denote by $\mathcal{L}_{loc}$ the class of all measurable and adapted processes f that satisfy

$$\mathbb{P}\left[\int_0^T f^2(s, \omega) ds < \infty\right] = 1 \quad \text{for all } T > 0.$$

5.1 The martingale representation theorem

We know that Itô integrals w.r.t. Brownian motion are local martingales with respect to the canonical filtration generated by the Brownian motion B . In this section we prove a converse statement, namely that every square integrable local martingale with respect to a Brownian filtration (this is important!) can be represented as a stochastic integral with respect to B .

We work on a finite time interval $[0, T]$ and assume w.l.o.g. that $\mathcal{F} = \mathcal{F}_T$. Our goal is to prove the following *Itô Representation Theorem*.

Theorem 5.1.1 *Let $F \in L^2(\mathcal{F})$. Then there exists a unique process $f \in \mathcal{L}_2$ such that*

$$F(\omega) = \mathbb{E}[F] + \int_0^T f(t, \omega) dB_t(\omega).$$

Before proving this theorem we state and prove the *Martingale Representation Theorem*.

Theorem 5.1.2 *Let (M_t) be an $(\mathcal{F}_t)$ -martingale with $M_t \in L^2$ for all $t \in [0, T]$. Then there exists a unique process $g \in \mathcal{L}_2$ such that*

$$M_t = \mathbb{E}[M_0] + \int_0^t g(s, \omega) dB_s \quad \mathbb{P}\text{-a.s. for all } t \geq 0.$$

PROOF: Applying Theorem 5.1.1 to the square integrable random variable M_t yields a process $f^{(t)}$ such that

$$M_t = \mathbb{E}[M_t] + \int_0^t f^{(t)} dB_s.$$

For $0 \leq t_1 \leq t_2 \leq T$ this yields

$$M_{t_1} = \mathbb{E}[M_{t_2} | \mathcal{F}_{t_1}] = \mathbb{E}[M_0] + \mathbb{E}\left[\int_0^{t_2} f^{(t_2)} dB_s | \mathcal{F}_{t_1}\right] = \mathbb{E}[M_0] + \int_0^{t_1} f^{(t_2)} dB_s$$

because Itô integrals are martingales. At the same time

$$M_{t_1} = \mathbb{E}[M_0] + \int_0^{t_1} f^{(t_1)} dB_s$$

so

$$\int_0^{t_1} (f^{(t_2)} - f^{(t_1)}) dB_s = 0.$$

Hence, by the Itô isometry

$$0 = \mathbb{E}\left[\left(\int_0^{t_1} (f^{(t_2)} - f^{(t_1)}) dB_s\right)^2\right] = \int_0^{t_1} \mathbb{E}[(f^{(t_2)} - f^{(t_1)})^2] ds$$

and so $f^{(t_1)} = f^{(t_2)}$ for almost every $(s, \omega) \in [0, t_1] \times \Omega$. This proves the assertion. $\square$

The martingale representation theorem yields a fairly abstract result. A closed form expression for the integrands can only be obtained in special cases.

Example 5.1.3 (i) *Let $F = B_1^2$ and consider the martingale*

$$M_t := \mathbb{E}[F | \mathcal{F}_t] = B_t^2 + (1 - t).$$

By Itô's formula

$$B_t^2 = t + 2 \int_0^t B_s dB_s$$

so that we obtain the following representation of (M_t) :

$$M_t = 1 + 2 \int_0^t B_s dB_s.$$

(ii) *Let $F = \int_0^T B_s ds - TB_T$. The integration by parts formula yields*

$$F = \int_0^T B_s ds - TB_T = - \int_0^T s dB_s - \int_0^T (T - t) dB_t.$$

(iii) Let $F = \exp(B_T)$. Since

$$d\left(\exp\left(B_t - \frac{1}{2}t\right)\right) = \exp\left(B_t - \frac{1}{2}t\right)dB_t$$

we have

$$\exp\left(B_T - \frac{1}{2}T\right) = 1 + \int_0^T \exp\left(B_t - \frac{1}{2}t\right)dB_t$$

and hence

$$F = \exp\left(\frac{1}{2}T\right) + \int_0^T \exp\left(B_t + \frac{1}{2}(T-t)\right)dB_t.$$

(iv) Let $F = \sin(B_T)$. We have that

$$d\left(e^{\frac{1}{2}t} \sin(B_t)\right) = e^{\frac{1}{2}t} \cos(B_t)dB_t.$$

or

$$e^{\frac{1}{2}T} \sin(B_T) = \int_0^T e^{\frac{1}{2}t} \cos(B_t)dB_t.$$

Hence

$$\sin(B_T) = \int_0^T e^{\frac{1}{2}(t-T)} \cos(B_t)dB_t.$$

Proof of Theorem 5.1.1.

In this subsection, we prove the Itô representation theorem. To this end, we first recall that if H is any Hilbert space and $X \subseteq H$ is a closed subset, then

$$H = X \oplus X^\perp$$

Let us now put

$$\mathcal{I} := \left\{ \int_0^T f dB : f \in \mathcal{L}_2 \right\}, \quad \mathcal{H} := \{H \in L^2 : \mathbb{E}H = 0\}.$$

Then $\mathcal{H} \subseteq L^2(\mathcal{F}_T)$ closed and hence a Hilbert space w.r.t. the usual inner product $\langle X, Y \rangle_{L^2} = \mathbb{E}[XY]$ on L^2 . Moreover, we have the following result, due to the Itô isomerie.

Lemma 5.1.4 *The set $\mathcal{I}$ is a closed subset of $\mathcal{H}$. In particular,*

$$\mathcal{H} = \mathcal{I} \oplus \mathcal{I}^\perp.$$

PROOF: Let $\{X_n\}$ be a sequence in $\mathcal{I}$ with corresponding integrands $\{f_n\}$. Let $X_n \rightarrow X$ in $L^2(\mathbb{P})$. Then $\{X_n\}$ is Cauchy in $L^2(\mathbb{P})$. By the Itô isomerie $\{f_n\}$ is Cauchy in $L^2(\lambda \otimes \mathbb{P})$ and hence converges to some f . Using the Itô isomerie again we see that $X_n \rightarrow \int_0^T f dB$ and hence $X \in \mathcal{I}$. $\square$

Our goal is to prove that $\mathcal{I}^\perp$ contains only the zero element. To this end, let $Z \in \mathcal{I}^\perp$. Then $\mathbb{E}[ZN] = 0$ for all $N \in \mathcal{I}$. We will show that this actually implies that

$$\mathbb{E}[ZH] = 0 \quad \text{for all } H \in \mathcal{H}. \quad (5.1)$$

Taking $H = Z$ in the above equation we conclude that $Z = 0$ a.s. As a result, $\mathcal{I}^\perp = \{0\}$, hence

$$\mathcal{H} = \mathcal{I}.$$

The proof of (5.1) proceeds in two steps. The first step uses the Stone-Weierstrass theorem.

Theorem 5.1.5 (*Stone-Weierstrass Theorem, algebraic version*) Let $\mathbb{K}$ be equal to $\mathbb{C}$ or $\mathbb{R}$. Let $\mathcal{A}$ be an algebra of continuous $\mathbb{K}$ -valued functions on a compact Hausdorff space X that contains the constant function 1, separates points (that is, for all $x, y \in X, x \neq y$, then there exists $a \in \mathcal{A}$ such that $a(x) \neq a(y)$), and is stable with respect to complex conjugation if $\mathbb{K} = \mathbb{C}$. Then, $\mathcal{A}$ is dense in $C(X, \mathbb{K})$ with respect to the topology of uniform convergence.

Let $Z \in \mathcal{I}^\perp$ and suppose that we can prove that for any $m \in \mathbb{N}$ there exists an algebra $\mathcal{A}_m$ of continuous functions $g : \mathbb{R}^m \rightarrow \mathbb{R}$ that separate points on $\mathbb{R}^m$ (and hence on any compact subset of $\mathbb{R}^m$) such that

$$\mathbb{E}[Zg(B_{t_1}, \dots, B_{t_m})] = 0, \quad (5.2)$$

for any $t_1, \dots, t_m \in [0, T]$. Then it follows from that Stone-Weierstrass theorem that

$$\mathbb{E}[Zh(B_{t_1}, \dots, B_{t_m})] = 0 \quad (5.3)$$

for any continuous function $h : \mathbb{R}^m \rightarrow \mathbb{R}$ with compact support, any $m \in \mathbb{N}$ and any $t_1, \dots, t_m \in [0, T]$.

The next lemma shows that the equality (5.3) extends to all random variables $H \in L^2(\mathcal{F})$ so that it remains to construct the point-separating algebra that satisfies (5.2).

Lemma 5.1.6 For any $T > 0$, the following set of random variables is dense in $L^2(\mathcal{F}_T)$:

$$\left\{ g(B_{t_1}, \dots, B_{t_n}) : t_i \in [0, T], i = 1, 2, \dots, n \text{ and } g \in C_0(\mathbb{R}^n), n \in \mathbb{N} \right\}.$$

PROOF: Let $\{t_i\}_{i=1}^\infty$ be a dense subset of $[0, T]$, and for each $n \in \mathbb{N}$ let $\mathcal{H}_n$ be the σ -field generated by $B_{t_1}, \dots, B_{t_n}$. Then

$$\mathcal{F}_T = \bigvee_n \mathcal{H}_n.$$

Let us now choose a random variable $g \in L^2(\mathcal{F}_T)$. Then

$$\left\{ \mathbb{E}[g|\mathcal{H}_n], \mathcal{H}_n \right\}_{n \in \mathbb{N}}$$

forms an L^2 -bounded, hence almost surely and in L^2 convergent martingale. In view of Exercise 13¹

$$\lim_{n \rightarrow \infty} \mathbb{E}[g|\mathcal{H}_n] = g \quad \text{a.s.}$$

By the Doob-Dynkin (“transformation”) lemma a random variable is measurable with respect to the σ -field generated by another random variable if and only if it can be represented as a Borel-measurable function of that variable. Thus, there exist Borel-measurable functions $g_n : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\mathbb{E}[g|\mathcal{H}_n] = g_n(B_{t_1}, \dots, B_{t_n}).$$

The next exercise allows us to conclude. □

Exercise 48 Let μ be a probability measure on $(\mathbb{R}^m, \mathcal{B}^m)$. Prove that the class $C_0(\mathbb{R}^m)$ of all continuous real-valued functions on $\mathbb{R}^m$ with compact support is norm dense in $L^2(\mathbb{R}^m, \mathcal{B}^m, \mu)$. Use this result to finish the proof of Lemma 5.1.6.

¹Using the notation of the exercise $g = Y$, $\mathcal{F}_\infty = \mathcal{F}_T$ and so $\lim_{n \rightarrow \infty} \mathbb{E}[g|\mathcal{H}_n] = \mathbb{E}[g|\mathcal{F}_T] = g$ a.s.

The point separating algebra

Let us now construct the desired point-separating algebras. The set $\mathcal{A}_m$ of linear combinations of the functions

$$g_\theta(x_1, \dots, x_m) := \exp\left(\sum_{j=1}^m \theta_j(x_j - x_{j-1})\right), \quad x_0 = 0$$

where $\theta = (\theta_j) \in \mathbb{R}^m$ forms a subalgebra of $C(\mathbb{R}^m, \mathbb{R})$ that separates points on $\mathbb{R}^m$ and hence on any compact subset. Hence, for any $m \in \mathbb{N}$ the algebra $\mathcal{A}_m$ satisfies the assumptions of the Stone-Weierstrass theorem (when restricted to compact sets). It thus remains to prove that that

$$\mathbb{E}[Zg_\theta(B_{t_1}, \dots, B_{t_m})] = 0. \quad (5.4)$$

For this we recall that we call two (right-continuous) L^2 -bounded local martingales M, N orthogonal if their cross variation process vanishes, which we proved to be equivalent to their product MN is a (uniformly integrable) martingale.²

For $Z \in \mathcal{I}^1$ we now consider the L^2 -bounded martingale with zero expected value

$$Z_t := \mathbb{E}[Z | \mathcal{F}_t], \quad 0 \leq t \leq T.$$

Since our filtration satisfies the usual condition, (Z_t) is right continuous. In fact, by backwards martingale convergence, a.s. and in L^1 ,

$$\mathbb{E}[Z | \mathcal{F}_{t+\frac{1}{n}}] \rightarrow \mathbb{E}[Z | \mathcal{F}_{t+}] = \mathbb{E}[Z | \mathcal{F}_t]$$

Hence $Z_{t+\frac{1}{n}} \rightarrow Z_t$ a.s. and in L^1 . Moreover, this martingale is strongly orthogonal to any stochastic integral processes as shown by the following result.

Lemma 5.1.7 *Any integral process*

$$N_t := \int_0^t f dB, \quad f \in \mathcal{L}_2,$$

on $[0, T]$ is strongly orthogonal to the martingale (Z_t) , i.e. $(Z_t N_t)$ is a martingale on $[0, T]$.

PROOF: An adapted, right-continuous, integrable process X on $[0, T]$ is a martingale starting in zero if and only if for any stopping time $\tau \leq T$ it holds that

$$\mathbb{E}[X_\tau] = 0.$$

The “if-part” is clear; for the “only if part” let $s < t \leq T$, let $A \in \mathcal{F}_s$ and define a stopping time $\tau = s\mathbf{1}_A + t\mathbf{1}_{A^c}$. Then, by assumption

$$\mathbb{E}X_t = 0 \quad \text{and} \quad \mathbb{E}[X_\tau] = \mathbb{E}[X_s\mathbf{1}_A + X_t\mathbf{1}_{A^c}] = 0.$$

But this implies that

$$\mathbb{E}[X_t\mathbf{1}_A] = \mathbb{E}[X_s\mathbf{1}_A].$$

²If the product MN is a martingale, then it is automatically a uniformly integrable martingale, due to Exercise 13. We have convergence $M_t \rightarrow M_\infty$ and $N_t \rightarrow N_\infty$ in L^2 and hence convergence of the product in L^1 .

In our case, for any stopping time $\tau \leq T$, the process

$$N_t^\tau := \int_0^{\tau \wedge t} f dB = \int_0^t f^\tau dB$$

belongs to $\mathcal{I}$. Since $Z = Z_T \in \mathcal{I}^\perp$, since $N_T^\tau \in \mathcal{I}$ and because $N_T^\tau = N_\tau$ it follows that

$$0 = \mathbb{E}[Z_T N_T^\tau] = \mathbb{E}[\mathbb{E}[Z_T N_\tau | \mathcal{F}_\tau]] = \mathbb{E}[Z_\tau N_\tau] = \mathbb{E}[(ZN)_\tau].$$

This shows that ZN is indeed a (uniformly integrable) martingale. $\square$

We now take a particular integral process, namely the solution to the SDE

$$dM_t = \theta M_t dB_t.$$

We have already seen that the process (M_t) admits the explicit representation

$$M_t := \exp\left(\theta B_t + \frac{1}{2}|\theta|^2 t\right).$$

Moreover, we conclude from Exercise 18 that $\theta M \in \mathcal{L}_2$ on $[0, T]$ and so

$$M - 1 \in \mathcal{I}.$$

By Lemma 5.1.7 the product ZM is a martingale, too. For $t > s$ the martingale property of ZM implies

$$Z_s M_s = \mathbb{E}\left[Z_t \exp\left(\theta B_t + \frac{1}{2}|\theta|^2 t\right) | \mathcal{F}_s\right].$$

Hence

$$Z_s \exp\left(-\frac{1}{2}|\theta|^2(t-s)\right) = \mathbb{E}\left[Z_t \exp\left(\theta(B_t - B_s)\right) | \mathcal{F}_s\right].$$

Let us now take a partition $0 \leq t_1 < \dots < t_m = T$ of $[0, T]$ and repeat this argument. Using that $Z_T = Z$ and that

$$\mathbb{E}[Z_{t_1}] = \mathbb{E}[Z] = 0 \quad \text{because} \quad Z \in \mathcal{I}^\perp \subseteq \mathcal{H},$$

we get that

$$\mathbb{E}\left[Z \exp\left(\sum_j \theta_j (B_{t_j} - B_{t_{j-1}})\right)\right] = \mathbb{E}\left[Z_{t_1} \exp\left(-\frac{1}{2} \sum_j |\theta_j|^2 (t_j - t_{j-1})\right)\right] = 0.$$

This shows that (5.4) holds.

5.2 Lévy characterisation of Brownian motion

We know that the quadratic variation of Brownian motion is the identity function on $\mathbb{R}_+$. Lévy's characterisation of the Brownian motion provides a characterisation of the Brownian motion by this property.

Theorem 5.2.1 (*Lévy characterisation*) *Let M be an n -dimensional continuous local martingale starting in 0 with respect to a filtration $(\mathcal{F}_t)$ that satisfies the usual conditions. If M satisfies*

$$\langle M \rangle_t = \text{Id}(n) \cdot t \text{ for all } t \geq 0 \text{ a.s.}$$

where $\text{Id}(n)$ denotes the $n \times n$ identity matrix, then M is an n -dimensional $(\mathcal{F}_t)$ Brownian motion.

It is important to notice that the Lévy characterisation assumes a *continuous* local martingale. The compensated Poisson process with unit intensity provides an example of a *discontinuous* square integrable martingale with $\langle M \rangle_t = t$.

We start with a time-change result that is of independent interest. This result, as well as the general time-change theorem stated later require the following properties of generalized inverses.

Exercise 49 *Let (T_t) be a continuous, non-decreasing function with $T_0 = 0$ and $\lim_{t \rightarrow \infty} T_t = \infty$, and let*

$$A_t := \inf\{s \geq 0 : T_s > t\}$$

be the generalized inverse. Show that the function A has the following properties.

(i) *A is non-decreasing and right-continuous.*

(ii) $(T \circ A)_t = t$

(iii) $(A \circ T)_t = \sup\{s \geq t : T_s = T_t\}$

(iv) *If $\psi : [0, \infty) \rightarrow \mathbb{R}$ is continuous with the property that*

$$T_t = T_s \quad \text{for some } 0 \leq s < t$$

implies $\psi_s = \psi_t$, then $s \mapsto \psi \circ A_s$ is continuous and

$$(\psi \circ A \circ T)_s = \psi_s.$$

(v) *For any $0 \leq t, s < \infty$ it holds that*

$$s < T_t \Leftrightarrow A_s < t$$

and

$$A_s \leq t \Rightarrow s \leq T_t.$$

Our first time-change result shows that continuous local martingales remain local martingales when considered on a time-scale generated by stopping times.

Lemma 5.2.2 *Let $(\mathcal{F}_t)$ be a filtration that satisfies the usual conditions and let $\tau = (\tau_t)$ be a family of stopping times such that the mapping $t \mapsto \tau_t$ are a.s. continuous and non-decreasing and $\tau_t \uparrow \infty$. Let (M_t) is a continuous local martingale w.r.t $(\mathcal{F}_t)$. Then*

$$N_t := M_{\tau_t}$$

is a local martingale w.r.t. the filtration

$$\mathcal{G}_t := \mathcal{F}_{A_t}.$$

It is a continuous local martingale if τ is continuous.

PROOF: Since τ_t is a stopping time for each $t \geq 0$, the filtration $(\mathcal{G}_t)$ is well-defined. Moreover, since M is continuous and adapted, it is progressively measurable and hence

$$M_\sigma \in \mathcal{F}_\sigma$$

for any stopping time σ . In particular, N is $(\mathcal{G}_t)$ -adapted.

To construct a localising sequence for N we fix such a sequence $\{\nu_n\}$ for M . Since M is continuous we can choose the sequence so that the stopped processes M^{ν_n} are bounded, hence uniformly integrable.

Let A denote the $(\omega$ -wise) generalized inverse of τ . Our goal is to show that

$$\mu_n := A_{\nu_n} = \inf\{t \geq 0 : \tau_t > \nu_n\} = \inf\{t \geq 0 : \tau_t \geq \nu_n\}$$

defines a localizing sequence for N . Here, the equality follows from the fact that τ is continuous and increasing. By Exercise 49(v),

$$\{\mu_n \leq t\} = \{\tau_t \geq \nu_n\} \in \mathcal{F}_{\tau_t} = \mathcal{G}_t.$$

Hence $\{\mu_n\}$ is a sequence of $(\mathcal{G}_t)$ -stopping times. Since $\nu_n \uparrow \infty$ and $\tau_t \uparrow \infty$ we also have that $\mu_n \uparrow \infty$. Moreover, Exercise 49(ii) yields

$$\tau_{A_{\nu_n}} = \nu_n$$

from which we deduce the key equality $\tau_{t \wedge \mu_n} = \tau_t \wedge \nu_n$. In fact, if $t < \mu_n$ then $\tau_t < \nu_n$ and if $t \geq \mu_n$, then $\tau_t \geq \nu_n$. As a result,

$$N_{t \wedge \mu_n} = M_{\tau_{t \wedge \mu_n}} = M_{\tau_t \wedge \nu_n} = M_t^{\nu_n}.$$

Since M^{ν_n} is a uniformly integrable martingale we can apply optional sampling:

$$\mathbb{E}[N_{t \wedge \mu_n} | \mathcal{G}_s] = \mathbb{E}[M_{\tau_t}^{\nu_n} | \mathcal{F}_{\tau_s}] = M_{\tau_s}^{\nu_n} = N_{s \wedge \mu_n}.$$

Hence $\{\mu_n\}$ is a localising sequence for N and thus N is a (continuous) local $(\mathcal{G}_t)$ -martingale (if A is continuous). $\square$

The next result states that continuous local martingales are Brownian motions after time-suitable time-changes. As a special case, we obtain the Lévy characterisation in dimension one. The result easily extends to high dimensions.

Theorem 5.2.3 (*Time change/Dumbis-Dubins-Schwarz theorem.*) *Let X be a continuous local martingale on $(\Omega, (\mathcal{F}_t), \mathcal{F}, \mathbb{P})$ where $(\mathcal{F}_t)$ satisfies the usual conditions with continuous quadratic variation $\langle X \rangle$ that satisfies*

$$\lim_{t \rightarrow \infty} \langle X \rangle_t = +\infty.$$

Let

$$\tau_t := \inf\{s \geq 0 : \langle X \rangle_s > t\}.$$

Then $B_t := X_{\tau_t}$ defines an $(\mathcal{F}_{\tau_t})$ Brownian motion, $(\mathcal{F}_{\tau_t})$ satisfies the usual conditions and a.s.

$$X = B_{\langle X \rangle}.$$

In particular, if $\langle X \rangle_t \equiv t$, then X is a Brownian motion.

PROOF: We start by showing that $X_{\tau_t} - X_{\tau_s} \sim N(0, t - s)$ and independent of $\mathcal{F}_{\tau_s}$. This will be achieved by considering the moment generating function of the increments. To this end, we consider the continuous complex-valued process

$$Z_t := \exp\left(i\theta X_t + \frac{1}{2}\theta^2 \langle X \rangle_t\right)$$

Using Itô's formula on the real and complex part we see that this defines a continuous complex-valued local martingale that satisfies

$$Z_t = 1 + i\theta \int_0^t Z_s dX_s.$$

By Lemma 5.2.2

$$\{\tau_s < t\} = \{\langle X \rangle_t > s\} \in \mathcal{F}_t.$$

Thus, τ_s is optional. Since the filtration $(\mathcal{F}_t)$ is complete, τ_s is a stopping time. We conclude from the above lemma that

$$(Z_{\tau_t}, \mathcal{F}_{\tau_t})$$

is a local martingale starting in one. By Exercise 49 (ii),

$$Z_{\tau_t} = \exp\left(i\theta X_{\tau_t} + \frac{1}{2}\theta^2 \langle X \rangle_{\tau_t}\right) = \exp\left(i\theta X_{\tau_t} + \frac{1}{2}\theta^2 t\right)$$

and it follows from Euler's formula that the local martingale is uniformly bounded on any bounded time interval. Hence, it is a (true) martingale. In particular, for any $0 \leq s \leq t$

$$\mathbb{E}\left[\frac{Z_{\tau_t}}{Z_{\tau_s}} \mid \mathcal{F}_{\tau_s}\right] = 1$$

from which we deduce that

$$\mathbb{E}\left[\exp(i\theta(X_{\tau_t} - X_{\tau_s})) \mid \mathcal{F}_{\tau_s}\right] = \exp\left(-(t-s)\frac{\theta^2}{2}\right).$$

By Lemma 5.2.4 below it follows that $(X_{\tau_t} - X_{\tau_s}) \perp \mathcal{F}_{\tau_s}$ and $N(0, t - s)$ distributed. It remains to prove the a.s. continuity of the sample paths of the time-changed process.

If the quadratic variation is a.s. strictly increasing, then (τ_t) is continuous and the result follows. Else, we need to show that for all ω in some set Ω^* of full measure

$$\langle X \rangle_{t_1}(\omega) = \langle X \rangle_t(\omega) \quad \text{for some } 0 \leq t_1 < t$$

implies

$$X_{t_1}(\omega) = X_t(\omega).$$

In this case, the continuity of $t \mapsto X_{\tau_t}$ and the equality $X = B_{\langle X \rangle}$ follow from Exercise 49(iv) with

$$T := \langle X(\omega) \rangle, \quad A = \tau, \quad \psi = X(\omega)$$

therein. By the continuity of the quadratic variation process it is enough to prove the above implication for a fixed $t_1 \in \mathbb{Q}_+$. Let

$$\sigma := \inf\{t > t_1 : \langle X \rangle_t > \langle X \rangle_{t_1}\};$$

this is a stopping time. Further, let

$$N_s := X_{(t_1+s) \wedge \sigma} - X_{t_1}; \quad s \geq 0;$$

this is a continuous local martingale w.r.t. $(\mathcal{F}_{t_1+s})$ and quadratic variation

$$\langle N \rangle_s = \langle X \rangle_{(t_1+s) \wedge \sigma} - \langle X \rangle_{t_1} = 0.$$

By Exercise 34 we conclude that for each $t_1 \in \mathbb{Q}_+$ there exists a set $\Omega(t_1)$ of full measure such that the implication holds for all $\omega \in \Omega(t_1)$. By continuity the implication holds for all $\omega \in \bigcap_{t \in \mathbb{Q}_+} \Omega(t)$. $\square$

Lemma 5.2.4 *Let $\mathcal{G} \subseteq \mathcal{F}$ be σ -fields. Let X be $\mathcal{F}$ -measurable and suppose that*

$$\mathbb{E}[\exp(i\theta X) | \mathcal{G}] = \exp\left(-\frac{\theta^2}{s} \sigma^2\right), \quad \theta \in \mathbb{R}.$$

Then $X \perp \mathcal{G}$ and $X \sim N(0, \sigma^2)$.

PROOF: Taking expectations on both sides of the equation yields

$$\mathbb{E}[e^{i\theta X}] = \exp\left(-\frac{\theta^2}{s} \sigma^2\right).$$

Hence normality of the distribution follows from the uniqueness of the Fourier transform. Moreover, for any $A \in \mathcal{G}$ by the definition of the conditional expectation³

$$\mathbb{E}[\mathbf{1}_A e^{i\theta X}] = \mathbb{E}\left[\mathbf{1}_A \exp\left(-\frac{\theta^2}{s} \sigma^2\right)\right] = \mathbb{E}[\mathbf{1}_A] \exp\left(-\frac{\theta^2}{s} \sigma^2\right) = \mathbb{E}[\mathbf{1}_A] \mathbb{E}[e^{i\theta X}].$$

This shows the independence. $\square$

Exercise 50 *If M is a continuous local martingale starting from zero, and $\langle M \rangle_t = f_t$ for some deterministic function f taking values in the non-negative real numbers. Show that M is a Gaussian Martingale, that is, M_t has a Gaussian distribution for almost all $t \geq 0$. In particular, the martingale M defined by $dM_t = f_t dB_t$ satisfies*

$$M_t \sim N\left(0, \int_0^t |f_s|^2 ds\right).$$

5.3 Change of measure - Girsanov's theorem

Changes of measures play a key role in many applications of stochastic calculus, notably in financial mathematics when pricing and hedging financial derivatives. In this section we discuss Girsanov's theorem. Loosely speaking it states that if we add a drift to a Brownian motion, then the distribution will not change dramatically. In fact, the law of the new process is absolutely continuous with respect to the law of the original process and we can explicitly calculate the density process.

Throughout, we fix a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, \mathbb{P})$ that satisfies the usual conditions and carries a standard Brownian motion B .

³We recall that Z is a version of $\mathbb{E}[X | \mathcal{G}]$ if and only if $\mathbb{E}[\mathbf{1}_A X] = \mathbb{E}[\mathbf{1}_A Z]$ for all $A \in \mathcal{G}$.

5.3.1 Review: absolutely continuous measures

We recall that a measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ is absolutely continuous with respect to $\mathbb{P}$ on $\mathcal{F} = \mathcal{F}_T$, denoted $\mathbb{Q} \ll \mathbb{P}$, if and only if there exists a random variable

$$Z := \frac{d\mathbb{Q}}{d\mathbb{P}}$$

called the *Radon-Nykodyn derivative* such that

$$\mathbb{Q}(A) = \int_A Z(\omega) \mathbb{P}(\omega), \quad A \in \mathcal{F}_T.$$

If Z is strictly positive, then $\mathbb{Q}$ and $\mathbb{P}$ are equivalent, written $\mathbb{Q} \sim \mathbb{P}$, and $\frac{d\mathbb{P}}{d\mathbb{Q}} = Z^{-1}$.

Exercise 51 Suppose that $\mathbb{Q}$ is absolutely continuous with respect to $\mathbb{P}$ on $\mathcal{F} = \mathcal{F}_T$. Show that the restriction $\mathbb{Q}|_{\mathcal{F}_t}$ of $\mathbb{Q}$ to $\mathcal{F}_t$ is absolutely continuous with respect to the restriction $\mathbb{P}|_{\mathcal{F}_t}$ of $\mathbb{P}$ to $\mathcal{F}_t$ and the density process

$$Z_t := \frac{d(\mathbb{Q}|_{\mathcal{F}_t})}{d(\mathbb{P}|_{\mathcal{F}_t})}$$

is a uniformly integrable $\mathbb{P}$ -martingale.

We denote by (Z_t) the (càdlàg version of the) density process of $\mathbb{Q}$ with respect to $\mathbb{P}$ and denote by $\mathbb{E}_{\mathbb{P}}$ and $\mathbb{E}_{\mathbb{Q}}$ the expectation w.r.t $\mathbb{Q}$ and $\mathbb{P}$, respectively. An easy argument using the martingale property of the density process (Z_t) and the definition of conditional expectation shows that

$$d\mathbb{Q}_t := Z_t d\mathbb{P} \quad \text{and} \quad d\mathbb{Q}_s := Z_s d\mathbb{P}$$

coincide on $\mathcal{F}_s$ for $s \leq t$. Next, we recall Bayes' formula for conditional expectations.

Lemma 5.3.1 (*Bayes' formula*) Let $\mathbb{Q} \ll \mathbb{P}$ with density Z . Then,

$$\mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] \mathbb{E}_{\mathbb{P}}[Z | \mathcal{F}_t] = \mathbb{E}_{\mathbb{P}}[HZ | \mathcal{F}_t] \quad \text{a.s. for any } H \geq 0 \text{ or } HZ \in L^1(\mathbb{P}). \quad (5.5)$$

PROOF: Since $H \geq 0$ or $HZ \in L^1(\mathbb{P})$ and hence $H \in L^1(\mathbb{Q})$ all the conditional expectations in what follows exist. Let $A \in \mathcal{F}_t$. Then, by the definition of conditional expectations

$$\int_A \mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] Z d\mathbb{P} = \int_A \mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] d\mathbb{Q} = \int_A H d\mathbb{Q} = \int_A HZ d\mathbb{P} = \int_A \mathbb{E}[HZ | \mathcal{F}_t] d\mathbb{P}.$$

On the other hand, using the tower property of conditional expectations we obtain that

$$\begin{aligned} \int_A \mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] Z d\mathbb{P} &= \mathbb{E}_{\mathbb{P}}[\mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] Z \mathbf{1}_A] \\ &= \mathbb{E}_{\mathbb{P}}[\mathbf{1}_A \mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] \mathbb{E}_{\mathbb{P}}[Z | \mathcal{F}_t]] \\ &= \int_A \mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] \mathbb{E}_{\mathbb{P}}[Z | \mathcal{F}_t] d\mathbb{P}. \end{aligned}$$

Hence

$$\int_A \mathbb{E}[HZ | \mathcal{F}_t] d\mathbb{P} = \int_A \mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] \mathbb{E}_{\mathbb{P}}[Z | \mathcal{F}_t] d\mathbb{P} \quad \text{for all } A \in \mathcal{F}_t.$$

Since the integrands are $\mathcal{F}_t$ -measurable we conclude that

$$\mathbb{E}[HZ | \mathcal{F}_t] = \mathbb{E}_{\mathbb{Q}}[H | \mathcal{F}_t] \mathbb{E}_{\mathbb{P}}[Z | \mathcal{F}_t] \quad \mathbb{P}\text{-a.s.}$$

□

The next exercise shows how local martingales under $\mathbb{P}$ and $\mathbb{Q}$ are related.

Exercise 52 Consider the stopping time $R := \inf \{t \geq 0 \mid Z_t = 0\}$. Then the following holds:

1. $R = +\infty$ a.s. under $\mathbb{Q}$.
2. For any $(\mathcal{F}_t)$ -adapted process Y , if YZ is local martingale under $\mathbb{P}$ up to time R , then Y is a $\mathbb{Q}$ -local martingale.

5.3.2 Motivation

To motivate the subsequent analysis let us assume that we are given a stock price process (S_t) whose (discounted) dynamics follows a geometric Brownian motion (“Black-Scholes model”):

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t$$

where $\mu \in \mathbb{R}$, $\sigma > 0$ and B is a standard Brownian motion defined on $(\Omega, \mathcal{F}, \mathbb{P})$. Let $(\mathcal{F}_t)$ be the filtration generated by B , augmented by the null sets.

A call option on S gives its holder the right but not the obligation to purchase one unit of the stock at a predetermined date T (“maturity”) at a predetermined price K (“strike”). Hence, its payoff at maturity is

$$C = (S_T - K)^+.$$

To find the fair/reasonable option price we assume that there is a probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F})$ that is equivalent to $\mathbb{P}$ such that

$$W_t := \frac{\mu}{\sigma}t + B_t$$

is a $\mathbb{Q}$ -Brownian motion. We emphasize that W and B generate the same filtration; in economic terms, we change the measure but not the information flow.⁴ Then,

$$\frac{dS_t}{S_t} = \sigma dW_t.$$

Hence S is a $\mathbb{Q}$ -martingale, $C \in L^2(\mathbb{Q})$ and the martingale representation theorem yields a process $\pi \in \mathcal{L}_2$ such that

$$C = \mathbb{E}_{\mathbb{Q}}[C] + \int_0^T \pi_t dW_t = \mathbb{E}_{\mathbb{Q}}[C] + \int_0^T \frac{1}{\sigma} \frac{\pi_t}{S_t} dS_t.$$

This suggests that if the option seller charges the amount $\mathbb{E}_{\mathbb{Q}}[C]$ and then follows the investment strategy of holding $\frac{1}{\sigma} \frac{\pi_t}{S_t}$ stocks at any time $t \in [0, T]$ then he/she generates an amount

$$\int_0^T \sigma^{-1} \frac{\pi_t}{S_t} dS_t$$

from trading the stock which, together with the option price, perfectly replicates the option payoff at maturity. As a result, the economically only reasonable option price is $\mathbb{E}_{\mathbb{Q}}[C]$.

This question is, then, if we can find the desired measure. Girsanov’s theorem states that this is indeed the case.

⁴This may no longer be true if μ, σ are stochastic processes!

5.3.3 Girsanov's theorem and the Novikov condition

We are now going to state and prove the change of measure theorem. Subsequently, we establish a sufficient condition for the involved exponential martingales to be true martingales.

Theorem 5.3.2 (*Girsanov's Theorem*) Let $I = [0, T]$ Let $\mu = (\mu_t)_{t \in I} \in \mathcal{L}_2$ and let $Y = (Y_t)_{t \in I}$ be defined by

$$Y_t := B_t + \int_0^t \mu_s ds.$$

Assume that the process

$$\mathcal{E}_t := \exp\left(-\int_0^t \mu_u dB_u - \frac{1}{2} \int_0^t \mu_u^2 du\right), \quad t \in I$$

is a (true) martingale and put

$$dQ_T = \mathcal{E}_T d\mathbb{P}.$$

Then $(Y_t)_{t \in I}$ is a Brownian motion on I under Q_T with respect to the original filtration $(\mathcal{F}_t)_{t \in I}$.

Definition 5.3.3 The process $\mathcal{E} = (\mathcal{E}_t)$ is called the stochastic exponential of the process $dX = -\mu_t dB_t$.⁵

PROOF OF GIRSANOV'S THEOREM. Since $(\mathcal{E}_t)$ is a martingale we have that

$$\mathbb{Q}(\Omega) = \mathbb{E}_{\mathbb{P}}[\mathcal{E}_T] = \mathbb{E}_{\mathbb{P}}[\mathcal{E}_0] = 1$$

so $\mathbb{Q}$ is indeed a probability measure. Let us then define the process (K_t) by

$$K_t := \mathcal{E}_t Y_t.$$

By Itô's formula $d\mathcal{E}_t = -\mu_t \mathcal{E}_t dB_t$ and so the differential of K is given by

$$\begin{aligned} dK_t &= \mathcal{E}_t dY_t + Y_t d\mathcal{E}_t + d\mathcal{E}_t dY_t \\ &= \mathcal{E}_t(\mu_t dt + dB_t) - \mu_t Y_t \mathcal{E}_t dB_t - \mathcal{E}_t \mu_t dt \\ &= \mathcal{E}_t(1 - \mu_t Y_t) dB_t. \end{aligned}$$

Hence (K_t) is a stochastic integral with respect to Brownian motion and thus a local $\mathbb{P}$ -martingale. Thus, it follows from Exercise 52 (2.) that (Y_t) is a local $\mathbb{Q}_T$ martingale (with respect to the same filtration). Similar arguments show that $(Y_t^2 - t)$ is also a local $\mathbb{Q}_T$ -martingale. In particular, $\langle Y \rangle_t \equiv t$. Since Y is continuous the Lévy characterization shows that (Y_t) is a $\mathbb{Q}_T$ -Brownian motion. $\square$

Exercise 53 (i) Let $Y(t) = t + B(t)$. For each $T > 0$ find a probability measure Q_T on $\mathcal{F}_T$ such that $\mathbb{Q}_T \sim \mathbb{P}$ and $\{Y(t)\}_{t \leq T}$ is Brownian motion w.r.t. $\mathbb{Q}_T$. Prove that there exists a probability measure $\mathbb{Q}$ on $\mathcal{F}_{\infty}$ such that

$$\mathbb{Q}|_{\mathcal{F}_T} = \mathbb{Q}_T \quad \text{for all } T > 0.$$

(ii) Show that

$$\mathbb{P}\left(\lim_{t \rightarrow \infty} Y(t) = \infty\right) = 1 \quad \text{while} \quad \mathbb{Q}\left(\lim_{t \rightarrow \infty} Y(t) = \infty\right) = 0.$$

Why does this not contradict the Girsanov theorem?

⁵See also Example 4.3.5. If we need to emphasize the dependence of $\mathcal{E}$ on X we write $\mathcal{E}_t(X)$ instead of $\mathcal{E}_t$.

We notice that stochastic exponentials are positive continuous local martingales and hence supermartingales but not martingales in general. A sufficient condition to guarantee that $(\mathcal{E}_t)$ is a martingale is the Novikov condition.

Theorem 5.3.4 (*Novikov Condition*) *The condition*

$$\mathbb{E} \left[\exp \left(\frac{1}{2} \int_0^T \mu_s^2 ds \right) \right] < \infty$$

implies that $\mathcal{E}$ is a martingale. In particular, $\mathcal{E}$ is a martingale if μ is bounded.

We prove the above theorem in a general martingale framework and consider a continuous local martingale M with quadratic variation $\langle M \rangle$. The stochastic exponential of M is given by the process

$$\mathcal{E}_t(M) = 1 - \int_0^t \mathcal{E}_s(M) dM_s.$$

This equation can be solved explicitly. In fact, an application of Itô's formula to $\ln \mathcal{E}_t(M)$ yields

$$\mathcal{E}_t(M) = \exp \left(-M_t - \frac{1}{2} \langle M \rangle_t \right), \quad t \in [0, \infty).$$

The following proposition provides a sufficient condition for the stochastic exponential to be an L^q -martingale; the result is closely related to Proposition 3.3.5 ("a local martingale of Class DL is a true martingale".)

Proposition 5.3.5 *Let M be a continuous local martingale with quadratic variation $\langle M \rangle$ and let $\mathcal{T}_T$ be the class of all stopping times bounded by T . If for some $p, q \in (1, \infty)$ with $\frac{1}{p} + \frac{1}{q} = 1$*

$$\sup \left\{ \mathbb{E} \left[\exp \left(-\frac{\sqrt{p}}{2(\sqrt{p}-1)} M_\tau \right) \right] : \tau \in \mathcal{T}_T \right\} < \infty \quad (5.6)$$

for all $T > 0$, then the stochastic exponential $\mathcal{E}(M)$ of M is an L^q -martingale, hence uniformly integrable.

PROOF: Using the fact that $\mathcal{E}(M)$ is a continuous local martingale along with Proposition 3.3.5 it is enough to prove that

$$\sup_{\tau \in \mathcal{T}_T} \mathbb{E} \mathcal{E}_\tau^q(M) < \infty, \quad \text{for all } T > 0.$$

The remaining arguments are based on a clever application of the Hölder inequality. Let

$$u = \frac{\sqrt{p}+1}{\sqrt{p}-1}, \quad v = \frac{\sqrt{p}+1}{2}.$$

Then $u > 1$ and $u^{-1} + v^{-1} = 1$ and

$$\left(q - \sqrt{\frac{q}{u}} \right) v = \frac{\sqrt{p}}{2(\sqrt{p}-1)}.$$

Using that $q = \frac{p}{p-1}$, that $(\sqrt{p}-1)(\sqrt{p}+1) = p-1$ and plugging in the definition of u , the last equation is equivalent to

$$\sqrt{\frac{p}{p-1}} - \sqrt{\frac{\sqrt{p}-1}{\sqrt{p}+1}} = \frac{1}{\sqrt{p-1}} \Leftrightarrow \sqrt{p} - \sqrt{(p-1) \frac{\sqrt{p}-1}{\sqrt{p}+1}} = 1 \Leftrightarrow \sqrt{p} - \sqrt{(\sqrt{p}-1)^2} = 1,$$

which shows that the equation is indeed correct. Since

$$\mathcal{E}_\tau^q(M) = \exp\left(-\sqrt{\frac{q}{u}}M_\tau - \frac{q}{2}\langle M \rangle_\tau\right) \exp\left(-(q - \sqrt{\frac{q}{u}})M_\tau\right)$$

the Hölder inequality yields

$$\begin{aligned} \mathbb{E}\mathcal{E}_\tau^q(M) &\leq \left(\mathbb{E}\exp\left(-\sqrt{qu}M_\tau - \frac{qu}{2}\langle M \rangle_\tau\right)\right)^{1/u} \left(\mathbb{E}\exp\left(-(q - \sqrt{\frac{q}{u}})vM_\tau\right)\right)^{1/v} \\ &= (\mathbb{E}\mathcal{E}_\tau(\sqrt{qu}M))^{1/u} \left(\mathbb{E}\exp\left(-\frac{\sqrt{p}}{2(\sqrt{p}-1)}M_\tau\right)\right)^{1/v}. \end{aligned}$$

Since $(\mathcal{E}_{\tau \wedge t}(\sqrt{qu}M))$ is a continuous positive local martingale by optional sampling it is a super-martingale (cf. Exercise 30). Thus,

$$\mathbb{E}\mathcal{E}_\tau(\sqrt{qu}M) \leq \mathbb{E}\mathcal{E}_0(\sqrt{qu}M) = 1.$$

The second term in the above estimate is uniformly bounded for all bounded stopping times by assumption. Hence we have established the desired uniform L^q -bound, so $\mathcal{E}(M)$ is a martingale. $\square$

Corollary 5.3.6 *Let M be a continuous local martingale that satisfies*

$$\mathbb{E}\left[\exp\left(-\frac{1}{2}M_\tau\right)\right] \leq C \quad \text{for all } \tau \in \mathcal{T}_T.$$

Then $\mathcal{E}(aM)$ is a martingale on $[0, T]$ for all $a \in (0, 1)$.

PROOF: The assertion follows from the previous lemma applied to aM by choosing $p > 1$ such that $\frac{\sqrt{p}}{\sqrt{p}-1} = a^{-1}$ in (5.6). $\square$

We are now ready to prove the Novikov condition. The proof uses the fact that a supermartingale X with constant expected value is a martingale: if $Y = X_t - \mathbb{E}[X_t|\mathcal{F}_s]$, then $Y \geq 0$ and $\mathbb{E}[Y] = 0$, hence $X_t = \mathbb{E}[X_s|\mathcal{F}_s] = 0$ a.s. for all $0 \leq s \leq t$.

Proposition 5.3.7 *Let M be a continuous local martingale with quadratic variation $\langle M \rangle$. If*

$$\mathbb{E}\exp\left(\frac{1}{2}\langle M \rangle_T\right) < \infty,$$

then $\mathcal{E}(M)$ is a martingale on $[0, T]$.

PROOF: We use again the fact that $\mathcal{E}(M)$ is a positive local martingale and hence a super-martingale. As a result, it is enough to prove that

$$1 \leq \mathbb{E}[\mathcal{E}_t(M)], \quad \text{for all } t \in [0, T]$$

as the converse equality holds by the super-martingale property and because super-martingales with constant expected values are martingales.

We are now going to verify the condition of the preceding corollary. To this end, let $\tau \leq T$ be a stopping time. Since

$$\begin{aligned} \exp\left(-\frac{1}{2}M_\tau\right) &= \exp\left(-\frac{1}{2}M_\tau - \frac{1}{4}\langle M \rangle_\tau\right) \exp\left(\frac{1}{4}\langle M \rangle_\tau\right) \\ &\leq \exp\left(-\frac{1}{2}M_\tau - \frac{1}{4}\langle M \rangle_\tau\right) \exp\left(\frac{1}{4}\langle M \rangle_T\right) \\ &= \sqrt{\mathcal{E}_\tau(M)} \exp\left(\frac{1}{4}\langle M \rangle_T\right) \end{aligned}$$

the super-martingale property of $\mathcal{E}(M)$ along with Hölder's inequality yields

$$\mathbb{E} \exp\left(-\frac{1}{2}M_\tau\right) \leq \left(\mathbb{E} \exp\left(\frac{1}{2}\langle M \rangle_T\right)\right)^{1/2} < \infty.$$

Since the above inequality holds uniformly in $\tau \in \mathcal{T}_T$ it follows from the preceding corollary that $\mathcal{E}(aM)$ is a martingale for all $a \in (0, 1)$. The idea is now to move a out of the stochastic exponential.

To this end, we again apply Hölder's inequality, this time with $p := 1/(1 - a^2)$ and $q := 1/a^2$. Since

$$\mathcal{E}_t(aM) = \mathcal{E}_t^{a^2}(M) \exp(a(a-1)M_t)$$

Hölder yields that

$$\begin{aligned} 1 &= \mathbb{E} \mathcal{E}_t(aM) \\ &\leq \left(\mathbb{E} \mathcal{E}_t^{a^2 q}(M)\right)^{1/q} \left(\mathbb{E} [\exp(a(a-1)pM_t)]\right)^{1/p} \\ &= (\mathbb{E} \mathcal{E}_t(M))^{a^2} \left(\mathbb{E} \left[\exp\left(-\frac{1}{2}M_t\right)^{2a/(1+a)}\right]\right)^{1-a^2}. \end{aligned}$$

For $u := 2a/(1+a) < 1$ it follows from Jensen's inequality that

$$\left(\mathbb{E} \exp\left(-\frac{1}{2}M_t\right)^u\right)^{1/u} \leq \mathbb{E} \exp\left(-\frac{1}{2}M_t\right)$$

and hence, using that $\frac{1}{u} = \frac{1-a^2}{2a(1-a)}$,

$$\left(\mathbb{E} \exp\left(-\frac{1}{2}M_t\right)^u\right)^{1-a^2} \leq \left(\mathbb{E} \exp\left(-\frac{1}{2}M_t\right)\right)^{2a(1-a)}.$$

Altogether,

$$1 \leq (\mathbb{E} \mathcal{E}_t(M))^{a^2} \left(\mathbb{E} \exp\left(-\frac{1}{2}M_t\right)\right)^{2a(1-a)}.$$

Since $\mathbb{E} \exp(-\frac{1}{2}M_t) < \infty$, the right hand side converges to $\mathbb{E} \mathcal{E}_t(M)$ as $a \rightarrow 1$. Hence

$$1 \leq \mathbb{E} \mathcal{E}_t(M).$$

□

Girsanov's theorem easily generalized to higher dimensions. For this we need a the generalization of Lévy's characterization of Brownian motion: an n -dimensional continuous local martingale M starting in zero is an n -dimensional (standard) Brownian motion if and only if

$$\langle M_i, M_j \rangle_t = \delta_{i,j} t.$$

Theorem 5.3.8 (Generalized Girsanov) *Let Y be an n -dimensional process of the form*

$$dY_t = \beta(t, \omega)dt + \theta(t, \omega)dB_t$$

where B is an m -dimensional Brownian motion, and β, σ are suitable $\mathbb{R}^n$ -valued, respectively, $\mathbb{R}^{n,m}$ -valued processes. If there exists a suitable processes u and α such that

$$\theta(t, \omega)u(t, \omega) = \beta(t, \omega) - \alpha(t, \omega)$$

and

$$\mathcal{E}_t := \exp\left(-\int_0^t u dB - \frac{1}{2} \int_0^t |u|^2 ds\right), \quad 0 \leq t \leq T$$

is a true martingale, then

$$\hat{B}_t := \int_0^t u ds + B_t$$

is a $d\mathbb{Q} = M_T d\mathbb{P}$ Brownian motion and

$$dY_t = \alpha(t, \omega)dt + \theta(t, \omega)d\hat{B}_t.$$

Exercise 54 *Let*

$$dY(t) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} dt + \begin{bmatrix} 1 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} dB_1(t) \\ dB_2(t) \end{bmatrix}; \quad t \leq T.$$

Find a probability measure $\mathbb{Q}$ on $\mathcal{F}_T$ such that $\mathbb{Q} \approx \mathbb{P}$ and such that

$$\tilde{B}(t) := \begin{bmatrix} -3t \\ t \end{bmatrix} + \begin{bmatrix} B_1(t) \\ B_2(t) \end{bmatrix}$$

is a Brownian motion w.r.t. $\mathbb{Q}$ and

$$dY(t) = \begin{bmatrix} 1 & 3 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} d\tilde{B}_1(t) \\ d\tilde{B}_2(t) \end{bmatrix}.$$

Chapter 6

Stochastic Differential Equations

A stochastic differential equation (SDE) is a mathematical model that describes how a system changes over time when its behavior is influenced by both predictable trends and random uncertainty. Unlike ordinary differential equations, which assume that a system evolves in a completely deterministic way, SDEs account for randomness, fluctuations, or external disturbances. They are widely used to model financial markets, physical processes, biological systems, engineering applications, and many other phenomena where uncertainty plays a significant role.

We consider stochastic differential equations in finite time intervals $[0, T]$. For a given n -dimensional stochastic process X , we denote by $[X]_t := (X_s)_{0 \leq s \leq t}$ the path of the process on $[0, t]$ and by $[X]$ the process on the whole time interval $[0, T]$. Let

$$b : [0, T] \times C([0, T], \mathbb{R}^n) \rightarrow \mathbb{R}^n \quad \text{and} \quad \sigma : [0, T] \times C([0, T], \mathbb{R}^n) \rightarrow \mathbb{R}^{n,d}$$

be given functions. For $\varphi = b, \sigma$ we assume throughout (in a sense to be made precise) that $\varphi(t, [x]) = \varphi(t, [x]_t)$. An equation of the form

$$\begin{aligned} dX_t &= b(t, [X])dt + \sigma(t, [X])dB_t \\ X_0 &= Z \end{aligned} \tag{6.1}$$

where B is a d -dimensional Brownian motion and Z is an independent random variable defined on some probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ that satisfies the usual conditions is called an n -dimensional *stochastic differential equation* (SDE) with coefficients $b(\cdot, \cdot)$, $\sigma(\cdot, \cdot)$ and initial condition Z . The vector

$$((\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P}), (Z, B))$$

is called a (probabilistic) *setup*. We consider two different solution concepts: strong solutions and weak solutions. If an SDE admits a strong solution it admits a solution on any setup; for a weak solution the setup, in particular the driving Brownian motion, is part of the solution.

6.1 Strong and pathwise solutions

We say that the SDE (6.1) admits a *strong solution* if, on any setup, there exists a path-wise continuous, adapted process X such that

- $X_0 = Z$ a.s.
- the integrals $\int_0^t b(s, [X]_s)ds$ and $\int_0^t \sigma(s, [X]_s)dB_s$ exist for all $t \in I$
- the process satisfies the integral equation

$$X_t = Z + \int_0^t b(s, [X])ds + \int_0^t \sigma(s, [X])dB_s, \quad \text{for all } t \in I, \text{ a.s.}$$

A continuous, adapted process X is called a *pathwise solution* on a given setup $((\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P}), (Z, B))$ if X is defined on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ satisfies the above.

Remark 6.1.1 We note that we assume that the coefficients of our SDE were deterministic functions. This assumption can essentially be made w.l.o.g. as we can always add the driving Brownian motion as an additional component to the vector X .

Example 6.1.2 Consider the one-dimensional SDE $dX_t = \alpha_t dB_t$ for some deterministic function $\alpha \in C^1$. Then

$$X_t = x + \int_0^t \alpha_s dB_s.$$

The integration by parts formula yields

$$X_t = x + \alpha_t B_t - \int_0^t B_s \alpha'_s ds$$

so X is a strong solution.

Example 6.1.3 For a one-dimensional Brownian motion B an Ornstein-Uhlenbeck process takes the form

$$dX_t = -\alpha X_t dt + \sigma dB_t, \quad X_0 = x$$

for some positive constants α and σ . Such a process fluctuates around zero (at least in the long run), due to the minus sign in front of the drift term. To solve this SDE we consider the process $Y_t = X_t e^{\alpha t}$ so

$$dY_t = e^{\alpha t} dX_t + \alpha e^{\alpha t} X_t dt.$$

Hence

$$dY_t = \sigma e^{\alpha t} dB_t.$$

Thus,

$$X_t = e^{-\alpha t} \left(x + \sigma \int_0^t e^{\alpha s} dB_s \right).$$

An Ornstein Uhlenbeck process is a Gauss process because X_t follows a normal distribution. Specifically, if we define $\gamma := \frac{\sigma}{\sqrt{2\alpha}}$ then the martingale property of stochastic integrals implies

$$X_t \sim N(xe^{-\alpha t}, \gamma^2(1 - e^{-2\alpha t})).$$

Replacing the constant x by a random variable X_0 that is independent of the Brownian motion and satisfies

$$X_0 \sim N(0, \gamma^2) \quad \text{yields} \quad X_t \sim N(0, \gamma^2).$$

Thus, unlike Brownian motion that asymptotically fluctuates between infinity and negative infinity, an Ornstein-Uhlenbeck process admits a stationary distribution. The covariance function of the stationary process is given by

$$\rho(s, t) = \gamma^2 e^{-|t-s|}.$$

Exercise 55 Let X be a stationary Ornstein-Uhlenbeck process. Prove that the path $t \rightarrow X_t(\omega)$ is almost surely nowhere differentiable.

Hint: Consider the rescaled process $Y_t = \sqrt{t} X_{\frac{1}{2} \log t}$.

6.1.1 Solutions as functions of Brownian paths

In the above examples, the solutions were functionals of the Brownian path and the initial condition. In what follows we show that this result is universal.

The canonical setup

If an SDE has a strong solution, then it has a solution on any setup, in particular, it has a solution on the canonical setup that we now introduce in analogy to the canonical model for Brownian motion. The *canonical sample space* is

$$\bar{\Omega} = \mathbb{R}^n \times C^d[0, T].$$

A typical element from this space is denoted $\bar{\omega} = (z, \omega)$. The canonical sample space is a Polish space, so we equip it with the σ -field $(\mathcal{F}_t)$ (“raw filtration”) generated by the projection mappings

$$Z(z, \omega) := z \quad \text{and} \quad B_t(y, \omega) := \omega_t$$

and define the measure

$$\mathbb{P}_\mu := \mu \otimes \mathbb{P}^*$$

where μ will denote the law of the initial state and $\mathbb{P}^*$ is the Wiener measure on $C^d[0, T]$. The $\mathbb{P}^*$ -completed filtration is denoted $(\bar{\mathcal{F}}_t)$. We note that the completed filtration is not a Borel- σ -field. The *canonical setup* is then given by

$$(\bar{\Omega}, \bar{\mathcal{F}}_T, (\bar{\mathcal{F}}_t), \mathbb{P}_\mu).$$

Remark 6.1.4 The space $C^d[0, T]$ equipped with the raw filtration is a Polish space, hence the Doob-Dynkin lemma applies. This does not hold for the completed filtration. The next exercise “remedies” this problem.

Exercise 56 Let X be a càdlàg stochastic process that is $(\bar{\mathcal{F}}_t)$ -adapted. Then X is indistinguishable from an $(\mathcal{F}_t)$ -adapted process X' . Conclude that this is also true to predictable processes.

Hint. Use conditional expectations and that the predictable σ -field is generated by the càdlàg processes.

Previsible path-functionals

The previsible σ -fields on $[0, T] \times \bar{\Omega}$ corresponding to $\mathcal{F}$ and $\bar{\mathcal{F}}$ are denoted $\mathcal{P}$ and $\bar{\mathcal{P}}$, respectively. We recall that $\mathcal{P}$ is the smallest σ -field w.r.t. which all $\mathcal{F}$ -adapted left-continuous processes are measurable.

Definition 6.1.5 A $\mathcal{P}$ -measurable mapping ϕ on $[0, T] \times C^d[0, T]$ will be called a $\mathcal{P}$ -previsible path-functional. A similar definition applies to $\bar{\mathcal{P}}$ -previsible path-functional.

Assumption 6.1.6 The coefficients b and σ are $\bar{\mathcal{P}}$ -previsible path-functionals.

The next exercise shows that if σ is a previsible path functional and X is an adapted, continuous process then the stochastic integral $\int_0^T \sigma(t, [X]) dB_t$ is well defined.

Exercise 57 Let $\phi : [0, T] \times \bar{\Omega} \rightarrow \mathbb{R}^m$ be a previsible path-functional and X be any continuous process defined on some probability space $(\tilde{\Omega}, \mathcal{G}, (\mathcal{G}_t), \mathbb{Q})$. Then $\phi(t, [X])$ is previsible relative to $(\mathcal{G}_t)$.

Hint. Show that the mapping $(t, \omega) \mapsto (t, [X])$ is previsible.

Solutions as Brownian path-functionals

We are now going to show that stochastic integrals can be represented as Brownian path-functionals.

Lemma 6.1.7 On the canonical setup, suppose that $\phi : [0, T] \times \bar{\Omega} \rightarrow \mathbb{R}^d$ is a $\bar{\mathcal{P}}$ -previsible path functional, and that

$$\mathbb{P} \left[\int_0^T |\phi(s, Z, [B])|^2 ds < \infty \right] = 1.$$

Then there exists a $\mathcal{F}$ -measurable function¹

$$\Phi : \Omega \rightarrow C^1$$

such that

$$\mathbb{P} \left[\int_0^\cdot \phi(s, Z, [B]) dB_s \equiv \Phi(Z, [B]) \right] = 1. \quad (6.2)$$

Moreover, on any setup $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}}, \tilde{Z}, \tilde{B})$ where $\tilde{Z}$ has law μ ,

$$\tilde{\mathbb{P}} \left[\int_0^\cdot \phi(s, \tilde{Z}, [\tilde{B}]) d\tilde{B}_s \equiv \Phi(\tilde{Z}, [\tilde{B}]) \right] = 1. \quad (6.3)$$

PROOF: In view of Exercise 56 the process ϕ is indistinguishable from a $\mathcal{P}$ -predictable process ϕ' and we may assume $\phi = \phi'$; this is where the canonical setup is used. By localizing we may furthermore assume that

$$\|\phi\|^2 := \mathbb{E} \left[\int_0^T |\phi(\dots)|^2 dt \right] < \infty.$$

If ϕ is elementary the asserting follows. If $\|\phi_n - \phi\| \rightarrow 0$ for some sequence of elementary processes, by passing to a subsequence if necessary, we may assume that (see the proof of Proposition 4.1.7)

$$\int_0^\cdot \phi_n(\dots) dB_s \rightarrow \int_0^\cdot \phi(\dots) dB_s \quad \text{uniformly a.s.}$$

¹Note that ϕ is a stochastic process, while Φ is a function. This motivates the different filtrations.

Let Φ_n be the function corresponding to ϕ_n and

$$\Phi(Z, [B]) := \lim_n \Phi_n(Z, [B])$$

be the uniform limit where it exists and zero else. Then (6.2) holds. The same approximation argument also yields (6.3). $\square$

Next, we prove that pathwise solutions on any setup can be represented in terms of functions of Brownian paths if a pathwise solution on the canonical setup exists..

Proposition 6.1.8 *Let $b : [0, T] \times C^n[0, T] \rightarrow \mathbb{R}^n$ and $\sigma : [0, T] \times C^n[0, T] \rightarrow \mathbb{R}^{n,d}$ be previsible path functionals and X be a pathwise solution to the SDE (6.1) on the canonical setup. Then there exists a function $F_\mu : \Omega \rightarrow C^n[0, T]$ such that*

$$X = F_\mu(Z, B) \quad \mathbb{P}_\mu\text{-a.s.}$$

Moreover, on any other setup $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \mathbb{P}), \tilde{Z}, \tilde{B})$ where $\tilde{Z}$ has law μ the process

$$\tilde{X} := F_\mu(\tilde{Z}, \tilde{B})$$

satisfies the SDE (6.1) with initial state $\tilde{Z}$ on that setup.

PROOF: In view of Exercise 56 the solution X has a version X' that is measurable w.r.t. the raw filtration $\mathcal{F}$. Since $\mathbb{R}^n \times C^d$ is a Polish space the existence of a measurable function $F_\mu : \Omega \rightarrow C^n[0, T]$ such that a.s.

$$X' = F_\mu(Z, B)$$

follows from the Doob-Dynkin lemma. To prove the second assertion we apply the preceding lemma (assuming w.l.o.g. we are in dimension one). By Exercise 57 the function defined by

$$\phi(s, Z, [B]) := \sigma(s, F_\mu(Z, [B]))$$

satisfies the conditions of the previous lemma (up to localization) and hence there exists a function Φ such that a.s.

$$\int_0^\cdot \phi(s, \tilde{Z}, [\tilde{B}]) d\tilde{B}_s \equiv \Phi(\tilde{Z}, [\tilde{B}]).$$

The second assertion is then equivalent to

$$F_\mu(\tilde{Z}, [\tilde{B}])_t = \tilde{Z} + \Phi(\tilde{Z}, [\tilde{B}])_t + \int_0^t b(s, F_\mu(\tilde{Z}, [\tilde{B}])) ds \quad \text{for all } t \in [0, T] \quad \tilde{\mathbb{P}}\text{-a.s.}$$

Since we already know that

$$F_\mu(Z, [B])_t = Z + \Phi(Z, [B])_t + \int_0^t b(s, F_\mu(Z, [B])) ds \quad \text{for all } t \in [0, T] \quad \mathbb{P}_\mu\text{-a.s.}$$

and because $(\tilde{Z}, [\tilde{B}])$ has the same law as $(Z, [B])$ the assertion follows; see also the following remark. $\square$

Remark 6.1.9 *The previous proof used that any pair of measurable functions F, G and any random variables (X, Y) defined on $(\Omega, \mathcal{F}, \mathbb{P})$ and (X', Y') defined on $(\Omega', \mathcal{F}', \mathbb{P}')$ with (X, Y) and (X', Y') being equal in law,*

$$\mathbb{P}(F(X) = G(Y)) = 1 \quad \text{implies} \quad \mathbb{P}'(F(X') = G(Y')) = 1$$

In fact, with $h(x, y) = \mathbf{1}_{F(x)=G(y)}$ we have that

$$1 = \mathbb{P}(F(X) = G(Y)) = \mathbb{E}[h(X, Y)] = \mathbb{E}'[h(X', Y')] = \mathbb{P}'(F(X') = G(Y'))$$

6.1.2 Existence and uniqueness of solutions

We move on to the following existence and uniqueness results for SDEs. To simplify notation we consider only diffusion SDEs, that is, SDEs where the coefficients depend on the path of the solution only through the current state. In this case, the “previsible path functional” property translates into the following assumptions on the coefficients

$$b : \Omega \times [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n \quad \text{and} \quad \sigma : \Omega \times [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^{n,d}$$

- for $\varphi = b, \sigma$ and $\omega \in \Omega$, the functions $(t, x) \mapsto \varphi(\omega, t, x)$ is Borel-measurable
- for $\varphi = b, \sigma$ and $x \in \mathbb{R}^n$ the function $(\omega, t) \mapsto \varphi(\omega, t, x)$ is progressively measurable

It is customary to drop the dependence of the coefficients on ω in the notation.

The existence and uniqueness proof is similar to the standard Picard-Iteration in the theory of ordinary differential equation.

Theorem 6.1.10 *Let $T > 0$ and let $b, \sigma : [0, T] \times \mathbb{R} \times \Omega \rightarrow \mathbb{R}$ satisfy the linear growth condition*

$$|b(t, x, \omega)| + |\sigma(t, x, \omega)| \leq C (\kappa_t(\omega) + |x|)$$

with $\kappa \in L^2(\lambda \otimes \mathbb{P})$ along with the Lipschitz condition

$$|b(t, x) - b(t, y)|^2 + |\sigma(t, x) - \sigma(t, y)|^2 \leq D|x - y|^2.$$

Let Z be a square integrable random variable that is independent of B . Then the SDE

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t \quad \text{on} \quad [0, T]$$

with initial condition $X_0 = Z$ has a unique strong solution X and

$$\sup_{0 \leq t \leq T} \mathbb{E}|X_t|^2 < \infty.$$

PROOF:

- **Existence.** The existence of a solution follows via the Picard iteration known from the theory of ordinary differential equations. Specifically, let us define $Y_t^0 := Z$ and afterwards recursively define a sequence of stochastic processes

$$Y_t^{k+1} = Z + \int_0^t b(s, Y_s^k)ds + \int_0^t \sigma(s, Y_s^k)dB_s.$$

We notice that the integrals are well defined, due to the continuity of Y^k and the linear growth condition on the coefficients. Using this iteration, one can show that $\{Y^k\}_{k \in \mathbb{N}}$ forms a Cauchy sequence in $L^2(\lambda \otimes \mathbb{P})$ by exploiting an induction argument. To this end, one has to find bounds on $\mathbb{E}[(Y_t^{k+1} - Y_t^k)^2]$. Specifically, we prove the bound

$$\mathbb{E}[(Y_t^{k+1} - Y_t^k)^2] \leq \frac{A^{k+1}t^{k+1}}{(k+1)!}$$

for all $k \geq 0$ where $A := 4(1+T)\max\{C^2\|\kappa\|_{L^2}\mathbb{E}[|Z|^2], D\}$.

For the induction start, we use the Itô isometry and Hölder's inequality along with the linear growth condition to obtain:

$$\begin{aligned} \mathbb{E}[(Y_t^1 - Y_t^0)^2] &= \mathbb{E}\left[\left(\int_0^t b(s, Z)ds + \int_0^t \sigma(s, Z)dB_s\right)^2\right] \\ &\leq 2\left(\mathbb{E}\left[\left(\int_0^t b(s, Z)ds\right)^2\right] + \mathbb{E}\left[\left(\int_0^t \sigma(s, Z)dB_s\right)^2\right]\right) \quad (\text{binomial formula}) \\ &\leq 2\left(\mathbb{E}\left[T \int_0^t b^2(s, Z)ds\right] + \mathbb{E}\left[\int_0^t \sigma^2(s, Z)ds\right]\right) \quad (\text{Hölder inequality, Itô isometry}) \\ &\leq 2C^2(1+T)\mathbb{E}\left[\int_0^t (\kappa_s + |Z|)^2ds\right] \quad (\text{linear growth condition}) \\ &\leq 4(1+T)C^2\|\kappa\|_{L^2}\mathbb{E}[|Z|^2]t \\ &\leq At. \end{aligned}$$

Similarly, Itô's isometry and Hölder's inequality along with Lipschitz continuity, Fubini's theorem and the induction hypothesis yield that

$$\begin{aligned} \mathbb{E}[|Y_t^{k+1} - Y_t^k|^2] &= \mathbb{E}\left[\left(\int_0^t (b(s, Y_s^k) - b(s, Y_s^{k-1}))ds + \int_0^t (\sigma(s, Y_s^k) - \sigma(s, Y_s^{k-1}))dB_s\right)^2\right] \\ &\leq 2\left(\mathbb{E}\left[\left(\int_0^t (b(s, Y_s^k) - b(s, Y_s^{k-1}))ds\right)^2\right] + \mathbb{E}\left[\left(\int_0^t (\sigma(s, Y_s^k) - \sigma(s, Y_s^{k-1}))dB_s\right)^2\right]\right) \\ &\leq 2\left(\mathbb{E}\left[T \int_0^t (b(s, Y_s^k) - b(s, Y_s^{k-1}))^2ds\right] + \mathbb{E}\left[\int_0^t (\sigma(s, Y_s^k) - \sigma(s, Y_s^{k-1}))^2ds\right]\right) \\ &\leq 4D(1+T) \int_0^t \mathbb{E}[(Y_s^k - Y_s^{k-1})^2]ds \\ &\leq A \int_0^t \frac{A^k s^k}{k!} ds \\ &= \frac{A^{k+1}t^{k+1}}{(k+1)!}. \end{aligned}$$

For $n, m \in \mathbb{N}$ we obtain from the this induction result that

$$\begin{aligned}
\|Y^m - Y^n\|_{L^2(\lambda \otimes \mathbb{P})}^2 &= \left\| \sum_{k=n}^{m-1} Y^{k+1} - Y^k \right\|_{L^2(\lambda \otimes \mathbb{P})}^2 \\
&\leq \sum_{k=n}^{m-1} \int_0^T \mathbb{E}[(Y_t^{k+1} - Y_t^k)^2] ds \\
&\leq \sum_{k=n}^{m-1} \int_0^T \frac{A^{k+1} t^{k+1}}{(k+1)!} dt \\
&= \sum_{k=n}^{m-1} \frac{A^{k+1} T^{k+2}}{(k+2)!}.
\end{aligned}$$

Due to the series expansion of the exponential function, the above expression must converge to zero for $m, n \rightarrow \infty$. Hence $\{Y^k\}_{k \in \mathbb{N}}$ forms a Cauchy sequence in $L^2(\lambda \otimes \mathbb{P})$. By completeness of the space, there exists a limiting element $X \in L^2(\lambda \otimes \mathbb{P})$. As the limit of adapted processes this limit is adapted.

It remains to show that $t \rightarrow X_t$ is continuous and that X does indeed solve our stochastic differential equation. To this end, we show that terms involved in the iteration scheme converge to their respective analogue with Y^k and Y^{k+1} replaced by X . We have already established that $\|Y^{k+1} - X\|_{L^2(\lambda \otimes \mathbb{P})} \rightarrow 0$. Moreover using again Hölder's inequality and the Lipschitz condition one has

$$\begin{aligned}
\left\| \int_0^\cdot b(s, Y_s^k) ds - \int_0^\cdot b(s, X_s) ds \right\|_{L^2(\lambda \otimes \mathbb{P})}^2 &= \mathbb{E} \left[\int_0^T \left(\int_0^t b(s, Y_s^k) ds - \int_0^t b(s, X_s) ds \right)^2 dt \right] \\
&\leq \mathbb{E} \left[\int_0^T t \cdot \int_0^t (b(s, Y_s^k) - b(s, X_s))^2 ds dt \right] \\
&\leq D \mathbb{E} \left[\int_0^T t \cdot \int_0^t (Y_s^k - X_s)^2 ds dt \right] \\
&\leq DT^2 \mathbb{E} \left[\int_0^T (Y_s^k - X_s)^2 ds \right] \\
&= DT^2 \|Y^k - X\|_{L^2(\lambda \otimes \mathbb{P})}^2 \rightarrow 0.
\end{aligned}$$

Similarly, using again Fubini's theorem, Itô's isometry and the Lipschitz condition, we obtain the convergence of the stochastic integral part:

$$\begin{aligned}
\left\| \int_0^\cdot \sigma(s, Y_s^k) dB_s - \int_0^\cdot \sigma(s, X_s) dB_s \right\|_{L^2(\lambda \otimes \mathbb{P})}^2 &= \mathbb{E} \left[\int_0^T \left(\int_0^t \sigma(s, Y_s^k) - \sigma(s, X_s) dB_s \right)^2 dt \right] \\
&= \int_0^T \mathbb{E} \left[\left(\int_0^t \sigma(s, Y_s^k) - \sigma(s, X_s) dB_s \right)^2 \right] dt \\
&= \int_0^T \mathbb{E} \left[\int_0^t (\sigma(s, Y_s^k) - \sigma(s, X_s))^2 ds \right] dt \\
&\leq D \int_0^T \mathbb{E} \left[\int_0^t (Y_s^k - X_s)^2 ds \right] dt \\
&\leq D^2 T \|Y^k - X\|_{L^2(\lambda \otimes \mathbb{P})}^2 \rightarrow 0.
\end{aligned}$$

Collecting all these convergences, we obtain from the Picard iteration scheme that

$$X = Z + \int_0^\cdot b(s, X_s) ds + \int_0^\cdot \sigma(s, X_s) dB_s.$$

Since the Riemann integral is continuous and the Itô integral allows a continuous modification, the above process is continuous in time and we can write for all $t \in [0, T]$

$$X_t = Z + \int_0^t b(s, X_s) ds + \int_0^t \sigma(s, X_s) dB_s.$$

We conclude that X is a continuous strong solution.

- **Uniqueness.** The uniqueness part of the proof is mainly based on Gronwall's lemma (which we recall below) and continuity. Let X and $\hat{X}$ be two continuous strong solutions starting in Z and $\hat{Z}$, respectively. We choose different initial conditions in order to better see how Gronwall's lemma is used. Later we set $Z = \hat{Z}$. As before, we use the Itô isometry, the Hölder's inequality and the Lipschitz continuity in order to obtain

$$\begin{aligned} & \mathbb{E}[(X_t - \hat{X}_t)^2] \\ &= \mathbb{E}\left[\left(Z - \hat{Z} + \int_0^t b(s, X_s) - b(s, \hat{X}_s) ds + \int_0^t \sigma(s, X_s) - \sigma(s, \hat{X}_s) dB_s\right)^2\right] \\ &\leq 3\left(\mathbb{E}[(Z - \hat{Z})^2] + t\mathbb{E}\left[\int_0^t (b(s, X_s) - b(s, \hat{X}_s))^2 ds\right] + \mathbb{E}\left[\int_0^t (\sigma(s, X_s) - \sigma(s, \hat{X}_s))^2 ds\right]\right) \\ &\leq 3\mathbb{E}[(Z - \hat{Z})^2] + 3D(1 + T) \int_0^t \mathbb{E}[(X_s - \hat{X}_s)^2] ds. \end{aligned}$$

By Gronwall's lemma this implies the existence of a constant $C > 0$ such that that

$$\mathbb{E}[|X_t - \hat{X}_t|^2] \leq C\mathbb{E}[(Z - \hat{Z})^2]$$

for all $t \in [0, T]$. Since $Z = \hat{Z}$ one has in particular $\mathbb{P}[X_t = \hat{X}_t] = 1$ for all $t \in [0, T]$ and hence,

$$\mathbb{P}[X_t - \hat{X}_t = 0 \text{ for all } t \in [0, T] \cap \mathbb{Q}] = 1.$$

Continuity of $X - \hat{X}$ then yields of solutions up to indistinguishability.

□

Lemma 6.1.11 (Gronwall lemma) *Let $L \geq 0$, let $u : [0, T] \rightarrow \mathbb{R}$ be continuous and $\alpha : [0, T] \rightarrow \mathbb{R}$ be integrable on $[0, T]$ and satisfy*

$$u(t) \leq \alpha(t) + L \int_a^t u(s) ds.$$

Then the following inequality holds on $[0, T]$:

$$u(t) \leq \alpha(t) + L \int_0^t \alpha(s) e^{L(t-s)} ds.$$

PROOF: We have that

$$\frac{d}{dt} \left(e^{-Lt} \int_0^t u(s) ds \right) \leq e^{-Lt} \alpha(t)$$

and so

$$\int_0^t u(s) ds \leq e^{Lt} \int_0^t \alpha(s) e^{-Ls} ds.$$

Using the assumption again, we see that

$$u(t) \leq \alpha(t) + L \int_0^t \alpha(s) e^{L(t-s)} ds.$$

□

Let us return to the proof of Theorem 6.1.10. We showed that the Picard iterations $\{Y^{(k)}\}_{k \in \mathbb{N}}$ satisfy

$$\|Y^m - Y^n\|_{L^2(\lambda \otimes \mathbb{P})}^2 \leq \sum_{k=n}^{m-1} \frac{A^{k+1} T^{k+2}}{(k+2)!}.$$

Using the same arguments together with Doob's inequality, we can actually prove that

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} (Y_t^m - Y_t^n)^2 \right] \leq \sum_{k=n}^{m-1} \frac{A^{k+1} T^{k+2}}{(k+2)!}$$

and hence that

$$\lim_{n, m \rightarrow \infty} \mathbb{E} \left[\sup_{0 \leq t \leq T} (Y_t^m - Y_t^n)^2 \right] = 0.$$

By Fatou's lemma

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} (X_t - Y_t^n)^2 \right] \leq \liminf_m \mathbb{E} \left[\sup_{0 \leq t \leq T} (Y_t^m - Y_t^n)^2 \right].$$

Hence, we have the following result.

Proposition 6.1.12 *Under the assumptions of Theorem 6.1.10 the SDE has a unique strong solution X that satisfies*

$$\mathbb{E} \left[\sup_{t \in [0, T]} |X_t|^2 \right] < C_T (1 + \mathbb{E}|Z|^2).$$

The previous result allows us to extend the global existence and uniqueness result to the case of locally Lipschitz continuous coefficients.

Proposition 6.1.13 *If the coefficients $b, \sigma : [0, T] \times \mathbb{R} \times \Omega \rightarrow \mathbb{R}$ are locally Lipschitz continuous and satisfy the linear growth condition of the previous theorem, then, for every initial condition Z our SDE admits a unique strong solution.*

PROOF: For each $R > 0$, we choose globally Lipschitz continuous coefficients b_R, σ_R that coincide with b, σ on $|x| \leq R$. By the existence and uniqueness theorem for globally Lipschitz coefficients, the corresponding truncated equation admits a unique strong solution $X^{(R)}$. Let

$$\tau_R := \inf \left\{ t \geq 0 : |X_t^{(R)}| \geq R \right\}.$$

If $S > R$, by pathwise uniqueness, $X^{(R)} = X^{(S)}$ on $[0, \tau_R]$. Hence $X := X^{(R)}$ on $[0, \tau_R]$ is well defined and solves the original equation on $[0, \tau_\infty]$ where

$$\tau_\infty := \lim_{R \rightarrow \infty} \tau_R.$$

Since the truncated coefficients satisfy the same linear growth condition with a constant independent of R , the above moment estimate yields, for every $T > 0$,

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} |X_t^{(R)}|^2 \right] \leq C_T (1 + \mathbb{E}|Z|^2),$$

where the constants C_T are independent of R . Consequently,

$$\mathbb{P}(\tau_R \leq T) = \mathbb{P}\left(\sup_{0 \leq t \leq T} |X_t^{(R)}| \geq R\right) \leq \frac{C_T(1 + \mathbb{E}|Z|^2)}{R^2} \rightarrow 0 \quad \text{for } R \rightarrow \infty.$$

This shows that

$$\mathbb{P}(\tau_\infty = \infty) = 1.$$

Hence X is defined on all of $[0, \infty)$. Finally, if X and Y are two strong solutions of the original equation with the same initial condition, then they coincide on $[0, \tau_R]$ for every $R > 0$ by the uniqueness of the truncated equations. Since $\tau_R \uparrow \infty$ almost surely, it follows that $X = Y$ almost surely. $\square$

6.1.3 Markov property

Our existence result was proved under the assumption that the SDE starts at time 0 in some state Z that is independent of the Brownian motion. Likewise, we get a unique strong solution $X^{t,\xi}$ to our SDE starting at time $t \in [0, T]$ in the random state $\xi \in L^2(\mathcal{F}_t)$, that is, $X_t^{t,\xi} = \xi$. To this end, we put

$$\hat{B}_u =: B_{u+t} - B_t, \quad u \geq 0$$

and recall that $\hat{B}$ is an $(\mathcal{F}_{t+u})_{u \geq 0}$ -Brownian motion that is independent of $\mathcal{F}_t$. Since ξ is $\mathcal{F}_t$ -measurable, the equation

$$X_{s+t}^{t,\xi} = \xi + \int_0^s b(u+t, X_{u+t}^{t,\xi}) du + \int_0^s \sigma(u+t, X_{u+t}^{t,\xi}) d\hat{B}_u, \quad s \in [0, T-t]$$

satisfies the assumptions of Theorem 6.1.10 and hence it has a unique strong solution $X^{t,\xi}$. Since stochastic integrals can be viewed as limits in probability of Riemann sums provided the integrands are continuous, we have that

$$\begin{aligned} X_s &= x + \int_0^s b(u, X_u) du + \int_0^s \sigma(u, X_u) dB_u \\ &= X_t + \int_0^{s-t} b(u+t, X_{u+t}) du + \int_0^{s-t} \sigma(u+t, X_{u+t}) d\hat{B}_u. \end{aligned}$$

By our uniqueness of path-wise solutions result for SDEs this shows that

$$X_s = X_{s-t}^{t, X_t} \quad \text{for all } s \in [t, T-t]$$

almost surely. Of course, it is more convenient to write $X^{t,\xi} = (X_u^{t,\xi})_{t \leq u \leq T}$. If the coefficients b and σ are autonomous, that is, depend only on x , then the preceding arguments also apply to stopping times τ instead of fixed times t , due to the strong Markov property of the Brownian motion.

Since the BM $\hat{B}$ is independent of $\mathcal{F}_t$ we have also shown that strong solutions have the Markov property.

Theorem 6.1.14 *Let the coefficients of the SDE*

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t$$

satisfy the conditions of Theorem 6.1.10. Then it holds for all $t_1 \leq t_2 \leq T$ that

$$\mathbb{P}[X_{t_2} \leq y | \mathcal{F}_{t_1}] = \mathbb{P}[X_{t_2} \leq y | X_{t_1}] \quad \mathbb{P}\text{-a.s.}$$

which is equivalent to

$$\mathbb{E}[f(X_{t_2}) | \mathcal{F}_{t_1}] = \mathbb{E}[f(X_{t_2}) | X_{t_1}]$$

for all bounded and measurable f .

6.1.4 Linear SDEs

As with ordinary differential equations, stochastic differential equations can rarely be solved in closed form. A closed form solution can be obtained for linear SDEs, i.e., for equations of the form

$$dX_t = (\alpha_t + \beta_t X_t)dt + (\gamma_t + \delta_t X_t)dB_t \quad (6.4)$$

where the coefficients are adapted stochastic processes and continuous in the time variable. The special case where $\alpha \equiv 0 \equiv \delta$ corresponds to a generalization of the Ornstein-Uhlenbeck process studied in Example 6.1.3. When $\alpha_t = 0$ and $\gamma_t = 0$ the linear SDE (6.4) simplifies to

$$dU_t = \beta_t U_t dt + \delta_t U_t dB_t. \quad (6.5)$$

This SDE is of the form

$$dU_t = U_t dY_t$$

where the Itô process (Y_t) is defined by

$$dY_t = \beta_t dt + \delta_t dB_t \quad \text{with} \quad Y_0 = 1.$$

The SDE (6.5) is called the *stochastic exponential* of Y . By direct calculation

$$U_t = U_0 \exp\left(Y_t - Y_0 - \frac{1}{2}\langle Y \rangle_t\right)$$

U_t is always positive if U_0 is. Since the bracket $\langle Y \rangle$ of Y is given by $\frac{1}{2} \int \delta_s^2 ds$ we obtain that

$$U_t = U_0 \exp\left(\int_0^t (\beta_s - \frac{1}{2}\delta_s^2)ds + \int_0^t \delta_s dB_s\right). \quad (6.6)$$

The case of general linear SDEs can be reduced to stochastic exponentials by a variation of constants formula. More precisely, we look for solutions of the form

$$X_t = U_t V_t \quad \text{where} \quad dV_t = a_t dt + b_t dB_t$$

and (U_t) satisfies (6.5). If we put $U_0 = 1$ and $V_0 = X_0$, take the differential of $U_t V_t$ and compare coefficients, we see that

$$b_t U_t = \gamma_t \quad \text{and} \quad U_t(a_t + \delta_t b_t) = \alpha_t.$$

Hence

$$b_t \frac{\gamma_t}{U_t} \quad \text{and} \quad a_t = \frac{\alpha_t - \delta_t \gamma_t}{U_t}.$$

Using the representation (6.6) for U_t , this determines the functions a_t and b_t , and

$$X_t = U_t \left(X_0 + \int_0^t \frac{\alpha_s - \delta_s \gamma_s}{U_s} ds + \int_0^t \frac{\gamma_s}{U_s} dB_s \right).$$

Example 6.1.15 Let X satisfy the SDE

$$dX_t = a_t X_t dt + dW_t.$$

To solve this equation using our general formula we first need to solve for the U process, i.e., solve

$$dU_t = \alpha_t U_t dt.$$

We obtain $U_t = \exp\left(\int_0^t a_s ds\right)$ so

$$X_t = e^{\int_0^t a_s ds} \left(X_0 + \int_0^t e^{-\int_0^s a_u du} dB_s \right).$$

Example 6.1.16 (Brownian bridge) A linear SDE of the form

$$dX_t = \frac{b - X_t}{T - t} dt + dB_t$$

starting in $X_0 = a$ is called a Brownian bridge and is of major importance in many financial market models of insider trading. With

$$\beta_t = -\frac{1}{T-t}, \quad \alpha_t = \frac{b}{T-t}, \quad \gamma_t = 1 \quad \text{and} \quad \delta_t = 0$$

the solution is given by

$$X_t = a \left(1 - \frac{t}{T}\right) + b \frac{t}{T} + (T-t) \int_0^t \frac{1}{T-s} dB_s.$$

By Fatou's lemma and the Itô isometry we get

$$\begin{aligned} 0 \leq \mathbb{E} \left[\liminf_{t \rightarrow T} \left(\int_0^t \frac{T-t}{T-s} dB_s \right)^2 \right] &\leq \liminf_{t \rightarrow T} \mathbb{E} \left[\left(\int_0^t \frac{T-t}{T-s} dB_s \right)^2 \right] \\ &= \liminf_{t \rightarrow T} \int_0^t \left(\frac{T-t}{T-s} \right)^2 ds = \liminf_{t \rightarrow T} \frac{t(T-t)^2}{T(T-t)} = 0 \end{aligned}$$

and hence

$$\int_0^t \frac{T-t}{T-s} dB_s \xrightarrow{t \rightarrow T} 0 \quad \mathbb{P} - \text{almost surely.}$$

That means, that almost surely so that $X_T = b$. Thus we do not only know the starting but also the end point of the process.

We notice that this is our second encounter of the Brownian bridge. In Example 2.1.5 we introduced the Brownian bridge as a Gauss process; this examples introduces it as scaled stochastic integral. Unlike the process in Example 2.1.5 the process considered here is adapted. A direct calculation shows that $\mathbb{E}X_t X_s = s \wedge t - \frac{st}{T}$ if $a = b = 0$, which is the same as in Example 2.1.5. Since the mean and the covariance function determine Gauss processes, we conclude that the process constructed in Example 2.1.5 and the one constructed here have the same distribution.

Exercise 58 Verify that the given processes solve the given stochastic differential equations (here, $\{B_t : t \geq 0\}$ is an $\mathbb{R}$ -valued standard Brownian motion):

(1) $X_t = e^{B_t}$ solves $dX_t = \frac{1}{2}X_t dt + X_t dB_t$;

(2) $X_t = B_t/(1+t)$ with $B_0 = 0$ solves

$$dX_t = -\frac{X_t}{1+t}dt + \frac{dB_t}{1+t};$$

(3) $X_t = \sin B_t$ with $B_0 \in [-\pi/2, \pi/2]$ solves

$$dX_t = -\frac{1}{2}X_t dt + \sqrt{1-X_t^2}dB_t, \quad t \leq \inf\{s \geq 0 : B_s \notin [-\pi/2, \pi/2]\};$$

(4) $(X_t, Y_t) = (t, e^t B_t)$ solves

$$\begin{pmatrix} dX_t \\ dY_t \end{pmatrix} = \begin{pmatrix} dt \\ Y_t dt \end{pmatrix} + \begin{pmatrix} 0 \\ e^{X_t} \end{pmatrix} dB_t.$$

Exercise 59 Show the exact expression for the unique strong solution $\{(X_t, Y_t) : t \geq 0\}$ to the following stochastic differential equation:

$$\begin{pmatrix} dX_t \\ dY_t \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} X_t \\ Y_t \end{pmatrix} dt + \begin{pmatrix} 0 & -a/b \\ b/a & 0 \end{pmatrix} \begin{pmatrix} X_t \\ Y_t \end{pmatrix} dB_t, \quad \begin{pmatrix} X_0 \\ Y_0 \end{pmatrix} = \begin{pmatrix} a \\ 0 \end{pmatrix}$$

where $a, b > 0$ and $\{B_t : t \geq 0\}$ is an $\mathbb{R}$ -valued Brownian motion. Usually, the solution $\{(X_t, Y_t) : t \geq 0\}$ is called *Brownian motion on the ellipse*

$$\left\{ (x, y) : \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \right\}.$$

Exercise 60 Let $\{B_t : t \geq 0\}$ be an $\mathbb{R}$ -valued Brownian motion, $b, \sigma \in C(\mathbb{R}, \mathbb{R})$ be bounded and Lipschitz, $\inf_{x \in \mathbb{R}} |\sigma(x)| > 0$, and

$$X_t = x + \int_0^t b(X_s) ds + \int_0^t \sigma(X_s) dB_s$$

Find a function $F : \mathbb{R} \mapsto \mathbb{R}$ such that $F(X)$ is a local martingale.

6.2 Weak solutions and the martingale problem

In this section we introduce weak solutions to SDEs. Unlike in the previous section we do not start with a stochastic basis. Instead, we construct a stochastic basis as part of the solution. The formal definition is as follows.

Definition 6.2.1 Let $b : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma_k : [0, T] \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be measurable for $k = 1, \dots, n$.² A weak solution of the SDE

$$dX_t = b(t, X_t)dt + \sum_{k=1}^d \sigma_k(t, X_t)dB_t^k, \quad X_0 = x \tag{6.7}$$

comprises a filtered probability $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ that satisfies the usual conditions and a pair of processes (B, X) defined on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ where B is a d -dimensional $(\mathcal{F}_t)$ Brownian motion and X is a continuous adapted process that satisfies the above SDE.

²Notice that these functions do not depend on ω .

Obviously, a strong solution is a weak solution and a weak solution is a strong solution on the stochastic basis and relative to the Brownian motion which is part of the solution. The most prominent example of an SDE that does not have a strong solution is the famous *Tanaka equation*

$$dX_t = \text{sign}(X_t)dB_t \quad \text{where} \quad \text{sign}(x) = \begin{cases} +1 & \text{if } x > 0 \\ -1 & \text{if } x \leq 0 \end{cases}.$$

We start by showing that strong uniqueness of solutions for this equation cannot hold. Indeed, suppose that X is a solution. Then X is a Brownian motion because

$$\int_0^t \text{sign}(X_s)^2 ds = t.$$

But then $-X$ is a solution as well because

$$-X_t = \int_0^t [-\text{sign}(X_s)]dB_s = \int_0^t \text{sign}(-X_s)dB_s$$

where the second equation follows from the Itô isometry together with the fact that Brownian level sets have measure zero: in fact, using that X is necessarily a Brownian motion if it is a strong solution in the last step, we get that

$$\begin{aligned} \mathbb{E} \left[\left(\int_0^t (-\text{sign}(X_s) - \text{sign}(-X_s))dB \right)^2 \right] &= \mathbb{E} \left[\int_0^t (-\text{sign}(X_s) - \text{sign}(-X_s))^2 ds \right] \\ &= 4\mathbb{E} \int_0^t \mathbf{1}_0(X_s)ds \\ &= 4 \int_0^t \mathbb{P}(X_s = 0)ds \\ &= 0. \end{aligned}$$

Next, we show that the Tanaka equation admits a weak solution exists. To this end, let

$$M_t := \int_0^t \text{sign}(B_s)dB_s.$$

By Lévy's theorem, M is a Brownian motion. Moreover,

$$\int_0^t \text{sign}(B_s)dM_s = \int_0^t \text{sign}^2(B_s)dB_s = B_t. \quad (6.8)$$

In other words, (B, M) is a weak solution to the Tanaka equation. If we are given another weak solution, then this is a strong solution on the stochastic basis and relative to the Brownian motion that is part of the solution and we conclude from the above arguments that the two solutions have the same the law. Hence the Tanaka equation satisfies uniqueness in law in the sense of the following definition.

Definition 6.2.2 *The SDE (6.7) satisfies uniqueness in law if any two weak solutions with the same initial condition have the same law.*

Finally, we prove that a strong solution cannot exist. To this end, let (B, X) be any weak solution. Then

$$\int_0^t \text{sign}(X_s)dX_s = \int_0^t \text{sign}^2(X_s)dB_s = B_t.$$

We now show that X is not measurable w.r.t. the filtration generated by B , and hence it is not a strong solution. In fact, since X is a Brownian motion (starting in x) w.r.t. some filtration we conclude from Exercise 39 that

$$|X_t| = x + \int_0^t \text{sign}(X_s) dX_s + L_t = B_t + L_t$$

where L denotes the local time of the Brownian motion X at zero. We see from Exercise 39 (ii) that L and hence B is adapted to the filtration generated by $|X|$, which is strictly smaller³ than the one generated by X .

The canonical setup

Our first goal is to show that a weak solution always induces a weak solution with the same law on the canonical path space

$$(\bar{\Omega}, \bar{\mathcal{B}}) = (C^d[0, T] \times C^n[0, T], \mathcal{B}^d \otimes \mathcal{B}^n)$$

so that we can always work with this setting (d -dim Brownian motion, n -dim SDE).

Proposition 6.2.3 *Let $((B, X), (\Omega, (\mathcal{F}_t), \mathbb{Q}))$ be a weak solution to our SDE. Then there exists a weak solution*

$$((W, Y), (\bar{\Omega}, (\bar{\mathcal{B}}_t), \mathbb{P}))$$

such that

$$\mathbb{Q} \circ (B, X)^{-1} = \mathbb{P} \circ (W, Y)^{-1}.$$

PROOF: For $\bar{\omega} = (\omega^w, \omega^y) \in C^d \times C^n$, let $Y_t(\bar{\omega}) := \omega_t^y$ and $W_t(\bar{\omega}) := \omega_t^w$ be the canonical projection mappings, and let $\mathbb{P}$ be the measure on $\bar{\Omega}$ defined by

$$\mathbb{P}(A) := \mathbb{Q}[(B, X) \in A].$$

Then the projection on the 2nd coordinate is the Wiener measure $\mathbb{P}^*$ and so W is a Brownian motion w.r.t. the completion of the Borel- σ -field. Moreover, the measures $\mathbb{P} \circ (W, Y)^{-1}$ and $\mathbb{Q} \circ (B, X)^{-1}$ on $C^d \times C^n$ are equivalent by construction. As a result, (W, Y) satisfies our SDE $\mathbb{P}$ -a.s. and so $((W, Y), (\bar{\Omega}, (\bar{\mathcal{B}}_t), \mathbb{P}))$ is a weak solution. $\square$

Pathwise uniqueness

Our next goal is to show that for any two weak solutions defined on possibly different probability spaces with the same initial condition there exist weak solutions on a common probability space with the same common law. To this end, we recall an important result on regular conditional probabilities.

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{G}$ be a sub- σ -field of $\mathcal{F}$. A function $Q : \Omega \times \mathcal{F} \rightarrow [0, 1]$ is called a *regular conditional probability (r.c.p.)* for $\mathbb{P}$ given $\mathcal{G}$ if:

- i) for each $\omega \in \Omega$, $Q_\omega(\cdot)$ is a probability measure on $(\Omega, \mathcal{F})$,
- ii) for each $A \in \mathcal{F}$, the mapping $\omega \mapsto Q_\omega(A)$ is $\mathcal{G}$ -measurable
- iii) for each $A \in \mathcal{F}$, $Q_\omega(A) = \mathbb{P}[A|\mathcal{G}](\omega)$ for $\mathbb{P}$ -a.s. $\omega \in \Omega$.

³The event $\{X_1 > 0\}$ is contained in $\sigma(X_s; s \leq 1)$ but not in $\sigma(|X_s|; s \leq 1)$

The following should be well-known from earlier stochastics courses (we emphasize the r.c.p.s only exist for Borel σ -fields!!).

Theorem 6.2.4 *Let Ω is a complete, separable metric space endowed with its Borel- σ -field $\mathcal{B}$, and let $\mathbb{P}$ be a probability measure on $(\Omega, \mathcal{B})$ and $\mathcal{G}$ a sub- σ -field. Then a regular conditional probability of $\mathbb{P}$ given $\mathcal{G}$ exists. Moreover, if S is another complete, separable metric space endowed with its Borel- σ -field $\mathcal{S}$, and if $X : \Omega \rightarrow S$ is a measurable mapping that induces the distribution $\mathbb{P}X^{-1}$ on $(S, \mathcal{S})$, then there exists a r.c.p. $Q : S \times \mathcal{B} \rightarrow [0, 1]$ for $\mathbb{P}$ given X that satisfies:*

- i) for each $x \in S$, $Q_x(\cdot)$ is a probability measure on $(\Omega, \mathcal{B})$,
- ii) for each $A \in \mathcal{B}$, the mapping $x \mapsto Q_x(A)$ is $\mathcal{S}$ -measurable,
- iii) for each $A \in \mathcal{B}$, $Q_x(A) = \mathbb{P}[A|X = x]$ for $\mathbb{P}X^{-1}$ -a.e. $x \in S$,
- iv) $Q_x(\{X = x\}) = 1$ for $\mathbb{P}X^{-1}$ -a.e. $x \in S$.

The space $C^n[0, T]$ of all $\mathbb{R}^n$ -valued continuous functions on $[0, T]$ is complete and separable when equipped with the topology of uniform convergence. The Borel- σ field on $C^n[0, T]$ is denoted $(\mathcal{B}_t^n)$. We put $\mathcal{B}^n := \mathcal{B}_T^n$. Regular conditional probabilities with respect to sub- σ fields of $\mathcal{B}^n$ exist.

Proposition 6.2.5 *Let $((B^j, X^j), (\Omega^j, (\mathcal{F}_t^j, \mathbb{Q}^j)))$, $j = 1, 2$, be weak solutions to our SDE with the same initial condition defined on possibly different probability spaces. Then there exist weak solutions with the same common distributions as (B^j, X^j) on a common probability space with a common Brownian motion.*

PROOF: By analogy to the proof of the preceding proposition let $\mathbb{P}^j$ be the probability measures on $\bar{\Omega}$ defined by

$$\mathbb{P}^j(A) := \mathbb{Q}^j[(B^j, X^j) \in A].$$

Then the projection on the 2nd coordinate is the Wiener measure $\mathbb{P}^*$. Let P^j be the regular conditional probability of $\mathbb{Q}^j$ given $\mathcal{B}^d$ ("the conditional probability of X^j given B^j "). Let

$$\hat{\Omega} := C^d[0, T] \times C^n[0, T] \times C^n[0, T],$$

again equipped with the Borel- σ -field and denote a typical element of $\hat{\Omega}$ by $\hat{\omega} = (\omega^w, \omega^{y_1}, \omega^{y_2})$. Let

$$Y_t^j(\hat{\omega}) := \omega_t^{y_j} \quad \text{and} \quad W_t(\hat{\omega}) := \omega_t^w$$

be the canonical projection mappings. We now define a measure on $\hat{\Omega}$ by

$$\hat{\mathbb{P}}(d\hat{\omega}) := P^1(\omega^w; d\omega^{y_1})P^2(\omega^w; d\omega^{y_2})\mathbb{P}^*(d\omega^w).$$

Then the distribution of (W, Y^j) under $\hat{\mathbb{P}}$ is the same as the distribution of (B^j, X^j) under $\mathbb{Q}^j$ and W is a Brownian motion with respect to the completed filtration. Using the same argument as in the proof of the above proposition (W, Y^1) , (W, Y^2) are weak solutions on a common probability space. $\square$

The preceding result shows that we can bring two weak solutions together on a common probability space. Weak solutions on a common probability space can be compared pathwise.

Definition 6.2.6 *We say that pathwise uniqueness holds for our SDE if any two weak solutions (X, B) and (Y, B) with a common Brownian motion B defined on a common probability space and common initial condition but possibly different filtrations are indistinguishable.*

The next result shows that whenever our SDE satisfies pathwise uniqueness, then it satisfies uniqueness in law. In other words, whenever we know that any two weak solutions on a common probability space are indistinguishable, then we know that any two weak solutions defined on possibly different probability spaces must coincide in law.

Proposition 6.2.7 *Pathwise uniqueness implies uniqueness in law. In particular, strong uniqueness implies uniqueness in law.*

PROOF: Let $((B^j, X^j), (\Omega^j, (\mathcal{F}_t^j), \mathbb{Q}^j))$ be weak solutions with common initial condition. In view of Proposition 6.2.5 we can bring the two solutions together, obtaining two weak solutions (W, Y^1) and (W, Y^2) on the common canonical probability space

$$(\hat{\Omega}, \hat{\mathcal{B}}, \hat{\mathbb{P}}).$$

Now, if pathwise uniqueness prevails, then $\hat{\mathbb{P}}[Y^1 = Y^2 \text{ on } [0, T]] = 1$ or, equivalently,

$$\hat{\mathbb{P}}[\hat{\omega} = (\omega^w, \omega^{y_1}, \omega^{y_2}) : \omega^{y_1} = \omega^{y_2}] = 1.$$

In view of the construction of the measure $\hat{\mathbb{P}}$, we conclude that

$$\mathbb{Q}^1[(B, X^1) \in A] = \hat{\mathbb{P}}[(W, Y^1) \in A] = \hat{\mathbb{P}}[(W, Y^2) \in A] = \mathbb{Q}^2[(B, X^2) \in A].$$

This proves the desired result. □

Weak solution via Girsanov's theorem

The principal method for creating weak solutions to SDEs is transformation of drift via the Girsanov theorem. The following result illustrates this method.

Proposition 6.2.8 *Assume that the SDE*

$$dX_t = b(t, X_t)dt + \sigma(t, X_t)dB_t, \quad X_0 = x$$

satisfies the assumptions of Theorem 6.1.10 on $[0, T]$. Let $f \in \mathcal{L}_2$ satisfy the Novikov condition and let

$$W_t = B_t + \int_0^t f_s ds, \quad t \in [0, T].$$

Then, under the measure Q_T defined by

$$\frac{dQ_T}{d\mathbb{P}} = \exp\left(-\int_0^T f_s dB_s - \frac{1}{2} \int_0^T f_s^2 ds\right)$$

the process X solves the SDE

$$dX_t = (b(t, X_t) - \sigma(t, X_t)f_t) dt + \sigma(t, X_t)dB_t, \quad X_0 = x.$$

PROOF: By Girsanov's theorem W is a Q_T Brownian motion. Hence (a bit formally)

$$\begin{aligned} dX_t &= \sigma(t, X_t)dB_t + b(t, X_t)dt \\ &= \sigma(t, X_t)dW_t + (b(t, X_t) - \sigma(t, X_t)f_t)dt. \end{aligned}$$

□

The philosophy behind this approach is that if we want to solve the SDE

$$dX_t = (b(t, X_t) - \sigma(t, X_t)f_t)dt + \sigma(t, X_t)dB_t, \quad X_0 = x$$

then we can construct a Brownian motion and a probability space such that the SDE can be solved on this probability space w.r.t. the constructed Brownian motion if b and σ are nice enough.

6.2.1 From weak solutions to martingale problems

We now discuss the link between weak solutions to SDEs and the so-called martingale problem. We start with the following simple result. It is an immediate application of Itô's formula.

Proposition 6.2.9 *Let $((X, B), (\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P}))$ be a weak solution to (6.7). Then for all $f \in C^2(\mathbb{R}^n)$*

$$M^f := f(X_t) - f(X_0) - \int_0^t \mathcal{A}_s f(X_s) ds \quad (6.9)$$

is a continuous local $(\mathcal{F}_t)$ -martingale starting in 0 where the operator $\mathcal{A}_t$ acts on $C^2(\mathbb{R}^d)$ according to

$$\mathcal{A}_t f(x) := \langle b(t, x), \nabla f \rangle + \frac{1}{2} \text{trace} \left(a(t, x) D^2 f(x) \right). \quad (6.10)$$

Here, $D^2 f$ denotes the Hesse matrix of f and $a(t, x) := (\sigma \sigma^t)(t, x)$, i.e.

$$a_{i,j}(t, x) := \sum_{k=1}^d (\sigma_k(t, x))_i (\sigma_k(t, x))_j.$$

If f has compact support, and b and σ are locally bounded, then M^f is a continuous L^2 -martingale.

The idea is now to characterize weak solutions as the solution to a (local) martingale problem. In what follows we fix an n -dimensional drift vector b and a symmetric $n \times n$ diffusion matrix a .

Definition 6.2.10 *A continuous $\mathbb{R}^n$ -valued process X on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is called a solution to the local martingale problem for (a, b) if for each $f \in C_c^\infty(\mathbb{R}^n)$ the process*

$$M^f := f(X_t) - f(X_0) - \int_0^t \mathcal{A}_s f(X_s) ds \quad (6.11)$$

is a continuous local martingale with respect to the filtration generated by X . In this case we say that X solves $\text{LMP}(a, b)$. If the process is a martingale for each $f \in C_c^\infty$, then X solves the martingale problem for (a, b) , in short, X solves $\text{MP}(a, b)$. We say that the solution to $\text{LMP}(a, b)$ or $\text{MP}(a, b)$ is unique if for any two solutions X and $\tilde{X}$ on possibly different probability spaces the laws of X and $\tilde{X}$ coincide.

Remark 6.2.11 *In view of Proposition 6.2.9 any weak solution to our SDE induces a solution to the martingale problem.*

The following observation will be important for our subsequent analysis. A solution X to LMP(a, b) and can always be transferred to the canonical setting. Indeed, let

$$(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}}) := (C^n[0, T], \mathcal{B}^n, (\mathcal{B}_t^n), \mathcal{L}(X))$$

where $\mathcal{L}(X)$ denotes the law $\mathbb{P} \circ X^{-1}$ of X . Let π be the canonical process on $\tilde{\Omega}$ defined by $\pi_t(\tilde{\omega}) = \tilde{\omega}_t$. Then π has the same law as X and $(\mathcal{B}_t^n)$ is the filtration generated by π . Thus, π solves LMP(a, b). As a result, we can identify X with the probability measure $\mathcal{L}(X)$ on the canonical path space. In particular, every weak solution induces a probability measure $\mathbb{P}$ on the canonical path space such that the process

$$f(\pi_t) - f(\pi_0) - \int_0^t \mathcal{A}_s f(\pi_s) ds \quad (6.12)$$

is a continuous local $\mathbb{P}$ -martingale w.r.t. $(\mathcal{B}_t^n)$. It is thus appropriate to say that the law of a weak solution solves the associated martingale problem.

6.2.2 From martingale problems to weak solutions

Our goal is now to prove the converse result, namely that the every solution to a local martingale problem as identified with its law on the canonical path space gives rise to a weak solution of the associated SDE (through the drift and diffusion matrix defining the differential operator $\mathcal{A}$) with that law. Our first result establishes the representation of local martingales with absolutely continuous covariation as stochastic integrals with respect to a Brownian motion on a possibly extended probability space. The extension is defined as follows.

Definition 6.2.12 *Let $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ be a filtered probability space and $d \in \mathbb{N}$. Let B be a d -dimensional Brownian motion on a filtered probability space $(\hat{\Omega}, \hat{\mathcal{F}}, (\hat{\mathcal{F}}_t), \hat{\mathbb{P}})$. Let $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}})$ be the corresponding product space. Let $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}})$ be the smallest filtered probability space that contains $(\hat{\Omega}, \hat{\mathcal{F}}, (\hat{\mathcal{F}}_t), \hat{\mathbb{P}})$ and that satisfies the usual conditions. The space $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}})$ called a d -extension of the original space. For any adapted process A on the original space we write $\tilde{A}_t(\tilde{\omega}) = \tilde{A}_t(\omega, \hat{\omega}) := A_t(\omega)$ and we define $\tilde{B}_t(\tilde{\omega}) := B_t(\hat{\omega})$. Then $\tilde{B}$ is an $(\tilde{\mathcal{F}}_t)$ Brownian motion. We drop the tildes when there is no danger of confusion.*

Theorem 6.2.13 *Let M be a continuous local $\mathbb{R}^n$ -valued martingale on $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ with $M_0 = 0$ such that*

$$\langle M^i, M^j \rangle_t = \sum_{k=1}^d \int_0^t V_s^{ik} V_s^{jk} ds = \int_0^t (V_s V_s^t)^{ij} ds, \quad t \geq 0$$

for some progressively measurable processes (V^{ik}) , $1 \leq i \leq n$, $1 \leq k \leq d$. Then, on any d -extension $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}})$ there exists an $\mathbb{R}^d$ -valued $(\tilde{\mathcal{F}}_t)$ Brownian motion B such that

$$M_t^i = \sum_{k=1}^d \int_0^t V_s^{ik} dB_s^k.$$

PROOF: For each $t \geq 0$ we can regard (the matrix) $V_t = (V_t^{ik})$ as a linear map from $\mathbb{R}^d$ to $\mathbb{R}^n$. Let $N_t \subset \mathbb{R}^d$ be its null space and $R_t \subset \mathbb{R}^n$ be its range. We recall that $\mathbb{R}^d = N_t \oplus N_t^\perp$ and that $\mathbb{R}^n = R_t \oplus R_t^\perp$. Let $V_t^{-1} : R_t \rightarrow N_t^\perp$ be the (bijective) inverse of the restriction of V_t to $N_t^\perp$.

We denote the orthogonal projection from a Euclidean space to a subspace U by π_U . Let W be the d -dimensional Brownian motion as in the previous definition. Then M and W are independent by the definition of a d -extension. Let

$$B_t := \int_0^t V_s^{-1} \pi_{R_s} dM_s + \int_0^t \pi_{N_s} dW_s, \quad t \geq 0$$

where we interpret $V_s^{-1} \pi_{R_s}$ as a linear map from $\mathbb{R}^n$ to $\mathbb{R}^d$. Then

$$d\langle B^k, B^l \rangle_t = \sum_{i,j=1}^n (V_t^{-1} \pi_{R_t})_{ki} (V_t^{-1} \pi_{R_t})_{lj} d\langle M^i, M^j \rangle_t + \sum_{\nu=1}^d (\pi_{N_t})_{k\nu} (\pi_{N_t})_{l\nu} dt.$$

Inserting the covariation $\langle M^i, M^j \rangle$ we see that $\langle B \rangle_t = tI_d$ where I_d denotes the $d \times d$ identity matrix. This shows that B is an d -dimensional Brownian motion.

It remains to prove that $dM_t = V_t dB_t$. To this end, we decompose M as

$$M_t = \int_0^t \pi_{R_s} dM_s + \int_0^t \pi_{R_s^\perp} dM_s.$$

The second term is a continuous local martingale starting in zero with quadratic variation

$$\left\langle \int_0^\cdot \pi_{R_s^\perp} dM_s \right\rangle_t = \int_0^t \pi_{R_s^\perp}^2 V_s V_s^T ds = 0.$$

Hence $M_t = \int_0^t \pi_{R_s} dM_s$. Using that $V_s \pi_{N_s} \equiv 0$ we get that

$$\begin{aligned} \int_0^t V_s dB_s &= \int_0^t V_s V_s^{-1} \pi_{R_s} dM_s + \int_0^t V_s \pi_{N_s} dW_s \\ &= \int_0^t \pi_{R_s} dM_s + 0 \\ &= M_t. \end{aligned}$$

This completes the proof. $\square$

We are now ready to establish a one-to-one correspondence between weak solutions and solutions to martingale problems.

Theorem 6.2.14 *Let $b : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $a : [0, \infty) \times \mathbb{R}^n \rightarrow \mathcal{S}^n$ where $\mathcal{S}^n$ denotes the class of all symmetric positive definite $n \times n$ matrices. Let X be a solution to LMP(a, b) defined on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let $n \in \mathbb{N}$ and $\sigma : [0, \infty) \times \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$ be a measurable function such that $a = \sigma \sigma^t$.⁴ Let $(\mathcal{F}_t)$ be the completed filtration generated by X . Then, on any d -extension $(\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}})$ of $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ there exists an $\mathbb{R}^d$ -valued $(\tilde{F}_t)$ Brownian motion B such that*

$$\left((B, X), (\tilde{\Omega}, \tilde{\mathcal{F}}, (\tilde{\mathcal{F}}_t), \tilde{\mathbb{P}}) \right)$$

is a weak solution to (6.7).

PROOF: We first show that

$$M_t^i := X_t^i - X_0^i - \int_0^t b_i(s, X_s) ds \tag{6.13}$$

⁴Such a function always exists when $d \geq n$.

is a continuous local martingale. This does not follow immediately from the definition of the solution of martingale problems as the identity function does not have compact support. Instead, we proceed as follows. For each $n \in \mathbb{N}$, we choose $f_n^i \in C_c^\infty(\mathbb{R}^d)$ with $f_n^i(x) = x^i$ for $|x| \leq n$ along with a localizing sequence $(T_m^n)_{m \in \mathbb{N}}$ for the continuous local martingale $M^{f_n^i}$. Then $(M_{t \wedge T_m^n}^{f_n^i})$ is a martingale for all $n, m \in \mathbb{N}$.

Let $\tau_n := \inf\{t \geq 0 : |X_t| \geq n\}$ and $\sigma_n := \tau_n \wedge T_n^n$ and. Assuming w.l.o.g. that $T_1^{n+1} \geq \max\{T_n^n, T_{n+1}^1\}$ we conclude that $\{\sigma_n\}$ is a sequence of stopping times with $\sigma_n \uparrow \infty$ a.s. and $(M_{t \wedge \sigma_n}^{f_n^i})$ is a martingale for each $n \in \mathbb{N}$. Since $M_{t \wedge \sigma_n}^{f_n^i} = M_t^i$ this shows that M^i and hence X is a continuous local martingale.

Next, we choose functions $f_n^{i,j} \in C_c^\infty(\mathbb{R}^d)$ such that $f_n^{i,j}(x) = x_i x_j$ if $|x| \leq n$. Following the same arguments as above, one can show that

$$M_t^{i,j} := X_t^i X_t^j - X_0^i X_0^j - \int_0^t \{X_s^i b_j(s, X_s) + X_s^j b_i(s, X_s) + a_{i,j}(s, X_s)\} ds$$

is a continuous local martingale. Using integration by parts and (6.13) we obtain that

$$\begin{aligned} M_t^{i,j} &= \int_0^t X^i dX_s^j + \int_0^t X^j dX_s^i + \langle X^i, X^j \rangle_t \\ &\quad - \int_0^t \{X_s^i b_j(s, X_s) + X_s^j b_i(s, X_s) + a_{i,j}(s, X_s)\} ds \\ &= \int_0^t X^i dM_s^j + \int_0^t X^j dM_s^i + \langle M^i, M^j \rangle_t - \int_0^t a_{i,j}(s, X_s) ds. \end{aligned}$$

As a result, $\langle M^i, M^j \rangle_t - \int_0^t a_{i,j}(s, X_s) ds$ is a continuous local martingale of bounded variation. Since it starts at 0 it is almost surely equal to zero, i.e.,

$$\langle M^i, M^j \rangle_t = \int_0^t a_{i,j}(s, X_s) ds.$$

Hence we have shown that the processes M^i are continuous local martingales starting at 0 such that their covariation is given as above. The assertion therefore follows from the preceding theorem. $\square$

Remark 6.2.15 *The previous theorem established a one-to-one correspondence between weak solution to SDEs and solutions to martingale problems up to the non-unique choice of the matrix σ when a is given. We notice that the law of a weak solution depends on σ only through $\sigma \sigma^t$.*

6.2.3 Existence of solutions

We have already stated an existence result for the solution to a local martingale problem in case $a = I_m$ using Girsanov's theorem. In this section we prove a similar statement for more general diffusion matrices. The proof uses Prohorov's tightness criteria (cf. Theorem 2.3.23).

Theorem 6.2.16 *If $a : \mathbb{R}^d \rightarrow \mathcal{S}^d$ and $b : \mathbb{R}^d \rightarrow \mathbb{R}^d$ are bounded and continuous, then $\text{LMP}(a, b)$ has a solution.*

PROOF: We give an outline of the proof. First, we are going to approximate a and b by smooth functions a_n and b_n such that we know that $\text{LMP}(a_n, b_n)$ has a solution $\mathbb{P}_n$ (a probability measure in C^d) and then show that the sequence $\{\mathbb{P}_n\}$ is tight. Any weak limit turns out to solve $\text{LMP}(a, b)$.

Specifically, let $b_n \in C^\infty(\mathbb{R}^d, \mathbb{R}^d)$ be such that $b_n \rightarrow b$ uniformly. In order to approximate a we first decompose a as $a = \sigma \sigma^t$, then choose $\sigma_n \in C^\infty(\mathbb{R}^d, \mathbb{R}^{d \times d})$ such that $\sigma_n \rightarrow \sigma$ uniformly and put $a_n := \sigma_n \sigma_n^t$. Without loss of generality all these are uniformly bounded by 1.

Let $\sigma_{n,k}$ be the k -th column of σ_n . Let X^n denote the unique strong solution to the SDE with the locally Lipschitz continuous coefficients $b_n, \sigma_{n,1}, \dots, \sigma_{n,d}$. In particular, the equation has a weak solution and LMP(a_n, b_n) has a solution $\mathbb{P}_n$, namely the law of X^n .

We now verify that the sequence $\{P_n\}$ is tight. In fact, the first condition in Exercise 22 is trivially satisfied because $X^n(0) = x$ a.s. for all $n \in \mathbb{N}$. In order to verify the second condition we apply Burkholder's inequality (multi-dimensional version): for any $p \geq 2$,

$$\begin{aligned} \mathbb{E}|X_t^n - X_s^n|^p &= \mathbb{E} \left[\left| \int_s^t b_n(X_u^n) du + \sum_{k=1}^d \int_s^t \sigma_{n,k}(X_u^n) dB_u^k \right|^p \right] \\ &\leq (d+1)^{p-1} \left(\mathbb{E} \left[\left| \int_s^t b_n(X_u^n) du \right|^p \right] + \sum_{k=1}^d \mathbb{E} \left[\left| \int_s^t \sigma_{n,k}(X_u^n) dB_u^k \right|^p \right] \right) \\ &\leq (d+1)^{p-1} \left((t-s)^p + C_p d^{1+p/2} (t-s)^{p/2} \right). \end{aligned}$$

Choosing $p > 2$ yields tightness. By Prohorov's theorem, the sequence $\{\mathbb{P}_n\}$ has a weak accumulation point $\mathbb{P}$. W.l.o.g. we may assume that $\mathbb{P}_n \rightarrow \mathbb{P}$ weakly.

Let X be the canonical process on $(C^d, \mathcal{B}^d, \mathbb{P})$. We claim that X solves LMP(a, b). Since a, b are bounded this is the same as claiming that X solves MP(a, b). Since the martingale property needs to hold w.r.t. the filtration generated by X , by the monotone class theorem we need to show that for any $0 \leq s < t < \infty$, any $f \in C_c^\infty(\mathbb{R}^d)$ and any bounded continuous function $g : C([0, s], \mathbb{R}^d) \rightarrow \mathbb{R}$,

$$\mathbb{E} \left[\left(f(X_t) - f(X_0) - \int_s^t \mathcal{A}_r f(X_r) dt \right) g(X_u, u \in [0, s]) \right] = 0.$$

Let $\pi_t : C^d \rightarrow C^d$ be defined by $(\pi_t(X))(s) = X_s$, $s \in [0, t]$. Then for any fixed $f \in C_c^\infty(\mathbb{R}^d)$ we can write the last equality in the form

$$\mathbb{E} [\phi_f(\pi_t \circ X)] = 0$$

for a suitable bounded and continuous (this is important) functional

$$\phi_f : C([0, t], \mathbb{R}^d) \rightarrow \mathbb{R}.$$

If ϕ_f^n is defined like ϕ_f with $\mathcal{A}_t$ replaced by $\mathcal{A}_t^n$, then the corresponding equation

$$\mathbb{E}_n [\phi_f^n(\pi_t \circ X)] = 0$$

has a solution. Since $\phi_n \rightarrow \phi$ uniformly the weak convergence $\mathbb{P}_n \rightarrow \mathbb{P}$ yields the desired result.⁵ $\square$

😊 — Done!! — 😊

⁵Let $\{f_n\}$ be a sequence of uniformly bounded continuous functions s.t. $f_n \rightarrow f$ uniformly. If $\mu_n \rightarrow \mu$ weakly, then $\int f_n d\mu_n \rightarrow \int f d\mu$.